# Munchausen numbers

What is known about numbers equal to the sum
of their digits raised to themselves

$$3435 = 3^3 + 4^4 + 3^3 + 5^5$$

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An exposition of the library in `docs/library`,
written with Claude Code

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# Preface

In base 10 the number 3435 has a curious property:

$$3435 = 3^3 + 4^4 + 3^3 + 5^5.$$

A number that equals the sum of its digits, each raised to its own power, is a *Munchausen number*. This document collects what is known about them in every base, and in some number systems that are not bases at all.

The material comes from a research library kept in the same repository, in `docs/library`. The library grew in rounds of questions and answers, so its 28 chapters follow the order in which things were found, and later chapters correct earlier ones. This document sorts the results by theme and states each one in its current form. It is written for a mathematics undergraduate who enjoys number theory and is willing to meet a few tools from beyond a first course; the appendices introduce those tools.

## How to read the boxes

Every claim sits in a box with a coloured badge that says what kind of claim it is.

| badge | meaning |
|---|---|
| PROVEN | the library has a proof; this document gives the proof or a sketch when it helps |
| CITED | a published theorem, used without proof, with a reference |
| CONDITIONAL | a proof that assumes a conjecture, such as abc |
| COMPUTATION | an exhaustive computation; its range is part of the statement |
| HEURISTIC | a prediction of a probabilistic model, with the data that test it |
| CONJECTURE | a conjecture, with its evidence |
| OPEN | an open problem |

A badge such as L8.3, or a pointer L8.3 in the text, refers to section 8.3 of the library, in the file `docs/library/8-barriers.md`. There the reader finds the full proof, the script that made a computation, and its data. A pointer such as Q64 refers to question 64 of `docs/library/questions.md`.

The difference between the badges matters in this subject more than in most. Many statements about Munchausen numbers can be checked by computer up to enormous bounds and are still not theorems. One of the stories of the library is a conjecture that looked safe: that under $0^0 = 1$ only finitely many bases have a solution with three digits. Every base up to $10^{51}$ had been searched, and a heuristic put the chance of a further solution below $10^{-70}$. Then an algebraic family turned up with a 3-digit solution in a base near $10^{108942}$ (Theorem 2.21). The heuristic was right about random coincidences and blind to identities. Chapter 2 tells this story.

## What is known, in brief

- A solution in base $b$ has at most $b$ digits, so every base has finitely many (Proposition 1.3). The search over all digit multisets is exact and has found every solution in bases 2 to 41 and 43 (Section 1.8, Appendix K).
- Some solutions come from identities between powers: twins, self-paying digits, swaps, the two-digit families, the binomial families and a second family of power identities (Chapter 2).

Under either convention every length $k \geq 3$ has infinitely many solutions (Proposition 2.20, Corollary 2.22).

- The other solutions behave like coincidences. A probabilistic model predicts about one solution per base and convention, a Poisson count, the length law 8/55/23%, and the doubling of odd bases under $0^0 = 0$ (Chapter 3). It fails in instructive ways in other number systems (Section 7.8, Section 7.9).
- The exact search costs about $1.9^b$. In the model no tree search can do better than about $1.64^b$, and a proof of any lower bound would settle P versus NP (Chapter 4). The digit-sum identity still gives a constant factor, at the nodes that miss only one digit (Section 4.1.6).
- For a fixed number of digits the question becomes one polynomial equation in the base. Local obstructions, Galois groups and a root heuristic describe the short solutions, and almost all bases have none (Chapter 5). For 3 digits, a 2-adic form of Baker's theorem excludes whole families with an even large digit (Section 5.7).
- The 3-digit solution $(1, 7, 1)$ in base 904 leads to the equation $b(b + x) = x^x + 1$ and its relatives, which need quadratic fields, linear forms in logarithms and the abc conjecture (Chapter 6).
- Negative, complex, rational, Pisot, factorial and function-field bases show which laws are universal (Chapter 7).
- Factorials behave differently from $x^x$. Since $x!$ is divisible by every prime up to $x$, congruences prove that 3-digit factorions with one large digit are finite, except for a case of Brocard's problem $n! + 1 = m^2$ (Section 7.7). Since $x!$ is almost never a power, factorials have no identity families (Section 2.7.1). Under abc, every 3-digit solution of $x^x$ with one dominant digit is a family member (Theorem 5.54).
- In the model the expected count is the trace of a stochastic matrix, $1 + C/b$ with $C = 1.1046$, and the parity factors come from one character modulo $b - 1$ (Section 3.2.2, Section 3.3.1). The largest remaining deviation, 12% under $0^0 = 1$, is not a new kind of correction. Taken together, the P values of the library show selection, and the tests named in advance agree with the model (Section 3.6.3).
- The root heuristic for short strings and the per-base model for long ones are one heuristic, averaged over the residue of the base in one case and not in the other (Section 5.5.4). One ideal gives the identity modulus of every number system (Section 7.1.1). And every base $b \geq 27$ provably has a near-solution whose sum agrees with it in about $\sqrt{b/e}$ top digits (Section 3.5.5).
- Some of the questions connect to other fields: anti-concentration in probability, the map $x \mapsto x^x$ modulo a prime, and sums of two squares (Section 8.3).
- Chapter 8 lists the open problems.

## A living document

New results keep arriving in the library. Each chapter here is a separate file, and every statement carries its own status and pointer, so a new result goes into the chapter of its theme without changes elsewhere. The file `docs/exposition/README.md` explains how. The index of results (Appendix N) builds itself from the boxes, and results that are formalized in Lean are marked automatically (Appendix M).

The library and this exposition were written with Claude Code, an AI coding agent, under the direction of the author. The computations were run on a laptop with the programs described in Appendix L.

Chapter 1

# Definitions and first facts

This chapter fixes the notation and proves the facts that every later chapter uses. Most of them are elementary. Two of them shape everything that follows. A solution in base $b$ has at most $b$ digits, so each base has finitely many solutions. And a fixed digit string is a solution in at most one base, so short solutions can be found for all bases at once.

## 1.1 The problem

Fix a base $b \geq 2$. A number $n$ with $k$ digits has the base-$b$ representation

$$n = s_{k-1}b^{k-1} + ... + s_1 b + s_0, \quad 0 \leq s_i \leq b-1, \quad s_{k-1} \neq 0.$$

Position $i$ holds the digit $s_i$, and position 0 is the last digit. We write the digit string most significant digit first, as $(s_{k-1}, ..., s_1, s_0)$, and we write each digit in decimal. So $(3, 4, 3, 5)$ is the number 3435 in base 10, and $(3, 3)$ in base 17 is $3 \cdot 17 + 3 = 54$.

**Definition 1.1** (Munchausen number).
Let $m \in \{0, 1\}$ be the value given to $0^0$. The *weight* of a digit is

$$w(x) = x^x \text{ for } x \geq 1, \quad w(0) = m.$$

The *Munchausen sum* of $n$ is $F(n) = \sum_i w(s_i)$, the sum of the weights of its base-$b$ digits. A *Munchausen number* in base $b$ is a number with $F(n) = n$. The numbers 1 (in both conventions) and 0 (under $0^0 = 0$) are the *trivial* solutions. In this document a *solution* means a nontrivial one, so it has at least two digits.

Three examples show the flavour.

- Base 10: $3435 = 3^3 + 4^4 + 3^3 + 5^5 = 27 + 256 + 27 + 3125$. It is the only solution in base 10 under $0^0 = 1$. Under $0^0 = 0$ base 10 has one more, $438579088 = 4^4 + 3^3 + 8^8 + 5^5 + 7^7 + 9^9 + 0 + 8^8 + 8^8$.
- Base 17: $54 = (3, 3)$ and $3^3 + 3^3 = 54$.
- Base 3: $(1, 2) = 5 = 1^1 + 2^2$ and $(2, 2) = 8 = 2^2 + 2^2$.

We write $F_0$ and $F_1$ for the sum under $0^0 = 0$ and $0^0 = 1$, and $N_m(b)$ for the number of solutions in base $b$ under the convention $m$. $N_{m,k}(b)$ counts those with exactly $k$ digits. For a digit string, $c_x$ is the number of times the digit $x$ occurs; the vector $c = (c_0, ..., c_{b-1})$ is the *digit multiset*. The sum $F$ depends only on the multiset, not on the order of the digits. This simple fact is the reason why a search can run over multisets instead of numbers (Chapter 4).

**Remark 1.2**.
The value of $0^0$ has no agreed meaning here. Throwing away every number with a digit 0 discards too much, and neither convention is more natural than the other. So the library keeps both, and so does this document. A number without the digit 0 is a solution under both conventions or

under neither (Proposition 1.8). The two conventions behave differently in interesting ways: $0^0 = 0$ admits large algebraic families, and $0^0 = 1$ is much more rigid.

## 1.2 How long a solution can be

A number with $k$ digits is at least $b^{k-1}$, while its Munchausen sum is at most $k(b-1)^{b-1}$. The first grows exponentially in $k$, the second linearly. So long numbers cannot be solutions.

**Proposition 1.3** (Length bound). PROVEN L1.2 LEAN
A solution in base $b$ has at most $b$ digits. In particular every base has finitely many solutions, and a solution is at most $b(b-1)^{b-1}$.
Lean: `length_digits_le`

*Proof.* Let $n$ have $k$ digits. Then $n \geq b^{k-1}$ and $F(n) \leq k(b-1)^{b-1}$. For $k = b+1$,

$$\frac{(b+1)(b-1)^{b-1}}{b^b} = \left(1+\frac{1}{b}\right)\left(1-\frac{1}{b}\right)^{b-1} = \left(1-\frac{1}{b^2}\right)\left(1-\frac{1}{b}\right)^{b-2} < 1.$$

So $F(n) < b^b \leq n$. Going from $k$ to $k+1$ multiplies the bound $k(b-1)^{b-1}$ by $(k+1)/k < 2 \leq b$ and the lower bound $b^{k-1}$ by $b$. So the inequality $F(n) < n$ persists for every $k \geq b+1$. □

**Example 1.4**.
In base 2 the only digits are 0 and 1, both of weight at most 1, so $F(n) \leq k$ while $n \geq 2^{k-1}$. The only solution is $(1,0) = 2$ under $0^0 = 1$. In base 10 a solution has at most 10 digits and is below $10 \cdot 9^9 \approx 3.9 \cdot 10^9$. In base 100 it has at most 100 digits, and the search space is about $4^{100}$ digit multisets. This is why base 100 is still out of reach (Section 4.9).

Van Berkel proved the weaker bound $n < 2b^b$ [1], and Kureš proved the bound of Proposition 1.3 [2]. The bound is sharp in the sense that solutions with exactly $b$ digits exist: base 15 and base 16 each have one under $0^0 = 0$ (Section 1.7).

For short solutions a second bound is more useful. A single digit $x$ already contributes $x^x$ to $F(n) = n$.

**Proposition 1.5** (Digit bound). PROVEN L1.2 LEAN
A $k$-digit solution in base $b$ uses only digits $x$ with $x^x < b^k$.
Lean: `pow_self_lt_of_mem_digits`

Write $X_k(b)$ for the largest such digit. Taking logarithms in $x \ln x \approx k \ln b$ gives

$$X_k(b) \sim \frac{k \ln b}{\ln(k \ln b)} \quad (b \to \infty, k \text{ fixed}).$$

So a solution with a fixed number of digits uses only small digits, of the order of $\ln b / \ln \ln b$. This is what makes short solutions cheap to find, and it is behind every finiteness argument for short solutions in Chapter 5.

## 1.3 One base per digit string

Fix a digit string $s = (s_{k-1}, \ldots, s_0)$ instead of a number. Its value in base $r$ is the *value polynomial*

$$P_s(r) = s_{k-1} r^{k-1} + \ldots + s_1 r + s_0,$$

and its Munchausen sum $F(s) = \sum_i w(s_i)$ does not depend on the base at all.

**Definition 1.6** (Munchausen polynomial).
The *Munchausen polynomial* of a digit string $s$ is

$$Q_s(r) = P_s(r) - F(s).$$

It has integer coefficients and degree $k-1$. The string $s$ is a solution in base $b$ exactly when $Q_s(b) = 0$ and $b$ exceeds every digit of $s$.

**Proposition 1.7** (One base per string). PROVEN L1.4 LEAN
A digit string with at least two digits is a solution in at most one base.
Lean: `IsSolution.base_unique`

*Proof.* The leading digit is at least 1, so $P_s(r)$ is strictly increasing for $r > 0$. The constant $F(s)$ is fixed. So $P_s(r) = F(s)$ has at most one positive root. □

This turns the search around. Instead of fixing the base and looking for digit strings, one can fix the digits and solve for the base. By Proposition 1.5 the digits of a $k$-digit solution are at most $X_k(b)$, so enumerating all strings with digits up to $M$ finds every $k$-digit solution in every base $b$ with $b^k \le (M+1)^{M+1}$. For three digits and $M = 80$ this covers all bases up to $10^{51.5}$ with half a million strings. Chapter 5 treats $Q_s$ as an algebraic object: its roots modulo primes, its Galois group, and its roots that are not ordinary bases.

## 1.4 The two conventions for $0^0$

**Proposition 1.8** (The two conventions). PROVEN L1.5 LEAN
$F_1(n) = F_0(n) + z(n)$, where $z(n)$ is the number of zero digits of $n$. So a number without the digit 0 is a solution under both conventions or under neither, and a number with a zero is a solution under at most one.
Lean: `F_one_eq`

The solutions without a zero are called *shared*. In bases 10 to 41, 35 of the 56 solutions under $0^0 = 1$ are shared, and this makes the two counts $N_0(b)$ and $N_1(b)$ correlated (Section 3.7).

## 1.5 Balance

Rewrite $F(n) = n$ as a sum over positions:

$$\sum_i (s_i b^i - w(s_i)) = 0.$$

The term $\beta(x, i) = xb^i - w(x)$ is the *balance* of the digit $x$ at position $i$. A number is a solution exactly when the balances of its placements add up to zero. Most placements have a large balance of either sign. A few have balance zero on their own.

Table 1.1. Placements with balance zero (L1.3).

| placement | why its balance is zero |
|---|---|
| a 0 anywhere, under $0^0 = 0$ | $0 \cdot b^i - 0 = 0$ |
| a 1 at position 0 | $1 - 1^1 = 0$ |
| a digit $a$ at position $i \ge 1$ with $a^{a-1} = b^i$ | $a \cdot b^i = a^a$ |

A digit of the third kind is called *self-paying*: it adds the same amount to $n$ and to $F(n)$. Chapter 2 builds solutions out of these placements. The point of view also explains why $0^0 = 0$ is the richer convention. Under $0^0 = 0$ zeros are free, so zero-balance placements can be combined at will. Under $0^0 = 1$ each zero costs 1, and something else in the number must pay for it.

## 1.6 The digit-sum identity and other congruences

Every integer is congruent to its digit sum modulo $b-1$, because $b \equiv 1 \pmod{b-1}$. For a solution, $n = F(n)$, and the digit sum of $n$ is $\sum_x x c_x$. This gives a condition on the multiset alone.

**Proposition 1.9** (Digit-sum identity). PROVEN L1.6 LEAN

Every solution satisfies

$$\sum_x c_x(w(x) - x) \equiv 0 \pmod{b-1},$$

that is, the sum of $w(x) - x$ over its digits is divisible by $b-1$.

Lean: `IsSolution.Fs_modEq_sum`

**Example 1.10**.

For 3435 in base 10 the terms are $3^3 - 3 = 24$, $4^4 - 4 = 252$, 24 again and $5^5 - 5 = 3120$. Their sum is $3420 = 9 \cdot 380$, divisible by $b - 1 = 9$ as it must be. A random multiset meets the identity with probability about $1/(b-1)$, so the identity alone removes most candidates. In odd bases under $0^0 = 0$ it holds about twice as often, as rule (2) below shows.

The parts of this identity modulo the prime powers dividing $b-1$ give simple rules. The basic fact is that $x^x \equiv x \pmod 2$ for every $x \geq 1$, since $x^x$ and $x$ have the same parity.

**Proposition 1.11** (Rules from the identity). PROVEN L1.6 LEAN

Let $n$ be a solution in base $b$.

1. If $b$ is odd and $0^0 = 1$, then $n$ has an even number of zeros.
2. If $b$ is odd and $0^0 = 0$, the identity holds modulo 2 for every multiset.
3. If $b \equiv 1 \pmod 4$ and $0^0 = 0$, the number of digits $x \equiv 2 \pmod 4$ is even.
4. If $b \equiv 1 \pmod 3$ and $0^0 = 0$, the number of digits $x \equiv 2 \pmod 6$ is a multiple of 3.
5. If $b$ is even, the last digit of $n$ is odd exactly when $n$ has an odd number of odd digits; under $0^0 = 1$, count the zeros together with the odd digits.

Lean: `IsSolution.even_count_zero`, `IsSolution.even_countP_mod_four`, `IsSolution.three_dvd_countP_mod_six`, `IsMunchausen.last_digit_parity`

*Proof.* (1) and (2): modulo 2 every term $w(x) - x$ with $x \geq 1$ vanishes, and a zero contributes $m$. So the identity modulo 2 reads $mc_0 \equiv 0$.

(3): modulo 4, $x^x \equiv x$ for odd $x$ (since $x^2 \equiv 1 \pmod 8$ and $x - 1$ is even) and for $x \equiv 0 \pmod 4$. For $x \equiv 2 \pmod 4$, $x^x \equiv 0$ while $x \equiv 2$. So the sum is $2c$ modulo 4, where $c$ counts the digits $\equiv 2 \pmod 4$.

(4): modulo 3, $x^x \equiv x$ unless $x \equiv 2 \pmod 3$ and $x$ is even, that is $x \equiv 2 \pmod 6$; then $x^x \equiv (-1)^x = 1$ and $w(x) - x \equiv -1$. So the sum is $-c$ modulo 3.

(5): in an even base $n \equiv s_0 \pmod 2$, and $F(n) \equiv \sum_i s_i^{s_i} \equiv$ (number of odd digits) $+ mz(n)$. □

The catalog of all known solutions obeys every rule, as it must. Rule (2) is the most important one: under $0^0 = 0$ a random multiset in an odd base meets the identity modulo 2 for free. Chapter 3 shows that this doubles the expected number of solutions in odd bases.

Can a cleverer congruence say more? The identity uses one residue of $n$ that the digit multiset determines. There are no others.

**Proposition 1.12** (The only invariant residue). PROVEN L10.4 LEAN

If $n \bmod M$ depends only on the digit multiset of $n$ for all $n$ with at least two digits, then $M$ divides $b-1$.

Lean: `residue_invariant_iff`

*Proof.* Take a number whose last two digits are $x$ and $y = x - 1$, and swap them. The multiset is unchanged, and the value changes by $(x - y)b + (y - x) = b - 1$. So $M$ divides $b - 1$. □

A related question asks for congruences that the map $n \mapsto F(n)$ itself preserves. Here there is exactly one case.

**Theorem 1.13** (Parity invariant). PROVEN L15.1 LEAN

Let $b \geq 3$ and $q \geq 2$. Then $F(n) \equiv n \pmod q$ for every $n \geq 1$ if and only if $q = 2$, $b$ is odd and $0^0 = 0$.

Lean: `F_modEq_self_iff`

*Proof.* Suppose the congruence holds for all $n$. The number $n = b + 1$ has digits $(1, 1)$, so $2 \equiv b + 1$ and $b \equiv 1 \pmod q$. The number $n = b$ has digits $(1, 0)$, so $1 + m \equiv b \equiv 1$, which forces $m = 0$. The digit 2 exists because $b \geq 3$, and $n = 2$ gives $4 \equiv 2$, so $q = 2$, and then $b$ is odd. Conversely, for odd $b$ every $n$ is congruent modulo 2 to its digit sum, and $x^x \equiv x \pmod 2$ with $w(0) = 0$. So $F(n) \equiv n \pmod 2$. □

So in an odd base under $0^0 = 0$ the even numbers and the odd numbers are two sets that $F$ maps into themselves. This has consequences for the cycles of $n \mapsto F(n)$ (Section 3.10).

## 1.7 Leading digits and solutions with $b$ digits

The largest digits of a number supply almost all of its Munchausen sum, so they decide its length and its leading digit. The leading digit of a $k$-digit solution is $\lfloor F(n)/b^{k-1} \rfloor$. In bases 20 to 38, for all 26 solutions under $0^0 = 1$, the two largest digit values alone give the leading digit.

The largest digit $b - 1$ weighs

$$(b-1)^{b-1} = b^{b-1}\left(1 - \frac{1}{b}\right)^{b-1} \approx \frac{b^{b-1}}{e}.$$

So $c$ copies of $b - 1$ contribute about $(c/e)b^{b-1}$. Since $\left(1 - \frac{1}{b}\right)^{b-1} > \frac{1}{e}$, three copies already exceed $b^{b-1}$. A solution with $b - 1$ digits therefore contains the digit $b - 1$ at most twice. One copy puts its leading digit near $b/e$, and two copies near $2b/e$.

**Proposition 1.14** (Solutions with $b$ digits). PROVEN L1.8

For $b \geq 7$, a $b$-digit solution contains the digit $b - 1$ at least three times.

*Proof.* If $b - 1$ occurs at most twice, then $F(n) \leq 2(b-1)^{b-1} + b(b-2)^{b-2}$. Divided by $b^{b-1}$ this is $2\left(1 - \frac{1}{b}\right)^{b-1} + \left(1 - \frac{2}{b}\right)^{b-2}$. Both terms decrease as $b$ grows, and at $b = 7$ the sum is 0.979. So $F(n) < b^{b-1} \leq n$. □

With three to five copies of $b - 1$ the sum is about $(c/e)b^{b-1}$, between 1.1 and 1.9 times $b^{b-1}$, so the leading digit of a $b$-digit solution is 1. Both known $b$-digit solutions fit this description exactly.

**Computation 1.15** (Solutions with $b$ digits). COMPUTATION L7.1 L8.6

Under $0^0 = 0$, base 15 has the $b$-digit solution $(1, 2, 4, 3, 14, 14, 13, 3, 4, 1, 9, 1, 1, 0, 14)$ and base 16 has $(1, 2, 3, 11, 2, 3, 0, 9, 15, 15, 5, 7, 2, 15, 6, 11)$. Both start with 1 and contain $b - 1$ exactly three times. No other base from 3 to 45 has a $b$-digit solution under $0^0 = 0$, and none of them has one under $0^0 = 1$.

The original conjecture was that no base $b > 2$ has a $b$-digit solution. The two examples refute it under $0^0 = 0$. Under $0^0 = 1$ it remains open. The probabilistic model of Chapter 3 at first expected a $b$-digit solution in about 8% of large bases, so the empty range 3 to 45 looked suspicious. A finer look (Section 3.5) shows that the model overestimates this length at these sizes, because the forced top digits of a $b$-digit sum use up copies of large digits. With that correction, calibrated on bases 10 to 30, the absence there is an ordinary event, with probability about 0.27.

## 1.8 The data at a glance

The exhaustive search of Chapter 4 has found every solution in every base from 2 to 41, and in base 43, under both conventions. Table 1.2 gives the counts. The full lists are in Appendix K.

Table 1.2. Nontrivial solutions per base. $N_0$ and $N_1$ count solutions under $0^0 = 0$ and $0^0 = 1$; "shared" counts the zero-free solutions, which belong to both; "longest" is the largest number of digits under either convention. Computed exhaustively (L4.2).

| base | $N_0$ | $N_1$ | shared | longest | base | $N_0$ | $N_1$ | shared | longest |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 0 | 1 | 0 | 2 | 23 | 1 | 0 | 0 | 22 |
| 3 | 2 | 2 | 2 | 2 | 24 | 1 | 1 | 1 | 23 |
| 4 | 3 | 2 | 2 | 3 | 25 | 5 | 2 | 2 | 24 |
| 5 | 2 | 0 | 0 | 4 | 26 | 1 | 2 | 0 | 25 |
| 6 | 2 | 2 | 2 | 5 | 27 | 0 | 0 | 0 | - |
| 7 | 1 | 1 | 1 | 5 | 28 | 0 | 3 | 0 | 27 |
| 8 | 2 | 0 | 0 | 3 | 29 | 1 | 0 | 0 | 28 |
| 9 | 6 | 3 | 3 | 8 | 30 | 0 | 1 | 0 | 29 |
| 10 | 2 | 1 | 1 | 9 | 31 | 3 | 2 | 1 | 30 |
| 11 | 5 | 2 | 2 | 8 | 32 | 4 | 0 | 0 | 31 |
| 12 | 1 | 1 | 1 | 10 | 33 | 3 | 2 | 1 | 32 |
| 13 | 5 | 4 | 4 | 12 | 34 | 1 | 1 | 1 | 33 |
| 14 | 1 | 4 | 1 | 13 | 35 | 0 | 0 | 0 | - |
| 15 | 7 | 5 | 2 | 15 | 36 | 0 | 1 | 0 | 35 |
| 16 | 4 | 2 | 2 | 16 | 37 | 3 | 3 | 3 | 36 |
| 17 | 4 | 4 | 3 | 16 | 38 | 2 | 3 | 2 | 37 |
| 18 | 2 | 3 | 1 | 17 | 39 | 2 | 2 | 1 | 38 |
| 19 | 0 | 0 | 0 | - | 40 | 3 | 1 | 1 | 39 |
| 20 | 2 | 2 | 2 | 17 | 41 | 1 | 1 | 1 | 40 |
| 21 | 1 | 2 | 1 | 20 | 43 | 1 | 2 | 1 | 42 |
| 22 | 1 | 1 | 1 | 20 | | | | | |

A few features stand out, and later chapters explain them.

- The counts are small and irregular: zero to seven per base. The probabilistic model of Chapter 3 predicts about one solution per base and convention on average, with Poisson fluctuations, and about twice as many in odd bases under $0^0 = 0$.
- The longest solution usually has $b - 1$ digits. That is the model's most likely length.
- Bases 19, 27 and 35 have no solution under either convention. They are the only such bases from 3 to 41; base 43 has solutions, and base 42 has not been searched in full. The model expects roughly one base in eight to be empty (Section 3.8).
- Perfect powers stand out under $0^0 = 0$. Base 25 has five solutions and base 32 has four. They come from self-paying digits (Chapter 2).

Bases beyond 43 are expensive: each base costs about 1.9 times the previous one. Base 44 was searched with an inexact rule that can miss solutions (Section 4.3). Under $0^0 = 1$ it found one solution with 43 digits,

(32, 32, 36, 30, 0, 0, 37, 23, 32, 33, 4, 5, 17, 4, 30, 7, 31, 25, 43, 25, 15, 21, 38, 19, 18, 7, 35, 10, 34, 39, 4, 30, 25, 33, 24, 43, 27, 26, 13, 1, 38, 24, 22),

and none under $0^0 = 0$. A search of bases 39 to 127 restricted to numbers with at most 30 digits adds the short solutions of those bases, such as $(3, 8, 6, 2, 0)$ in base 48 and $(6, 1, 5)$ in base 91.

## 1.9 Earlier work

Munchausen numbers belong to a family of recreational problems about digits. The best known relatives are the *narcissistic numbers*, equal to the sum of their digits raised to the number of digits, such as $153 = 1^3 + 5^3 + 3^3$ [3], and the *factorions*, equal to the sum of the factorials of their digits, such as $145 = 1! + 4! + 5!$ [4, 5]. Both appear again in Chapter 7 as test cases for the probabilistic model.

For Munchausen numbers themselves the published record is short.

- OEIS A166623 [6] lists the solutions under $0^0 = 1$ for bases 2 to 35. The table comes from one search by Karl W. Heuer in 2011, with an unpublished method. The library's search agrees with it in every base. For $0^0 = 0$ no published list beyond about base 17 was found, and no published search of bases 36 to 41 under either convention. (These two points come from the repository's top-level `README.md`.)
- Van Berkel [1] studied 3435 and proved that a solution is below $2b^b$.
- Kureš [2] proved the length bound of Proposition 1.3 and described the two-digit family $(1, c)$ of Section 2.4.
- Knuth's program back-pdi [7] searches for decimal perfect digital invariants. It uses the same leading-digit bounds as the search of Chapter 4, with exact pruning.

Almost everything else in this document comes from the library in `docs/library`, built by a long series of computations and arguments. Each statement carries a pointer such as L8.3 to the section of the library where its proof, its computation or its data can be found.

### 1.9.1 A map of this document

The chapters are organized by theme, not by the order of discovery.

- Chapter 2 collects the solutions that follow from identities: twins, self-paying digits, swaps, the two-digit families, and the binomial families that give infinitely many solutions of every length.
- Chapter 3 presents the probabilistic model. It treats a solution as a coincidence and predicts the counts, the lengths and the parity effect.
- Chapter 4 describes the exhaustive search, its cost, faster searches that accept a small miss rate, and the complexity barriers.
- Chapter 5 treats short solutions through the Munchausen polynomial: local obstructions, coverings, Galois groups, and a heuristic for integer roots.
- Chapter 6 studies the exponential equation $b(b + x) = x^n + c$ behind the three-digit solutions $(1, x, 1)$, such as $(1, 7, 1)$ in base 904.
- Chapter 7 moves the problem to other number systems and other weights, and asks what is universal.
- Chapter 8 lists the open problems.

The appendices give short primers on the tools that go beyond a first course, tables of all known solutions, a description of the software and data, and an index of every result with its status.

Chapter 2

# Constructions and families

Most solutions in large bases look random, and Chapter 3 treats them as coincidences. This chapter is about the ones that do not look random: families that follow from an identity between powers and need no search. They explain why some bases have many solutions, why every length occurs, and why some integers are solutions in many bases at once. They also include the most surprising result of the library, the binomial families of Section 2.6, which give infinitely many solutions of every length under $0^0 = 1$.

All the constructions below were checked with exact integer arithmetic (`analysis/verify_constructions.py` and the scripts named in each library section), and the exhaustive searches find exactly the predicted members in their ranges.

## 2.1 Twins

The simplest construction changes one digit.

**Proposition 2.1** (Twins). PROVEN L2.1 LEAN
Under $0^0 = 0$, a number $n$ whose last digit is 0 is a solution if and only if $n + 1$ is a solution.
Lean: `isSolution_twin`

*Proof.* The number $n + 1$ has the digits of $n$ with the final 0 replaced by a 1. So $F(n + 1) = F(n) + 1$. □

So under $0^0 = 0$ every solution that ends in 0 or 1 has a twin. Bases 2 to 41 contain 8 twin pairs, for example $(4, 0, 0)$ and $(4, 0, 1)$ in base 8, where $4 \cdot 64 = 256 = 4^4$. Under $0^0 = 1$ there are no twins: a final 0 and a final 1 both weigh 1, so $n$ and $n + 1$ have the same sum and cannot both equal it.

## 2.2 Self-paying digits

A digit $a \geq 2$ at a position $k \geq 1$ is *self-paying* if $ab^k = a^a$, that is $a^{a-1} = b^k$. Its balance is zero (Section 1.5): it adds the same amount to $n$ and to $F(n)$. The base 64 is the standard example: $4 \cdot 64 = 4^4$ and $16 \cdot 64^{10} = 2^{64} = 16^{16}$, so the digit 4 pays for itself at position 1 and the digit 16 at position 10.

**Proposition 2.2** (Which digits pay for themselves). PROVEN L2.2 LEAN
Write $b = r^t$ with $r$ not a perfect power. The self-paying digits of base $b$ are the digits $a = r^s$ with $1 \leq s < t$ and $t \mid s(r^s - 1)$, at position $k = s(r^s - 1)/t$. In particular only perfect powers have self-paying digits.
Lean: `selfPaying_of_dvd, SelfPaying.exists_pow`

*Proof.* If $a^{a-1} = b^k$, then $a$ and $b$ are multiplicatively dependent, so both are powers of one integer. Since $r$ is not a perfect power, $b = r^t$ and $a = r^s$ for some $s$. The equation becomes $r^{s(r^s-1)} = r^{tk}$, that is $s(r^s - 1) = tk$, and $a < b$ means $s < t$. □

Every odd square $r^2$ has one: the digit $r$ at position $(r-1)/2$, because $r \cdot r^{r-1} = r^r$. Some perfect powers have none, for example 4, 16, 27, 36 and 100. The next statement says exactly which.

**Proposition 2.3** (Bases with a self-paying digit). PROVEN L15.3 LEAN
Write $b = r^t$ with $r$ not a perfect power. Base $b$ has a self-paying digit if and only if $t$ has a prime factor that does not divide $r$.

Lean: `prop15_8`

*Proof.* Write $t = UV$, where $V$ collects the prime powers of $t$ whose primes divide $r$, and $U$ the rest. Since $r^s - 1$ is prime to $r$, the condition $t \mid s(r^s-1)$ forces $V \mid s$. If $U = 1$, this means $t \mid s$, impossible for $1 \le s < t$. If $U > 1$, take $s = V \cdot \mathrm{ord}_U(r)$, where $\mathrm{ord}_U(r)$ is the multiplicative order of $r$ modulo $U$. Then $U \mid r^s - 1$ and $V \mid s$, so $t \mid s(r^s-1)$, and $s \le V\varphi(U) < UV = t$. □

So every base $r^p$ with $p$ prime and $p$ not dividing $r$ has a self-paying digit, while $4 = 2^2$, $16 = 2^4$, $27 = 3^3$, $36 = 6^2$ and $100 = 10^2$ do not. There are 47 bases up to 4096 with at least one self-paying digit (`analysis/selfpaying.py`). A base can have several, but they are spread out.

**Proposition 2.4** (Self-paying digits are far apart). PROVEN L8.3 LEAN
If $r^s$ and $r^u$ with $s < u$ are self-paying digits of base $b = r^t$, at positions $k_s$ and $k_u$, then $k_u > rk_s \ge 2k_s$. So base $b$ has fewer than $\log_2 b$ self-paying digits, and no two of them sit at adjacent positions.

Lean: `SelfPaying.position_gt, card_selfPaying_lt_log`

*Proof.* By Proposition 2.2, $k_u/k_s = u(r^u-1)/(s(r^s-1))$. With $u \ge s+1$ this is at least $(r^{s+1}-1)/(r^s-1) > r$. The digits are $r^s$ with $s < t = \log_r b \le \log_2 b$. □

Under $0^0 = 0$ self-paying digits combine freely with any solution, because the zeros around them cost nothing.

**Proposition 2.5** (Structure under $0^0 = 0$). PROVEN L2.2 LEAN
Under $0^0 = 0$, setting a self-paying digit to 0 turns a solution into a solution, and so does putting a self-paying digit on a position that holds

0. Every solution splits uniquely into

- a *core*: a solution, or 0, with no self-paying digit and with a last digit other than 1;
- a set of self-paying digits on positions where the core has a 0;
- optionally a final 1, if the core ends in 0.

Every such combination is a solution.

Lean: `isSolution_selfPaying_iff, isMunchausen_iff_core`

*Proof.* Each operation adds or removes one placement of balance zero, so it keeps the total balance at zero. The core is what remains after removing all self-paying digits and a final 1. □

**Corollary 2.6** (Solutions from self-paying digits). PROVEN L2.2 LEAN
If base $b$ has $C$ self-paying digits, then $N_0(b) \ge 2^{C+1} - 2$.

Lean: `corollary_2_4`

*Proof.* Take the core 0, any subset of the self-paying digits and an optional final 1, and leave out the two trivial solutions 0 and 1. □

Under $0^0 = 1$ a zero weighs 1, so removing a self-paying digit leaves a number whose sum is too large by one. Self-paying digits still help when they sit directly above a solution of the right length.

**Proposition 2.7** (A self-paying digit on top). PROVEN L2.2 LEAN
Let $a$ be self-paying at position $k$ in base $b$, and let $n' < b^k$ have $p$ zeros between its leading digit and position $k$. Then $ab^k + n'$ is a solution if and only if $F(n') + pw(0) = n'$.
Lean: `isSolution_selfPaying_cons_iff`

**Example 2.8**.
$(3, 1)$ in base 9 is the self-paying 3 at position 1 above the trivial solution 1. $(5, 1, 3)$ in base 25 is the self-paying 5 at position 2 ($5 \cdot 25^2 = 5^5$) above the 2-digit solution $(1, 3)$ of base 25, since $25 + 3 = 1 + 3^3$. Both are solutions under either convention.

## 2.3 Swaps and chains

Two digits can also pay for each other. Put a digit $a$ at position $k$ and the digit $c = a^a$ at position 0. If $ab^k = c^c$, then the value of each digit equals the weight of the other: $ab^k = w(c)$ and $c = w(a)$. With zeros in between and $0^0 = 0$, the number $(a, 0, ..., 0, a^a)$ is a solution. The condition is

$$b^k = a^{a^{a+1}-1} \quad \text{and} \quad a^a < b.$$

**Example 2.9** (Swaps).
- $a = 2$, $k = 1$: $(2, 4)$ in base 128, since $2 \cdot 128 = 2^8 = 4^4$.
- $a = 3$: $b^k = 3^{80}$, so $(3, 0, ..., 0, 27)$ is a solution in the bases 81, 243, $3^8$, $3^{10}$, $3^{16}$, $3^{20}$, $3^{40}$ and $3^{80}$. In base $3^t$ it has $80/t + 1$ digits.
- Every odd $a$ gives the 3-digit swap $(a, 0, a^a)$ in base $a^{(a^{a+1}-1)/2}$.
- $a = 4$: $(4, 0, ..., 0, 256)$ with 187 digits in base 2048, since $4^{1023} = 2^{2046} = 2048^{186}$.

Swaps combine with self-paying digits under $0^0 = 0$. Base 81 has $(3, 0, ..., 0, 9, 0, 0, 0, 27)$ with 21 digits: the swap of 3 and 27, and the self-paying 9 at position 4 ($9 \cdot 81^4 = 3^{18} = 9^9$).

Under $0^0 = 1$ the zeros must be paid for. Base 64 has an example that uses every mechanism at once:

$$(16, 0, 0, 0, 0, 0, 1, 0, 0, 4, 8).$$

The digits 16 (position 10) and 4 (position 1) pay for themselves. The 1 at position 4 is worth $64^4 = 2^{24} = 8^8$, the weight of the final 8. The final 8 is worth 8, which pays for the weight of the 1 and of the seven zeros. A general construction behind this example follows.

**Theorem 2.10** (Chains and cycles). PROVEN L15.5
Let $b = r^t$. For $0 \le u < t$ put $\varphi(u) = ur^u \bmod t$ and $\pi(u) = \lfloor ur^u/t \rfloor$. Choose $u_0 \ge 1$ and follow $u_{i+1} = \varphi(u_i)$ until $u_j = 0$. Place $c = r^{u_0}$ at position 0 and the digit $r^{u_{i+1}}$ at position $\pi(u_i)$ for $0 \le i < j$. Add any set of cycles $(v_1, ..., v_l)$ of $\varphi$ on $\{1, ..., t-1\}$, with $r^{v_i}$ at position $\pi(v_{i-1})$, indices modulo $l$. If all positions differ and the number has exactly $c - 1$ zeros below its leading digit, it is a solution under $0^0 = 1$.

*Proof.* The digit $r^{\varphi(u)}$ at position $\pi(u)$ has the value $r^{\varphi(u)+t\pi(u)} = r^{ur^u} = w(r^u)$. So each chain digit pays for the weight of its predecessor in the chain, and each cycle digit pays for the weight of the previous digit of its cycle. What is left to pay is the weight 1 of the last chain digit $r^0 = 1$ and the weight $c - 1$ of the zeros. Their sum is $c$, the value of the digit at position 0. The digits are below $b$ because $u < t$. □

Cycles of length 1 are the self-paying digits, and cycles of length 2 are the swaps. In base $64 = 2^6$ the chain $3 \to 0$ puts $c = 8$ at position 0 and the digit 1 at position $\pi(3) = \lfloor 24/6 \rfloor = 4$; the fixed

points $u = 2$ and $u = 4$ of $\varphi$ give the self-paying 4 and 16. The family $(a, 1)$ in base $a^{a-1}$ is the degenerate case without zeros.

**Computation 2.11** (Chain solutions). COMPUTATION L15.5 L33.5
For all bases $r^t$ with $2 \le r \le 40$, $r$ not a perfect power, and $2 \le t \le 300$, the construction of Theorem 2.10 gives 43 solutions. 42 of them are $(a, 1)$ in base $a^{a-1}$, and the other one is the 11-digit solution of base 64. No base gets more than 2. A later search, for $r \le 40$ with $t \le 20{,}000$ and for $r \le 7$ with $t \le 10^5$, finds no further solution.

Could the construction give long solutions, with a length that grows like $\log b$ or faster (Q99)? Two facts limit it L33.5. First, the count of zeros fixes the length. A solution with a chain of $j$ steps and a set $S$ of self-paying digits has $c + j + |S|$ digits, and $|S|$ is at most about $\log_r b$. So the length is $c + O(\log b)$, where $c = r^{u_0}$ is the last digit, and some position of a chain digit or a self-paying digit must land within $O(\log b)$ of $c$. Second, for the shortest chains this coincidence happens only once.

**Proposition 2.12** (One-step chains). PROVEN L33.5
The solutions of Theorem 2.10 whose chain has one step, $u_0 \to 0$, are finitely many. The only one is the 11-digit solution of base 64.

*Proof sketch.* With one step the digit 1 sits at position $u_0 c/t < c$, so the top position belongs to a self-paying digit $r^v$, and the zero count gives $v(r^v - 1)/t = c + |S|$. Comparing this with $|S| \le v$ and reducing modulo $c$ gives $c \le (u_0 + 1)v$ and $r^v \le 2u_0(u_0 + 1)^2 v^2$. These leave $u_0 \le 7$, $v < 25$ and $t < 1000$, all inside the search above. □

For longer chains each solution needs a position to land in a window of width $O(\log b)$ around $c$, with a chance of about $\log b/(c \ln r)$. Summed over the bases this converges (heuristic). So the library expects no infinite family of long structured solutions under $0^0 = 1$ from this construction.

Swaps also show that the cores of Proposition 2.5 can sit in bases with self-paying digits in infinitely many ways.

**Theorem 2.13** (Swaps are cores in self-paying bases). PROVEN L15.3 LEAN
For every $a \ge 2$ that is not a perfect power,

$$(a^{a+1}, 0, ..., 0, a, a^a) \quad (a + 2 \text{ digits})$$

is a solution in base $a^{a^{a+1}-1}$ under $0^0 = 0$. For $a = 2$ it is $(8, 0, 2, 4)$ in base 128.
Lean: `theorem15_9`

*Proof sketch.* The swap $(a, a^a)$ is a 2-digit solution in $B = a^{a^{a+1}-1}$. The digit $a^{a+1}$ is self-paying at position $a + 1$, because $(a^{a+1})^{a^{a+1}-1} = B^{a+1}$. Position $a + 1 \ge 3$ is free, so Proposition 2.5 applies. □

## 2.4 Two-digit solutions

Two-digit solutions are completely described by a divisibility condition. Write $g(c) = w(c) - c$.

**Proposition 2.14** (Two-digit solutions). PROVEN L2.4 LEAN
The string $(a, c)$ is a solution if and only if $a$ divides $g(c)$ and $b = (a^a + g(c))/a$ exceeds $a$ and $c$. The base is then $b$.
Lean: `isSolution_two_iff`

*Proof.* The equation $ab + c = a^a + w(c)$ solved for $b$. □

Infinite families follow at once.

- $a = 1$: $(1, c)$ in base $c^c - c + 1$, for every $c \geq 2$. For example $(1, 2)$ in base 3 and $(1, 3)$ in base 25. Under $0^0 = 1$ also $(1, 0)$ in base 2.
- $a = c$: $(c, c)$ in base $2c^{c-1} - 1$, since $c \mid c^c - c$. For example $(2, 2)$ in base 3, $(3, 3)$ in base 17 and $(4, 4)$ in base 127.
- $c = 1$, and under $0^0 = 0$ also $c = 0$: $(a, 1)$ and $(a, 0)$ in base $a^{a-1}$, for every $a \geq 3$. Here $a$ is self-paying.
- Any $a$ with $c \equiv 0$ or $1 \pmod a$, because then $a \mid c^c - c$. For example $(2, c)$ in base $(c^c - c + 4)/2$ for every $c \geq 2$.

**Computation 2.15** (Two-digit solutions with small digits). COMPUTATION L2.4

With digits up to 60 there are 1,347 two-digit solutions under $0^0 = 0$ and 1,290 under $0^0 = 1$. Under $0^0 = 0$, 59 have the form $(1, c)$, 59 the form $(c, c)$, 116 end in 0 or 1, and the other 1,113 are pairs with $a \mid c^c - c$. Under $0^0 = 1$ the counts are 60, 59, 58 and 1,113. Under each convention about 35% of the 2-digit strings with digits up to 60 are solutions in some base, about 21% up to 200 and about 12.5% up to 600.

For $c \geq 1$ the condition $a \mid c^c - c$ says that $c$ is a *fixed point* of the map $x \mapsto x^x$ modulo $a$. This map has a literature of its own, mostly for a prime modulus, and it describes the last digits that go with a given leading digit.

**Proposition 2.16** (Density of the last digits). PROVEN L30.2

Let $p$ be prime. The integers $c \geq 1$ with $c^c \equiv c \pmod p$ are periodic modulo $p(p-1)$, with density

$$f(p) = \frac{1}{p}\left(1 + \sum_{d \mid p-1} \frac{\varphi(d)}{d}\right).$$

*Proof.* By Fermat's little theorem, $c^c \bmod p$ depends only on $u = c \bmod p$ and $v = c \bmod (p-1)$. Since $p$ and $p-1$ are coprime, $(u, v)$ runs over all pairs as $c$ runs over one period. If $u = 0$, both sides are 0: that gives $p - 1$ pairs. If $u \neq 0$, the condition is $u^{v-1} \equiv 1$, that is, the order of $u$ divides $v - 1$. This holds for $(p-1)/\operatorname{ord}(u)$ values of $v$, and there are $\varphi(d)$ units of order $d$. So the count is $(p-1)\left(1 + \sum_{d \mid p-1} \varphi(d)/d\right)$. Divide by $p(p-1)$. □

For $p = 3$ the density is $5/6$. On average over primes, $f(p)$ is about $\ln p / p$. For a digit pair $(p, c)$ with $2 \leq c \leq p - 1$ the base $(p^p + c^c - c)/p$ is larger than both digits. So such a pair is a solution exactly when $c$ is a fixed point modulo $p$, and the theorems on fixed points apply directly.

**Corollary 2.17** (Small last digits). CITED L30.2

Call $(p, c)$, with $p$ prime and $2 \leq c \leq p - 1$, a small pair. Under both conventions:

1. For almost all primes $p$ some small pair $(p, c)$ is a 2-digit solution.
2. If $p \equiv 3 \pmod 8$ and $(p-1)/2$ is prime, no small pair $(p, c)$ is a solution.
3. At most $p^{1/3+o(1)}$ small pairs $(p, c)$ are solutions.

These are theorems of Kurlberg, Luca and Shparlinski on the fixed points of $x \mapsto x^x$ modulo $p$ [8], and Felix and Kurlberg sharpen the count of the exceptional primes [9]. Part 2 has a short proof. A fixed point has $c^{c-1} \equiv 1$, so the order of $c$ divides both $p - 1 = 2q$ and $c - 1$. Order 1 gives $c = 1$. Order 2 gives $c = p - 1$, but then $c - 1$ is odd. Order $2q$ needs $c - 1 \geq p - 1$. Order $q$ forces $c = q + 1 = (p+1)/2$, which is $1/2$ modulo $p$. Then $c^{c-1} \equiv 2^{-q}$, which is the Legendre symbol $(2/p) = -1$, not 1.

**Computation 2.18** (Fixed points and the census). COMPUTATION L30.2

Of the 2,261 odd primes below 20,000, 146 have no small pair that is a solution, and 90 of these are of the form in part 2. (The list of such primes printed by Kurlberg, Luca and Shparlinski starts with 7 by mistake: $4^4 \equiv 4 \pmod 7$.) Under $0^0 = 1$, the share of the pairs $1 \le a \le Y$, $0 \le c \le Y$ that are solutions, against the mean density of the fixed points modulo $a \le Y$:

| $Y$ | 20 | 60 | 200 |
|---|---|---|---|
| census share | 0.524 | 0.353 | 0.212 |
| mean density | 0.576 | 0.374 | 0.217 |

So the census of Computation 2.15 is a mean-value problem for the fixed points of $x \mapsto x^x$. Its asymptotic is open (Q121).

So two-digit solutions are plentiful: the condition $a \mid g(c)$ holds with probability about $1/a$, and $\sum 1/a$ diverges. But each base has very few. The next theorem, proven in L15.2, settles the question how many.

**Theorem 2.19** (Two 2-digit solutions). PROVEN L15.2 LEAN

Under $0^0 = 1$, base 3 is the only base with two 2-digit solutions, $(1, 2)$ and $(2, 2)$. Under $0^0 = 0$ the bases with two 2-digit solutions are base 3 and the bases $a^{a-1}$ with $a \ge 3$, which have the twins $(a, 0)$ and $(a, 1)$. No base has three.

Lean: `theorem15_7 (partial: the case 0^0 = 1)`

*Proof sketch.* Let $(a, c)$ and $(A, C)$ be solutions in base $b$: $ab = a^a + g(c)$ and $Ab = A^A + g(C)$. If $a = A$ then $g(c) = g(C)$, and $g$ is injective except for $g(0) = g(1) = 0$ under $0^0 = 0$, which gives the twins. So let $a < A$. Then $b \ge A^{A-1}$, and with $b = A^{A-1} + u$ the equations become $g(c) = aA^{A-1} - a^a + au$ and $g(C) = Au$. Four elementary steps follow. The first equation makes $g(c)$ larger than $g(A-1)$, so $c \ge A$. The fast growth $g(c+1) \ge 2(c+1)g(c)$ rules out $C > c$. Comparing the two equations bounds $g(c)$ below $g(A+1)$, so $c = A$. Finally an inequality of the form $A^A(A-2a) + A(a^a + a - A) \le 0$ leaves only $a = 1$, $A = 2$, and then $c = C = 2$ and $b = 3$. □

An exact computation with $a \le 200$ and $c \le 3000$, which covers bases up to about $10^{10400}$, confirms the statement (`analysis/dyn_two_digit.py`).

## 2.5 Every length under $0^0 = 0$

Self-paying digits give solutions of every length under $0^0 = 0$.

**Proposition 2.20** (Every length under $0^0 = 0$). PROVEN L2.5 LEAN

Under $0^0 = 0$, every length $k \ge 2$ has infinitely many bases with a $k$-digit solution.

Lean: `isSolution_prop27`

*Proof.* Take $t \ge 2$, $r = t(k-1) + 1$ and $b = r^t$. Then $r^{r-1} = r^{t(k-1)} = b^{k-1}$, so $r$ is self-paying at position $k-1$, and $r < b$. So $(r, 0, \ldots, 0)$ and $(r, 0, \ldots, 0, 1)$ are $k$-digit solutions. □

With $t = 2$ this puts a $k$-digit solution in base $(2k-1)^2$. In the bases up to 41 the smallest base with a $k$-digit solution under $0^0 = 0$ is usually $k+1$ or $k+2$; the exceptions $k = 11$ (base 64) and $k = 21$ (base 81) are perfect powers (L9.9).

## 2.6 Binomial families: every length under $0^0 = 1$

Under $0^0 = 1$ zeros cost 1 each, and the constructions above give nothing beyond two digits. For a long time the library conjectured that each length $k \ge 3$ has only finitely many solutions under

$0^0 = 1$. The evidence was strong. Every 3-digit solution in every base up to $10^{51.5}$ is known, and there are six. A heuristic that treats the base as the root of a random polynomial (Section 5.5) gives a finite expected number for every $k \geq 3$. The conjecture is false. The counterexamples come from one identity, the binomial theorem.

**Theorem 2.21** (Binomial families). PROVEN L22.3 LEAN
Let $j \geq 2$, $c \geq 1$, $e \geq 1$, and put

$$t_i = c^{j-i}\binom{j}{i}e^i \quad (0 \leq i \leq j-1), \quad x = e^j + \sum_{i=0}^{j-1} t_i^{t_i}.$$

Suppose that $x^x = y^j$ for an integer $y$, that $c$ divides $y - e$, and that $b = (y-e)/c$ exceeds $x$ and every $t_i$. Then $(t_0, \ldots, t_{j-1}, x)$ is a $(j+1)$-digit solution in base $b$ under both conventions.
Lean: `isSolution_binomial`

*Proof.* The $t_i$ are the coefficients of $(cr + e)^j$ without its constant term. So the value of the string is

$$P(b) = (cb + e)^j - e^j + x = y^j - e^j + x = x^x - e^j + x.$$

All digits are positive, so under both conventions $F = \sum_i t_i^{t_i} + x^x$. The definition of $x$ says exactly that $P(b) = F$. □

**Corollary 2.22** (Every length under $0^0 = 1$). PROVEN L22.3 LEAN
For every $k \geq 3$ there are infinitely many $k$-digit solutions under $0^0 = 1$, and under $0^0 = 0$.
Lean: `isSolution_corollary_22_7, infinite_solutions_of_length`

*Proof.* Let $j = k - 1$, $c = j$ and $e = ju$ with $u \geq 1$. Every $t_i$ is a positive multiple of $j$, since it contains the factor $c^{j-i}$ with $j - i \geq 1$. So $t_i^{t_i} \equiv 0$ and $e^j \equiv 0 \pmod{j}$, hence $j \mid x$. Then $x^x = y^j$ with $y = x^{x/j}$, and $j \mid y$ because $j \mid x$, so $c = j$ divides $y - e$. Finally $x \geq t_0^{t_0} = (j^j)^{j^j}$, so $y \geq x^2$ and $b = (y-e)/j > x$. Different $u$ give different digit strings. □

A second infinite family, with zeros inside, gives smaller solutions of lengths 4 to 6; it is described in Section 2.7.

For $j = 2$ the condition on $x$ is that $x$ is even or a square. With $c = 1$ this holds for every odd $e$, and $e = 1$ gives the smallest member, $(1, 2, 6)$ in base $6^3 - 1 = 215$: here $(b+1)^2 = 216^2 = 6^6$. This solution had been known for years as one of the six 3-digit solutions. Only the family behind it was new. The next member of the family is short enough to check by hand. It is also the next 3-digit member of any power identity (Theorem 2.36).

**Example 2.23** (The second member for $k = 3$).
Take $j = 2$, $c = 1$, $e = 3$. Then $(r+3)^2 = r^2 + 6r + 9$, so the digits are 1 and 6, and $x = 9 + 1^1 + 6^6 = 46666$. Since $x$ is even, $y = 46666^{23333}$ and $b = 46666^{23333} - 3$, about $10^{108942}$. In this base

$$P = b^2 + 6b + 46666 = (b+3)^2 - 9 + 46666 = 46666^{46666} + 46657,$$

$$F = 1 + 6^6 + 46666^{46666} = 46666^{46666} + 46657.$$

Table 2.1. The smallest known member of the binomial family for each length, after $(1, 2, 6)$ in base 215 (L22.3). The search covered $c \leq 6$ and $e \leq 12$. For 4 to 6 digits the family of Theorem 2.30 gives smaller solutions (Table 2.2).

| $k$ | $(c, e)$ | digits | $x$ | base $b$ |
|---|---|---|---|---|
| 3 | $(1, 3)$ | $(1, 6, x)$ | 46,666 | $46666^{23333} - 3 \approx 10^{108942}$ |
| 4 | $(1, 2)$ | $(1, 6, 12, x)$ | 8,916,100,494,921 | $x^{x/3} - 2 \approx 10^{3.85 \cdot 10^{13}}$ |
| 5 | $(2, 2)$ | $(16, 64, 96, 64, x)$ | $\approx 10^{190.3}$ | $\approx 10^{10^{192.0}}$ |
| 6 | $(1, 4)$ | $(1, 20, 160, 640, 1280, x)$ | $\approx 10^{3977}$ | $\approx 10^{10^{3980}}$ |
| 7 | $(1, 1)$ | $(1, 6, 15, 20, 15, 6, x)$ | $\approx 1.05 \cdot 10^{26}$ | $\approx 10^{10^{26.7}}$ |

| $k$ | $(c, e)$ | digits | $x$ | base $b$ |
|---|---|---|---|---|
| 8 | $(3, 1)$ | $(2187, 5103, 5103, 2835, 945, 189, 21, x)$ | $\approx 10^{18921}$ | $\approx 10^{10^{18924}}$ |
| 9 | $(2, 2)$ | $(256, 2048, 7168, ..., 2048, x)$ | $\approx 10^{76220}$ | $\approx 10^{10^{76224}}$ |

Why did the heuristic miss these? It treats the real root of $Q_s$ as a random number, and asks for the chance that it is an integer. For the binomial strings $Q_s(r) = (cr + e)^j - x^x$ exactly, so the root is an integer whenever $x^x$ is a $j$-th power. An identity has probability one where it holds, and the heuristic gives it weight near zero. The lists of Section 5.1 stay complete in their ranges, but they are not the end of the story: the lengths 9, 18, 21, 22, 24, 26 and 28, which have no solution in any searched base (Section 3.9), do have solutions in astronomically large bases.

For three digits the binomial families are the whole story of a certain equation. Completing the square turns $pb^2 + qb + r = F$ into $(2pb + q)^2 = 4pF + q^2 - 4pr$. When one digit $x$ is large, the dominant term of the right side is $4px^x$. Write the right side as $4px^x + R$.

**Proposition 2.24** (When the remainder vanishes). PROVEN L22.3 LEAN

Let one digit $x$ of a 3-digit string vary and fix the other two, under $0^0 = 1$. Then $R = 0$ only if $x$ is the last digit and $x = p^p + q^q + q^2/(4p)$. If moreover $x$ is even, then $p = c^2$ is a square, $q = 2ce$, and the solutions are exactly the members of Theorem 2.21 with $j = 2$.

Lean: `prop22_8`

**Remark 2.25**.

The library once proved finiteness for 3-digit solutions in the "square class", where $4px^x$ is a perfect square, with the other two digits at most 20 (L8.2, Corollary 8.4). The argument bounded $|R|$ from below and so silently assumed $R \neq 0$. Proposition 2.24 shows that $R = 0$ happens exactly for the binomial families. With the two fixed digits at most 20 there are 9 such strings: $(1, 2, 6)$ and eight more, $(1, 6, x)$, $(1, 10, x)$, $(1, 14, x)$, $(1, 18, x)$, $(4, 8, x)$, $(4, 16, x)$, $(9, 6, x)$ and $(9, 18, x)$, all in bases above $10^{108942}$. With these eight added the corollary holds. The lesson is general: a heuristic, or an inequality, that treats a quantity as generic must check that no identity makes it vanish.

Two near-misses of the same kind appear in the short-string data of L19.2. The strings $(1, 0, 0, x)$ and $(1, 3, 3, x)$ with $x^x$ a cube satisfy $P_s(b) - F(s) = x - 3$ and $x - 56$ in suitable bases, so they are near-solutions with a fixed defect, not solutions.

## 2.7 Power identities and a second family

The binomial families raise a natural question (Q64): which identities can turn a short string into a solution, and where can the large digit sit? The library answers it in L25.1. The answer is short. The large digit is always last, a linear identity gives exactly two families, and every other mechanism needs one more coincidence. Throughout this section $0^0 = 1$.

**Definition 2.26** (Power identity). L25.1

Let $L$ be a nonempty set of positions of a digit string $s$ whose digits are at least 2, the *large* digits, and let $W$ be the sum of their weights. A *power identity* for $(s, L)$ is an equation

$$Q_s(r) = \lambda A(r)^j - W$$

of polynomials in $\mathbb{Q}[r]$, with $j \geq 2$, a rational $\lambda > 0$ and a nonconstant $A$. Equivalently $P_s(r) - E = \lambda A(r)^j$, where $E$ is the sum of the weights of the digits outside $L$.

With such an identity, $s$ is a solution in base $b$ exactly when $W = \lambda A(b)^j$ and $b$ exceeds the digits. The root heuristic treats $Q_s(b) = 0$ as a coincidence. An identity replaces it by the condition $W = \lambda A(b)^j$, which can hold for a structural reason: $x^x$ is a $j$-th power whenever $j$ divides $x$.

**Theorem 2.27** (The large digit is last). PROVEN L25.1 LEAN
In every power identity the last position belongs to $L$.
Lean: `theorem25_2`

*Proof.* Suppose the last digit $s_0$ is not large, so it is one of the digits outside $L$, and $E \geq w(s_0) \geq s_0$. If $E > s_0$, the polynomial $P_s(r) - E$ has a positive leading coefficient, nonnegative middle coefficients and a negative constant term. By Descartes' rule of signs it has exactly one positive root, counted with multiplicity, so that root is simple. A positive root exists, since the polynomial is negative at 0 and positive for large $r$. But every root of $A$ is a root of $\lambda A^j$ of multiplicity at least $j \geq 2$. If $E = s_0$, then $s_0 = 1$ is the only digit outside $L$. Then $A(0) = 0$, so $r^j$ divides $P_s(r) - E$, and the coefficient of $r$ vanishes. But that coefficient is a large digit, at least 2. □

**Corollary 2.28** (One large digit). PROVEN L25.1
Let $s$ have one large digit $x$ with $Q_s(r) = \lambda A(r)^j - x^x$. Then $x$ is the last digit. Writing $s = (t, x)$,

$$rP_t(r) + C = \lambda A(r)^j, \quad x = F(t) + C$$

for an integer $C$, so the digits of $t$ alone fix $j$, $A$, $\lambda$, $C$ and $x$. The string is a solution in base $b$ exactly when $x^x = \lambda A(b)^j$ and $b > x$.

So an identity never puts the large digit in front or in the middle. The simplest case is a linear $A = cr + e$ with $c \geq 1$ and $\gcd(c, e) = 1$. Then $j = k - 1$, the digits are the coefficients $t_i = \lambda c^{j-i}\binom{j}{i}e^i$, and $x = F(t) + \lambda e^j$.

**Theorem 2.29** (Linear identities). PROVEN L25.1
For a linear identity as above:

1. If $\lambda = \mu^j$ is a $j$-th power, the members are those of Theorem 2.21 (rescaled, for $e \geq 1$) and of Theorem 2.30 below (for $e = 0$). When $j$ divides $x$ the conditions are congruences on the parameters, so the members come in infinite families.
2. If $\lambda$ is not a $j$-th power, then $j$ does not divide $x$, and the part of $x$ prime to $\lambda$ is a perfect $m$-th power with $m = j/\gcd(j, x) \geq 2$.
3. If $\lambda$ is a $j$-th power but $j$ does not divide $x$, then $x$ is a perfect $m$-th power, with $m$ as before.

*Proof sketch.* Part 1 is Theorem 2.21 after rescaling $A$. For part 2, if $j$ divides $x$ then $x^x$ is a $j$-th power, and so is $\lambda = x^x/y^j$. For a prime $l$ not dividing $\lambda$, $j$ must divide $v_l(x^x) = xv_l(x)$, and with $g = \gcd(j, x)$ this forces $m = j/g$ to divide $v_l(x)$. Part 3 is the same argument with no prime of $\lambda$. □

Types 2 and 3 need $x$, a number of size about the weight of the largest prefix digit, to be nearly a perfect power. Heuristically that happens with probability about $x^{1/m-1}$, a convergent sum, so these members are coincidences. For 3 digits with $p, q \leq 2000$, 6,143 shapes need $x/\lambda$ to be a square; the expected number of squares is 0.89, and none occurs.

The case $e = 0$ gives nothing for three digits (Proposition 2.24). For more digits it gives a new family. The zeros now weigh 1 each, and the large last digit pays for them.

**Theorem 2.30** (A second infinite family). PROVEN L25.1 LEAN
Let $j \geq 3$ and $c \geq 2$ be such that every prime factor of $c$ divides $j - 1$ and $c^{jc^j} \equiv 1 \pmod{j}$. Put

$$x = c^{jc^j} + j - 1, \quad b = x^{x/j}/c.$$

Then $(c^j, 0, ..., 0, x)$, with $j-1$ zeros, is a $(j+1)$-digit solution in base $b$ under $0^0 = 1$. The conditions hold for $c = (j-1)^a$ with any $a \geq 1$.

Lean: `theorem25_5`

*Proof.* First $x \equiv c^{jc^j} - 1 \equiv 0 \pmod{j}$, so $y = x^{x/j}$ is an integer. A prime $l$ dividing $c$ divides $j-1$, so $x \equiv j-1 \equiv 0 \pmod{l}$ and $v_l(y) = (x/j)v_l(x) \geq x/j > c \geq v_l(c)$. So $c$ divides $y$, and $b = y/c$ is an integer with $b > x > c^j$. Then

$$P = c^j b^j + x = y^j + x = x^x + x,$$

$$F = w(c^j) + (j-1)w(0) + w(x) = c^{jc^j} + j - 1 + x^x = x + x^x.$$

For $c = (j-1)^a$ we have $c \equiv (-1)^a \pmod{j}$, and the exponent $jc^j$ is even because $j(j-1)$ is even, so $c^{jc^j} \equiv 1$. □

**Example 2.31** (The smallest member, $k = 4$).
With $j = 3$ and $c = 2$: $x = 2^{24} + 2 = 16777218$, and $(8, 0, 0, 16777218)$ is a solution in base $16777218^{5592406}/2$, about $10^{4.04 \cdot 10^7}$. Here $8 \cdot b^3 = y^3 = x^x$, and the value $x$ of the last digit pays for $w(8) = 2^{24}$ and the two zeros. The library checked it with exact integers.

**Remark 2.32**.
Under $0^0 = 0$ the zeros are free, and $\left(c^j, 0, ..., 0, c^{jc^j}\right)$ is the swap of Section 2.3 with $a = c^j$. So Theorem 2.30 is the swap under $0^0 = 1$: the last digit is raised by $j-1$ to pay for the zeros. Since $c = (j-1)^a$ works for every $a$, it gives a second proof that every length $k \geq 4$ has infinitely many solutions under $0^0 = 1$ (Corollary 2.22).

Table 2.2. The smallest known solution of each length beyond the complete lists, from the two linear families (L25.1). The family of Theorem 2.30 wins for 4, 5 and 6 digits, the binomial family for 7, 8 and 9.

| $k$ | Theorem 2.30: $c$, $x$, base | Theorem 2.21: smallest, base | smaller |
|---|---|---|---|
| 4 | 2, $2^{24}+2$; $10^{4.04 \cdot 10^7}$ | $(1, 6, 12, x)$; $10^{3.85 \cdot 10^{13}}$ | zeros |
| 5 | 3, $81^{81}+3 \approx 10^{154.6}$; $10^{10^{156.2}}$ | $(16, 64, 96, 64, x)$; $10^{10^{192.0}}$ | zeros |
| 6 | 2, $2^{160}+4 \approx 10^{48.2}$; $10^{10^{49.1}}$ | $x \approx 10^{3977}$; $10^{10^{3980}}$ | zeros |
| 7 | 5, $5^{93750}+5 \approx 10^{65528}$; $10^{10^{65532}}$ | $(1, 6, 15, 20, 15, 6, x)$; $10^{10^{26.7}}$ | binomial |
| 8 | 6, $x \approx 10^{1.52 \cdot 10^6}$ | $x \approx 10^{18921}$; $10^{10^{18924}}$ | binomial |
| 9 | 7, $x \approx 10^{3.90 \cdot 10^7}$ | $x \approx 10^{76220}$; $10^{10^{76224}}$ | binomial |

For 10 digits the zero family with $c = 2$ gives $x = 2^{4608} + 8$, about $10^{1387}$; the binomial table stops at 9 digits.

Other identities exist, but each needs a second coincidence.

**Proposition 2.33** (Nonlinear identities). PROVEN L25.1
Let $s = (t, x)$ have a power identity with $\deg A \geq 2$, where $A$ is not a constant times a power of a linear polynomial. Then $s$ is a solution only if $A(b) = y$ for the integer $y$ with $x^x = \lambda y^j$. This is one polynomial equation in $b$ for a number $y$ of size about $x^{x/j}$ that $t$ fixes.

A number of size $y$ is a value of a fixed polynomial of degree $d \geq 2$ with probability about $y^{1/d-1}$, so such members are coincidences. The other mechanisms that one might try reduce to the same cases (L25.1):

- a decomposable value polynomial $P_s = G(h(r))$ gives a nonlinear $A$;
- a factorization of a 3-digit equation into two algebraic factors is a square identity; the Aurifeuillian pair behind $(1, 7, 1)$ in base 904 holds for $x = 7$ only, not as an identity in $x$ (Section 6.2);

- several large digits need a sum such as $u^u + v^v$ to be nearly a $j$-th power, a Fermat-type coincidence; equal large digits leave finitely many small strings for each $j$;
- a self-paying digit above a tail, as $(5, 1, 3)$ in base 25, needs the tail to be a solution of that base, again a coincidence.

**Computation 2.34** (Census of identities). COMPUTATION L25.1

For every prefix with digits up to 120, 60, 30, 16 and 10 (for 3, 4, 5, 6 and 7 digits) the library found every power identity and decided its two conditions. The only members are linear with $\lambda$ a $j$-th power and $j$ dividing $x$: 65 three-digit members of Theorem 2.21, and three 4-digit members, $(1, 6, 12, x)$ and $(8, 24, 24, x)$ of Theorem 2.21 and $(8, 0, 0, x)$ of Theorem 2.30. 108 nonlinear shapes pass the power test, and each fails modulo a prime below 60.

**Conjecture 2.35** (Families explain the rest). CONJECTURE L25.1

For each $k \geq 3$, all but finitely many $k$-digit solutions under $0^0 = 1$ belong to the two linear families of Theorem 2.21 and Theorem 2.30. Every $k$-digit solution in a base above the largest base of the complete list of that length belongs to them.

Under $0^0 = 0$ the conjecture fails, because self-paying digits and swaps give further identity families. For three digits the families are completely understood.

**Theorem 2.36** (The next 3-digit member). PROVEN L25.2

After $(1, 2, 6)$ in base 215, the next 3-digit member of a power identity is $(1, 6, 46666)$ in base $46666^{23333} - 3$.

*Proof sketch.* A member with $x \leq 46666$ has $w(p), w(q) \leq x$, so $p, q \leq 6$. The few shapes $(\lambda c^2, 2\lambda ce)$ in that range give $x$ in $\{6, 261, 262, 513, 46666, 46686\}$; all but 6 and 46666 fail the parity or square condition, or exceed 46666. The shapes with $e = 0$ give nothing. □

Whether $(1, 6, 46666)$ is the next 3-digit solution of any kind (Q65) is only partly settled. An exhaustive sieve finds no 3-digit solution in the bases from 905 to $10^{3301}$ other than the known ones, and the strings that remain below $46666^{23333} - 3$ have two digits above 200 and a digit above 2000; the root heuristic gives them an expected $10^{-3294}$ solutions (Section 5.1, Section 5.6). Under an explicit form of the abc conjecture the strings with two digits at most 200 are settled in every base (Corollary 5.55).

### 2.7.1 Which weights have families

The binomial families exist because $x^x$ is a $j$-th power whenever $j$ divides $x$. Factorials are almost never powers, and they have no such families. The library shows that this is the whole difference L33.3, Q96. The construction of Theorem 2.21 works for any weight $w$ on the digits.

**Theorem 2.37** (A sufficient condition for families). PROVEN L33.3

Let $w$ be a weight and $j \geq 2$. Suppose there is an integer $M \geq 1$ such that for all large $x$ divisible by $M$:

1. $w(x) = y^j$ for an integer $y = y(x)$ divisible by $M$;
2. $M$ divides $w(d)$ for every $d \geq 1$ divisible by $M$;
3. $y > M\big(x + x^{1/j}\big) + 1$.

For an integer $u \geq 1$ put $t_i = M^{j-i}\binom{j}{i}(Mu)^i$ for $0 \leq i < j$ and $x = (Mu)^j + \sum_i w(t_i)$. Then for every large $u$ the string $\big(t_0, \ldots, t_{j-1}, x\big)$ is a solution with $j + 1$ digits in base $b = (y(x) - Mu)/M$, under either value of $w(0)$. So length $j + 1$ has infinitely many solutions.

*Proof.* Every $t_i$ is a positive multiple of $M$, so by (ii) $x$ is a multiple of $M$, and (i) gives $w(x) = y^j$ with $M \mid y$. So $b$ is an integer and $Mb + Mu = y$. By the binomial theorem the value of the string is

$$\sum_{i<j} t_i b^{j-i} + x = (Mb + Mu)^j - (Mu)^j + x = y^j + \sum_i w(t_i),$$

which is the sum of the weights of its digits. Condition (iii) and $Mu \le x^{1/j}$ give $b > x$, so every digit is below the base. □

For $x^x$ take $M = j$ and $y = x^{x/j}$; this is Theorem 2.21. The weights $x^{x+t}$ with $j \mid t$ work the same way, and so do the powers $x^d$ when $j$ divides $d$ and $j < d$. For factorials condition (i) fails for every $j$.

**Proposition 2.38** (Factorials are almost never powers). PROVEN L33.3
Let $x! = \lambda y^j$ with $j \ge 2$ and positive rationals $\lambda$ and $y$. Then every prime in $(x/2, x]$ divides the numerator or the denominator of $\lambda$. In particular $x < 2P$, where $P$ is the largest such prime.

*Proof.* By Bertrand's postulate there is a prime $\ell$ with $x/2 < \ell \le x$ for $x \ge 2$. It divides $x!$ exactly once, since $2\ell > x$. In $\lambda y^j$ the exponent of $\ell$ is the exponent in $\lambda$ plus a multiple of $j$. So the exponent in $\lambda$ is not a multiple of $j$, and in particular not 0. □

So for each $\lambda$ only finitely many $x!$ lie in $\lambda$ times the $j$-th powers, while for $x^x$ a whole progression does. A census of the linear identity shapes of Computation 2.34 confirms the contrast.

**Computation 2.39** (Identity members for five weights). COMPUTATION L33.3
For the linear shapes $rP_t(r) + C = \lambda(cr + e)^{k-1}$ with prefix digits up to 60 ($k = 3$) and 20 ($k = 4$), the number of shapes with $w(x) = \lambda y^{k-1}$, and of those that give solutions:

| weight | $k = 3$ | $k = 4$ |
|---|---|---|
| $x^x$ | 38 / 31 | 3 / 2 |
| $x^{x+1}$ | 73 / 65 | 14 / 14 |
| $x!$ | 0 / 0 | 0 / 0 |
| $x^2$ | 83 / 0 | 19 / 0 |
| $x^3$ | 62 / 62 | 4 / 0 |

For $x^2$ the base is never above $x$, as condition (iii) predicts. For $x^3$ and three digits the members are mostly $(a, 0, a^3)$ in base $a^4$ under $w(0) = 0$: here $ab^2 = a^9 = w(a^3)$, so the large digit is itself a weight and a cube by construction.

The converse direction needs a conjecture, and it reaches only solutions with one dominant digit.

**Theorem 2.40** (One dominant digit under abc). CONDITIONAL L33.4
Assume the abc conjecture. Let $w$ be $x^x$ or $x!$, under either convention, and let $k \ge 3$. There is a constant $C_k$ such that every $k$-digit solution whose largest digit $x$ exceeds $C_k \max(2, B)$, where $B$ is the second largest digit, comes from a power identity: $x$ is the last digit, and the other digits are the coefficients of $\lambda(cr + e)^{k-1}$ apart from the constant term. Under $w(0) = 0$ the self-paying shapes $(x, 0, \ldots, 0, s)$ with $w(s) = s$ are also possible.

*Proof sketch.* Let $d = k - 1$. Completing the $d$-th power, as one completes the square for 3 digits, writes the equation as $Y^d + E = Cw(x)$ with $Y$ linear in $b$, a constant $C$, and a remainder $E$ that is small compared with $Y^d$. If $E \ne 0$, abc applied to this triple bounds $w(x)$ in terms of $w(B)$, because the radical of $w(x)$ is tiny: at most $x$ for $x^x$, and about $e^x$ for $x!$, against a size of about $x^x$. This gives $x \le C_k B$. If $E = 0$, the polynomial of the string is a power identity. □

For factorials, $E = 0$ needs $Cx! = Y^d$, which Proposition 2.38 rules out for large $x$. So under abc each length has only finitely many factorions with one dominant digit, and for $x^x$ every solution with one dominant digit is a family member. This proves the criterion of Q96 in the dominant-digit range, under abc: a weight has such families of length $k$ exactly when the values $x$ produced by the binomial shapes meet the set where $w(x)$ is $\lambda$ times a $(k-1)$-th power infinitely often. Strings whose digits are all of similar size are outside every method here.

## 2.8 Self-paying blocks

A self-paying digit is one placement that pays for itself. A longer string can do the same. Split a $k$-digit string $s = (t, u)$ into a top part $t$ of $L$ digits and a tail $u$ of $m = k - L$ digits.

**Lemma 2.41** (Splits). PROVEN L22.1 LEAN

Put $R_m = b^m P_t(b) - F(t)$. Then $s = (t, u)$ is a solution in base $b$ exactly when $R_m = F(u) - P_u(b)$. In particular $R_m = 0$ exactly when the tail pays for itself, $P_u(b) = F(u)$, where leading zeros of $u$ weigh $w(0)$.

Lean: `split_iff`

*Proof.* $P_s(b) = b^m P_t(b) + P_u(b)$ and $F(s) = F(t) + F(u)$. □

**Definition 2.42** (Self-paying block). L22.2

A *self-paying block* is a triple $(t, b, m)$: a string $t = (t_1, ..., t_L)$ with $t_1 \geq 1$ and $t_L \geq 1$, a shift $m \geq 1$ and a base $b > \max(t)$, with

$$b^m P_t(b) = F(t).$$

For $L = 1$ these are the self-paying digits.

Under $0^0 = 0$ a block is the same as the solution $(t, 0, ..., 0)$ with $m$ zeros. Under $0^0 = 1$ that string instead has $P - F = -m$, because each trailing zero costs 1. A block followed by any tail that pays for itself is a solution (Lemma 2.41). Under $0^0 = 1$ the only one-digit tail that pays for itself is (1), and a two-digit tail must be a 2-digit solution of the base.

**Computation 2.43** (All blocks with small digits). COMPUTATION L22.2

With $L \geq 2$, digits up to 40 to 150, and bases in the complete range of each search, there are exactly four self-paying blocks: $(1, 3)$ in base 4 and $(1, 7)$ in base 904, both at shift 1; $(6, 6, 5)$ in base 11 at shift 2; and $(3, 3, 6, 6)$ in base 13 at shift 1.

Each of them explains known solutions. $(1, 3)$ and $(1, 7)$ followed by the digit 1 give $(1, 3, 1)$ in base 4 and $(1, 7, 1)$ in base 904. The block $(3, 3, 6, 6)$ followed by 1 gives the 5-digit solution $(3, 3, 6, 6, 1)$ in base 13:

$$13 \cdot P_{(3,3,6,6)}(13) = 13 \cdot 7182 = 93366 = 3^3 + 3^3 + 6^6 + 6^6.$$

The block $(6, 6, 5)$ gives $(6, 6, 5, 0, 0)$ and $(6, 6, 5, 0, 1)$ in base 11 under $0^0 = 0$, from $6 \cdot 11^4 + 6 \cdot 11^3 + 5 \cdot 11^2 = 96437 = 6^6 + 6^6 + 5^5$.

How often should blocks occur? A block is one equation in $L + 1$ unknowns, while a solution of $L + m$ digits is one equation in $L + m + 1$ unknowns. The coincidence model of Chapter 3 gives the rate.

**Heuristic 2.44** (Rate of blocks). HEURISTIC L22.2

The expected number of blocks of $L$ digits at shift $m$ in base $b$ is smaller than the expected number of solutions with $L+m$ digits by a factor of about $(k/L)X^m$, where $k = L+m$ and $X = X_k(b)$ is the largest usable digit. Both rates fall like $b^{-(k-1)}$ up to powers of $\log b$, so the expected total over all bases is finite for every $L \geq 2$.

The exact rates computed in L22.2 follow this law within a factor 1.5, and a base-by-base calibration finds 2 blocks from base 13 on, against 0.73 expected. Blocks are coincidences, like solutions, only rarer. The self-paying digits ($L = 1$) are the exception: $xb^m = x^x$ is an identity that holds whenever $x^{x-1}$ is an $m$-th power.

Blocks were studied because of three 5-digit solutions in bases 13 to 99 that seemed too many for the root heuristic (Section 5.5). Table 2.3 shows their structure.

Table 2.3. The 5-digit solutions in bases 13 to 99 under $0^0 = 1$ and their remainders $R_m$ at each split (L22.1).

| base | digits | $n$ | largest weights | $R_1$ | $R_2$ | $R_3$ | $R_4$ |
|---|---|---|---|---|---|---|---|
| 13 | $(3,3,6,6,1)$ | 93,367 | $6^6+6^6$: 99.94% | 0 | 46,578 | 92,220 | 85,656 |
| 14 | $(2,6,0,3,6)$ | 93,344 | $6^6+6^6$: 99.97% | 46,650 | 46,635 | 46,636 | 76,828 |
| 48 | $(3,8,6,2,0)$ | 16,823,904 | $8^8$: 99.72% | 1 | −91 | 32,741 | 15,925,221 |

The first is an exact block followed by the digit 1. The third is a block with remainder 1 followed by a 0, which weighs 1; by Lemma 2.41 this is the same pattern with the last digit 0 instead of 1. The second has no structure at all. None has a self-paying digit, since 13, 14 and 48 are not perfect powers, and none belongs to a binomial family. Blocks are already counted by the root heuristic: the polynomial of $(t,1)$ is the polynomial of the block $(t,b,1)$. So blocks do not explain the excess, which remains a fluctuation with probability about 0.005, or 0.015 after the choice of the cell (Section 5.5).

## 2.9 Many solutions in one base

How large can $N_m(b)$ be? Under $0^0 = 0$ self-paying digits make it unbounded. The cleanest family uses towers $n^{n-1}$.

**Proposition 2.45** (Unbounded counts under $0^0 = 0$). PROVEN L4.1 L9.8 LEAN

For every $n \geq 3$, base $n^{n-1}$ has at least $2^{n-1}-2$ nontrivial solutions under $0^0 = 0$. So $N_0(b)$ is unbounded.

Lean: `theorem9_2, prop4_2`

*Proof.* For $1 \leq s \leq n-2$, the digit $a = n^s$ is below $b = n^{n-1}$ and $a^{a-1} = n^{s(n^s-1)}$. Since $n-1$ divides $n^s-1$, this is $b^k$ with $k = s(n^s-1)/(n-1)$. So base $b$ has at least $n-2$ self-paying digits, and Corollary 2.6 gives the count. □

Here $\ln b = (n-1)\ln n$, so $n$ grows like $\ln b / \ln\ln b$ and the count is $b^{(\ln 2 + o(1))/\ln\ln b}$. That is faster than any power of $\log b$ and slower than any power of $b$. Powers of 2 give another family: base $2^{L(2^L-1)}$ has the self-paying digits $2^{jL}$ for $1 \leq j \leq 2^L-2$ (L4.1).

Table 2.4. The smallest base with at least $C$ self-paying digits, and the resulting lower bound for $N_0(b)$ (L9.8). Most records are the towers $(C+2)^{C+1}$; the others are powers of 2, 3, 5 and 6 whose exponents have many divisors.

| $C$ | smallest base | size | solutions, at least |
|---|---|---|---|
| 1 | $2^3$ | $10^{0.9}$ | 2 |
| 2 | $2^6$ | $10^{1.8}$ | 6 |
| 3 | $5^4$ | $10^{2.8}$ | 14 |
| 5 | $7^6$ | $10^{5.1}$ | 62 |
| 8 | $10^9$ | $10^{9.0}$ | 510 |

| $C$ | smallest base | size | solutions, at least |
|---|---|---|---|
| 14 | $2^{60}$ | $10^{18.1}$ | 32,766 |
| 23 | $5^{48}$ | $10^{33.6}$ | 16,777,214 |
| 30 | $2^{155}$ | $10^{46.7}$ | 2,147,483,646 |
| 39 | $41^{40}$ | $10^{64.5}$ | 1,099,511,627,774 |

These bases are rare.

**Proposition 2.46** (Rich bases are rare). PROVEN L9.8 LEAN
The number of bases up to $X$ with $c$ or more self-paying digits is at most $X^{1/(c+1)}(1+o(1))$ as $X \to \infty$.[1]
Lean: `selfPaying_base_pow, card_selfPaying_le (partial: the shape r^t with t >= c + 1)`

*Proof.* By Proposition 2.2 the self-paying digits of $b = r^t$, with $r$ not a perfect power, are $r^s$ with $s < t$. So $c$ of them need $t \geq c+1$. For each $t$ there are at most $X^{1/t}$ numbers $r^t \leq X$, and $t \leq \log_2 X$. So the count is at most

$$\sum_{t \geq c+1} X^{1/t} \leq X^{1/(c+1)} + (\log_2 X) X^{1/(c+2)} = X^{1/(c+1)}(1+o(1)).$$

□

So the structure that produces many solutions lives on a set of density zero. The average count, about 1.5 per base under $0^0 = 0$ in the model, comes from coincidences with a Poisson law and no heavy tail (Chapter 3). The extremes come from a different source than the average.

Under $0^0 = 1$ the picture is different, and the question whether $N_1(b)$ is bounded is open. Every placement of balance zero is now a self-paying digit or a final 1, since a zero costs 1. A solution made only of such placements needs a self-paying digit at every position above 0, and by Proposition 2.4 no two of them are adjacent. So the only such solutions are $(a, 1)$ in base $a^{a-1}$ (L8.3). Self-paying digits help only together with near-solutions.

**Proposition 2.47** (Self-paying digits under $0^0 = 1$). PROVEN L8.4
Under $0^0 = 1$, consider strings of a fixed length $L$, leading zeros allowed, in which every zero weighs 1. The solutions of length $L$ with $j$ self-paying digits correspond exactly to the pairs made of a string $n$ with $F(n) = n + j$ (a near-solution with defect $j$) and a choice of $j$ self-paying positions where $n$ has a 0, such that the result does not start with 0.

*Proof.* Each self-paying digit placed on a 0 raises $n$ and $F(n)$ by the same amount, minus the weight 1 of the zero it replaces in $F$. So it lowers the defect $F(n) - n$ by exactly 1. □

Base 64 shows the mechanism. Removing the self-paying 16 and 4 from $(16, 0, 0, 0, 0, 0, 1, 0, 0, 4, 8)$ leaves $(0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 8)$, whose weight $9 + 1 + 8^8 = 16777226$ exceeds its value $64^4 + 8 = 16777224$ by 2. A construction like Proposition 2.45 would need near-solutions of every small defect in one base, and none is known. The chain construction of Theorem 2.10 could in principle give $\binom{n}{k}$ solutions in one base, if a chain reached a position above $n$ self-paying digits, but the search of Computation 2.11 found no instance.

## 2.10 Integers with several bases

By Proposition 1.7 a digit string is a solution in at most one base. An integer is another matter: it has a different digit string in each base, and it can be a solution in several.

[1] Such a base need not be a perfect $(c+1)$-th power: base $8 = 2^3$ has the self-paying digit 4 but is not a square. So the bound needs the sum over all exponents $t \geq c+1$.

**Proposition 2.48** (Many bases under $0^0 = 0$). PROVEN L15.6
Under $0^0 = 0$, let $c = r^u$ with $r$ not a perfect power. Then $N = c^c + 1$ is a solution in every base $r^t$ with $u < t$ and $t \mid u(c-1)$, and in base $c^c - c + 1$. So one integer can be a solution in arbitrarily many bases.

*Proof.* In base $r^t$ the digit $c$ is self-paying at position $k = u(c-1)/t$, so $(c, 0, ..., 0, 1)$ with $k+1$ digits is a solution, with value $c^c + 1$. In base $c^c - c + 1$ the string $(1, c)$ is a solution with the same value. The number of divisors of $u(r^u - 1)$ is unbounded. □

**Example 2.49**.
- For $c = 256 = 2^8$, $8 \cdot 255 = 2040$ has 25 divisors above 8. So $2^{2048} + 1$ is a solution in 26 bases. A complete search confirms that these are all.
- $7^7 + 1 = 823544$ is a solution in five bases under $0^0 = 0$: 49, 343 and $7^6 = 117649$ (the self-paying 7), 823537 (the string $(1, 7)$), and 904 (the string $(1, 7, 0)$).
- $28 = 3^3 + 1$ is a solution in bases 9 and 25, as $(3, 1)$ and $(1, 3)$, and also as $(1, 3, 0)$ in base 4 and $(1, 0, 3)$ in base 5.

Under $0^0 = 1$ the two-digit representations are rigid.

**Proposition 2.50** (Two-digit representations under $0^0 = 1$). PROVEN L15.6 LEAN
Under $0^0 = 1$, an integer $N \geq 3$ has at most two 2-digit representations as a solution. If it has two, they are $(a, c)$ and $(c, a)$ with $N = a^a + c^c$, $a \mid c^c - c$ and $c \mid a^a - a$.
Lean: `prop15_16`

*Proof.* A 2-digit solution has $N = w(a) + w(c)$. If $X^X + Y^Y = U^U + V^V$ with $X \geq Y$, $U \geq V$ and $X > U$, then $X^X \geq (U+1)^{U+1} > 2U^U$, which is impossible. So the pair of weights is unique. The weight 1 belongs to both digits 0 and 1, but $(a, 0)$ is a solution only if $a \mid w(0) - 0 = 1$, which gives $(1, 0)$ in base 2 with $N = 2$, excluded by $N \geq 3$. So the two orders of the pair give at most two strings. □

So $N = c^c + 1$ has the two bases of $(1, c)$ and $(c, 1)$ for every $c \geq 3$ (for $c = 2$ the string $(2, 1)$ would need base 2), and a third base needs a representation with three or more digits. To find every base of a given integer, one needs a bound on where its base can lie. The leading digit and the second digit pin it down.

**Lemma 2.51** (Window for the base). PROVEN L15.6 L20.1
Let $n$ be a solution in base $b$ with $k \geq 2$ digits, leading digit $p$ and largest digit $M$. Then

$$pb^{k-1} \leq n < pb^{k-1} + (M+1)b^{k-2}$$

and so

$$b \leq (n/p)^{1/(k-1)} < b + \frac{M+1}{p(k-1)}.$$

*Proof.* The digits after the second one add at most $b^{k-2} - 1$, and the second digit is at most $M$. For the second claim put $u = (M+1)/(pb)$, so that $n/p < b^{k-1}(1+u)$, and use Bernoulli's inequality $(1+u)^{1/(k-1)} \leq 1 + u/(k-1)$. □

All 3,135 known solutions satisfy both inequalities, and the window is close to sharp. Under $0^0 = 1$ an integer with a 2-digit representation has very constrained longer representations.

**Lemma 2.52** (Rigid representations). PROVEN L18.3

Under $0^0 = 1$, let $N = a^a + w(c)$ with $0 \le c \le a$ and $a \ge 2$. Every representation of $N$ as a solution in some base contains the digit $a$ exactly once, unless it is $(a, a)$, and its other digits have weights adding up to $w(c)$. In a representation with 3 or more digits these other digits are all below $c$; the 2-digit representation is $(a, c)$ itself. So if $c \le 1$ or $c = a$, every representation of $N$ has 2 digits, and if $2 \le c < a$, every longer one has between $c + 2$ and $a + 1$ digits.

**Corollary 2.53** (Three bases). PROVEN L18.3

An integer with three bases under $0^0 = 1$ either has at most one 2-digit representation and at least two longer ones, or it is $N = a^a + c^c$ with $2 \le c < a$, $c \mid a^a - a$, $a \mid c^c - c$, and a third representation of $c + 2$ to $a + 1$ digits: one digit $a$ and digits below $c$ whose weights add up to $c^c$.

**Computation 2.54** (Integers with several bases). COMPUTATION L15.6 L18.3

Under $0^0 = 1$: of the 2,117 integers with a 2-digit representation with digits up to 90, 1,779 have one base, 338 have two, and none has three or a representation with 3 or more digits. None of the 483 integers $a^a + c^c$ with two 2-digit bases and $a, c \le 150$ has a third base. Under $0^0 = 0$: among all known solutions and all $a^a + w(c)$ with $a, c \le 40$, 428 integers have two or more bases; in all but two of those with a longer representation, every such representation comes from self-paying digits or swaps. The exceptions are 28 and 823544. The maximum is 26 bases, for $2^{2048} + 1$.

So under $0^0 = 0$ the number of bases of an integer is unbounded, through self-paying digits, while under $0^0 = 1$ no integer with three bases is known. A third base would need a coincidence of the kind that Chapter 5 studies, and no reason is known why one cannot exist.

Chapter 3

# The probabilistic model

The constructions of Chapter 2 explain a few solutions, mostly short ones and mostly in special bases. The long solutions near $b$ digits show no structure at all. This chapter treats them as coincidences. A digit multiset has some small chance to reproduce itself, and the chance can be computed. The resulting model predicts how many solutions a base has, how long they are, and why odd bases have more of them. It survives every test the library has tried on it, with one exception in the factorial number system and one in function fields (Section 7.9, Section 7.8).

Two warnings before we start. First, everything in this chapter marked "heuristic" is a statement about a random model, not about $x^x$. The model is tested against data, and the tests are the evidence. Second, the data are small: a few dozen bases with one or two solutions each. So the chapter often asks whether a deviation is chance. The answer depends on how the question was chosen, a point that Section 3.11 explains.

## 3.1 The counts per base

The exhaustive search of Chapter 4 has found every solution in bases 2 to 41 and in base 43, under both conventions. The full table is in Section 1.8 and Appendix K. Here are the facts that the model must explain.

- Most bases have one to three nontrivial solutions under each convention. The largest count up to base 43 is 7, in base 15 under $0^0 = 0$.
- The longest solution usually has $b - 1$ digits: in 23 of the 29 bases from 10 to 41 that have a solution L4.3. Only bases 15 and 16 have a $b$-digit solution, both under $0^0 = 0$.
- Odd bases have more solutions under $0^0 = 0$. Over bases 10 to 41, odd bases average 2.56 solutions and even bases 1.56. Under $0^0 = 1$ the averages are 1.81 and 1.69 L4.3.
- The counts of the two conventions are correlated, 0.56 over bases 10 to 41. A solution without the digit 0 counts under both conventions (Proposition 1.8), and 35 of the 56 solutions under $0^0 = 1$ in these bases are zero-free.
- Bases 19, 27 and 35 have no solution under either convention. They are the only empty bases from 3 to 41, and base 43 has solutions.

If each base had a Poisson number of solutions with the mean of the refined model of Section 3.3, bases 10 to 39 would show the distribution of Table 3.1. The counts follow the Poisson shape roughly. The upper tail is heavier, because twins and constructed solutions come in groups and small bases have extra solutions.

Table 3.1. Bases 10 to 39 with exactly $n$ nontrivial solutions, observed against a Poisson law with the mean of the refined model L4.4.

| | $n = 0$ | 1 | 2 | 3 | 4 | 5 | 6+ |
|---|---|---|---|---|---|---|---|
| $0^0 = 0$, expected | 6.3 | 8.9 | 7.1 | 4.2 | 2.1 | 0.9 | 0.4 |
| $0^0 = 0$, observed | 6 | 9 | 5 | 3 | 3 | 3 | 1 |
| $0^0 = 1$, expected | 8.6 | 10.7 | 6.7 | 2.8 | 0.9 | 0.2 | 0.1 |
| $0^0 = 1$, observed | 6 | 7 | 9 | 4 | 3 | 1 | 0 |

Some counts are forced by structure. Under $0^0 = 0$ solutions ending in 0 or 1 come in twins (Proposition 2.1), so $N_0(b)$ has the parity of the number of solutions whose last digit is at least 2. Bases with self-paying digits always have solutions under $0^0 = 0$ (Proposition 2.5), and the count is unbounded along special bases (Proposition 2.45). The model ignores all of this. It describes the generic coincidences, and it should be compared with the data after the families are set aside or in large bases where they are rare.

## 3.2 The basic model and the large-base limit

Fix a length $k$ and a digit multiset $c$ with $k$ digits. Its weight $S(c) = \sum_x c_x w(x)$ is one fixed number. The multiset gives a solution exactly when the base-$b$ digits of $S(c)$ form the multiset $c$. The model replaces the digits of $S(c)$ by random digits.

**Definition 3.1** (Basic model). L5.1

For each multiset $c$ of $k$ digits, the digits of $S(c)$ are drawn uniformly among the $k$-digit strings with a nonzero leading digit, independently for different multisets. The multiset is a *model solution* if the drawn string has the multiset $c$.

The chance that a random $k$-digit string has a given multiset $c$ is the same for the drawn string as for a random string $n$. Summing over all multisets $c$ whose weight has $k$ digits gives a formula that is easy to estimate by sampling.

**Proposition 3.2** (Expected count per length). PROVEN L5.1

In the basic model, the expected number of $k$-digit solutions is

$$E_k = P(F(n) \text{ has exactly } k \text{ digits}),$$

where $n$ is a uniform random $k$-digit number.

*Proof.* Let $p(c)$ be the probability that a uniform $k$-digit string has multiset $c$. A multiset $c$ whose weight has $k$ digits is a model solution with probability $p(c)$, and one whose weight has another length is never one. So the expected count is the sum of $p(c)$ over the multisets with $k$-digit weight. That is the probability that a uniform $n$ has such a multiset, and $F(n) = S(c)$. □

The formula says something striking. Sum over all lengths. The events "$F(n)$ has $k$ digits" for a $k$-digit random $n$ are different events for different $k$, but in a large base they behave like a partition of one event. So the expected total is close to 1.

Let $b$ grow with $J = b - k$ fixed. The digit counts of a random $k$-digit string become independent Poisson variables with mean $k/b \to 1$. A digit $b - y$ weighs

$$(b-y)^{b-y} = b^{b-y}(1 - y/b)^{b-y} \approx e^{-y} b^{b-y},$$

so the largest digit present decides the length of $F(n)$. One copy of $b - 1$ weighs about $b^{b-1}/e$, so $c$ copies give $F \approx (c/e)b^{b-1}$. This crosses $b^{b-1}$, the smallest $b$-digit number, when $c \geq 3$.

**Heuristic 3.3** (The length law). HEURISTIC L5.2 L10.5

In the large-base limit a base has on average one solution per convention before the parity effect of Section 3.3. Its length is $b$, $b-1$, $b-2$, $b-3$, $b-4$ with probabilities about 8.0%, 55.2%, 23.2%, 8.6%, 3.2%, and each further digit of shortfall costs a factor of about $e$. The expected number of solutions with $J = b - k \geq j$ is about $e^{-(j-1)}$ for $j \geq 1$.

*Proof sketch.* The cases are disjoint and cover almost everything. Write $c_x$ for the Poisson(1) counts.

- $k = b$: the sum needs at least $b$ digits, so at least 3 copies of $b-1$. The probability is $P(\text{Poisson}(1) \geq 3) = 1 - (5/2)e^{-1} = 0.080$.
- $k = b-1$: one or two copies of $b-1$ put $F$ between $b^{b-2}$ and $b^{b-1}$, with probability $(3/2)e^{-1} = 0.552$.
- $k = b-j$ for $j \geq 2$: no digit above $b-j$ and at least one copy of $b-j$. This has probability $e^{-(j-1)}(1-e^{-1})$, which gives 0.233, 0.086, 0.032 for $j = 2, 3, 4$.

The five numbers add up to 0.983, and the tail adds the rest to 1. In each case the event "$F(n)$ has $k$ digits" has the stated probability, and Proposition 3.2 turns it into the expected count. The third case is slightly loose: more than $e^j$ copies of $b-j$ push $F$ up one length, and more than $e^{j+1}$ copies of $b-j-1$ push it up to $b-j$ digits. These rare cases move the shares by less than $10^{-5}$. Theorem 3.7 below uses the exact events. □

The law holds well at finite $b$. Table 3.2 shows the exact model at bases 50 to 800 L10.5. For $J$ up to about $\ln b$ the values match the limit. Beyond that they are about three times the limit, a bump that Section 3.9 explains.

Table 3.2. Expected number of solutions with $J = b - k$ in the basic model L10.5.

| $J$ | base 50 | base 100 | base 200 | base 800 | limit |
|---|---|---|---|---|---|
| 0 | 0.078 | 0.080 | 0.081 | 0.078 | 0.080 |
| 1 | 0.554 | 0.554 | 0.546 | 0.553 | 0.552 |
| 2 | 0.237 | 0.234 | 0.233 | 0.231 | 0.233 |
| 4 | 0.036 | 0.034 | 0.033 | 0.031 | 0.032 |
| 8 | 0.0029 | 0.0021 | 0.0018 | 0.0010 | 0.0006 |
| 12 | 0.00030 | 0.00009 | 0.00002 | 0.00003 | 0.00001 |

**Remark 3.4.**
The total of 1 does not depend on the details of $x^x$. It only needs the length of $F(n)$ to be nearly independent of the length of $n$ near the critical length, so that $\sum_k P(\ell(F_k) = k)$ is close to $\sum_\ell P(\ell(F) = \ell) = 1$. This is why "about one solution per base" holds in every number system of Chapter 7 where digits can move freely. The length shares, in contrast, depend on how high the top weight sits between two powers of the base, and they change from system to system (Section 7.10).

### 3.2.1 The length shares are a theorem in the model

The sketch of Heuristic 3.3 replaces the digit counts by independent Poisson variables and the weights by their limits. Both steps can be made exact, so in the basic model the shares are a theorem about random strings. Whether real bases follow the model is a separate question; the heuristic label of Heuristic 3.3 refers to that.

Fix $j \geq 0$. For $y \geq 1$ let $\gamma(y) = (e^y - \lfloor e^y \rfloor)/e^y$. It is positive because $e^y$ is not an integer, and it measures how far $e^y$ is from the integer below it. Let $\gamma_j$ be the smaller of $\gamma(j)$ and $\gamma(j+1)$ (only $\gamma(1)$ when $j = 0$). Write $N_x$ for the number of copies of the value $x$ in a random string and $y_0$ for the distance of the largest digit from $b$, so that the largest digit is $b - y_0$. The limit share is

$$s_j = \sum_{1 \leq k < e^j} \frac{e^{-j}}{k!} + \sum_{k > e^{j+1}} \frac{e^{-(j+1)}}{k!}.$$

The first sum is "the largest digit is $b-j$, with fewer than $e^j$ copies". The second is "the largest digit is $b-j-1$, with more than $e^{j+1}$ copies"; it adds $3.8 \cdot 10^{-6}$ to $s_1$ and less elsewhere. So $s_0 = e^{-1}(e - 5/2)$, and $s_0, \ldots, s_5$ are 0.0803, 0.5518, 0.2325, 0.0855, 0.0315, 0.0116.

The proof has three steps. The first makes the weights precise.

**Lemma 3.5** (Top weights). PROVEN L29.1
For $0 \le y < b$,

$$e^{-y} b^{b-y} \le (b-y)^{b-y} \le e^{-y} b^{b-y} \exp\left(\frac{y^2}{2(b-y)}\right).$$

*Proof.* Put $t = y/b$. Then $\ln (b-y)^{b-y} - (b-y)\ln b = b(1-t)\ln(1-t)$, and $(1-t)\ln(1-t) = -t + \sum_{k\ge 2} t^k/(k(k-1))$, a series whose terms after the first are positive. So the left side is at least $-y$, and at most $-y + (y^2/2b)\sum_{m\ge 0} t^m = -y + y^2/(2(b-y))$. □

The second step compares the top counts with Poisson variables. It uses the total variation distance of Definition A.2 and the coupling of Section A.4.1.

**Lemma 3.6** (Poisson coupling of the top counts). PROVEN L29.1
Let $d = b - j$ and $K = j + 1$. The vector $(N_{b-1}, ..., N_{b-K})$ of a random $d$-digit string is within total variation distance $((j+1)^2 + 1)/b$ of $K$ independent Poisson(1) variables.

*Proof sketch.* Replacing the leading digit by a uniform digit costs $1/b$. For $d$ uniform digits, given the total of the $K$ counts, the counts are multinomial with equal cells, and so are independent Poisson variables given their total. So only the totals need to be compared: a binomial total against a Poisson total costs at most $K/b$ by Theorem A.9, and moving the Poisson mean from $dK/b$ to $K$ costs at most $1 - e^{-Kj/b} \le Kj/b$. □

The third step shows that the length of $F(n)$ depends only on these counts, unless some top value occurs more than $\sqrt{b}$ times, which is very unlikely.

**Theorem 3.7** (Length shares in the basic model). PROVEN L29.1 L13.2
Fix $j \ge 0$ and let $b_0(j) = \max\left(e^{j+2}, j + 1 + 2(j+1)^2/\gamma_j, 64/\gamma_j^2\right)$. For every base $b \ge b_0(j)$, under either convention, the expected number $E_{b-j}(b)$ of $(b-j)$-digit solutions in the basic model satisfies

$$|E_{b-j}(b) - s_j| \le \frac{(j+1)^2 + 1}{b} + \frac{j+2}{\left(\lfloor \sqrt{b} \rfloor\right)!}.$$

In particular $E_{b-j}(b) \to s_j$.

*Proof sketch.* Let $M = \lfloor \sqrt{b} \rfloor$ and let $H$ be the event that each of the values $b-1, ..., b-j-2$ occurs at most $M$ times. A value occurs more than $M$ times with probability at most $1/M!$, so $P(\text{not } H) \le (j+2)/M!$. Suppose $H$ holds and the largest digit is $b - y$ with $k$ copies, $y \in \{j, j+1\}$. By Lemma 3.5, $F/b^{b-y}$ lies between $ke^{-y}$ and $ke^{-y}$ times a factor close to 1, plus at most $(M+2)/b$ for the lower digits. The condition $b \ge b_0(j)$ keeps all these error terms below $\gamma_j$, the gap between $ke^{-y}$ and 1. So $F$ has $b - y + 1$ digits if $k > e^y$ and $b - y$ digits if $k < e^y$. A largest digit above $b - j$ makes $F$ too long, and one below $b - j - 1$ makes it too short. So on $H$ the event "$F$ has $b - j$ digits" depends only on the top $j + 1$ counts, and under independent Poisson(1) counts its probability is exactly $s_j$, because $P(y_0 = y, N_{b-y} = k) = e^{-y}/k!$. Lemma 3.6 bounds the change from the true counts to the Poisson ones. □

The constant $b_0(j)$ is crude: $b_0(0) = 919$ and $b_0(1) = 23,000$. It only keeps the threshold $e^y$ away from the integers. The exact values below show the convergence already at base 30.

**Corollary 3.8** (One solution per base). PROVEN L29.1
In the basic model, $\sum_{d=1}^{b} E_d(b) \to 1$ as $b \to \infty$, and the share of the length $b-j$ tends to $s_j$.

*Proof sketch.* The events of Theorem 3.7 for $j = 0, 1, 2, \ldots$ split the Poisson outcomes, so $\sum_j s_j = 1$. For the tail, on the event that $F$ has $b-j$ digits every digit $x$ satisfies $x^x < b^{b-j}$, which forces $x < b - 0.75j$ for $b \geq 21$. Each of the about $b$ digits must avoid $0.75j - 1$ values, so the share of the lengths with $J \leq j \leq b/2$ is at most $5.6e^{-J/3}$. The lengths below $b/2$ contribute $O\big(1/\sqrt{b}\big)$, because then every digit is below $\max\big(2d, \sqrt{b}\big)$. Let $b$ and then $J$ tend to infinity. □

**Computation 3.9** (The shares at finite bases). COMPUTATION L29.1
Exact values of $E_{b-j}(b)$ from the law of the top counts, and the scaled error $b\big(E_{b-j} - s_j\big)$:

| $b$ | $E_b$ | $E_{b-1}$ | $E_{b-2}$ | $E_{b-3}$ | $b(E-s)$ for $j = 0, 1, 2, 3$ |
|---|---|---|---|---|---|
| 30 | 0.07735 | 0.55501 | 0.24203 | 0.09693 | $-0.089, +0.096, +0.285, +0.341$ |
| 100 | 0.07939 | 0.55275 | 0.23536 | 0.08879 | $-0.091, +0.093, +0.282, +0.324$ |
| 1000 | 0.08021 | 0.55191 | 0.23282 | 0.08587 | $-0.092, +0.092, +0.281, +0.318$ |
| 100000 | 0.08030 | 0.55182 | 0.23254 | 0.08555 | $-0.092, +0.092, +0.281, +0.317$ |

A Monte Carlo of the model program itself agrees within 1.5 standard errors at bases 30 and 100.

The last column converges, so the error is $c_j/b$ to first order. The library computes the constants exactly: $E_{b-j}(b) = s_j + c_j/b + O(1/b^2)$ with $c_0 = -1/(4e) = -0.092$ and $c_1, \ldots, c_5 = 0.092, 0.281, 0.317, 0.227, 0.136$ (Proposition 29.7 of L29.1). The base-100 numbers of Heuristic 3.24 are these values. Every later use of "about one solution per base" in the basic model rests on Corollary 3.8.

### 3.2.2 The expected count is a trace

The shares and their total of 1 have a compact form in the language of matrices. Let $M_b$ be the $b \times b$ matrix whose entry $M_b(d, d')$ is the chance that $F(n)$ has $d'$ digits when $n$ is a uniform $d$-digit string. A multiset with at most $b$ digits has $F < b^b$ (Proposition 3.74), so $F(n)$ has between 1 and $b$ digits, and every row of $M_b$ is a probability law. So $M_b$ is a stochastic matrix (Section A.5). Its diagonal entry $M_b(d, d)$ is $E_d$ of Proposition 3.2.

**Proposition 3.10** (The count is a trace). PROVEN L34.3
The expected count of the basic model over the lengths 2 to $b$ is $\operatorname{tr} M_b - M_b(1,1)$. The first entry is $M_b(1,1) = E_1 = N_1(b)/(b-1)$, where $N_1(b)$ is the number of digits $x \geq 1$ with $x^x < b$. So $N_1(b) = 2$ for $5 \leq b \leq 27$, 3 for $28 \leq b \leq 256$ and 4 for $257 \leq b \leq 3125$.

*Proof.* The diagonal entries are the $E_d$, and the count over the lengths 2 to $b$ is their sum without $E_1$. A 1-digit string $x$ has $F = x^x$, which has one digit exactly when $x^x < b$, and $x$ is uniform on $1, \ldots, b-1$. □

Length 1 is an artefact of the model: whatever $E_1$ says, the only 1-digit number with $n = F(n)$ is the trivial $n = 1$. So the library counts from length 2. The entry $E_1$ is also almost an eigenvalue of $M_b$, because a string of length $d \geq 2$ has $F < b$ only if all its digits are tiny. At $b = 100$ the eigenvalue is 0.0312 against $E_1 = 0.0303$. A first computation that kept length 1 found a second eigenvalue "about $3/b$". That was this entry, with $N_1 = 3$ L34.3.

The theorem on the shares separates into rows and columns.

**Theorem 3.11** (The rows converge). PROVEN L34.3
Fix $j, j' \geq 0$. For every base $b \geq b_0(j')$, the constant of Theorem 3.7,

$$|M_b(b-j, b-j') - s_{j'}| \leq \frac{(j+1)(j'+1)+1}{b} + \frac{j'+2}{\left(\lfloor\sqrt{b}\rfloor\right)!}.$$

So for each fixed $J$, the $J \times J$ corner of $M_b$ at the longest lengths tends to the matrix whose rows all equal $(s_0, ..., s_{J-1})$.

*Proof sketch.* This is the proof of Theorem 3.7 with the row and the column kept apart. The column $b-j'$ decides which event counts: on the event $H$ the length $b-j'$ depends only on the top $j'+1$ counts. The row $b-j$ enters only through the coupling of Lemma 3.6, applied to the top $K = j'+1$ counts of a string of $b-j$ digits, which costs $(1+K+Kj)/b$. Under the Poisson law the chance of the event is $s_{j'}$, whatever $j$ is. □

**Corollary 3.12** (One law for every row). PROVEN L34.3
For each fixed $j$, the law of the length of $F(n)$, for $n$ a uniform string of $b-j$ digits, tends to the law $(s_0, s_1, ...)$ at the distances $0, 1, ...$ from $b$. The shares $s_j$ of Theorem 3.7 are the diagonal of this limit, and they add to 1 because the limit is a probability law.

In the language of Section A.5, the corner of $M_b$ is close to a matrix with equal rows, whose trace is 1. That is the reason behind "about one solution per base" (Remark 3.4): the length of $F(n)$ barely depends on the length of $n$.

How far is the trace from 1 at a finite base? A string with $b-j$ digits has on average $(b-j-1)/b$ copies of each value among its free digits, a little fewer than 1, and fewer for shorter strings. So its top counts behave like Poisson variables with mean $\mu \approx 1-(j+1)/b$ in place of 1. Write $s_{j'}(\mu)$ for the share of the length $b-j'$ when the top counts are independent Poisson($\mu$) variables, and $\sigma_{j'} = -s'_{j'}(1)$ for minus its derivative at $\mu = 1$.

**Proposition 3.13** (The first correction to the trace). PROVEN L34.3
For fixed $j$ and $j'$,

$$M_b(b-j, b-j') = s_{j'} + \frac{(j+1)\sigma_{j'} + a_{j'}}{b} + O\left(\frac{1}{b^2}\right),$$

where $a_{j'}$ does not depend on $j$. Both $\sigma_{j'}$ and $a_{j'}$ sum to 0 over $j'$. For every fixed $J$, the diagonal satisfies

$$\sum_{j<J} M_b(b-j, b-j) = \sum_{j<J} s_j + \frac{1}{b}\sum_{j<J} c_j + O_J\left(\frac{1}{b^2}\right), \quad c_j = (j+1)\sigma_j + a_j.$$

The sum of all the $c_j$ is

$$C = \sum_j c_j = -\frac{d}{d\mu} E_\mu[J'] \,|_{\mu=1} = \frac{e}{(e-1)^2} + \frac{1}{2e} + 2.2 \cdot 10^{-5} = 1.104636,$$

where $J'$ is the distance of the length of $F$ from $b$ when the top counts are independent Poisson($\mu$) variables.

*Proof sketch.* The term $(j+1)\sigma_{j'}/b$ is the first-order change of $s_{j'}(\mu)$ when $\mu$ moves from 1 to $1-(j+1)/b$. The term $a_{j'}/b$ collects the leading digit, the step from binomial to Poisson counts and the correction of Lemma 3.5, none of which depends on $j$. Since $\sum_{j'} s_{j'}(\mu) = 1$ for every $\mu$, the

$\sigma_{j'}$ sum to 0, and since the rows of $M_b$ sum to 1, so do the $a_{j'}$. Summing the diagonal over all $j$ gives $\sum_j\big((j+1)\sigma_j + a_j\big) = \sum_j j\sigma_j = -d/d\mu \sum_j j s_j(\mu)$, minus the derivative of the mean of $J'$. For the closed form, the distance $y_0$ of the largest digit from $b$ has $P(y_0 = y) = e^{-\mu(y-1)}(1-e^{-\mu})$, so $E_\mu[y_0] = 1/(1-e^{-\mu})$, whose derivative at 1 is $-e/(e-1)^2$. The length is one more than $b - y_0$ when the largest digit has more than $e^{y_0}$ copies, and this adds the small terms $1/(2e)$ (from $y_0 = 1$) and $2.2 \cdot 10^{-5}$. □

The proposition is proved for each fixed $J$. Letting $J$ grow with $b$ needs a bound on $E_{b-j} - s_j - c_j/b$ that is uniform in $j$, which is open (Q141). The tail bound in the proof of Corollary 3.8 is not small enough. A computation closes the gap. It also shows that the whole first-order excess is a single eigenvalue.

**Heuristic 3.14** (One eigenvalue carries the excess). HEURISTIC L34.3

To first order the corner of $M_b$ is $\mathbf{1}u^T + (1/b)v\sigma^T$ with $u = s + a/b$ and $v_j = j+1$. By Section A.5 its eigenvalues are 1, $(\sigma \cdot v)/b = C/b$ and zeros. So $\operatorname{tr} M_b - E_1 = 1 + C/b + o(1/b)$, and the excess over 1 is one eigenvalue, $\lambda_2 \approx C/b$.

**Computation 3.15** (Trace and second eigenvalue). COMPUTATION L34.3

The exact length matrix (both conventions agree to 6 digits). The last column is $b\lambda_2$ for the matrix restricted to the lengths $2, \ldots, b$:

| $b$ | $\operatorname{tr} M_b$ | $E_1$ | $b(\operatorname{tr} M_b - E_1 - 1)$ | $b\lambda_2$ |
|---|---|---|---|---|
| 10 | 1.5417 | 0.2222 | 3.195 | 2.30 |
| 20 | 1.2676 | 0.1053 | 3.246 | 1.52 |
| 40 | 1.1236 | 0.0769 | 1.869 | 1.34 |
| 100 | 1.0448 | 0.0303 | 1.446 | 1.20 |
| 200 | 1.0214 | 0.0151 | 1.260 | 1.14 |
| 300 | 1.0175 | 0.0134 | 1.229 | 1.14 |

Both columns approach $C = 1.105$. At small bases the shorter lengths add more, which is why the excess at base 20 is about 3 times $C/b$. The constant has a plain meaning: it measures how much the length of $F$ follows the length of the string. A string with one digit fewer has fewer large digits, so its $F$ tends to be a little shorter too, and the diagonal gains. With length 1 included, the trace is $1 + (N_1(b)b/(b-1) + C)/b + o(1/b)$, so $b(\operatorname{tr} M_b - 1)$ grows like $\ln b / \ln\ln b$ and has no limit.

## 3.3 The digit-sum identity and the parity effect

The basic model forgets one fact. Every integer satisfies $n \equiv$ (digit sum of $n$) $(\bmod b-1)$, because $b \equiv 1$. For a solution this becomes the identity of Proposition 1.9: the multiset must satisfy $\sum_x c_x(w(x) - x) \equiv 0 \pmod{b-1}$.

**Definition 3.16** (Refined model). L5.3

The digits of $S(c)$ are drawn uniformly among the $k$-digit strings whose digit sum is congruent to $S(c)$ modulo $b-1$.

A multiset that meets the identity becomes $b-1$ times as likely to be a solution, and a multiset that fails it becomes impossible. On average nothing changes, unless the identity holds more often than one time in $b-1$. It does, because of the prime 2.

**Heuristic 3.17** (Parity factors). HEURISTIC L5.3 L17.3

In the large-base limit the refined model multiplies the expected count of the basic model by these factors:

| | even base | odd base |
|---|---|---|
| $0^0 = 0$ | 1 | 2 |
| $0^0 = 1$ | 1 | $1 + e^{-2} \approx 1.135$ |

More generally, with $w(0) = m$, the odd-base factor is 2 for even $m$ and $1 + e^{-2}$ for odd $m$.

*Proof sketch.* For every $x \geq 1$, $x^x \equiv x \pmod 2$. In an odd base $b - 1$ is even. Under $0^0 = 0$ every term $w(x) - x$ is then even, so the identity holds modulo the 2-part of $b - 1$ for every multiset. Modulo the odd part it holds with the usual chance. So the share of multisets that meet the identity is twice the random share, and the count doubles. Under $0^0 = 1$ a zero contributes $1 - 0 = 1$, so the identity modulo 2 says that the number of zeros is even. The zero count is close to Poisson(1), and $P(\text{Poisson}(1) \text{ even}) = (1 + e^{-2})/2$. Twice that is the factor. For an odd prime $p$ dividing $b - 1$, the residues of $x^x - x$ modulo $p$ are spread, and a sum of about $b$ of them is close to uniform, so the factor is 1. In an even base $b - 1$ is odd and nothing changes. □

So a large odd base expects two solutions under $0^0 = 0$ and about 1.135 under $0^0 = 1$, and a large even base expects one under each. Averaged over all bases, the model predicts 1.5 solutions per base under $0^0 = 0$ and about 1.07 under $0^0 = 1$ L7.4.

The factor 1 for odd primes is the one assumption of this section that is not obvious. It says that the residues of $x^x$ modulo the odd prime powers in $b - 1$ spread evenly over the multisets whose weight has the right length. Section 3.7 calls it Assumption R. It reduces to one exponential sum per base, which a computation bounds for every base up to 6000; beyond that it is open (Section 3.7.2). A weaker form, enough for the Poisson law in the limit, holds in every base (Section 3.7.3). For 2-digit narcissistic numbers the doubling of odd bases is a theorem (Proposition 7.40).

### 3.3.1 The refined count as a sum over characters

The refined model has a trace form too, and it shows where the parity factors come from. Put $q = b - 1$ and use the additive characters $e(t) = e^{2\pi i t}$ of Section 3.7.2. For a residue $t$ modulo $q$ let

$$\tau_t(b) = \sum_{d=2}^{b} E\left[\mathbf{1}\{F(n) \text{ has } d \text{ digits}\} e\left(t\frac{F(n) - n}{q}\right)\right],$$

with $n$ a uniform $d$-digit string in each term.

**Proposition 3.18** (A sum over characters). PROVEN L34.3

The expected count of the refined model over the lengths 2 to $b$ is $\sum_{t \bmod q} \tau_t(b)$. The term $t = 0$ is the basic count $\operatorname{tr} M_b - E_1$.

*Proof.* The $d$-digit numbers are $qb^{d-1}$ consecutive integers, so $n$ is exactly uniform modulo $q$. Since $n$ is congruent to its digit sum, a multiset meets the identity of Proposition 1.9 exactly when $F(n) \equiv n \pmod q$, and the refined model makes such a multiset $q$ times as likely. So the refined count of length $d$ is $qP(F(n)$ has $d$ digits and $F(n) \equiv n)$. The indicator of $F(n) - n \equiv 0$ is $(1/q)\sum_t e(t(F(n) - n)/q)$. Take expectations and sum over $d$. □

The term $t = 0$ counts every multiset with the right length, as the basic model does. The other terms correct for the residue. For odd $b$ the character with $t = q/2$ is $e(m/2) = (-1)^m$, a parity, and there the prime 2 acts exactly.

**Corollary 3.19** (Two closed classes). PROVEN L34.3 L15.1
For odd $b$ under $0^0 = 0$, $\tau_{q/2}(b) = \tau_0(b)$ exactly. The matrix of the classes (length, residue modulo $q$) splits into two closed blocks, the even and the odd numbers, so its eigenvalue 1 has multiplicity 2. The refined count is $2\tau_0$ plus the other characters. This is the only case in which a character other than $t = 0$ is constant on $F(n) - n$.

*Proof.* For odd $b$, $n$ is congruent to its digit sum modulo 2, and $x^x \equiv x \pmod 2$ for $x \geq 1$. So under $0^0 = 0$ the number $F(n) - n$ is always even, and $(-1)^{F(n)-n} = 1$. The last sentence is the theorem of the library on invariant residues L15.1. □

**Theorem 3.20** (The character of order 2 under $0^0 = 1$). PROVEN L34.3
For odd $b$ under $0^0 = 1$, $\tau_{q/2}(b) \to e^{-2}$ as $b \to \infty$.

*Proof sketch.* Now $F(n) - n \equiv z(n) \pmod 2$, where $z(n)$ is the number of zeros, so $\tau_{q/2} = \sum_d E\big[\mathbf{1}\{F \text{ has } d \text{ digits}\}(-1)^{z(n)}\big]$. For $d = b - j$ with $j$ fixed, the length event depends only on the top counts (Theorem 3.7). The coupling of Lemma 3.6, with the count of zeros as one more coordinate, makes all these counts independent Poisson(1) variables up to an error $O(1/b)$. Under that law the zeros are independent of the length, and $E[(-1)^Z] = e^{-2}$ for $Z$ Poisson(1). So the term of length $b - j$ tends to $e^{-2}s_j$. The tail bound of Corollary 3.8 handles the other lengths, and $\sum_j s_j = 1$. □

So the factor $1 + e^{-2}$ of Heuristic 3.17 is, in the model, a theorem about one character. The factor 2 is Corollary 3.19, and an even base has no character of order 2. What is left is the claim that the remaining characters contribute nothing in the limit.

**Conjecture 3.21** (The other characters vanish). CONJECTURE L34.3
$\sum_{t \notin \{0, q/2\}} \tau_t(b) \to 0$ as $b \to \infty$. Then the refined count tends to 1 in even bases, to 2 in odd bases under $0^0 = 0$ and to $1 + e^{-2}$ in odd bases under $0^0 = 1$.

This needs the sums $\sum(x^x - x)$ to be equidistributed modulo $q$ with a small relative error, a strong form of Assumption R. The anti-concentration of Section 3.7.3 gives only an upper bound for each residue, which is not enough.

**Computation 3.22** (The characters at finite bases). COMPUTATION L34.3
The refined count over the lengths 2 to $b$, split by characters:

| $b$ | $0^0$ | refined | $\tau_0$ | $\tau_{q/2}$ | other characters |
|---|---|---|---|---|---|
| 13 | 0 | 2.892 | 1.265 | 1.265 | +0.362 |
| 13 | 1 | 1.731 | 1.265 | 0.267 | +0.200 |
| 25 | 0 | 2.404 | 1.107 | 1.107 | +0.191 |
| 31 | 1 | 1.228 | 1.071 | 0.172 | −0.015 |
| 63 | 1 | 1.166 | 1.026 | 0.149 | −0.009 |
| 64 | 0 | 1.087 | 1.025 | | +0.062 |
| 100 | 0 | 1.014 | 1.015 | | −0.001 |

Over the bases 10 to 64 the other characters add −0.23 to +0.36, and at most 0.11 in absolute value for $25 < b \leq 128$. Under $0^0 = 1$ the ratio $\tau_{q/2}/\tau_0$ is 0.211 at $b = 13$ and 0.145 at $b = 63$, close to $(1 - 2/b)^{b-2}$, which tends to $e^{-2} = 0.135$.

The largest contributions of the other characters come at $b = 13$, 11, 25, 22, 29, 14, 37 and 43, with no simple rule yet (Q142).

### 3.3.2 The model against the data

**Computation 3.23** (Totals in bases 10 to 38). COMPUTATION L5.4

Nontrivial solutions in bases 10 to 38, summed over all lengths:

| convention | bases | basic model | refined model | observed |
|---|---|---|---|---|
| $0^0 = 0$ | even | 17.2 | 17.3 | 22 |
| $0^0 = 0$ | odd | 16.0 | 32.5 | 38 |
| $0^0 = 1$ | even | 17.2 | 17.4 | 26 |
| $0^0 = 1$ | odd | 16.0 | 19.1 | 26 |

The refined model explains the large difference between odd and even bases under $0^0 = 0$: the basic model predicts the same for both, the data show nearly twice as many in odd bases (38 against 22), and the refined model predicts a factor of 1.88. The remaining surplus sits in small bases, which contain the algebraic families of Chapter 2. For bases 20 to 38 under $0^0 = 1$ the basic model expects 20.6 and 26 occur, 1.2 standard deviations.

The model was built before bases 39 to 41 were searched, so these bases are a fair test. The model expected 5.19 solutions under $0^0 = 0$ and 3.38 under $0^0 = 1$; the search found 6 and 4 L5.7. All six new solutions have $b - 1$ digits, the most likely length.

### 3.3.3 Predictions

**Heuristic 3.24** (Base 100). HEURISTIC L5.6 L6.6

Base 100 is even, so no parity factor applies. The model expects 1.01 nontrivial solutions per convention: 0.55 with 99 digits, 0.24 with 98, 0.09 with 97, 0.08 with 100 and 0.06 with fewer.

The search has excluded every base-100 solution with at most 50 digits (Section 4.9). The model gives such solutions a total chance of about 0.002, so this is no test. A search of base 100 itself is out of reach (Section 4.2).

## 3.4 Testing the model by changing the weights

A model built to fit 40 bases could fit them by accident. The strongest tests change the weights and ask whether the model follows. The engine of Chapter 4 accepts other weight tables (Section 4.1), so the same exact search runs in worlds that the model describes differently.

### 3.4.1 Random weights keep the count and lose the parity effect

With `--perturb`, each weight $x^x$ with $x \geq 2$ gets a random extra amount of up to a quarter of its size. With `--keep-residues` the extra amount is a multiple of $2(b-1)$, so the residues of the weights stay the same.

**Computation 3.25** (Perturbed weights). COMPUTATION L9.6

Solutions with at least $b - 6$ digits in bases 16 to 28, three random seeds per base:

| weights | $0^0 = 0$ | $0^0 = 1$ |
|---|---|---|
| refined model | 20.1 | 14.9 |
| true weights | 16 | 16 |
| random perturbation | 13.7 | 13.3 |
| perturbation that keeps residues | 19.6 | 14.9 |

Random weights lose the extra solutions of odd bases under $0^0 = 0$, and the total falls to the value of the basic model. Weights that keep their residues bring the effect back, at exactly the size of the refined model. So the residues of $x^x$ cause the parity effect, and nothing else in the arithmetic of

$x^x$ matters for the counts. The perturbed worlds also reproduce the emptiness law of Section 3.8: in 240 runs on even bases 16 to 27, 47 were empty against 44.8 predicted, and in 240 runs on odd bases, 16 against 20.6 L9.7.

### 3.4.2 Forced residues: a factor of seventeen

The sharpest test forces the residues. The option `--set-residue r` raises each weight $w(x)$, $x \geq 1$, by less than $b-1$, so that $w(x) \equiv x + r \pmod{b-1}$. Let the digit 0 weigh $z$.

**Proposition 3.26** (Forced residues). PROVEN L17.2

Suppose $w(x) \equiv x + r \pmod{b-1}$ for every digit $x \geq 1$ and $w(0) = z$. A $k$-digit solution with $c_0$ zeros satisfies $r(k - c_0) + zc_0 \equiv 0 \pmod{b-1}$. In particular:

| $r$ | $z$ | condition on a solution | allowed lengths |
|---|---|---|---|
| 0 | 0 | none | all |
| 0 | 1 | $c_0 \equiv 0 \pmod{b-1}$ | all, with no zero (or $b-1$ zeros) |
| 1 | 1 | $k \equiv 0 \pmod{b-1}$ | only $k = b-1$ |
| 1 | 0 | $k - c_0 \equiv 0 \pmod{b-1}$ | $k = b-1$ without zeros, or $k = b$ with one zero |

*Proof.* This is Proposition 1.9 with $w(x) - x \equiv r$ for nonzero digits and $w(0) - 0 = z$. □

With $r = 0$ and $z = 0$ the identity holds for every multiset, so the refined model predicts $b-1$ times as many solutions as for random residues: in bases 12 to 24, about 17 times as many. With $r = 1$ and $z = 1$ only solutions with exactly $b-1$ digits survive, and the model predicts $(b-1)$ times the 55.2% share of that length.

**Computation 3.27** (Forced residues in bases 12 to 24). COMPUTATION L17.2

Bases 12 to 24, seeds 1 to 8 (104 runs per weight set, $\sum(b-1) = 1768$), solutions with at least $b-6$ digits, against the refined model computed with the exact weights of each run:

| weights | convention | observed | model | per $b-1$ | limit |
|---|---|---|---|---|---|
| `--perturb` | $0^0 = 0$ | 98 | 111.8 | | |
| `--perturb` | $0^0 = 1$ | 109 | 111.8 | | |
| `--set-residue 0` | $0^0 = 0$ | 1955 | 1891.4 | 1.106 | 1.000 |
| `--set-residue 0` | $0^0 = 1$ | 845 | 789.8 | 0.478 | 0.368 |
| `--set-residue 1` | $0^0 = 1$ | 1030 | 984.7 | 0.583 | 0.552 |
| `--set-residue 1` | $0^0 = 0$ | 454 | 453.1 | 0.257 | 0.233 |

Under $0^0 = 0$, forcing $w(x) \equiv x$ multiplies the count by $1955/98 = 19.9$; the model predicts $1891.4/111.8 = 16.9$, and the mean of $b-1$ is 17.0. With $w(x) \equiv x + 1$ under $0^0 = 1$, all 1030 solutions have exactly $b-1$ digits, as the identity demands. All four forced cases lie within 7% of the model. A seventeen-fold change in the count, predicted to a few percent by a model with no fitted parameter, is the best evidence that the model captures the mechanism.

Two details differ from a Poisson law, and both are structural. The counts per run scatter twice as much as Poisson counts, because forced residues fix the small weights and create families: in base 21 with `--set-residue 1`, $w(2) - w(1) = w(3) - w(2) = b$, so a digit 1, 2 or 3 in the second-to-last place gives three solutions at once. And short solutions become four times too common under `--set-residue 0`, for an algebraic reason: writing $w(x) = x + k_x(b-1)$, a 2-digit string $(a, c)$ is a solution exactly when $a = k_a + k_c$, and $k_1 = 0$, $k_2 = 1$ for every $b \geq 4$, so $(1, 2)$ and $(2, 2)$ are solutions in every base and for every seed L17.2.

### 3.4.3 The value of $0^0$ as a free parameter

Let the digit 0 weigh $m$ under the second convention, for $m$ from $-2$ to 3. For $m$ other than 0 and 1 there are no twins and no self-paying families, so these worlds test the model without the

families. Heuristic 3.17 predicts an odd-base factor of 2 for even $m$ and $1 + e^{-2}$ for odd $m$; at finite bases the model gives 2.00 to 2.02 and 1.19 to 1.22.

**Computation 3.28** (The convention $0^0 = m$). COMPUTATION L17.3

Bases 10 to 30, solutions with at least one zero, pooled by the parity of $m$:

| $m$ | bases | observed | model | $P$(at least observed) |
|---|---|---|---|---|
| even $(-2, 0, 2)$ | even | 16 | 21.7 | 0.91 |
| even $(-2, 0, 2)$ | odd | 42 | 39.5 | 0.37 |
| odd $(-1, 1, 3)$ | even | 28 | 21.7 | 0.11 |
| odd $(-1, 1, 3)$ | odd | 11 | 10.9 | 0.53 |

The parity factor follows the parity of $m$, as predicted. Zero-free solutions are set aside in this table because they are solutions for every $m$ at once and would be counted six times. There are 24 of them in these bases, against 16.5 expected L17.3.

## 3.5 Near-solutions and the defect spectrum

A base with one solution gives one data point. The defect spectrum turns each base into hundreds of data points.

**Definition 3.29** (Defect spectrum). L11.1

For a base $b$ and an integer $D$, let $N_D(b)$ be the number of integers $n$ with at least two digits and $n - F(n) = D$. The numbers counted by $N_D(b)$ with $D \neq 0$ are *near-solutions* with offset $D$.

The search finds $N_D(b)$ exactly by starting the interval of Section 4.1 at $D$ in place of 0. If the model is right, $D = 0$ is not special. An offset $D$ in the residue class of 0 modulo $2(b-1)$ keeps the digit-sum identity and the parity, so it is another draw of the same coincidence, with the true weights $x^x$ and all their arithmetic. The mean of $N_D(b)$ over such offsets estimates the expected number of solutions of base $b$ without any model. Call it $\lambda_m(b)$.

Short numbers need care, because $n = F(n) + D$ needs $n > D$. So the comparison counts only numbers with at least $b - 6$ digits, where every offset below $10^6$ sees the same lengths.

**Computation 3.30** (Solutions against their near-solutions). COMPUTATION L11.1

Bases 12 to 32, offsets $D = j(b-1)$ with even $j$ up to 120, numbers with at least $b - 6$ digits:

| group | solutions ($D = 0$) | predicted by the offsets |
|---|---|---|
| $0^0 = 0$, even bases | 13 | 10.9 |
| $0^0 = 0$, odd bases | 20 | 22.2 |
| $0^0 = 1$, even bases | 15 | 12.8 |
| $0^0 = 1$, odd bases | 15 | 14.0 |

Within a base the counts $N_D$ at single offsets have variance-to-mean ratio 0.98 (median over 42 cases). Across bases, $\sum (N_0 - \lambda)^2/\lambda = 44.6$ over 42 cases with the clustered calibration of Section 3.6 ($P = 0.36$; 48.8 with the naive one).

So solutions are ordinary draws. The count at a single offset is Poisson. Across bases the number of solutions scatters around the base's own mean as a Poisson draw would. One caution, which Section 3.6 makes precise: the offsets are not independent of each other, so the mean $\lambda$ is less precise than 60 independent offsets would make it. Odd bases under $0^0 = 0$ have twice as many near-solutions as even ones, at every offset in the class: the parity effect, now seen without the model. A base does not "know" whether it has a solution. The empty base 19 has an ordinary

spectrum, with $\lambda = 1.02$ under $0^0 = 1$ and 1.35 under $0^0 = 0$, inside the range 0.88 to 2.17 of the other bases.

### 3.5.1 How well the model predicts a single base

The ratio $\lambda/$ model averages 0.98 under $0^0 = 0$ and 1.12 under $0^0 = 1$ over bases 12 to 32. So the model is right on average. Its spread from base to base was first measured at about 20%, twice the sampling error of the offsets L11.2a. With the clustered variance of Section 3.6 about half of that excess disappears; it was an artefact of treating the offsets as independent. Part of the rest comes from the forced top digits. The top digits of $F(c)$ are fixed by the counts of the largest digits, and those top digits must also be in the multiset. The model treats them as random. A model that reads the top $t$ digits off $F$ and draws only the rest follows the bases more closely.

**Computation 3.31** (Forced top digits and the scatter). COMPUTATION L23.5 L26.4

Bases 12 to 32, lengths $b-6$ to $b$, 400,000 sampled multisets per base and length. Chi-square of $\lambda/$ model on 21 bases, with the naive variance of the offset means and with the clustered one:

| top digits forced, $t$ | $0^0 = 0$: naive | clustered | $0^0 = 1$: naive | clustered |
|---|---|---|---|---|
| 0 (refined model) | 97.0 | 41.4 ($P = 0.005$) | 95.7 | 43.3 ($P = 0.003$) |
| 3 | 68.2 | 30.2 ($P = 0.09$) | 75.7 | 34.7 ($P = 0.03$) |
| 4 | 59.6 | 28.0 ($P = 0.14$) | 72.4 | 34.5 ($P = 0.03$) |

With the clustered variance and the top three or four digits forced, the remaining scatter under $0^0 = 0$ is noise, and under $0^0 = 1$ it is marginal. Two items remain. Under $0^0 = 1$ the model is 12% low on average in both parities, at 3.0 standard deviations; Section 3.5.2 finds that this does not sit in the zeros and is about 2 standard deviations once the scatter between worlds is counted. Under $0^0 = 0$ base 19 has 0.67 times the model's rate of near-solutions; this looked like 4.6 standard errors with the naive variance and is 2.9 with the clustered one, one case among 42, so it is no longer a separate anomaly. Base 13 is high because its lengths are short and local factors of the kind in Section 5.5 raise the count. Section 3.11 collects the verdicts.

### 3.5.2 Where the 12% excess sits

The library knows two corrections to the coincidence model: the residues modulo $b-1$ and the forced top digits. The 12% excess under $0^0 = 1$ could be a third one. Two exact facts help to locate it.

**Lemma 3.32** (Zeros and the two conventions). PROVEN L34.1

1. A string without the digit 0 has the same $F$ under both conventions.
2. $F_1(n) = F_0(n) + z(n)$. So $n$ is a near-solution at offset $D$ under $0^0 = 1$ with $z$ zeros exactly when it is a near-solution at offset $D + z$ under $0^0 = 0$ with $z$ zeros.

*Proof.* A zero weighs $m$, so the two sums differ by $z(n)$. And $n - F_1(n) = D$ is the same as $n - F_0(n) = D + z(n)$. □

So the zero-free near-solutions are the same events under both conventions, and so is their expectation in the model. The library computes the refined model exactly, split by the number of zeros. It walks over the digit values from $b-1$ down, stops as soon as the length of $F$ is fixed, and handles the residue with the characters of Section 3.3.1 (Proposition 34.1 of L34.1). This replaces the sampling of earlier chapters, and it agrees with it within the sampling error.

**Computation 3.33** (The excess by number of zeros). COMPUTATION L34.1

Bases 12 to 26, offsets $2j(b-1)$ with $j = 1, ..., 60$, numbers with at least $b-6$ digits, model with the top 3 digits forced. Ratio of near-solutions to the model, by the number of zeros of the string:

| bases | $0^0$ | zero-free | 1 zero | 2 zeros | 3 or more | all |
|---|---|---|---|---|---|---|
| even | 0 | 0.99 | 0.93 | 0.80 | 0.90 | 0.93 |
| even | 1 | 1.00 | 1.30 | 1.07 | 0.39 | 1.09 |
| odd | 0 | 1.26 | 1.06 | 0.79 | 1.01 | 1.10 |
| odd | 1 | 1.26 | none | 0.94 | 0.64 | 1.16 |

The cells 1.26 have $z = +3.4$ and the cell 1.30 has $z = +2.9$, with the clustered variance of Section 3.6.

The excess is two different things. In odd bases it is a surplus of zero-free near-solutions, the same events under both conventions. Under $0^0 = 1$ the parity rule removes the strings with an odd number of zeros, so 72% of the expectation is zero-free, against 42% under $0^0 = 0$. The same surplus therefore lifts the $0^0 = 1$ ratio more, to 1.16 against 1.10. Base 13, whose lengths are short, carries almost half of it. In even bases the zero-free class fits, and the gap between the conventions sits in one cell, the strings with one zero under $0^0 = 1$. By the lemma these are the near-solutions of $0^0 = 0$ at the offsets $2j(b-1)+1$. A second search at the offsets $2j(b-1)+1$ and $+2$ shows cells of the same size elsewhere, in other zero classes and under both conventions. So the one-zero cell is one cell of a general scatter, not a property of the zeros.

Perturbed worlds give a control. Replace each weight $w(x)$, $x \geq 2$, by $w(x)$ plus a random amount below $w(x)/4$. This keeps the lengths and destroys every residue structure. Then compare the near-solutions of that world with the exact refined model of the same weights.

**Computation 3.34** (The two conventions in perturbed worlds). COMPUTATION L34.1

Five perturbed worlds, bases 12 to 26. The gap between the pooled ratios under $0^0 = 1$ and $0^0 = 0$ is $-0.039$, $-0.022$, $-0.021$, $-0.032$ and $-0.039$, with mean $-0.031 \pm 0.022$. With the true weights it is $+0.09 \pm 0.04$. The odd zero-free cell takes the values 0.79 to 1.23 across the five worlds, against 1.26 for the true weights. With the top digits forced, one cell of the table above varies between worlds with a standard deviation of 0.08 to 0.09, while its clustered sampling error is 0.065.

So the clustered variance of Section 3.6 understates how much a whole world can deviate from the model, by a factor of about 1.3 to 1.5 in standard deviation. The cause of this extra variance is open (Q139).

**Heuristic 3.35** (No third correction). HEURISTIC L34.1

The 12% excess under $0^0 = 1$ is not a correction tied to the zeros.

1. In odd bases it is a surplus of zero-free near-solutions, common to both conventions, which weighs more under $0^0 = 1$.
2. In even bases it is one cell, and the shifted offsets show cells of the same size in other places.
3. Perturbed worlds show no gap between the conventions, so the model treats the zeros correctly. The gap of the true weights is 2.5 standard errors above theirs, so a small effect of $x^x$ is not excluded.
4. With the extra variance between worlds, the $z = 3.0$ of the excess becomes about 2.0 to 2.3.

Fresh odd bases would decide whether the zero-free surplus is real (Q140).

### 3.5.3 The zeros of the long solutions

The long solutions under $0^0 = 0$ seemed to have too few zero digits (Q71). The exact model gives the law of the number of zeros of a solution of given base and length, with the residue condition in place.

**Computation 3.36** (Zero digits of the long solutions). COMPUTATION L34.2

Solutions with at least $b - 6$ digits in bases 12 to 43, and the zeros expected by the exact model:

| $0^0$ | bases | solutions | zeros | model |
|---|---|---|---|---|
| 0 | even | 19 | 10 | $16.7 \pm 4.0$ |
| 0 | odd | 30 | 19 | $26.4 \pm 5.0$ |
| 1 | even | 21 | 17 | $18.6 \pm 4.2$ |
| 1 | odd | 25 | 18 | $15.7 \pm 5.0$ |

Under $0^0 = 0$ that is 29 zeros against $43.1 \pm 6.4$, $z = -2.2$. Forcing the top three digits raises the expected zeros by 5% in even bases and lowers them by 1% in odd bases, so it does not close the gap.

**Computation 3.37** (Zero digits of the near-solutions). COMPUTATION L34.2

Bases 12 to 26, the offsets of Computation 3.33. Mean number of zeros per near-solution, against the model:

| $0^0$ | bases | observed | model | $z$ |
|---|---|---|---|---|
| 0 | even | 0.823 | 0.840 | −0.33 |
| 0 | odd | 0.720 | 0.849 | −3.11 |
| 1 | even | 0.842 | 0.842 | −0.09 |
| 1 | odd | 0.451 | 0.594 | −2.62 |

The near-solutions supply thousands of digits. In even bases they have exactly the zero rate of the model, so the even-base shortfall of the solutions (10 against 16.7) is chance. In odd bases they are short of zeros, but the zero classes of Computation 3.33 show why: there are too many zero-free near-solutions, not too few zeros in the others. This is the surplus found above, seen from the other side. So the zeros do not follow a law of their own.[2]

### 3.5.4 Why the integer digits are almost uniform

In the function-field world of Section 7.8 the model fails, because Frobenius makes the digits of $F$ strongly structured: whole positions are always zero, and zeros are 4 to 700 times more common than the model assumes. One can ask whether the integer digits of $F$ hide a similar structure. They do not, apart from the top.

**Computation 3.38** (Zero digits of $S(c)$ in the integers). COMPUTATION L24.6

For 100,000 random $(b-1)$-digit strings $c$ per base with a $(b-1)$-digit weight, in bases 11 to 41 under $0^0 = 1$: the top two or three digits of $S(c)$ are almost never zero, or zero far more often than $1/b$. Below them, the rate of zero digits is $1/b$ within 3%, at the top and at the bottom positions alike.

The forced top digits move the integer predictions by a few percent only: they change the chance that the zero counts of $c$ and $S(c)$ match by less than 2%, raise the parity factor under $0^0 = 1$ by 2% to 4%, and lower the total by about 1% through the $b$-digit share L24.6.

[2] Library section 24.6.2 says that the forced top digits account for 2 to 4 of the 13.5 missing zeros. The exact model does not support this; see L34.2.

There is a second lesson in the comparison. In $\mathbb{F}_q[T]$ the near-solution check of the library first agreed with the census, but only because adding $(B-1)g$ to a solution there gives a near-solution with the same class sums: the offsets mapped solutions to near-solutions one to one, so they were not independent draws. The library first argued that this cannot happen in the integers, because changing digits changes $F$ by $w(d_0-j)-w(d_0)+w(d_1+j)-w(d_1)\neq 0$. That argument overlooks rearrangements. Swapping two digits keeps the multiset, so it keeps $F$, and it moves $n$ by a multiple of $b-1$. So a multiset can be a near-solution at several offsets at once, and two thirds of the integer near-solutions share their multiset with another offset (Section 3.6). The integer offsets are therefore weaker evidence than independent draws, but they are not copies of the solutions as in $\mathbb{F}_q[T]$, and the test of Computation 3.30 stands with wider error bars.[3]

**Remark 3.39**.
Some solutions contain a block that pays for itself, such as $(3,3,6,6,1)$ in base 13, where $13\cdot P_{(3,3,6,6)}(13)=3^3+3^3+6^6+6^6$ (Section 2.8). In the model a solution ends in such a self-paying tail with probability $T_m(b)/b^m$, where $T_m(b)$ counts the $m$-digit tails that pay for themselves. Under $0^0=1$, $T_1(b)=1$ (the digit 1) and $T_2(b)$ is the number of 2-digit solutions, usually 0. So the block route is a share of about $1/b$ of the solutions, already counted. In bases 2 to 43 under $0^0=1$, 2 of the 59 solutions with at least 4 digits have an exact split at the last digit, against 3.38 expected L22.5.

### 3.5.5 Near-solutions exist in every base

The model puts many near-solutions with $|n-F(n)|$ of size $b$ in every base. Nothing of this is proved. But a weaker statement is: every large base has an $n$ whose sum $F(n)$ agrees with $n$ in about $\sqrt{b/e}$ top digits. It is the first existence statement of the library that is not an identity family (Q100).

**Theorem 3.40** (Near-solutions in every base). PROVEN L35.5
Let $b\geq 3$ and $T\geq 1$ satisfy $T+5\leq b\big(1-e^{-1+1/b}\big)$, $2(T+4)\leq b$ and

$$1+T\left(e(T+4)\frac{b}{b-T-5}+1\right)+\frac{eb}{b-2}+1+T\leq b-2.$$

Then under each convention some $n$ with $b-1$ digits has $0<|n-F(n)|<b^{b-1-T}$, and $n\geq (b/e-1)b^{b-2}$.

*Proof sketch.* Start with one digit $b-1$; its weight $(b-1)^{b-1}\approx b^{b-1}/e$ has $b-1$ digits and a top digit near $b/e$. Then steer the digits 2 to $T+1$ of $F$ one by one. If digit $j$ lies in a forbidden zone of $z\leq T+4$ values near $b-1$, add copies of the largest digit whose weight is at most $(b-z-1)b^{b-1-j}$. One copy moves digit $j$ by less than one unit, or by a jump that clears the zone, so after at most about $eT$ copies it leaves the zone, and the carry into the earlier digits is at most 1. The conditions keep the number of copies below the length. Finally add the matched top digits $a_1,\dots,a_T$ of $F$ to the multiset and fill with zeros. Their weights are too small to change the top $T$ digits, so the arrangement that starts with $a_1,\dots,a_T$ agrees with its own $F$ there. If it is an exact solution, swapping the digit $b-1$ with a zero gives a near-solution L35.5. □

**Corollary 3.41** (How close). PROVEN L35.5
Let $T(b)$ be the largest $T$ allowed. It does not decrease with $b$, and $T(b)=\sqrt{b/e}-O(1)$: it is 1 at $b=27$, 3 at 100, 16 at 1000 and 603 at $10^6$. So every base $b\geq 27$ has a near-solution with $|n-F(n)|<nb^{-t}$, $t=T(b)-\log_b(eb/(b-e))$, and $t\geq 2$ in every base $b\geq 77$.

[3]Library section 24.6.3 calls an integer near-solution "a genuine second draw". Section 26.4 corrects this.

The thresholds 27 and 77 are computed and extend to every larger base because the conditions only get easier as $b$ grows. The proof never needs the base-$b$ digits of the real numbers that the top digits of $F$ approximate, which question Q100 feared. The limit $\sqrt{b}$ comes from the zone that the matched digits must avoid.

**Computation 3.42** (The construction, run). COMPUTATION L35.5

The construction in exact integers, with steering only when a digit falls into its zone, stopped at the first $T$ that does not fit:

| $b$ | $T(b)$ | largest $T$ reached | copies used | $t$ reached |
|---|---|---|---|---|
| 30 | 1 | 15 | 11 | 14.76 |
| 100 | 3 | 24 | 47 | 23.95 |
| 200 | 6 | 47 | 76 | 47.22 |
| 500 | 11 | 134 | 361 | 134.38 |
| 1000 | 16 | 127 | 583 | 127.01 |
| 1500 | 20 | 255 | 1229 | 255.17 |

Every base from 10 to 200 reaches $T \geq 0.17b$, and the sampled bases 300 to 1500 reach $0.13b$ to $0.27b$.

Most digits never fall into their zone, so the construction does much better than the bound. Whether $t \geq cb$ can be proved is open (Problem 8.12).

## 3.6 Near-solutions as evidence

Almost every "chance" verdict in this document rests on near-solutions: a count at $D = 0$ is compared with its mean over many offsets in the class of 0. The library first treated the offsets as independent draws. They are not, and the reason is elementary.

**Lemma 3.43** (Swap lemma). PROVEN L26.1 LEAN

Let $n$ and $n'$ have the same digit multiset in base $b$, both without a leading zero. Then $F(n') = F(n)$ and

$$n' - F(n') = (n - F(n)) + (n' - n), \quad n' - n \equiv 0 \pmod{b-1}.$$

If $n'$ is $n$ with the digits $x$ at position $i$ and $y$ at position $j > i$ exchanged, then $n' - n = (x - y)(b^j - b^i) = (x - y)b^i(b-1)R_{j-i}$, where $R_s = 1 + b + \ldots + b^{s-1}$. For the last two digits this is $(x - y)(b - 1)$.

Lean: `swap_lemma, swap_lemma_perm, swap_last_two`

*Proof.* $F$ depends only on the multiset. Every power of $b$ is 1 modulo $b - 1$, so a number is congruent to its digit sum, and the two digit sums agree. For the transposition only positions $i$ and $j$ change. □

So the near-solutions of one multiset form a *cluster*: a multiset $c$ is a near-solution at every offset $D = n' - S(c)$ for which some arrangement $n'$ of $c$ puts $D$ in the offset set. A single coincidence, a weight $S(c)$ close to one of its own arrangements, is counted at several offsets. The same holds in every positional world where $F$ depends only on the multiset: negative bases, factorions, the factorial system and the Pisot base L26.1.

**Proposition 3.44** (What the clusters change). PROVEN L26.1

Assume that distinct multisets are independent, as the model does. Then the count $N_D$ at one offset is still Poisson, up to clumps such as twins at that offset, and the mean $\lambda$ of the counts over $J$ offsets is still unbiased. But the total $T$ over the offsets has variance $\varphi ET$ with the *design effect* $\varphi = E[K^2]/E[K]$, where $K$ is the number of offsets at which a near-solution's

multiset appears. So $J$ offsets are worth only about $J/\varphi$ independent ones. Finally $N_0$ and $T$ are positively correlated, because a solution and its rearrangements lie in one cluster.

The design effect is the familiar quantity of cluster sampling (Appendix A): a survey that interviews whole households learns less than one that interviews the same number of unrelated people.

**Computation 3.45** (Design effects). COMPUTATION L26.2 L26.3

Integer bases 12 to 26, offsets $2j(b-1)$, $j = 1, ..., 60$, numbers with at least $b-6$ digits. Between 43% and 88% of the near-solutions share their multiset with another offset. The design effect is 1.5 to 3.5 (pooled 2.48 under $0^0 = 0$ and 2.14 under $0^0 = 1$), so 60 offsets are worth 26 to 29 independent ones. The partner pairs sit at the moves of Lemma 3.43: 43% at a swap of the last two digits, 20% at positions 0 and 2, 18% at positions 1 and 2. Of the 52 solutions in these bases, 29 have a rearrangement that is a near-solution in the window. A simple count of the expected partners under uniform bottom digits predicts the design effect within a few percent under $0^0 = 1$. Factorions, negative bases, near-2-cycles and the Pisot bases have design effects of 1.6 to 6.4; offset families with steps divisible by many small numbers have none.

A better offset family avoids the problem. Choose offsets $D = k(b-1)$ such that no two differ by the move of a single transposition. This *transposition-free* family cuts the shared share to about 15%, keeps the mean within 4%, and is worth about 51 of 60 offsets. For a single count $N_0$ against its offsets, the right P value mixes the Poisson law over the uncertainty of $\lambda$, with the variance inflated by $\varphi$ and with the clumps at one offset (twins, factorion triples) built in L26.3.

### 3.6.1 Every verdict revisited

The library recomputed every P value that rests on offsets. Table 3.3 lists the main ones. The pattern is clear. Verdicts about $D = 0$ barely move, because the Poisson noise of a single count of 3 to 60 dominates the calibration error. What moves are statements about the precision of $\lambda$ itself: per-base comparisons with the model and comparisons between offset families.

Table 3.3. Verdicts that rest on near-solutions, with the naive and the clustered calibration L26.4.

| statement | old | new | verdict |
|---|---|---|---|
| integers, 4 groups of Computation 3.30 | $P$ 0.30 to 0.43 | 0.31 to 0.43 | unchanged |
| empty base 19, union mean 1.9 | 0.15 | 0.15 | unchanged |
| no $b$-digit solution under $0^0 = 1$, 0 against 1.30 | 0.27 | 0.27 | unchanged |
| per-base scatter, chi-square on 21 bases | 97, 96 | 41, 43 | half an artefact |
| base 19 near-solutions, $0^0 = 0$ | $z = -4.6$ | $z = -2.9$ | one case of 42 |
| model 12% low under $0^0 = 1$ | $z = 4.4$ | $z = 3.0$ | weaker, survives |
| odd-base factorions, per family | 0.003 to 0.02 | 0.004 to 0.025 | unchanged |
| factorions, combined with fresh bases | about 0.05 | about 0.09 | weaker |
| factorions, after the choice of 16 groups | about 0.1 | 0.13 | unchanged |
| 5-digit excess, bases 13 to 99, window calibration | 0.0047 | 0.0055 | unchanged |
| the same after the choice of 10 cells | 0.015 | 0.018 | unchanged |
| negative bases, last digit kept, 42 against 52 | 0.09 | 0.11 | unchanged |
| negative multiples of 3, 10 against 17.3 | 0.044 | 0.056 | unchanged |
| step-$b$ offsets against step offsets, negative bases | about 4 SD | 2.9 SD | weaker |
| 2-cycles in even bases, 2 against 5.25 | 0.10 | 0.11 | unchanged |

### 3.6.2 A pooled test across worlds

Each world of Chapter 7 with a near-solution calibration gives one comparison of $D = 0$ with its offsets. Pooling them tests whether $D = 0$ is special anywhere.

**Computation 3.46** (Eight worlds pooled). COMPUTATION L26.5

Integers, odd-base and even-base factorions, negative bases, rational bases, the Pisot bases, Eisenstein bases and 2-cycles: 284 solutions against 297 to 304 expected from their offsets. The worlds disagree more than chance allows: the heterogeneity statistic is 17.2 to 23.3 on 7 degrees of freedom ($P = 0.002$ to 0.016), depending on the Pisot calibration. Two worlds drive it, the odd-base factorions (ratio 0.33) and the Pisot bases (ratio 1.34 to 1.56). Without them the other six fit one common ratio of 0.91, with $P = 0.60$ for heterogeneity.

So outside two named worlds there is no sign that $D = 0$ is special. The odd-base factorions are the shortfall found after looking at 16 groups (Section 3.11). The Pisot excess is mostly a calibration fault: for even $x$ the offsets $jx$ mix two classes modulo $2x$, and the defect spectrum peaks near $D = 0$; against a local calibration in the right class, 25 solutions against 14.8 expected has $P \approx 0.05$ (Section 7.5) L26.6.

### 3.6.3 A ledger of P values

Each verdict above was judged on its own. A ledger judges them together (Section A.9). The library lists 241 statements of its chapters 3 to 32 that compare data with the model. For each one it records the test, the P value, the data used, and whether the hypothesis was named before the data were seen. Of the 221 statements that are still current, 133 have a P value. Tests that share data are merged into 56 groups, each with the smallest P value of the group after the Sidak correction.

**Computation 3.47** (The P values of the library). COMPUTATION L34.4

| groups | number | below 0.05 | expected | median | Fisher $P$ |
|---|---|---|---|---|---|
| all | 56 | 14 | 2.8 | 0.23 | $2 \cdot 10^{-9}$ |
| named in advance | 12 | 0 | 0.6 | 0.59 | 0.74 |
| all others | 44 | 14 | 2.2 | 0.18 | $3 \cdot 10^{-11}$ |
| without $\mathbb{F}_q[T]$ and the factorial system | 40 | 8 | 2.0 | 0.24 | $2 \cdot 10^{-4}$ |
| integers only | 9 | 3 | 0.45 | 0.23 | 0.005 |

The 14 groups below 0.05 are the census of $\mathbb{F}_q[T]$ (6 groups), the empty bases $\equiv 3 \pmod 8$, the negative-base offsets that keep the last digit, the 6-digit narcissistic numbers in even bases, the 5-digit excess, the odd-base factorions, the zeros of the long solutions, the $(b-1)$-digit excess, the Pisot class and the 12% excess.

Every one of the 14 is already named in this document as an anomaly, a pattern chosen after the data, or a known failure of the model.

**Heuristic 3.48** (Selection, not a hidden flaw). HEURISTIC L34.4

The P values of the library are not uniform, and the cause is selection.

1. The 12 groups named in advance (fresh bases, the forced and shifted weights, fresh factorion bases, base 18 for 2-cycles) look like uniform draws: none below 0.1, Kolmogorov-Smirnov $P = 0.56$. Twelve groups rule out only gross flaws, such as a model that is 30% off in every world.
2. If each group were the smallest of $k$ looks, 14 of 56 below 0.05 would need $k$ about 6, and the median 0.23 would need $k$ about 3. The lower tail is heavier than any single $k$ gives: a few groups are real failures ($\mathbb{F}_q[T]$, the factorial system), and the rest look like modest selection.
3. There is no lean toward too few solutions. Among the 25 groups with $P < 0.2$, 17 are excesses and 8 deficits.

4. Several groups share data, for example the two conventions of one base. So the Fisher values overstate the evidence. The conclusion does not rest on them.

So the "chance" verdicts cannot all be closed at once by a test of uniformity, because most tests were chosen after their data. The tests named in advance agree with the model. More of them, written down before fresh bases are searched, are the cheapest way to strengthen the case (Q104, Q143).

### 3.6.4 Are neighbouring bases independent?

The model treats different bases as independent. The second Borel-Cantelli lemma behind Conjecture 3.72 needs this, and so does every variance of a count pooled over bases. But the weights $S(c)$ are the same numbers in every base, and only their digits change. A multiset with largest digit $X$ is a candidate in every base above $X$.

**Computation 3.49** (Neighbouring bases). COMPUTATION L34.5

For bases 12 to 32, the residual $(\lambda - \text{model})/\text{ se}$ of each base, with the mean of each parity removed. Correlation of neighbouring bases, with a P value from 20,000 permutations within each parity:

| model | $0^0$ | lag-1 correlation | $P$ |
|---|---|---|---|
| refined | 0 | −0.31 | 0.060 |
| refined | 1 | −0.41 | 0.067 |
| top 3 digits forced | 0 | −0.09 | 0.65 |
| top 3 digits forced | 1 | −0.17 | 0.46 |

The two conventions of one base correlate at +0.85 (refined) and +0.78 (top digits forced). No near-solution multiset of bases 12 to 26 is a near-solution in a second base, against 0.02 expected. The forced top digits of one multiset in bases $b$ and $b+1$ have correlation between −0.033 and +0.044.

The negative correlation of the refined model disappears once the top digits are forced: it was the top-digit factor, which changes smoothly with $b$. With 20 pairs the test excludes only correlations above about 0.4. The strong correlation of the two conventions is Lemma 3.32 again: the zero-free near-solutions count under both. A pooled test must not add the two conventions as independent counts. The library's pooled test of Computation 3.46 did, which makes the $z$ of its integer row too large by a factor of about 1.3; its conclusion stands L34.5.

**Heuristic 3.50** (Independence across bases). HEURISTIC L34.5

For the second Borel-Cantelli lemma it is enough, by a theorem of Kochen and Stone, that the chances of solutions in two different bases $b, b'$ add up to at most $(1 + o(1))$ times the square of the sum of the single chances. In the model the only link between two bases is a multiset that is a candidate in both. A single multiset of $d$ digits is a solution with chance at most about $d!/b^{d-1}$, of order $b^{3/2}e^{-b}$ for $d = b$, so the link is negligible. Conjecture 3.72 needs no more independence than the model already has, and the data show none.

### 3.6.5 Do the low digits of $F$ mix?

One can ask the model's assumption directly. Draw random strings $c$ whose weight has the right length, remove from $c$ the top three digits of $F(c)$, and ask whether the low digits of $F(c)$ have exactly the remaining copies of each value, compared with random low digits.

**Computation 3.51** (A mixing statistic). COMPUTATION L26.8
The low digits of $F(c)$ agree with $c$ at the rate of random digits within 1.6% per value in the integers (base 20), in base $-20$, for factorions and in the factorial system, and within 3.7% in the Pisot base $x = 6$. $\mathbb{F}_5[T]$ is the one world with a clear excess, 2.2% to 3.4% per value.

The statistic does not follow the model's error. It flags $\mathbb{F}_q[T]$ with the same sign under both conventions, where the model error changes sign, and it misses the factorial failure, which sits in the top digits. Both places where the model fails have one narrow cause, the top digits in the factorial system and the zero count in $\mathbb{F}_q[T]$ under $0^0 = 1$, and an average over all digit values dilutes both. The negative answer still confirms the picture of this chapter: apart from the top, the integer digits of $F$ look random.

## 3.7 A Poisson limit theorem

The data say that the number of solutions in a base looks like a Poisson draw. In the model this can be proved, with the dependencies between solutions built in. The proof uses the Chen-Stein idea in its simplest form; Appendix A has the background.

### 3.7.1 The coupled model

To state a joint law for both conventions, the model must say how the events depend on each other. Split a multiset $c$ into its counts $c_0$, $c_1$ of zeros and ones and its upper part $u = (c_2, ..., c_{b-1})$. Put $V(u) = \sum_{x \geq 2} c_x x^x$. Then $F_m(c) = V(u) + c_1 + mc_0$.

**Definition 3.52** (Coupled model). L13.2
For each upper part $u$, let $Y_u$ be a random integer with as many digits as $V(u)$: uniform in the basic version, and uniform on the integers of that length congruent to $V(u)$ modulo $b - 1$ in the refined version. The $Y_u$ are independent. The multiset $c$ is a model solution under convention $m$ if the digits of $Y_u + c_1 + mc_0$ form $c$.

The model replaces the large number $V(u)$ by a random number of the same length and, in the refined version, the same residue. It keeps the exact additions of the small digits. So three dependencies are exact: twins under $0^0 = 0$, the shared zero-free solutions, and the common residue of all completions of one upper part.

Fix a minimal length $\ell \geq 6$. An index $\alpha$ is a multiset with $\ell \leq k \leq b$ and at least two digits above 1, together with a convention, where the two conventions are one index when $c_0 = 0$. The condition on the upper part leaves out the self-paying families $(x, 0, ..., 0, 1)$, which have tiny mass in the model. The *cluster* of an index is the set of indices with the same upper part. Indices in different clusters are independent.

**Lemma 3.53** (Twins are the only coupling). PROVEN L13.2
If two distinct indices in one cluster are both model solutions, then both belong to convention 0 (or are shared), and one string is the other plus 1, with last digits 0 and 1.

*Proof sketch.* The two strings are $s = Y_u + j$ and $s' = Y_u + j'$ with $|j - j'| \leq b$, and they have the same upper part. A length change would force many digits $b - 1$ into one of them and change the upper part, so the lengths agree. Then compare digit sums: adding $\delta > 0$ with $\kappa$ carries changes the digit sum by $\mathrm{ds}(\delta) - \kappa(b - 1)$, and $\mathrm{ds}(\delta) \leq \delta$ with equality only for $\delta < b$. If both are of convention 0, the digit sum changes by exactly the change in the number of ones, which forces $\delta = \pm 1$ and no carry: a twin pair. If one is of convention 1 with $c_0' \geq 1$ zeros, the same computation makes $c_0'$

a positive multiple of $b-1$, so the string has at most one nonzero digit, which is excluded. Two type-1 indices have $\delta = 0$. □

**Theorem 3.54** (Poisson approximation). PROVEN L13.2

Let $\lambda_s$, $\lambda_0$, $\lambda_1$ be the expected numbers of shared, convention-0-only and convention-1-only model solutions, and let $Z_s$, $Z_0$, $Z_1$ be independent Poisson variables with these means. Then

$$d_{\mathrm{TV}}((N_0, N_1), (Z_s + Z_0, Z_s + Z_1)) \le 2b_1 + b_2,$$

where $b_1 = \sum_u \sigma_u^2$ is the sum over clusters of the squared expected cluster counts, and $b_2 = 2E[T]$ with $T$ the number of model solutions under convention 0 that end in 0. For $\ell = \lceil b/2 \rceil$ and $b \ge 12$, the basic coupled model gives

$$d_{\mathrm{TV}} \le 14.2/b + 16b^2(b+1)\ell!/b^\ell = O(1/b).$$

The refined model gives the same with $14.2\kappa/b$ in the first term and an extra factor $b-1$ in the second term, under Assumption R below. For $12 \le b \le 6000$ the refined case holds without the assumption (Corollary 3.58). In every base it holds without the assumption with a weaker error term, which still tends to 0 (Theorem 3.64).

*Proof sketch.* Compare the family of indicators $X_\alpha$ with independent Poisson variables $Z_\alpha$ of the same means; the counts $(N_0, N_1)$ are sums of these. Clusters are independent in both families, and total variation is subadditive over independent blocks, so it suffices to bound one cluster. Inside a cluster, compare $X$ first with independent Bernoulli variables $B_\alpha$, then $B$ with $Z$. Le Cam's inequality gives $d_{\mathrm{TV}}(B, Z) \le \sum p_\alpha^2 \le \sigma_u^2$. For $X$ against $B$, split the probabilities by the number of ones in the outcome vector: the outcomes with at least two ones have probability at most $P(K \ge 2) + \sigma_u^2/2$, where $K$ is the number of solutions in the cluster, and the outcomes with zero or one ones differ by at most $E[K(K-1)] + \sigma_u^2$ in total. Summing over clusters, $\sum E[K(K-1)]$ is the expected number of ordered pairs of solutions in one cluster, and by Lemma 3.53 these are the twin pairs, so the sum is $2E[T]$. The error terms come from two estimates. A cluster has at most $2(b+1)$ relevant shifts $j$, and each has probability at most $2D!/b^{D-1}$ to produce the upper part, which gives $\sigma_u \le 4b(b+1)\ell!/b^\ell$. And a convention-0 solution ending in 0 has its last digit fixed, which costs a factor $1/b$, while the chance that the other digits fit is summable over the lengths; this gives $E[T] \le 7.1/b$. □

**Heuristic 3.55** (Assumption R). HEURISTIC L13.2 L29.3

There is a constant $\kappa$ such that for every length $k \ge b/2$ and every residue $r$, among uniform strings of $k-1$ digits whose weight has $k$ digits, the share with $\sum c_x(x^x - x) \equiv r \pmod{b-1}$ is at most $\kappa/(b-1)$.

Assumption R is about the residues of $x^x$, not about the model's randomness. It is the same assumption that gives the factor 1 for odd primes in Heuristic 3.17. In odd bases the prime 2 forces $\kappa \ge 2$. The proof of the theorem needs it only for $k \ge b/2$. The library first stated it for every length, and that version is false: in base 10 the only one-digit string whose weight has 2 digits is the digit 3, since $3^3 = 27$, so one residue class modulo 9 gets the whole share. For a fixed short length and a large base the same happens, because few digits are small enough.

### 3.7.2 Assumption R reduced to one sum

Write $m = b-1$ and $g(x) = x^x - x$ for $x \ge 1$, with $g(0) = 0$. A residue condition modulo $m$ can be written with the additive characters $e(t) = e^{2\pi i t}$. For an integer $S$,

$$\frac{1}{m} \sum_{h \bmod m} e\left(h\frac{S-r}{m}\right) = \begin{cases} 1 \text{ if } S \equiv r \pmod m \\ 0 \text{ otherwise,} \end{cases}$$

because the sum of all $m$-th roots of unity vanishes. Take expectations: $P(S \equiv r) \le (1/m)\sum_h |Ee(hS/m)|$. When $S$ is a sum of independent digits, $Ee(hS/m)$ is a product with one factor per digit. So Assumption R follows once these factors are bounded away from 1 in modulus.

A few $h$ always give modulus 1. Every $g(x)$ is even and $g(2) = 2$, so the set $H_0$ of $h$ with $hg(x) \equiv 0$ for all $x$ is $\{0, m/2\}$ for even $m$ and $\{0\}$ for odd $m$. These $h$ are the prime 2, the source of the parity factor. For the other $h$ and a digit range of length $L$ put

$$\psi_L(h) = \frac{1}{L}\sum_{x<L} e\left(h\frac{g(x)}{m}\right), \quad \rho(b) = \max_{\lceil b/2\rceil \le L \le b, h\notin H_0} |\psi_L(h)| \ .$$

**Theorem 3.56** (Assumption R from one number). PROVEN L29.3
Let $b \ge 12$ and $\ell = \lceil b/2\rceil$. The refined case of Theorem 3.54 holds without Assumption R, with

$$\kappa(b) = |H_0| + (b-1)\rho(b)^{\ell-2}.$$

So it suffices that $1 - \rho(b) \ge 3\ln b/b$.

*Proof sketch.* The proof of Theorem 3.54 needs the residue only on the event that the weight of the top $k-1$ digits has $k$ digits, with $k = b - j \ge \ell$. On that event every digit is below $L = b - \theta j$ for a constant $\theta \ge 0.71$, because a larger digit $x$ has $x^x \ge b^k$. Replace the length event by the larger event "every digit is below $L$". On it the digits are independent and uniform on $\{0, ..., L-1\}$, the leading one on $\{1, ..., L-1\}$. So each nonleading digit contributes the factor $\psi_L(h)$, and the characters give at most $(1/m)\big(|H_0| + m\rho(b)^{k-2}\big)$ for the share of one residue. The probabilities of the larger events still sum to at most 7.1, as in the original proof. □

So the conditioning on the length does no harm, and Assumption R becomes a statement about one number per base.

**Computation 3.57** (The number $\rho(b)$). COMPUTATION L29.3
For every base $b$ from 12 to 6000, every $L$ from $\lceil b/2\rceil$ to $b$ and every $h$ modulo $b-1$:

| bases | largest $\rho(b)$ | where | $\kappa(b) - \lvert H_0\rvert$ at most |
|---|---|---|---|
| 12 to 19 | 0.8197 | $b = 13, 16$, $h = m/3$ | 4.6 |
| 20 to 49 | 0.7954 | $b = 22$ | 2.7 |
| 50 to 99 | 0.7807 | $b = 52$ | 0.134 |
| 100 to 999 | 0.7725 | $b = 100$, $h = 33$ | 0.00041 |
| 1000 to 6000 | 0.7646 | $b = 1000$, $h = 333$ | $10^{-55}$ |

**Corollary 3.58** (The refined Poisson theorem up to base 6000). PROVEN L29.3
For $12 \le b \le 6000$ the refined case of Theorem 3.54 holds unconditionally, with $\kappa \le |H_0| + 5$, and with $\kappa \le |H_0| + 0.0005$ for $100 \le b \le 6000$. So $\kappa$ is 1 in even bases and 2 in odd bases, up to $5 \cdot 10^{-4}$: the factors of Heuristic 3.17.

From base 100 on, the worst $h$ is $m/3$ or $m/6$ in every base with $3 \mid b-1$, and $\rho(b) \le 0.537$ in all other bases. The prime 3 is the reason.

**Proposition 3.59** (The 3-adic limit). PROVEN L29.3 L15.4
If $3 \mid b-1$ and $h = m/3$, then $\psi_L(h) \to 5/6 + (1/6)e(2/3)$ as $L \to \infty$. The limit has modulus $\sqrt{7/12} = 0.7638$.

*Proof.* Modulo 3, $g(x)$ is 2 for $x \equiv 2 \pmod 6$ and 0 otherwise. A share $1/6 + O(1/L)$ of the $x < L$ has $x \equiv 2 \pmod 6$, and each of these contributes $e(2/3)$; the others contribute 1. □

So the computed $\rho(b)$ approaches 0.7638 from above, which supports the following conjecture.

**Conjecture 3.60** ($\rho(b)$ for large bases). CONJECTURE L29.3
$\rho(b) \le 0.77$ for every $b \ge 100$. For $b-1=p$ prime, $\max_{h\neq 0}|\sum_{x\le p} e(h(x^x-x)/p)| = O(\sqrt{p\log p})$.

For $b-1=p$ prime with $1000 \le p \le 20,000$, the largest sum lies between 0.84 and 1.40 times $\sqrt{p\ln p}$, which is the size of the largest of $p$ sums of $p$ random unit vectors. For small prime factors of $b-1$ there is a proof, which uses that $x^x$ modulo $p$ depends only on $x$ modulo $p(p-1)$.

**Proposition 3.61** (Small primes spread). PROVEN L29.3
Let $p$ be an odd prime dividing $b-1$, let $3p \le L \le b$ and let $p \nmid h$. Then

$$\left|\frac{1}{L}\sum_{x<L} e\left(h\frac{g(x)}{p}\right)\right| \le \frac{3(1+\tau(p-1))}{\sqrt{p}} + \frac{p^{3/2}(1+\ln p)}{L},$$

where $\tau$ counts divisors. So $g$ is close to uniform modulo every prime $p \le b^{2/3-\varepsilon}$ that divides $b-1$.

*Proof sketch.* Write $x = u + kp$. Since $p \equiv 1 \pmod{p-1}$, $x^x \equiv u^{u+k} \pmod p$. For fixed $u$ the sum over $k$ runs along the powers of $u$, a subgroup of $\mathbb{F}_p^*$. Sums of $e(ay/p)$ over a subgroup, twisted by a character, are at most $\sqrt{p}$, because they expand into Gauss sums of modulus $\sqrt{p}$. Grouping the $u$ by their order gives the bound. □

This leaves out the joint residue modulo all of $b-1$ and the case of a prime factor $p > b^{2/3}$; at most one prime factor can be that large. The literature has bounds for the number of solutions of $x^x \equiv \lambda \pmod p$ [10, 11], and these bound the sums $\sum_{x<p} e(ax^x/p)$ on average over $a$. No bound below $p$ for a single $a$ seems to be known, and that is what $\rho(b)$ needs. The needed statement is weaker than $o(p)$: the sum must stay below $p - 3\ln p$ for every $a$ and every initial segment of length at least $p/2$ (Q112).

### 3.7.3 Assumption R in the limit: anti-concentration

The Poisson limit needs less than Assumption R. Let $\rho^*(b)$ be the largest share of one residue class modulo $b-1$ among the strings that the proof of Theorem 3.54 uses, over all lengths $k \ge b/2$. The proof gives $b_2 \le 14.2\rho^*(b)$ in the refined model. Assumption R asks for $\rho^*(b) \le \kappa/(b-1)$, which gives the rate $O(1/b)$. For a limit theorem it is enough that $\rho^*(b) \to 0$, at any rate. This is an *anti-concentration* statement: a sum of independent random steps should not sit on one value with large probability. The Littlewood-Offord problem asks exactly this (Section A.6). The weights $x^x$ give a clean case, because the small digits have exact integer weights that grow very fast.

**Lemma 3.62** (Small digits are dissociated). PROVEN L30.1
Let $g(x) = x^x - x$. For every $z \ge 3$, $g(z) \ge 2zg(z-1)$. Let $2 \le z_0 \le y$ and $n \le 2z_0 + 1$. If two multisets of $n$ digits from $\{z_0, ..., y\}$ have the same sum of $g$, they are equal. If moreover $ng(y) < Q$, the same holds for the sums modulo $Q$.

*Proof.* Since $(z/(z-1))^{z-1} \ge 2$, we have $z^z \ge 2z(z-1)^{z-1}$, and the first claim follows. For the second, suppose the multisets differ, and let $w$ be the largest digit with different counts, $c_w > c'_w$. Remove the common digits above $w$. If $w = z_0$, the sums differ by $(c_w - c'_w)g(w) \neq 0$. If $w > z_0$, the first sum is at least $(c'_w + 1)g(w)$, and the second is at most $c'_w g(w) + ng(w-1)$. But $n < 2w$, so $ng(w-1) < g(w)$ by the first claim. Modulo $Q$, both sums lie in $[0, ng(y)]$, an interval shorter than $Q$, so congruent sums are equal. □

So if only a few digits fall into the window $\{z_0, ..., y\}$, their sum tells which digits they are. A fixed multiset is unlikely, so the sum is unlikely to hit a given residue. Chernoff bounds for the number of digits in the window make this quantitative.

**Theorem 3.63** (Anti-concentration for sums of $x^x - x$). PROVEN L30.1
Let $Q \geq 2$, let $y \geq 3$ satisfy $(y+3)y^y < Q$, and put $z_0 = \lceil (y+1)/2 \rceil$ and $J = y - z_0 + 1$. Let $X_1, ..., X_k$ be independent and uniform on $\{0, ..., M\}$ with $M \geq y$, and let $\mu = kJ/(M+1)$ be the mean number of them in $\{z_0, ..., y\}$. If $J \geq 11$ and $J/2 \leq \mu \leq J$, then for every residue $r$

$$P\left(\sum_i g(X_i) \equiv r \pmod{Q}\right) \leq 2e^{-J/5}.$$

*Proof sketch.* Let $N$ be the number of $X_i$ in the window. Condition on which indices these are and on all other values. If $N = n \leq 2z_0 + 1$, the sum over the window lies in $[0, Q)$, so at most one value of it qualifies, and by Lemma 3.62 at most one multiset. A fixed multiset of $n$ uniform values from $J$ has probability at most $n!/J^n$. The Chernoff bounds of Theorem A.14 cover $N < J/8$ and $N > 2z_0 + 1$, and Stirling's formula bounds $n!/J^n$ for $n$ in between. □

As $Q$ grows, the largest admissible $y$ is about $\ln Q / \ln \ln Q$, so $J$ is about $\ln Q/(2 \ln \ln Q)$.

**Theorem 3.64** (The refined Poisson law in the limit). PROVEN L30.1
Let $b \geq 12$ and $\ell = \lceil b/2 \rceil$. In the refined coupled model,

$$d_{\mathrm{TV}}((N_0, N_1), (Z_s + Z_0, Z_s + Z_1)) \leq 14.2\rho^*(b) + 16b^2(b^2-1)\ell!/b^\ell,$$

and $\rho^*(b) \leq 2\exp(-(1/10 + o(1)) \ln b / \ln \ln b)$. So the refined coupled model has the bivariate Poisson law in the limit, without Assumption R.

*Proof sketch.* As in Theorem 3.56, replace the length event for $k$ digits by the larger event that every digit is at most $M_k$, the largest digit with $M_k^{M_k} < b^k$. On it the $k-2$ digits after the leading one are independent and uniform on $\{0, ..., M_k\}$. For $k \geq b/2$ and $b \geq 12$ one checks $(M_k+1)/2 \leq k - 2 \leq M_k + 1$, so the mean $\mu$ of Theorem 3.63 lies in $[J/2, J]$. Apply that theorem with $Q = b - 1$. □

The rate is very slow (Table 3.4). The bound says nothing below $b = 10^{10}$ and little below $10^{30}$. So the two routes to Assumption R meet only in the limit. The computation of Computation 3.57 gives the strong form for bases 12 to 6000. This theorem gives a weak form for very large bases. In between only Conjecture 3.60 speaks.

Table 3.4. The bound of Theorem 3.63, optimized over the cut-off, for $Q = 10^E$ L30.1.

| $E$ | $y$ | $J$ | bound, $\mu = J/2$ | bound, $\mu = J$ |
|---|---|---|---|---|
| 10 | 9 | 5 | 0.90 | 0.89 |
| 30 | 21 | 11 | 0.090 | 0.036 |
| 100 | 56 | 28 | $5.7 \cdot 10^{-4}$ | $8.6 \cdot 10^{-6}$ |
| 300 | 138 | 69 | $9.1 \cdot 10^{-9}$ | $3.2 \cdot 10^{-13}$ |
| 1000 | 385 | 193 | $1.1 \cdot 10^{-23}$ | $2.1 \cdot 10^{-33}$ |

The theorem never looks at $x^x$ modulo a large prime, and it works for any modulus. So it also covers a composite $b - 1$, whose joint residue Proposition 3.61 leaves open. The same proof works for the weights $x^{x+t}$ and $x!$, whose ratios $g(z)/g(z-1)$ also grow. For the weights $x^d$ of narcissistic numbers the residue law is exact for a simpler reason. Modulo a prime $p$, $x^d - x$ depends only on $x \bmod p$, and the residue factor is $p$ when $(p-1) \mid (d-1)$ and 1 otherwise. For $x^x$ the residue modulo $p$ depends also on $x \bmod (p-1)$, which ties the primes of $b-1$ together. That coupling is what makes Assumption R hard.

### 3.7.4 Assumption R as a collision count

The strong form of Assumption R is needed for the rate $O(1/b)$. Already two digits give a bound by collisions.

**Proposition 3.65** (Concentration is bounded by collisions). PROVEN L30.2

Let $X_1, \dots, X_k$, $k \geq 2$, be independent and uniform on $\{0, \dots, M\}$. For every $Q \geq 2$ and every residue $r$,

$$P\left(\sum_i g(X_i) \equiv r \pmod{Q}\right) \leq N_M \frac{Q}{(M+1)^2},$$

where $N_M(Q)$ is the number of pairs $(x, y)$ with $0 \leq x, y \leq M$ and $g(x) \equiv g(y) \pmod{Q}$.

*Proof.* Condition on $X_3, \dots, X_k$. Let $\nu$ be the law of $g(X_1) \bmod Q$. The chance that $g(X_1) + g(X_2)$ hits a given class $v$ is $\sum_a \nu(a)\nu(v-a) \leq \sum_a \nu(a)^2$ by the Cauchy-Schwarz inequality, and $\sum_a \nu(a)^2 = N_M(Q)/(M+1)^2$. □

So if $N_{b-1}(b-1) \leq Cb$, Assumption R holds with $\kappa \leq 4C$, and a bound $b^{2-\delta}$ gives $\rho^*(b) \leq 4b^{-\delta}$ L30.2. For $b-1 = p$ prime this is a question about the map $x \mapsto x^x - x$ modulo $p$. The map $x \mapsto x^x$ has a literature of its own. Balog, Broughan and Shparlinski [10] and Cilleruelo and Garaev [11] bound its collisions by $p^{23/12+o(1)}$. Holden and Moree [12] predict their number, and Kurlberg, Luca and Shparlinski [8] count its fixed points. These methods write $x^x \equiv a$ as a condition on discrete logarithms. The shift by $x$ mixes addition with multiplication, so they do not transfer.

**Computation 3.66** (Collisions of $x^x$ and of $x^x - x$). COMPUTATION L30.2

For every prime $11 \leq p < 10^5$, with $1 \leq x, y \leq p-1$: the pairs with $x^x \equiv y^y$ number about $2.64(p-1)$, within 1.2% of the Holden-Moree prediction for every $p > 50,000$. The pairs $x \neq y$ with $x^x - x \equiv y^y - y$ number $1.002(p-1)$ on average for $50,000 < p < 10^5$, and between $0.981(p-1)$ and $1.052(p-1)$ for each such prime: the value for a random map. For the 426 bases $12 \leq b \leq 3000$ with $b-1$ prime, Proposition 3.65 bounds $(b-1)$ times the largest share by 2.57 to 4.18.

**Conjecture 3.67** (Shifted collisions). CONJECTURE L30.2

The number of pairs $1 \leq x, y \leq p-1$ with $x^x - x \equiv y^y - y \pmod{p}$ is $(2 + o(1))(p-1)$ as $p \to \infty$. This gives Assumption R with $\kappa \leq 8 + o(1)$ for every base with $b-1$ prime.

A weaker bound $p^{2-\delta}$ would already give a power rate (Q118). For composite $b-1$ two digits do not mix the residues modulo small primes, and the exponential sums of Section 3.7.2 are the better tool. A third route would be an inverse Littlewood-Offord theorem for steps drawn from a fixed set of residues (Q119).

### 3.7.5 The correlation of the two conventions

Under the theorem, $(N_0, N_1)$ is approximately the bivariate Poisson pair $(Z_s + Z_0, Z_s + Z_1)$. Within one base its correlation is $\rho(b) = \lambda_s / \sqrt{(\lambda_s + \lambda_0)(\lambda_s + \lambda_1)}$. In the large-base limit a model solution is zero-free with probability tending to $e^{-1}$, so with the parity factors:

**Heuristic 3.68** (Shared solutions in the limit). HEURISTIC L13.2

| | $\lambda_s$ | $\lambda_0$ | $\lambda_1$ | $EN_0$ | $EN_1$ | $\rho$ |
|---|---|---|---|---|---|---|
| even base | $e^{-1} = 0.368$ | 0.632 | 0.632 | 1 | 1 | 0.368 |
| odd base | $2e^{-1} = 0.736$ | 1.264 | 0.399 | 2 | 1.135 | 0.488 |

In odd bases zero-free strings meet the parity rule of $0^0 = 1$ automatically, so they keep the factor 2, and the shared share rises. The library's correlation of 0.56 is pooled over bases 10 to 41, so it has two sources: shared solutions within a base, and the parity of the base, which lifts both means in odd bases. Drawing the 32 pairs from the bivariate Poisson laws with each base's own means, 200,000 times, gives a mean sample correlation of 0.47, with 5% to 95% range 0.19 to 0.71. A correlation of at least 0.563 has probability 0.30. So the observed 0.56 is an ordinary draw L13.2.

## 3.8 Empty bases

A base is empty if it has no nontrivial solution under either convention. Bases 19, 27 and 35 are the only empty bases from 3 to 41, and base 43 has solutions.

**Heuristic 3.69** (The emptiness law). HEURISTIC L9.7

Let $E_0$ and $E_1$ be the expected numbers of solutions under the two conventions and $E_s$ the expected number of zero-free solutions, which count under both. If solutions occur independently, a base is empty with probability $\exp(-(E_0 + E_1 - E_s))$. For bases 10 to 41 this is 14% to 18% for even bases and 4% to 9% for odd bases; at base 40, 17% and 8%.

The law holds in the perturbed worlds of Section 3.4.[4] Odd bases are rarely empty because the parity effect doubles $E_0$. If the model is right, about one base in eight near base 40 is empty, and about one in seven in the large-base limit, so infinitely many bases are empty.

**Computation 3.70** (Empty bases in the data). COMPUTATION L9.7 L11.2

Bases 10 to 41: the model expects 2.62 empty even bases and 1.13 empty odd bases. The data have 0 and 3 (19, 27, 35). With each base's own spectrum $\lambda_m(b)$ from Section 3.5, bases 12 to 32 expect 11.6 bases with no long solution under one convention or the other and have 11; even bases have fewer than expected, odd bases more.

The chance of no empty even base among 10 to 41 is 0.057 under the model, and base 44, also even, has a solution. The per-base spectra put the deficit in perspective: the total over both parities is normal, the residue modulo $b$ is the one residue the offsets do not match, and offsets $D = 2jb(b-1)$ that also match it predict slightly fewer solutions, not more. So no mechanism tied to $D = 0$ is visible, and the even-base deficit counts as chance.

Can one prove that infinitely many bases are empty? The library argues that the tools in this document cannot L8.5. Showing that one base is empty is a finite computation. But a congruence modulo $q$ removes only a fixed share of candidates, about $1/q$ of all numbers satisfy $n \equiv F(n) \pmod q$, and almost all of those are not solutions. The constructions of Chapter 2 only show that bases have solutions. And in the model each base is empty with probability about 20% (even) or 9% (odd) in the large-base limit, independently of the others; a proof would have to control these chance events in infinitely many bases at once. The opposite statement, that every large base has a solution, is equally far out of reach, and the model says it is false.

What can be said for certain is short. A base has a solution when one of the families applies: under $0^0 = 0$, every perfect power with a self-paying digit (Proposition 2.2); under both conventions, every base of a 2-digit solution (Proposition 2.14). These families cover few bases. For example, 55 bases up to $10^9$ have a 2-digit solution under $0^0 = 0$ L9.7.

[4] The library quotes two versions of these rates. Sections 7.4 and 9.7 give 17% and 8% at base 40, with all lengths of a finite base. Section 4.5 gives 20% and 9%, the limits as $b \to \infty$ of the coupled model (Section 3.7), where the short lengths no longer add solutions. The numbers in Heuristic 3.69 are the finite-base ones.

## 3.9 Length against base

Write $J = b - k$ for a $k$-digit solution in base $b$. Proposition 1.3 gives $J \geq 0$, and Heuristic 3.3 says that $J$ is usually 1 or 2.

**Computation 3.71** (The law of $J$ in the data). COMPUTATION L9.9

Nontrivial solutions in bases 10 to 41, by $J = b - k$:

| | $J = 0$ | 1 | 2 | 3 | 4 to 9 | 12 or more |
|---|---|---|---|---|---|---|
| $0^0 = 0$ | 2 | 30 | 10 | 7 | 8 | 9 |
| $0^0 = 1$ | 0 | 30 | 8 | 6 | 6 | 6 |

The solutions with a large $J$ are short: under $0^0 = 1$ they are the short coincidences of Chapter 5, such as 3435 in base 10 with $J = 6$.

### 3.9.1 The bump beyond $\ln b$

The limit law says that the expected number with $J \geq j$ falls like $e^{-(j-1)}$. At finite $b$, Table 3.2 shows values about three times the limit once $J$ passes $\ln b$. The cause is the largest usable digit. A digit $x$ can appear in a $k$-digit solution only if $x^x < b^k$ (Proposition 1.5). Now $(b - J + 1)^{b-J+1} \approx b^{b-J+1}e^{-(J-1)}$, which drops below $b^{b-J}$ once $J > \ln b + 1$. From then on a digit above $k$ becomes possible and adds solutions; after that the decay is again about $e^{-1}$ per step. So the expected number of solutions in base $b$ with $J \geq j$ is $Ce^{-j}$ with a bounded $C$, and with $j = \ln b + a \ln\ln b$ it is about $C/(b(\ln b)^a)$.

### 3.9.2 The window

**Conjecture 3.72** (Length window). CONJECTURE L10.5

Under $0^0 = 1$ and for every $\varepsilon > 0$:

- only finitely many solutions with $k \geq 3$ digits, apart from algebraic families such as those of Theorem 2.21, lie in a base $b > k + \ln k + (1 + \varepsilon)\ln\ln k$;
- infinitely many lie in a base $b > k + \ln k + (1 - \varepsilon)\ln\ln k$.

The reason is the pair of Borel-Cantelli lemmas (Appendix A). The expected number of solutions in base $b$ with $J > \ln b + a \ln\ln b$ is about $C/(b(\ln b)^a)$. The sum over $b$ is finite for $a > 1$, so by the first lemma only finitely many bases have such a solution. It is infinite for $a \leq 1$, and in the model the bases are independent, so by the second lemma infinitely many do. That independence is an assumption of the model. Section 3.6.4 tests it and finds no dependence between neighbouring bases. This is the same law as for the longest run of heads in $n$ coin flips, which is $\log_2 n + O(\log\log n)$.

The library first stated a cruder window, $b \leq k + (1 + \varepsilon)\ln k$, and drew from it that each length has only finitely many solutions under $0^0 = 1$. The binomial families of Corollary 2.22 refute that: they give infinitely many solutions of every length $k \geq 3$, in bases far above $k$, for an algebraic reason the model cannot see. Hence the exception for algebraic families in the conjecture.

**Computation 3.73** (Exceptions to the window). COMPUTATION L10.5

Solutions under $0^0 = 1$ with $k \geq 3$ and $J > \ln b + 2$:

| region | digits | expected (model) | observed |
|---|---|---|---|
| bases 10 to 41 and 43, every length | 3 to 5 | 0.96 | 5 |
| | 6 to 10 | 0.86 | 0 |
| | 11 or more | 1.21 | 2 |
| bases 44 to 127, up to 30 digits | 3 to 5 | 0.05 | 2 |
| | 11 or more | 0.00 | 1 |

| region | digits | expected (model) | observed |
|---|---|---|---|
| bases 128 to 5000 | 3 to 5 | 0.01 | 2 |

The long exceptions agree with the model: 3 observed against about 1.2 expected, and one of the three lies in base 64, which has self-paying digits. The short exceptions do not: 9 against 1.0. That is expected, because the per-base model is the wrong tool for short strings. For a short string the base is the root of a polynomial (Proposition 1.7), and the right question is whether that root is an integer. Section 5.5 develops that heuristic. It explains part of the short solutions in the searched ranges, and it too misses the algebraic families. The 0.96 is the basic model. The exact refined model, computed at the short lengths, expects 1.35 of the 5 under $0^0 = 1$, one of which is a family member; four generic ones against 1.35 have probability 0.048 (Section 5.5.4).

For a fixed large length $k$, the model puts a $k$-digit solution in base $k$ with probability 8%, $k+1$ with 55%, $k+2$ with 23%, and so on. The expected number of bases with a $k$-digit solution is about 1, so a length has no base with probability about $e^{-1} = 37\%$ and two or more with probability about 26%. Under $0^0 = 1$, 7 of the 29 lengths from 2 to 30 have no solution in any base up to 127, which is 24% L3.4. The missing lengths are missing only near the length: the binomial families supply every length in astronomically large bases.

## 3.10 Cycles of $n \mapsto F(n)$

Solutions are the fixed points of $n \mapsto F(n)$. The other cycles of this map are a natural second test, because a random mapping has a known number of cycles of each length.

**Proposition 3.74** (Cycles live on multisets). PROVEN L15.1 LEAN

Let $\mathcal{M}_b$ be the set of digit multisets with 1 to $b$ digits and at least one nonzero digit. The map $c \mapsto$ (multiset of $S(c)$) sends $\mathcal{M}_b$ into itself, and its cycles correspond one to one to the cycles of $n \mapsto F(n)$ on the integers $n \geq 1$.

Lean: `F_lt_pow_self (partial: F maps numbers below b^b to themselves)`

*Proof.* A multiset with at most $b$ digits has $S(c) \leq b(b-1)^{b-1} < b^b$, so $S(c)$ has at most $b$ digits, and it has a nonzero digit because $S(c) \geq 1$. $F$ depends only on the multiset, so a cycle of integers gives a cycle of multisets of the same length. Two members of an integer cycle with the same multiset have the same image, and a cycle is a bijection on its members, so they are equal. Conversely, a multiset cycle gives the cycle of the integers $S(c)$. □

$|\mathcal{M}_b| = \binom{2b}{b} - b - 1$,[5] so a full census is a walk over about $4^b/\sqrt{\pi b}$ points. The engine walks each multiset once, with 2 bits of state per multiset; base 17 takes 670 seconds per convention, and base 18 about an hour L15.1, L23.3.

The parity invariant of Theorem 1.13 splits the map in odd bases under $0^0 = 0$: even numbers map to even numbers and odd ones to odd ones. A map from a finite nonempty set to itself has a cycle, so every odd base has a cycle of even numbers, and so a cycle besides the fixed point 1 L15.1.

### 3.10.1 A random mapping with the invariant

A uniform random mapping on $N$ points has about Poisson$(1/L)$ cycles of length $L$ [13]. The map $F$ is not uniform: the length of $S(c)$ and its residue modulo $b-1$ are fixed by $c$. The class-lumped model keeps exactly this information. A class is a pair $K = (k, r)$: the $k$-digit multisets with digit sum $r$ modulo $b-1$. In the model the image of $c$ is a random multiset of the class of $S(c)$, weighted by the number of strings with that multiset. Let $M$ be the matrix of class transitions.

[5] The multisets with at most $b$ digits over the values $0, \ldots, b-1$ number $\binom{2b}{b}$, counting the empty one. Removing the empty multiset and the $b$ multisets made of zeros only leaves $\binom{2b}{b} - b - 1$.

**Proposition 3.75** (Expected cycle counts). PROVEN L15.1

In the class-lumped model the expected number of cycles of length $L$ is $(1/L)$ times the sum, over $L$-tuples of distinct multisets, of the product of their transition probabilities. Up to the terms with a repeated multiset this is $(1/L)\ \mathrm{tr}(M^L)$.

The spectrum of $M$ explains the parity effect for cycles. In odd bases under $0^0 = 0$, $M$ splits into an even and an odd block, the eigenvalue 1 has multiplicity 2, and $E_L \approx 2/L$. Under $0^0 = 1$ the parity flips with the parity of the number of zeros; $E[(-1)^z] = (1 - 2/b)^k \approx e^{-2}$, so $M$ has the eigenvalues 1 and about $e^{-2}$, and $E_L \approx (1 + e^{-2L})/L$. For $L = 1$ this is the factor $1 + e^{-2}$ of Heuristic 3.17, seen from a new side.

**Computation 3.76** (Cycles in bases 3 to 18). COMPUTATION L15.1 L23.3

Cycles of length 2 in bases 4 to 18 against the class-lumped model:

| group | 2-cycles | model $E_2$ | $P(X \le \text{obs})$ | near-2-cycles per offset |
|---|---|---|---|---|
| even, $0^0 = 0$ | 2 | 4.02 | 0.24 | 5.25 |
| even, $0^0 = 1$ | 4 | 4.30 | 0.57 | 4.97 |
| odd, $0^0 = 0$ | 6 | 7.23 | 0.42 | 8.32 |
| odd, $0^0 = 1$ | 4 | 3.88 | 0.65 | 3.92 |

The total number of cycles of all lengths in bases 3 to 17 matches the model in all four groups (for example 92 against 94.4 in odd bases under $0^0 = 0$). Fixed points are in excess by a factor 1.4 to 1.8, because of the constructions and the small bases.

### 3.10.2 The image of $F$

A random mapping also predicts how many points are hit. The image of $F$ is much smaller than that prediction.

**Proposition 3.77** (Coincident sums). PROVEN L26.7 LEAN

The image of the multiset map is the same under both conventions. Under $0^0 = 0$, multisets that differ only in their zeros have the same weight; under $0^0 = 1$, multisets that differ only in the split of $c_0 + c_1$ have the same weight.

Lean: `prop26_10_one, prop26_10_two`

*Proof.* Under $0^0 = 0$ a zero weighs nothing, and under $0^0 = 1$ zeros and ones both weigh 1. So under both conventions the image is the set of multisets of $S_0(e)$ for the zero-free multisets $e$ of 1 to $b$ digits. □

**Computation 3.78** (The image share). COMPUTATION L26.7

The share of multisets with at least one preimage is 0.21 in base 10 and falls to 0.14 in base 16. The class-weighted random mapping predicts 0.38 to 0.31. Counting the distinct weights, not the multisets, closes 55% to 66% of the gap, and keeping the true top three digits of each weight closes about two thirds of the rest. The in-degrees are overdispersed: variance-to-mean 3.6 to 6.3 against 2.3 to 3.8 in the model.

So the forced top digits show up once more: the weights $S(c)$ share their leading digits far more than random integers do, and their multisets collide. The cycle counts above are not affected much, because they depend on the trace of the class matrix, not on the image.

An early observation said that every odd base from 3 to 11 has a 2-cycle under $0^0 = 0$ and no even base from 4 to 10 does. That is the parity doubling of the invariant plus chance: bases 13 and 15 have no 2-cycle, base 16 has one, base 18 has one. The deficit in even bases was tested with

near-2-cycles, $F(F(n)) = n - j(b-1)$, which keep every residue the map preserves. They come at the normal rate, 5.25 per offset against 4.02 in the model, so there is no second invariant L23.3.

## 3.11 Anomalies and how they were resolved

A model tested on many groups of data will show a few deviations by chance. The library found several, and each one taught something. Before going through them, one statistical point.

> **Remark 3.79** (The look-elsewhere effect).
> A deviation found by looking at the data is weaker evidence than its P value suggests. If you split the data into 14 groups and pick the most extreme one, a P value of 0.02 for that group is not surprising: the chance that at least one of 14 independent groups reaches 0.02 is $1 - 0.98^{14} \approx 0.25$. So the library corrects each P value for the number of groups it looked at, and it values fresh data, collected after a pattern was noticed, far above old data. A pattern that weakens in fresh data was probably chance.

**The $(b-1)$-digit excess** In bases 10 to 38 under $0^0 = 1$, solutions with $b-1$ digits were 1.6 times as common as the refined model predicts: 27 against 17.3, with $P = 0.019$ L5.5. But this was the most extreme of about 14 groups, so the corrected chance was about 23%. Correlation tests found nothing: the top 10 digits and the bottom 8 digits of $S(c)$ fit the multiset no more often than random digits do. In fresh bases 39 to 43 the excess continued weakly (5 against 2.45, $P = 0.10$). The resolution came from the perturbed weights of Section 3.4: over all lengths from $b-6$ to $b$, the true counts fit the model, 16 against 14.9 in bases 16 to 28 and 17 against 14.6 in bases 29 to 41 L9.6. The excess at $b-1$ digits is a fluctuation in how the solutions spread over the lengths; their total is normal.

**Empty bases $\equiv 3 \pmod 8$** Bases 19, 27 and 35 are all $\equiv 3 \pmod 8$, and so is 11, which has solutions. Three empty bases among the four such bases from 11 to 35 has probability about $10^{-3}$ in the model. But the residue class was noticed after the data, and among all 15 odd bases from 11 to 39 three empty ones has probability about 6%. Base 43, the next base $\equiv 3 \pmod 8$, has a solution L4.5, L8.5.

**Missing $b$-digit solutions** No base from 3 to 45 has a $b$-digit solution under $0^0 = 1$, against 3.6 expected in the refined model. This was the original conjecture of the library: no $b$-digit solutions at all. Under $0^0 = 0$ it is false, with bases 15 and 16. The defect spectrum showed that the model overestimates this length. Restricted to exactly $b$ digits, the offsets in bases 10 to 30 predict 1.30 solutions under each convention, against 2.33 and 1.74 from the model L11.2. The cause is the forced top digits. A $b$-digit sum needs at least three copies of $b-1$, its leading digit is then 1 and the next one is about $0.10b$, and these digits must also lie in the multiset, whose counts are depressed by the three copies of $b-1$. The exact correction for the top one to three digits is 0.6 to 0.91 in bases 16 to 30, and it shrinks like $1/b$. With the spectrum's calibration, no $b$-digit solution under $0^0 = 1$ in bases 10 to 30 has probability 0.27. So the model still predicts a $b$-digit solution under $0^0 = 1$ in about 8% of large bases, and the search has simply not reached one.

**The even-base empty deficit** No even base from 10 to 41 is empty, against 2.62 expected ($P = 0.057$). Section 3.8 explains why it counts as chance.

**Factorions in odd bases** With the weights $x!$, long factorions are rare in odd bases: 3 and 4 against about 11.9 per convention in bases 14 to 36. Offsets that keep every residue up to $b+1$ and modulo $b^3$ give normal near-factorion counts, so no congruence explains it. Fresh odd bases 37 to 47 give 3 per convention against 6.2. With the clustered calibration of Section 3.6 the combined evidence has $P \approx 0.09$, and 0.13 after the choice among 16 groups. The library counts it as an unexplained fluctuation (Section 7.6).

**Base 19** The empty base 19 also has few near-solutions under $0^0 = 0$, 0.67 times the model's rate. With the naive variance this was 4.6 standard errors, a real anomaly. With the clustered variance it is 2.9, one case among 42 bases, and no longer a separate anomaly L26.4.

Some deviations remain open. Each is a question in Section 8.2.

- Under $0^0 = 1$ the per-base rate of near-solutions is 12% above the refined model, in odd and even bases alike. It survives the clustered calibration at 3.0 standard deviations Q70. It is a zero-free surplus in odd bases and one cell in even bases, and perturbed worlds scatter enough to bring it to about 2 standard deviations (Heuristic 3.35).
- Base 19 is low in its near-solutions, now 2.9 standard errors Q68. The question has lost most of its basis.
- The long solutions under $0^0 = 0$ in bases 12 to 43 contain fewer zero digits than the model predicts: 29 against 43.1, about 2.2 standard deviations Q71. The forced top digits do not explain this. The near-solutions show no deficit in even bases, and in odd bases their deficit is the zero-free surplus (Section 3.5.3). So the parallel with the function-field world (Section 7.8) is weak.
- The model's per-base predictions seemed to scatter about 20% beyond sampling error. With the clustered variance and the top digits forced, the scatter is noise under $0^0 = 0$ and marginal under $0^0 = 1$ ($P = 0.03$) Q44.
- In even Pisot bases, offsets $jx$ with even $j$ carry 1.5 to 2.4 times as many near-solutions as those with odd $j$ under $0^0 = 0$. The identity behind this class factor modulo $2x$ is unknown L26.6.

Chapter 4

# Searching: algorithms and their limits

Every base has finitely many solutions (Proposition 1.3), so a computer can in principle list them all. In practice the cost grows like $1.9^b$, and the search reaches base 43 in a few hours. This chapter describes the exact search, measures its cost, shows how much a search that may miss a few solutions can save, and proves that no search of this kind can do much better. It then explains why a proof that the problem is hard is out of reach, and why short solutions, in contrast, are cheap to find in every base at once.

## 4.1 The prefix-pruned tree

### 4.1.1 Search over multisets

$F(n)$ depends only on how often each digit appears, not on their order. So the search does not visit numbers. For each length $k$ from 2 to $b$ it visits the digit counts $(c_0, \ldots, c_{b-1})$ with $\sum c_x = k$, computes the weight $S = \sum c_x w(x)$, and checks two things: $S$ has exactly $k$ digits, and the digit counts of $S$ are the count vector. If both hold, $S$ is a solution, and every solution arises this way.

There are $\binom{b+k-1}{k}$ count vectors with $k$ digits, and over all $k$ about $\binom{2b}{b} \approx 4^b/\sqrt{\pi b}$. The first search of the project, in Python, checked about $3.9^b$ of them L6.1. The search below builds the count vectors one digit value at a time and abandons a partial vector as soon as no completion can work. It visits about $1.9^b$ of them. Appendix L describes the programs.

### 4.1.2 The state of a node

The tree fixes $c_{b-1}$ first, then $c_{b-2}$, and so on down. A node at digit value $v$ has fixed the counts of all digits from $v$ to $b-1$. It stores the sum $t$ of the weights fixed so far, the number $r$ of positions still free, and the digits already known to stand at the front of $S$, the known prefix. The root for length $k$ has $t = 0$, $r = k$ and an empty prefix, and starts at the largest digit $v$ with $v^v < b^k$ (Proposition 1.5).

The free positions hold digits below $v$. So the final sum lies in an interval:

$$\text{lo} = t \le S \le \text{hi} = t + z + r \cdot (v-1)^{v-1},$$

where $z$ is the number of zeros in the known prefix, explained below. The search clips the interval to $[b^{k-1}, b^k - 1]$, the $k$-digit numbers. If nothing is left, the node fails.

### 4.1.3 The leading-digit test

All numbers in $[\text{lo}, \text{hi}]$ start with the same digits, up to the first position where lo and hi differ. $S$ lies in the interval, so it starts with those digits too, and a solution must contain them. Each new leading digit $x$ must pass one test.

- If $x \ge v$, its count is already fixed. If the prefix holds more copies of $x$ than $c_x$, the node fails.
- If $x < v$, its count is still open, and the digit uses up one free position. If no free position is left, the node fails.

A leading digit $x < v$ must be in the multiset even though $c_x$ is not fixed yet. The search counts it as placed: it adds $w(x)$ to $t$ at once. This narrows the interval, because lo grows by $w(x)$ and the

bound for hi loses a free position worth $(v-1)^{v-1}$. A narrower interval can expose more leading digits, which can force more digits, and the search repeats the test until nothing new appears. This repetition pays: a build that stops after the first round visits 1.46 to 1.47 times as many nodes in every base from 20 to 33 L31.3. A last test looks at the first unknown digit: it lies between the digits of lo and hi at that position, and if every value in that range is at least $v$ and already used up, the node fails. Knuth's program for perfect digital invariants uses the same test [7] it removes about 7% of the nodes.

At a node for digit $v$, the children try $c_v = p + e$ for $e = 0, 1, ..., r$, where $p$ is the number of copies of $v$ already in the prefix. At $v = 1$ the free positions hold ones and zeros, and the leaf tries every number of extra ones and checks the exact sum against the count vector.

**Example 4.1** (3435 in base 10).

The search for $k = 4$ starts at $v = 5$, because $5^5 = 3125 < 10^4 \leq 6^6$.

1. One copy of 5: $t = 3125$, $r = 3$, hi $= 3125 + 3 \cdot 4^4 = 3893$. The interval $[3125, 3893]$ has the leading digit 3, which is below 5, so it is forced: $t = 3152$, $r = 2$, hi $= 3152 + 2 \cdot 256 = 3664$.
2. One copy of 4: $t = 3408$, $r = 1$, hi $= 3408 + 3^3 = 3435$. The interval $[3408, 3435]$ has the leading digits 34, within the one copy of 4.
3. One more 3 beyond the forced one: $t = 3435 =$ lo $=$ hi. The new digits 3 and 5 fit the counts.
4. Digits 2 and 1 get no copies. The leaf finds $S = 3435$ with the digits $\{3, 3, 4, 5\}$: a solution under both conventions.

Two siblings in step 1 fail at once. With no 5, the interval $[0, 1024]$ clips to $[1000, 1024]$, which forces the digits 1 and 0, and then hi $= 514 < 1000$. With two 5s, $[6250, 6762]$ starts with 6, and the count of 6 is 0.

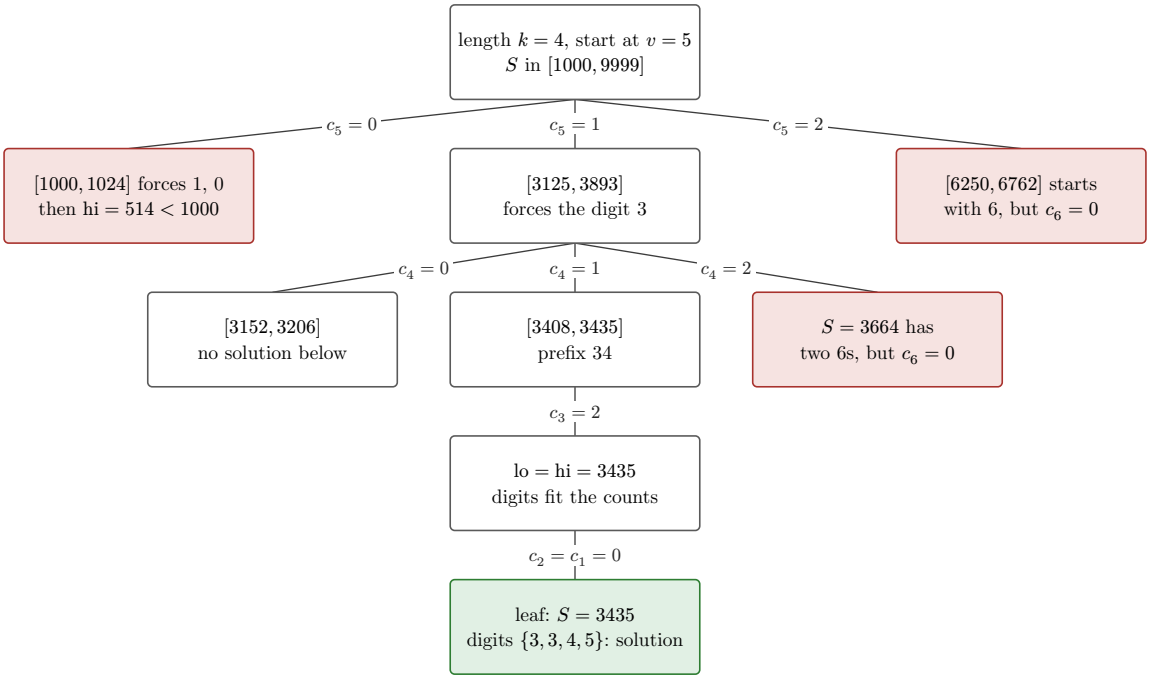



Figure 4.1. Part of the search tree for 4-digit numbers in base 10. Each node fixes one more digit count, from the largest digit down, and shows the interval for the sum $S$. Red nodes fail the leading-digit test; the green leaf is 3435. The subtree under $c_4 = 0$ is not drawn.

Figure 4.1 shows the walk. Most of the tree in a large base looks like the middle column: nodes that pass the tests for many levels because their small digits are still free.

### 4.1.4 Both conventions in one pass

The conventions differ only in what a zero adds to the sum. The interval above is wide enough for both: $\mathrm{lo} = t$ is the smallest sum under $0^0 = 0$, and hi adds 1 for each known zero, which covers $0^0 = 1$. The extra width is at most $k$, tiny next to $(v-1)^{v-1}$ except at the bottom of the tree. So one tree serves both conventions, and the leaves check each convention exactly. A combined run of base 28 visits 19,362,129 nodes, a run for one convention 19,361,945 (`docs/algorithm.md`).

**Proposition 4.2** (The search misses nothing). PROVEN

The prefix-pruned search finds every solution of every length in base $b$, under both conventions.

*Proof.* Each rule removes only count vectors that cannot be solutions. The interval contains every sum that a completion of the node can reach. Every leading digit of the interval is a digit of every reachable sum, so a solution contains it. Forcing a digit removes only completions that leave it out. The first unknown digit of every reachable sum lies in the range the last test checks. A count vector reaches a leaf only if it passed every check, and the leaf compares the exact sum with the vector. The residue rule of Section 4.1.6 removes a node only when no completion can satisfy the digit-sum identity. □

The implementation in Rust stores each number as one byte per base-$b$ digit in 256-bit vector registers, so addition in base $b$ needs no conversion and no bignum library. Carries resolve with a carry-lookahead identity on 64-bit masks, and one addition costs about 15 instructions. Two registers cover 62 digits and bases up to 127. The tree splits into about $10^5$ work units in a fixed order, which threads share by work stealing and a journal records, so long runs survive interruptions. Before a solution is printed, the program recomputes its sum with arbitrary-precision integers. The same engine accepts other weight tables, which Section 3.4 and Section 7.6 use. Appendix L has the details.

### 4.1.5 The interval is as narrow as it can be

Could a cleverer bound on the free positions shrink the interval further? Not without looking at the digits of the completions. Let $r$ be the number of free positions and $v$ the current digit value, as above, with $v \geq 2$.

**Lemma 4.3** (The interval is exact). PROVEN L31.3

Both ends of the interval are sums of completions of the node. The low end $t$ is the sum when every free position holds 0 under $0^0 = 0$. The high end $t + z + r(v-1)^{v-1}$ is the sum when every free position holds $v-1$ under $0^0 = 1$.

*Proof.* Under $0^0 = 0$ a zero adds nothing, so the first completion has sum $t$. In the second, each of the $z$ known zeros adds 1 and each free position adds $(v-1)^{v-1}$. □

So the interval is the smallest one that contains every sum the known counts allow. The forcing loop of the leading-digit test already iterates it to a fixed point. A narrower range must use more than the counts: the digits of the completions, which the levels below test one count at a time, or a residue, which the next rule uses L31.3.

### 4.1.6 The residue rule

The interval looks only at the size of $S$. The digit-sum identity (Proposition 1.9) gives a test of a different kind: the sum of $w(x) - x$ over the digits of a solution is divisible by $b-1$. The identity involves every digit, so at most nodes it says nothing: the unknown digits can make up any residue. But many nodes know almost everything.

Call a free position a *free slot*. A node with exactly one free slot knows its whole multiset except one digit $y < v$. In base 28, 43% of the nodes of the exact tree have one free slot and 25% have two L31.1. Their subtrees are tiny: a node with one free slot has 0.23 descendants on average. So the tree does most of its work in the middle levels, not at the leaves, and a test that needs all but one digit can still act on almost half of the nodes.

**Theorem 4.4** (The residue rule). PROVEN L31.2

Write $q = b - 1$ and let $R$ be the sum of $w(x) - x$ over the known digits of a node, modulo $q$, under the convention $0^0 = m$. If the node has exactly one free slot and no digit $y < v$ satisfies $R + w(y) - y \equiv 0 \pmod q$, then no completion of the node is a solution under $0^0 = m$.

*Proof.* A completion is the known multiset plus one digit $y < v$. By Proposition 1.9 a solution has $R + w(y) - y \equiv 0 \pmod q$. □

The engine keeps $R$ for both conventions in every node and updates it as digits are added. For each level a table of 128 bits lists the residues $-(w(y) - y)$ for $y < v$, so the test is two shifts and an "or". A node is dropped only when it fails under every convention the run reports. The rule removes nodes only by the theorem, so the search still misses nothing (Proposition 4.2).

How much does it remove? The values $w(y) - y$ for $y$ below the typical level $0.57b$ fill only 20% to 48% of the residue classes. The reason is the residue laws of Section 5.3: $x^x \equiv x$ modulo 2 always, and modulo 3 and 5 the values of $x^x - x$ miss whole classes. So the rule removes most when $b - 1$ has the factors 2, 3 and 5 (bases 25, 31, 37), and least when $b - 1$ is prime (bases 24, 30, 32).

**Computation 4.5** (What the residue rule saves). COMPUTATION L31.2 L31.6

With both conventions in one run, the rule removes 12% to 25% of the nodes in every base from 15 to 35, with no trend in the base: base 28 drops from 18.0 to 13.6 million nodes, base 35 from 1.51 to 1.22 billion. With one convention (`-m 0` or `-m 1`) it removes 13% to 30%. The solutions found are the same. The time falls less than the nodes, because each node now carries two residues: 1.11 to 1.22 times fewer cycles in bases 24 to 31 with both conventions, 1.22 to 1.29 times with one, and 1.18 times on the GPU at base 30. Under `--zscore 3.4 --zlaw cond` the rule removes 9% to 13% of the nodes, since the z-score rule already drops many of the same nodes.

The gain is a constant factor, as Proposition 1.12 predicts: the residue modulo $b - 1$ is the only one the multiset fixes, and it can only remove a share of the nodes with one free slot. The same test with two free slots removes almost nothing, because two digits reach 87% to 100% of the classes. With no free slot the test only repeats what the interval says. Both versions cost more cycles than they save L31.4.

## 4.2 The cost of the exact search

**Computation 4.6** (Growth of the exact search). COMPUTATION L6.1 L9.1

The prefix-pruned tree grows by a factor of about 1.87 to 1.90 per base: 1.87 from base 16 to 23 and 1.90 from base 30 to 43. Base 37 takes 205 seconds on the test laptop with the build of 17 October 2026, so base 40 takes about 25 minutes; the first build measured 36 minutes for base 40 on a machine shared with other jobs. An exact search of base 44 would take about 7 hours on 8 threads.

Where does the tree come from? At the top the search fixes the counts of the largest digits. The leading digits of the interval are then mostly small digits, and a small digit only adds a lower

bound on its own count. So the top levels prune almost nothing. In base 21 and $k = 20$, a node at the widest level has 15 children on average L6.2.

Many of these children are atypical. They give the large digits far more copies than a random number has, and such a node survives until its free positions run out. In base 21 the nodes at the widest level carry 12.5 large digits on average, against 7.6 for a random number. A measurement that drops every node with more than $s$ large digits above the random expectation (not a valid search, only a measurement of the tree) shows that the typical part grows like $1.60^b$ while the full tree grows like $1.87^b$; at base 23 the atypical nodes are 83% of the tree.

### 4.2.1 A formula for the tree

At the level where the counts of the $m$ largest digit values are fixed, let $L$ be the number of unknown digits of $n$ below the known prefix. Exactly $\binom{L+m}{m}$ count vectors for the large digits fit any given prefix. Summing these over the levels predicts the node count within a constant factor: 3.5 to 3.9 times the measured count, for bases 30 to 43 L9.1. The widest level sits near $v = 0.55b$.

**Heuristic 4.7** (Three growth rates). HEURISTIC L10.1

Let the level $v$ grow with $b$ at $v = \kappa b$. In the large-base limit:

- the exact tree has about $\binom{b}{\kappa b}$ nodes at that level, largest at $\kappa = 1/2$, so it grows like $2^b$;
- the likely part, the nodes that lie above a model solution with nonnegligible probability, has entropy $H = (1-\kappa)b \cdot H(\text{Poisson}(\kappa))$, largest at $\kappa \approx 0.37$, where it equals $0.4933b$; so it grows like $e^{0.4933} = 1.638$ per base.

At finite bases the two rates are 1.88 and 1.55 at base 40, 1.92 and 1.56 at base 100, and 1.96 and 1.59 at base 10,000.

### 4.2.2 Work units and time estimates

The engine cuts the tree into work units at depth 4. Their sizes are very uneven: in bases 24 to 34 the largest 1% of the units hold 47% to 81% of the work, and the pooled size law has a tail like a power law. The type of a unit, its length, free slots and known digits, explains 93% to 97% of the variance of the log of its size. Within a type the law is light-tailed, and each free slot multiplies the size by about 1.93, the growth of the whole search per base. So the heavy tail comes from mixing the types L27.5.

On one machine this costs nothing: on 8 threads the scheduling finishes within 0.1% of the ideal time. It matters for many shards, where a shard of 64 can get 2.6 times the mean work at depth 4; deeper units fix that. So does a better deal: `--shard-by size` predicts the size of each unit from its type and gives the largest units out first, each to the shard with the least predicted work. In base 30 the largest of 64 shards drops from 2.12 to 1.35 times the mean at depth 4, and from 1.21 to 1.14 at depth 6 L31.5. The prediction error within a type, a factor of about 1.6, is what is left. The best estimate of a run's cost before it starts is the previous base times 1.9 (the ratio of consecutive bases from 26 to 34 is 1.77 to 2.16). During a run the progress line weights each unit by its type, which is off by at most 0.10 where the plain fraction of finished units was off by up to 0.49.

### 4.2.3 The limit

The exact search therefore gets slightly worse in larger bases. The gap between the two rates, $(2/1.638)^b = 1.22^b$ in the limit, is all that skipping unlikely nodes can gain. Section 4.4 proves that in the model no tree search can beat $1.638^b$.

## 4.3 Inexact search with a measured miss rate

An exact search cannot drop the atypical nodes: every node is consistent with all the information the search has, so a solution below it is not impossible, only unlikely. A search that is allowed to miss a few solutions can drop them. The engine accepts rules that do so L9.2.

Table 4.1. Rules of the inexact search L9.2, L10.3.

| option | the node is dropped if |
|---|---|
| `--cap c` | one digit value has more than $c$ copies |
| `--band z` | the number of digits $\geq v$ is more than $z$ standard deviations from its mean in a random $k$-digit number |
| `--deficit z` | the known prefix holds too few digits $\geq v$, by more than $z$ standard deviations (a hypergeometric law) |
| `--ratio r` | the expected number of solutions below the node is less than $r$ times that of its parent |
| `--zscore c` | the log of that expected number is more than $c$ standard deviations below its mean (Section 4.4) |

The expected number of solutions below a node, used by the last two rules, is a multinomial probability:

$$E = (p!) \cdot \frac{v^f}{\prod_x \text{need}_x! \cdot f! \cdot b^p},$$

where $p$ is the number of unknown positions, $f$ the number of free digits below $v$, and $\text{need}_x$ the copies of a large digit $x$ that still have to appear. A fixed threshold on $E$ fails, because $E$ is tiny at every node of a large base. Comparing a child with its parent works. The search adds $E$ of every dropped node to a total, so each run reports its own expected number of missed solutions.

**Computation 4.8** (Speed against misses). COMPUTATION L9.3

Growth fitted over bases 24 to 32; misses counted against the 93 nontrivial solutions of bases 10 to 32:

| rules | growth per base | nodes at base 32 | missed |
|---|---|---|---|
| none (exact) | 1.891 | $2.31 \cdot 10^8$ | 0 |
| `--ratio 1e-3` | 1.828 | $1.16 \cdot 10^8$ | 0 |
| `--deficit 2` | 1.764 | $8.0 \cdot 10^7$ | 1 |
| `--deficit 2 --band 2.5` | 1.712 | $4.4 \cdot 10^7$ | 3 |
| `--ratio 1e-2 --deficit 2` | 1.693 | $2.8 \cdot 10^7$ | 3 |
| `--deficit 1 --band 2` | 1.616 | $7.3 \cdot 10^6$ | 19 |

At a few percent of missed solutions the growth drops from 1.89 to about 1.71 per base. That saves a factor 5.3 at base 32, and extrapolated about 85 at base 60 and 1,700 at base 90.

How often does a rule miss in a base that no one has searched? The path from the root to a solution is fixed by the solution itself, so one can replay it and apply the rules on the way, with no search. On all 93 solutions of bases 10 to 32 the replay drops exactly the solutions that the search misses. The replay also works on *planted* solutions: random strings whose weight has the right length, drawn with the length law of Heuristic 3.3. For a deficit rule with a fixed threshold the planted miss rate grows with $b$: for `--deficit 2` it is 2.5% at base 30, 9.2% at base 60 and 11.5% at base 100 L9.4.

The worst standardized deficit on a path, $z_{\min}$, behaves like the minimum of $N$ nearly independent Gaussians, which moves like $-\sqrt{2 \ln N}$. Its median fits $2.40 - 1.63\sqrt{\ln b}$ within 0.15 for bases 20 to 300. So a threshold $c(b) = O\left(\sqrt{\ln b}\right)$ keeps the miss rate below any fixed level, and such a threshold keeps the count vectors within $O\left(\sqrt{b \ln b}\right)$ of their mean, which does not change the exponential rate of the typical tree L9.5.

**Conjecture 4.9** (Fixed miss rate). CONJECTURE L9.5
For every $\delta > 0$ there is a search that, under the model, misses each solution with probability at most $\delta$ and whose cost grows like $(g + o(1))^b$, where $g$ is the growth rate of the typical tree. The measurements put $g$ between 1.6 and 1.7.

Section 4.4 shows that $g = 1.638$ is the right value and gives a rule that comes close.

**Example 4.10** (Base 44).
Base 44 had never been searched. With `--deficit 2 --band 2.5`, both conventions, the search took 31 minutes on 8 threads and visited $2.7 \cdot 10^{10}$ nodes, against about $5.5 \cdot 10^{11}$ nodes and 7 hours for the exact search. It found one solution under $0^0 = 1$, with 43 digits and two zeros, and none under $0^0 = 0$. The model gave $0^0 = 0$ an expected 1.08 solutions, and the rules miss about 15% at this size, so base 44 has no solution under $0^0 = 0$ with probability about 0.85. Only an exact search can settle it L9.3.

## 4.4 An entropy bound for tree searches

The likely part of the tree grows like $1.638^b$. Is that a limit for every search, or only for the rules tried so far? It is a limit for every search that walks the tree from the top, and the proof is a short probabilistic argument. Entropy and its fluctuation, the varentropy, are reviewed in Appendix A.

**Theorem 4.11** (Entropy bound). PROVEN L10.2
Fix a level of the tree. In the model, let $P$ be the law of the node at that level that lies above the solution, given that a solution exists, and let $H$ and $V$ be the entropy and the varentropy of $P$. A search that tries a set $S$ of nodes at that level and contains the right node with probability $p$ satisfies

$$|S| \geq (p - V/t^2)e^{H-t} \quad \text{for every } t > 0.$$

Since $V$ grows linearly in $b$, taking $t = \sqrt{2V/p}$ gives $|S| \geq (p/2)e^{H-O(\sqrt{b})}$. At the peak level $e^H$ grows like $1.638^b$, so every such search that finds a solution with probability $1/2$ visits about $1.638^b$ nodes in large bases.

*Proof.* Let $X$ be the random node with law $P$. Call a node $x$ heavy if $P(x) \geq e^{-H+t}$. A heavy node has $-\ln P(x) \leq H - t$, and by Chebyshev's inequality $P(-\ln P(X) \leq H - t) \leq V/t^2$, because $-\ln P(X)$ has mean $H$ and variance $V$. So the heavy nodes together carry probability at most $V/t^2$. Every other node in $S$ has probability below $e^{-H+t}$, so together they carry at most $|S|\ e^{-H+t}$. Hence $p \leq V/t^2 + |S|\ e^{-H+t}$, which is the claim. □

**Computation 4.12** (The bound at finite bases). COMPUTATION L10.2
At the peak level: base 40, level 17, $H = 17.3$, $\sqrt{V} = 3.7$, $e^H = 3.2 \cdot 10^7$; base 100, level 42, $H = 43.9$, $e^H = 1.1 \cdot 10^{19}$; base 1000, level 406, $H = 453.1$, $e^H = 6.3 \cdot 10^{196}$. At base 40 the bound is 1,300 times smaller than the exact tree.

The theorem is a statement about the model, in which the copies of the large digits in the unknown suffix are random. For the real weights this is the assumption that Chapter 3 tests.

### 4.4.1 Every order of deciding the counts

The tree decides the counts from the largest digit down. Could another order, or an order chosen node by node, do better? No. Let a tree reveal at level $m$ the counts of some set $R$ of $m$ digit values, and let $u$ be the largest value outside $R$. The free positions then hold values at most $u$,

so the interval for $S$ has width about $rw(u)$, and the unknown part of $n$ has about $L(u) = (\ln r + u \ln u)/\ln b + 1$ digits. In the model the node law then has entropy $H(R) = mh(L(u)/b)$, where $h(\lambda)$ is the entropy of a Poisson($\lambda$) variable.

**Lemma 4.13** (Top-down is best). PROVEN L27.2
For every set $R$ of $m$ values, $H(R) \geq H(\text{top } m)$, the entropy of the top-down tree at level $m$.

*Proof.* The complement of $R$ has $b - m$ values, so its largest value is $u \geq b - 1 - m$, with equality only for the top $m$ values. $L(u)$ does not decrease with $u$, and $h$ increases, because Poisson($\lambda + \mu$) is the law of a sum $X + Y$ of independent Poisson variables and $H(X + Y) \geq H(X + Y \mid Y) = H(X)$. □

**Theorem 4.14** (The bound for every order). PROVEN L27.2
In the model, take a tree that reveals one count per level, in an order that may depend on the node. If a set $S$ of its level-$m$ nodes contains the node above the solution with probability $p$, then $|S| \geq (p - V/t^2)e^{H_m - t}$ for every $t > 0$, where $H_m$ is the top-down entropy at level $m$ and $V$ the largest varentropy over the sets $R$ of size $m$. So every such tree needs about $1.638^b$ nodes, and only the top-down order can reach this floor.

*Proof sketch.* The path determines the revealed set, so $P(X = x) \leq P(X = x \mid R = R(x))$. Given $R$, the quantity $-\ln P(X \mid R)$ has mean $H(R) \geq H_m$ and variance at most $V$. The rest is the proof of Theorem 4.11. □

Other orders are far worse: bottom-up, middle-out and random orders peak near $1.30b$ nats, about $3.6^b$, because as long as the count of $b - 1$ is unknown the interval says nothing. Revealing the odd values first peaks at about $0.70b$ nats, $2.0^b$ L27.2. In bases 5 to 10 the best adaptive tree, computed exactly, branches on the largest unrevealed value at 78% to 86% of its nodes and is only 0.83 to 0.91 times the top-down tree.

### 4.4.2 Certificates without a model

Theorem 4.11 and Theorem 4.14 need the model. A weaker statement holds without it. Consider any exact method that learns about $S$ only through queries of the following kind: fix the counts of a set of values, and ask the interval test of Section 4.1 whether the completions can contain a solution. To report "no solution with $k$ digits", such a method must have collected a *certificate of emptiness*: a set of failing queries whose completions cover all count vectors. Otherwise some vector was never excluded, and a weight table with that vector as its only solution would give the same answers L27.3.

**Theorem 4.15** (Certificates are as large as the tree). PROVEN L27.3
For every level $v$, every certificate of emptiness has at least $N_v$ queries, where $N_v$ counts the distinct restrictions to the values $\geq v$ of count vectors whose sums are consistent with their own leading digits far enough down. For bases 5 to 21, $\max_v N_v$ is at least 0.86 times the widest level of the exact tree.

*Proof sketch.* A fooling-set argument. Two vectors that agree with their own sums to many leading digits cannot lie in one failing query unless the query fixes all their large counts: a query that leaves a large value free has an interval too wide to expose a contradiction. So vectors with distinct large counts need distinct queries. □

**Computation 4.16** (Smallest certificates). COMPUTATION L27.3

For $k = b - 1$ under $0^0 = 1$, solved exactly as set-cover problems: in base 10 the smallest certificate has 414 queries, the best adaptive tree 682 and the top-down tree's own certificate 794. In bases 8 to 10 a certificate that is not a tree saves a factor 1.6 to 1.9; an adaptive tree saves 9% to 20%.

So no method of this kind beats the exact tree by more than a polynomial factor, if the witness share stays bounded as it does up to base 21. A faster exact method must use what the interval hides: the low digits of $S$, that is, the arithmetic of $x^x$. Section 4.5 tests the two natural ways to do that.

### 4.4.3 A rule that reaches the bound

The proof says which nodes to keep: those whose probability is close to $e^{-H}$. In the model, the expected count $E$ of Section 4.3 is the probability of the observed multinomial outcome, so $\ln E$ has mean $-H$ and standard deviation $\sqrt{V}$. The rule `--zscore c` drops a node if $\ln E < -H - c\sqrt{V}$, with $H$ and $V$ precomputed per level. Near the top of the tree $\ln E$ has a long left tail, so the rule applies only at the lower three quarters of the levels.

The first version of the rule took $H$ and $V$ from independent Poisson cells. That gets the mean right but overstates the standard deviation by 1.15 times at the top of the tested range and up to 3.8 times at the bottom, because the log probability and the total number of unknown positions are correlated. Conditioning on the total, by linear regression, fixes it: the option `--zlaw cond` matches a direct simulation within 5%. The numbers below first describe the original rule, then the corrected one L28.1.

**Computation 4.17** (The `--zscore` rule, original law). COMPUTATION L10.3

| rule | growth, bases 30 to 38 | nodes at base 38 | missed |
|---|---|---|---|
| exact | 1.89 | about $10^{10}$ | 0 |
| `--zscore 3` | 1.75 | $1.17 \cdot 10^9$ | 0 of 112, bases 10 to 38 |
| `--zscore 2` | 1.70 | $4.5 \cdot 10^8$ | 6 of 112 |

For planted solutions the miss rate of `--zscore 3` stays between 1% and 3% from base 30 to base 300. On the 54 real solutions with at least $b - 6$ digits in bases 20 to 43 it misses 2.

**Computation 4.18** (The corrected law). COMPUTATION L28.1

On planted solutions in bases 30 to 200, the z-score under `--zlaw cond` is close to a standard normal variable at every tested level, with mean 0.03 to 0.25 and standard deviation 0.89 to 1.01. At a fixed threshold its planted miss rate does not depend on the base: 2.0% to 2.4% at threshold 3.35 and 0.6% to 1.0% at 3.82, from base 30 to base 200. Under the original law the miss rate at threshold 3 falls from 2.0% at base 30 to 0.7% at base 200, so it spends ever more nodes per miss. At equal planted miss rate (1% or 2%), `--zlaw cond` visits 1.10 to 1.34 times fewer nodes in bases 30 to 40, with the gain growing with the base, and runs 1.07 to 1.18 times faster. The thresholds 3.4 for 2% and 3.8 for 1% hold in every base of the table. With `--zscore 3.4 --zlaw cond` the search finds all 122 nontrivial solutions of bases 10 to 41.

One run measures a whole curve. If each node records the smallest z-score on its path, a run at a loose threshold $c_0$ gives the node count of every stricter threshold $c \le c_0$ exactly, because the tree of $c$ is the set of nodes of the tree of $c_0$ whose smallest z-score is at least $-c$ L28.1. The library used this to compare rules at equal miss rate. Testing every level, letting the threshold change with the level, and a cost-aware rule that keeps the nodes with the largest ratio of $E$ to subtree size all

reach the same gain or less. Rules that judge one node at a time by $E$ seem to lie on one frontier, bounded by Theorem 4.11; the calibrated rule loses only the width of its band.

**Computation 4.19** (Length windows do not pay). COMPUTATION L28.2
Under the z-score rule each length costs about its share of the solutions: $J = b - k = 1$ holds 51% to 57% of the nodes and 55% of the model solutions, while $J = 0$ costs 1.4 to 1.8 times its share of 8%. At base 40, a window on $J$ saves 1.03 to 1.38 times the nodes at 2 to 21 extra points of planted miss rate, while a lower threshold with the same miss rate saves 1.4 to 4.4 times. So every window is dominated by a lower threshold.

The gain of this rule is speed at a fixed miss rate. At the thresholds that miss a few percent at base 30, the miss rate of the deficit rule grows with $b$: for `--deficit 2` it is 2.5% at base 30 and 11.5% at base 100, and $P(z_{\min} < -2) = 0.317$ at base 300. A deficit threshold of 3 also stays near 1%, but it keeps 77% to 92% of the exact tree in bases 24 to 32, while `--zscore 3` grows like 1.75 per base L9.4, L9.5. The measured growth, 1.75, lies above the bound of 1.55 for these bases because the kept band has width $c\sqrt{V}$, which grows like $\sqrt{b}$ and adds a factor of about $e^{c\sqrt{V}/(2b)}$ per base, 1.15 at base 35. That factor tends to 1, so the rule's growth tends to the bound. In the limit it saves $1.22^b$ against the exact search with a fixed miss rate.

## 4.5 Ideas that do not help

Theorem 4.11 covers searches that walk the tree from the largest digit down. A larger gain needs information that the tree does not use. The library tried the candidates below L6.3, L10.4. None changes the exponential rate.

**Meet in the middle on digit values** Split the digits into large and small values. The small digits only change the low part of $S = A + L$, where $A$ comes from the large counts and $L$ from the small ones. But the digits of a sum are not a function of $A$ plus a function of $L$, so there is no key on which to join two lists. An exact join exists if the bottom digits of $n$ are enumerated as a string, but they must cover the weights of the small digits, about half of $n$, which needs about $b^{b/2}$ strings. Keying on the residue $A \bmod b^L$ instead costs a recomputation of the small side for each residue.

**Meet in the middle on positions** Splitting $n$ into a top string $T$ and a bottom string $B$ gives the exact equation $Tb^q - F(T) = F(B) - B$. Each side has $b^{k/2}$ strings. Strings cost far more than multisets: for $b = 100$ and $k = 99$ a table of at least $100^{50}$ entries, against $1.64^{100} \approx 10^{21.5}$ tree nodes L22.5.

**Lattice reduction** The equation $\sum(w(s_i) - s_i b^i) = 0$ is a subset-sum problem of density about $b/\log_2 b > 1$. At that density lattice methods fail, because the relaxed problem has many solutions [14].

**Congruences** $n$ is linear in the counts modulo $b^j$, and $n \equiv$ digit sum $(\bmod b - 1)$. Both constraints involve every count. By Proposition 1.12, $b - 1$ is the only modulus whose residue the multiset determines. It pays at the nodes with one free slot (Theorem 4.4), but only as a constant factor of 1.1 to 1.3. The other results of the library that restrict digit counts, such as the bound on the copies of $b - 1$ in a $b$-digit solution, are already implied by the interval test L31.4.

**Sharing work across bases** A multiset could be built once and tested in every base where its sum has the right length. That window holds about $\ln b$ bases for a short multiset and one base for a full-length one, and the pruning differs from base to base, so the trees of two bases overlap only near the root. A sweep of bases costs about $1/(1 - 1/1.89) \approx 2.1$ times its largest base, so even perfect sharing saves at most a factor 2.1 on a sweep and nothing on one base L32.3. The short-string engine of Section 4.8 already works across all bases, which is where the idea pays.

**Iterating $F$** Solutions are the fixed points of $n \mapsto F(n)$. From random starting points the orbits run for about $1.85^b$ steps before they repeat, and in bases 14 to 22 none of 600 orbits ended at a solution. A random map on the $4^b$ multisets has orbits of length about $2^b$, and a fixed point attracts a vanishing share of starting points (Section 3.10).

**The representation technique** The best algorithms for random subset sum split a solution into two halves in many ways and filter by a residue [15, 16]. Here the target $n$ is not fixed; it is the unknown itself. Guessing its residue costs as much as the filter saves, and the library's estimate for a full multiset search is $e^{1.28b}$, worse than the tree.

**Constraint solvers** A SAT or constraint solver that ends with "no further solution" has proven it, so it is an exact search that need not be a tree, and its learned clauses could in principle cut across branches. The library encoded the problem column by column for CP-SAT and as a circuit for the SAT solver CaDiCaL. Both list the solutions with $b-1$ digits correctly, in bases 8 to 20 and 8 to 18. Their conflicts grow 1.95 to 1.96 per base, as fast as the tree, at 2 to 5 (CP-SAT) and 57 to 153 (CaDiCaL) conflicts per node of the tree, and $10^4$ to $10^5$ times the time. Without redundant constraints that expose the interval, CP-SAT grows about 4 times per base. A learned clause excludes about one subtree of the tree, no more L27.1.

**Bottom digits in bases $2^k$** In base $b$, $v^v$ is divisible by $b^L$ when $v \cdot v_p(v) \geq L v_p(b)$ for every prime $p \mid b$. So in bases $2^k$ the even digits contribute nothing to the last few digits of $S$, and those digits are known once the counts of the odd values, of 0 and of a few small even values are known: at least $b/2+2$ values. The top-down tree learns them only at its leaves, so a check of the bottom digits never prunes it. Orders that reveal the odd counts early pay for it: in base 32 a hybrid tree visits 13.3 times as many nodes as the plain tree, and an odd-first tree about $10^{11}$ to $10^{12}$ against $2.5 \cdot 10^8$ L27.4. There is one place where a top-down node knows its low digits: a node with one free slot, where each choice of the last digit gives the exact sum. Checking that the last digits of that sum are still available removes another 17% of the nodes at base 26, but costs 2% more cycles, because these nodes are already cheap L31.4.

**Quantum backtracking** A quantum computer can search a backtracking tree of $T$ nodes in about $\sqrt{T}$ steps [17]. That gives about $1.41^b$ for the exact tree and $1.28^b$ for its likely part. It is the only known way past Theorem 4.11.

In short, the leading digits are the only information that a partial assignment of counts reveals cheaply, and the entropy bound and its extension to every order say what that information is worth. A large classical speedup would need a new source of information about the low digits of $n$. The library's view, which it marks as an argument and not a proof, is that a polynomial-time exact search would have to rule out large groups of multisets without looking at them, and nothing in the data hints at a structure that allows this L6.4.

## 4.6 Practical searches: the GPU and recommended settings

The engine also runs on a GPU, alone (`--gpu`) or together with the CPU threads (`--hybrid`). The GPU path handles bases up to 64, runs the z-score rule with every law option and accepts `--shard`, so a base can be split over machines, each with its own GPU, and merged afterwards. The GPU evaluates the rule in fixed point, with values rounded to multiples of $2^{-20}$.

**Lemma 4.20** (The GPU decides as the CPU does). PROVEN L28.3
The GPU and the CPU make the same decision for every child whose $\ln E$ differs from its threshold by more than $(k+3)2^{-21}$.

*Proof.* The GPU sum adds at most $k/2+2$ rounded terms, each off by at most $2^{-21}$; the terms with $\ln 0!$ and $\ln 1!$ are exact. All values stay far below $2^{11}$, so the 32-bit integers do not overflow. □

In tests on bases 10 to 26 and runs at bases 28 to 33 the node counts agree exactly with the CPU. With the z-score rule, `--hybrid` ran 2.6 to 2.9 times faster than 8 CPU threads at base 38, on a laptop shared with other jobs L28.3.

Table 4.2. Recommended settings for a large base L28.2, L28.3.

| goal | setting |
|---|---|
| exact answer | `fast search b`, with `--journal` for long runs; both conventions in one run, which visits fewer nodes than two runs with `-m 0` and `-m 1` |
| miss rate about 2% per solution | `--zscore 3.4 --zlaw cond` |
| miss rate about 1% per solution | `--zscore 3.8 --zlaw cond` |
| length window | none: a lower threshold saves more at the same miss rate |
| a machine with a GPU, base at most 64 | add `--hybrid`; a GPU run cannot use `--journal` |
| several machines | `--shard i/n` on each, then `fast merge`; for 16 shards or more add `--unit-depth 6 --shard-by size` |

## 4.7 Complexity

Can one prove that no fast exact search exists? The answer is a precise "not with current knowledge". The background on NP, #P and sparse sets is in Appendix H.

### 4.7.1 The real problem

Let $L_m$ be the set of strings $1^b$, the base written in unary, such that base $b$ has a nontrivial solution under convention $m$. The input has size $b$, so "polynomial time" means time polynomial in $b$.

**Proposition 4.21** (The real problem is in NP). PROVEN L8.1

$L_m$ is in NP: a solution is a certificate that can be checked in time polynomial in $b$.

*Proof.* A solution has at most $b$ digits, each below $b$ (Proposition 1.3), so it takes $O(b \log b)$ bits. Each weight $x^x$ has at most $b \log_2 b$ bits, so computing $n$ and $F(n)$ takes polynomial time. □

So if no algorithm decides $L_m$ in polynomial time, then P $\neq$ NP. A proof that Munchausen numbers are hard to find would settle the P versus NP problem. The usual way to give evidence of hardness fails as well. $L_m$ is a tally language: it has at most one string of each length. Berman proved that if a tally language is NP-complete, then P = NP [18], and Mahaney extended this to all sparse languages [19]. Book proved that NP contains a tally set outside P exactly when NE $\neq$ E [20].

**Corollary 4.22** (Two barriers). PROVEN L8.1 L13.1

- A proof that $L_m$ is not in P would prove NE $\neq$ E, and so P $\neq$ NP.
- A proof that $L_m$ is NP-hard would prove P = NP.
- The counting function $b \mapsto N_m(b)$ is in #P, and it is not #P-complete under parsimonious reductions unless P = NP.

Neither direction is within reach. The question "is there a polynomial-time search?" is out of reach of current methods, and Section 4.5 gives the heuristic reasons to expect that the answer is no.

### 4.7.2 Arbitrary weights

The barrier comes from the fixed weights, which leave one instance per base. Let the weights vary and the problem becomes hard in the usual sense.

**Definition 4.23** (The problem GEN). L13.1

Input: a base $b \geq 2$ in unary and nonnegative integers $w(0), \ldots, w(b-1)$ in binary. Question: is there an integer $n \geq b$ with $F_w(n) = n$, where $F_w(n)$ is the sum of $w(x)$ over the base-$b$ digits $x$ of $n$?

GEN is in NP: if $W$ is the largest weight, a solution with $D$ digits has $b^{D-1} \le DW$, so it has $O(\log W)$ digits and is a short certificate L13.1.

**Theorem 4.24** (Arbitrary weights). PROVEN L13.1
GEN is NP-complete, also when $w(0) = 0$, the weights increase strictly and $W < b^{b/2}$. Counting the solutions of GEN is #P-complete under parsimonious reductions.

*Proof sketch.* The reduction is from SUBSET-SUM: given positive integers $a_1, ..., a_k$ and a target $T$, is there a set $S$ with $\sum_{i \in S} a_i = T$? Let $B = \max(T, a_i)$. Choose a prime $p$ and a base $b = p + 1$, so $p \mid b - 1$, and an offset $L$ with $b^{L+1} > D^* pB$, where $D^* = L + 2k + 4$ bounds the length of any solution. All of these are polynomial in the input size by Bertrand's postulate. The digits get three roles.

- Item digits $x_i = b - 1 - i$ get the weight $x_i b^{P_i} + pa_i$ with $P_i = L + 2i$: each pays for itself at position $P_i$, plus an error $pa_i$.
- The mandatory digit $z = b - 2 - k$ gets the weight $zb^{P_z} - pT$ with $P_z = L + 2k + 2$.
- Every other nonzero digit is prohibited and gets the weight $y + 1$.

For a set $S$ the number $n_S = zb^{P_z} + \sum_{i \in S} x_i b^{P_i}$ has $F(n_S) = n_S + p\left(\sum_{i \in S} a_i - T\right)$, so it is a solution exactly when $S$ hits the target. The converse is the work, in five steps. A length bound keeps every solution below $D^*$ digits. The digit-sum congruence modulo $p \mid b - 1$ counts the prohibited digits, since each has $w(y) - y = 1$ while designated digits have $w(x) - x \equiv 0 \pmod p$; their number is below $p$ and divisible by $p$, so it is 0. A negative total error would borrow through the empty position $L + 1$ and leave the prohibited digit $b - 1$ there. Two copies of a designated digit, which all lie above $b/2$, carry a small nonzero digit into the next position, which is prohibited. So every count is 0 or 1, the error is 0, and the mandatory digit must be present, because without it the error would be positive. The solutions correspond one to one to the sets $S$, which makes the reduction parsimonious; #SUBSET-SUM is #P-complete under parsimonious reductions through the textbook chain from #SAT [21]. A relabeling of the roles gives strictly increasing weights. □

The library checked the reduction by brute force on 1,890 weight tables in bases 18 to 30: in every case the solutions are exactly the numbers $n_S$ L13.1. It also records two gadgets that fail, which explain the design. Prohibited digits with weight $y$ pay for themselves at position 0, and with weight 0 they are free wildcards; the weight $y + 1$ and the congruence modulo $p$ are what remove them. Errors $a_i$ that are not multiples of $p$ let a low prohibited digit absorb them.

What does this mean for the real problem? Every algorithm that treats the weights as a black box is subject to Theorem 4.24: the prefix-pruned tree needs only increasing weights, and the splits and joins of Section 4.5 need nothing more. On the hard instances the weights increase and stay below $b^{b/2}$, the size range of $x^x$. So no algorithm that is correct for all such tables runs in time polynomial in $b$, unless P = NP. A fast algorithm for $L_m$ itself is not excluded, but it would have to use the arithmetic of $x^x$. The congruences of Section 1.6 and the local obstructions of Section 5.2 are of this kind, and so far they give constant factors only. The hard instances are also unlike the real ones: their solutions have at most $b/2$ digits, while real solutions have about $b - 1$. Whether GEN stays hard when every solution must be long is open L13.1.

### 4.7.3 Powers as weights

Between the free weights and $x^x$ sits the problem with weights $w(x) = x^{e(x)}$, the exponents given in unary. It is in NP, and in P when all exponents are bounded by a constant, because then a solution has at most $K + 1$ digits L13.1. Whether it is NP-hard is open Q40. The reduction above cannot be copied, and linear forms in logarithms (Appendix E) show why.

**Proposition 4.25** (Powers cannot imitate the gadgets). PROVEN L18.6
Let $b \geq 3$, $2 \leq x < b$, $e \geq 2$, $P \geq 1$ and $L \geq 0$ with $x^e \neq xb^P$ and $|x^e - xb^P| < b^{L+1}$. Unless $x$ and $b$ are powers of one integer and $x^{e-1} = b^P$,

$$P - L - 1 \leq 25.2\max(\ln b' + 0.38, 10)^2 \max(\ln x, 1), \quad b' = \frac{e-1}{\ln b} + \frac{P}{\max(\ln x, 1)}.$$

Consequently, a weight table $x^{e(x)}$ with $k$ item digits at distinct positions, as the reduction needs, has $k \leq 25.2\max(\ln(2e_{\max}) + 0.38, 10)^2 \ln b + \log_2 b$. If $b$ is polynomial in $k$, the largest exponent must be at least $\exp\left(c\sqrt{k/\log k}\right)$, which is not polynomial.

*Proof sketch.* An item digit needs $x^{e-1}$ close to $b^P$: $|(e-1)\ln x - P\ln b| < b^{L+1-P}$. If $x$ and $b$ are multiplicatively independent, Laurent's lower bound for linear forms in two logarithms [22] bounds this form from below, which bounds $P - L - 1$. If they are powers of one integer, the form is a multiple of the log of that integer, so it vanishes and the item is exact; an exact item $x^{e-1} = b^P$ forces $x = r^s$ with $s < t$, where $b = r^t$ and $r$ is not a perfect power, so a base has fewer than $t \leq \log_2 b$ exact items L18.6. Items at distinct positions above $L+1$ need $P - L - 1$ at least their number minus one, and $P \leq e$. □

So the hardness of the power variants is a question about the digits of powers, the same arithmetic that Chapter 5 and Chapter 6 meet. One candidate for a gadget remains: digits near $b$, where $(b-c)^e = \sum_i \binom{e}{i}(-c)^{e-i}b^i$ has explicit digits for small $c$ and $e$. No one has found a gadget of that kind, or an argument that excludes one.

## 4.8 Short solutions are cheap

For a fixed number of digits the picture changes completely. Proposition 1.7 says that a digit string is a solution in at most one base. So the search can be turned around: fix the digits, and solve for the base.

### 4.8.1 Solving for the base

For a string $s$ with $k$ digits, $P_s(b) = F(s)$ has at most one root $b$, because $P_s$ increases. The program solves it with exact integers by Newton's method from above, which cannot skip the root because $P_s$ is convex and increasing on the positive axis. By Proposition 1.5, a $k$-digit solution in base $b$ uses only digits $x$ with $x^x < b^k$. So enumerating all strings with digits up to $M$ finds every $k$-digit solution in every base $b$ with $b^k \leq (M+1)^{M+1}$ L3.1.

**Proposition 4.26** (Few solutions of a fixed length). PROVEN L4.1 L10.5 LEAN
$N_{m,k}(b) \leq 2(X_k(b)+1)^{k-1}$, a power of $\log b$ for fixed $k$. The number of bases up to $B$ with a $k$-digit solution is at most $X(X+1)^{k-1}$ with $X = X_k(B)$.

Lean: `prop4_1, prop10_4`

*Proof.* Every digit is at most $X_k(b)$. Fix the last $k-1$ digits, in at most $(X_k(b)+1)^{k-1}$ ways. The leading digit $a$ must satisfy $a^a - ab^{k-1} = R$ with $R$ fixed by the other digits, and the left side is strictly convex in $a$, so at most two values work. For the second bound, every string with digits up to $X$ is a solution in at most one base. □

So the cost per base is a power of $\log b$, and a single enumeration covers all bases up to an enormous bound. Section 5.8 turns the counting argument into a theorem about almost all bases.

### 4.8.2 The sieve

The enumeration is fast, but most of its time goes into exact root computations with numbers of hundreds of digits. A cheap test removes almost all strings first. If $Q_s = P_s - F(s)$ has no root

modulo some integer $q$, then $s$ is a solution in no base (Proposition 5.9). The test splits over positions: for a fixed prefix of $k-1$ digits, the values of the prefix part of $Q_s$ modulo $q$, over all residues $r$, form a bit mask, and the last digit $x$ passes when $w(x) - x$ lies in the mask. So the test costs a few bit operations per string and modulus L11.5.

The first version tested the 18 primes up to 61. It passed about 3 strings in 10,000 for $k = 3$ (140 of 524,880) and 6 to 12 in 10,000 for $k = 4$ to 6. It reproduced the lists of Section 5.1 and ran about 150 times faster than the Python enumeration. The second version uses one power of each prime: a composite modulus with two prime factors tests nothing new by the Chinese remainder theorem, and a root modulo $p^e$ reduces to a root modulo $p^f$ for $f < e$. The powers of 2 and 3 gain most: modulo 2 the last digit moves $Q_s$ by even amounts only, because $x^x \equiv x$, while modulo 32 the test sees much more, and modulo 27 it sees the coverings of Theorem 5.13 L19.1.

**Computation 4.27** (The prime-power sieve). COMPUTATION L19.1

The default moduli are 32, 27, 25, 49 and the primes 11 to 127. Masks are built lazily, only while some last digit is alive, and residues are added as bytes without division. Against the prime sieve, on 4 threads:

| $k$ | $M$ | survivors, old | new | time, old | new |
|---|---|---|---|---|---|
| 4 | 150 | 615,113 | 2,089 | 19.5 s | 1.80 s |
| 5 | 60 | 507,922 | 716 | 49.0 s | 3.59 s |
| 6 | 30 | 588,869 | 961 | 76.7 s | 5.83 s |
| 7 | 18 | 596,001 | 1,318 | 125.1 s | 10.3 s |

The output is identical on all lists. The speedup of about 11 comes almost entirely from the code; the prime powers and the larger primes cut the survivors by a factor of 300 to 800.

**Computation 4.28** (How far the short lists reach). COMPUTATION L3.1 L11.5 L19.1

Every solution with $k$ digits, under both conventions, is known in every base up to the bound below.

| $k$ | digits up to | complete for bases up to | time |
|---|---|---|---|
| 3 | 80 | $10^{51.5}$ | below 0.1 s |
| 4 | 300 | $10^{186.5}$ | 24 s |
| 5 | 80 | $10^{30.9}$ | 20 s |
| 6 | 40 | $10^{11.0}$ | 39 s |
| 7 | 32 | $10^{7.2}$ | 565 s |
| 8 | 24 | 23,364 | 2,059 s |

The lists themselves are in Section 5.1 and Appendix K. The 7-digit and 8-digit runs found no new solution under $0^0 = 1$. Under $0^0 = 0$ they found new members of the self-paying families of Chapter 2. For longer lengths the tree search is the better tool when the base is small: no base up to 100,000 has a 9-digit solution under $0^0 = 1$, and no base up to 10,000 has one with 18, 21, 22, 24, 26 or 28 digits L3.4. Where the two methods cross over for lengths 8 to 12 is open Q29.

## 4.9 Base 100

Base 100 is the natural first target beyond the reach of the exact search. The engine handles bases up to 127 for numbers with at most 62 digits.

**Computation 4.29** (Base 100 up to 50 digits). COMPUTATION L6.6

No number with at most 50 digits in base 100 is a nontrivial solution under either convention.

| digits searched | nodes | time |
|---|---|---|
| up to 30 | 141,449 | 0.055 s |
| up to 40 | 15,778,515 | 0.88 s |
| up to 45 | 531,426,476 | 21 s |
| up to 50 | 3,611,595,932 | 139 s |

A run up to 55 digits finished 98% of its work units in 100 minutes without a solution and was stopped.

The model expects about 0.002 solutions with at most 50 digits, so the result says little. The expected solutions have 97 to 100 digits (Heuristic 3.24). At 1.9 per base from base 40, base 100 needs about $10^{20}$ seconds. Even the likely-tree rate of 1.6 per base would leave about $10^{15}$ seconds.

### 4.9.1 Solutions with $b$ digits

The original conjecture of the library said that no base above 2 has a $b$-digit solution. Under $0^0 = 0$ it is false (bases 15 and 16). Under $0^0 = 1$ the model expects a $b$-digit solution in about 8% of large bases, a little less at finite bases because of the forced top digits (Section 3.11). The search can look for $b$-digit numbers alone with `--min-digits b`, at about a quarter of the cost of a full base.

**Computation 4.30** (No $b$-digit solution under $0^0 = 1$). COMPUTATION L8.6
No base from 3 to 45 has a $b$-digit solution under $0^0 = 1$. Bases 42, 44 and 45 were searched for $b$-digit numbers only; base 42 took 29 minutes.

If the model is right, a counterexample is a matter of searching far enough; a proof of the conjecture would have to show that a whole family of coincidences never happens, which meets the barrier of Section 3.8.

Chapter 5

# Short solutions and the Munchausen polynomial

A solution with few digits is a different object from a solution with about $b$ digits. For a long solution the base is fixed first and the digits must conspire. For a short one it is natural to fix the digits first and ask for the base. By Proposition 1.7 a digit string $s$ with at least two digits is a solution in at most one base, namely a root of one polynomial. This chapter studies that polynomial.

Fix a digit string $s = (s_1, ..., s_k)$, most significant first. Its value polynomial and its *Munchausen polynomial* are

$$P_s(r) = s_1 r^{k-1} + s_2 r^{k-2} + ... + s_k, \qquad Q_s(r) = P_s(r) - F(s).$$

The string $s$ is a solution in base $b$ exactly when $Q_s(b) = 0$ and $b$ exceeds every digit of $s$. So $Q_s$ is a polynomial of degree $k-1$ with integer coefficients, and the short solutions are its integer roots above the largest digit. Everything in this chapter comes from that one sentence: complete lists found by solving for the base, local obstructions modulo primes and prime powers, Galois groups, a heuristic for how often a root is an integer, and the roots that are not ordinary bases.

## 5.1 The complete lists

Solving for the base turns the search around. Enumerate digit strings, and for each one find the integer root of $Q_s$, if any. The root is found exactly with integer arithmetic: $P_s$ is increasing and convex for $r \geq 0$, so Newton's method from above cannot jump over it. By Proposition 1.5 a $k$-digit solution in base $b$ only uses digits $x$ with $x^x < b^k$. So the strings with digits up to $M$ contain every $k$-digit solution in every base $b$ with $b^k \leq (M+1)^{M+1}$. The range of a list therefore grows like $M^{M/k}$, very fast in $M$.

The Rust program `kdigit` does this with a sieve that rejects almost every string before the exact step (Section 5.2 and Section 4.8). The lists below are exhaustive within their ranges.

**Computation 5.1** (Short solutions under $0^0 = 1$). COMPUTATION L3.2 L11.5 L19.1
Under $0^0 = 1$, the solutions with 3 to 8 digits in all bases up to the stated bound are exactly those of Table 5.1.

Table 5.1. All solutions with 3 to 8 digits under $0^0 = 1$. A list with digits up to $M$ is complete for every base $b$ with $b^k \leq (M+1)^{M+1}$. For 3 digits a sharper argument gives a larger range (Computation 5.3).

| $k$ | digits up to | complete for bases up to | solutions (base: digits) |
|---|---|---|---|
| 3 | 2000 | $10^{3301}$ | 4: (1, 3, 1), (3, 1, 3); 25: (5, 1, 3); 91: (6, 1, 5); 215: (1, 2, 6); 904: (1, 7, 1) |
| 4 | 300 | $10^{186.5}$ | 10: (3, 4, 3, 5); 20: (6, 5, 3, 4) |
| 5 | 80 | $10^{30.9}$ | 6: (2, 2, 3, 5, 2), (2, 3, 4, 5, 2); 7: (1, 3, 4, 5, 4); 13: (3, 3, 6, 6, 1); 14: (2, 6, 0, 3, 6); 48: (3, 8, 6, 2, 0) |

| $k$ | digits up to | complete for bases up to | solutions (base: digits) |
|---|---|---|---|
| 6 | 40 | $10^{11.0}$ | 9: $(1,5,6,2,6,2)$ |
| 7 | 32 | $10^{7.2}$ | 9: $(1,6,5,6,5,4,7)$ |
| 8 | 24 | $23\,364 \approx 10^{4.4}$ | 11: $(1,8,4,5,3,2,7,8)$, $(1,8,4,5,3,4,8,7)$ |

So 3435, the famous example, is one of exactly two 4-digit solutions in all bases up to $10^{186}$. The other is $(6,5,3,4)$ in base 20: $6 \cdot 20^3 + 5 \cdot 20^2 + 3 \cdot 20 + 4 = 50\,064 = 6^6 + 5^5 + 3^3 + 4^4$. A separate search over 3-digit strings with two digits up to 1000 and the third up to 30 reaches bases near $10^{1500}$ and finds the same six 3-digit solutions L16.1.

Under $0^0 = 0$ the lists are infinite, because of the self-paying digits of Section 2.2. A digit $x$ with $x^{x-1} = b^{k-1}$ gives the two solutions $(x,0,\ldots,0,0)$ and $(x,0,\ldots,0,1)$ in base $b$. Apart from these families the lists are short.

**Computation 5.2** (Short solutions under $0^0 = 0$). COMPUTATION L3.3 L19.1

Under $0^0 = 0$, in the ranges of Table 5.1, the solutions with 3 to 8 digits are the self-paying families and the strings of Table 5.2.

Table 5.2. Solutions with 3 to 8 digits under $0^0 = 0$ that are not of the form $(x,0,\ldots,0,e)$ with $x$ self-paying. Same ranges as Table 5.1, except that the 3-digit list is complete only for bases up to $10^{51.5}$ (digits up to 80). Swaps and other constructions of Chapter 2 are marked.

| $k$ | solutions (base: digits) |
|---|---|
| 3 | 4: $(1,3,0)$, $(1,3,1)$, $(3,1,3)$; 5: $(1,0,3)$; 25: $(5,1,3)$; 91: $(6,1,5)$; 215: $(1,2,6)$; 904: $(1,7,0)$, $(1,7,1)$; $3^{40}$: $(3,0,27)$ (swap) |
| 4 | 5: $(2,0,2,4)$; 10: $(3,4,3,5)$; 20: $(6,5,3,4)$; 128: $(8,0,2,4)$ (self-paying 8 on the swap $(2,4)$); $2^{682}$: $(4,0,0,256)$ (swap; this base, about $10^{205.3}$, lies outside the complete range) |
| 5 | 6: $(2,2,3,5,2)$, $(2,3,4,5,2)$; 7: $(1,3,4,5,4)$; 11: $(6,6,5,0,0)$, $(6,6,5,0,1)$; 13: $(3,3,6,6,0)$, $(3,3,6,6,1)$; $3^{20}$: $(3,0,0,0,27)$ (swap) |
| 6 | 9: $(1,5,6,2,6,2)$; 11: $(5,1,7,5,0,3)$; $3^{16}$: $(3,0,0,0,0,27)$ (swap), $(3,0,0,0,9,27)$ (swap with the self-paying 9) |
| 7 | 9: $(1,6,4,7,0,6,3)$, $(1,6,5,6,5,4,7)$ |
| 8 | 9: $(3,4,6,6,4,0,8,4)$; 11: $(1,8,4,5,3,2,7,8)$, $(1,8,4,5,3,4,8,7)$ |

Appendix K lists every string, the families included. Two-digit solutions are plentiful under both conventions and have a complete description (Proposition 2.14), so this chapter starts at three digits.

### 5.1.1 Three digits up to $10^{3301}$

For three digits the range of the list is much larger, because a sieve with many primes is cheap and a direct bound replaces the general one.

**Computation 5.3** (The 3-digit list). COMPUTATION L25.2

Under $0^0 = 1$:

1. every 3-digit solution in a base up to $10^{3301}$ is one of the six of Table 5.1;
2. every 3-digit solution in a base below $46666^{23333} - 3$ with at most one digit above 200 is one of these six.

So a 3-digit solution in a base strictly between 904 and $46666^{23333} - 3$ has two digits above 200, one digit above 2000, and a base above $10^{3301}$.

The method is the completed square of Section 5.6. A string $(p,q,r)$ is a solution exactly when $D = 4p(F-r) + q^2$ is a square $s^2$ and $b = (s-q)/(2p)$ is an integer above the digits. A program tests $D$ for quadratic residues modulo the 108 odd primes below 600, with tables of $w(d)$ modulo each prime, and checks the survivors exactly. Part 1 takes all strings with digits up to 2000: a solution with a digit $M > 2000$ has $b^2 > M^M/(M+1) \geq 2001^{2001}/2002$, so every base below $10^{3301.2}$ is

covered. Part 2 takes all strings with two digits up to 200 and one up to 46,700. In all $1.36 \cdot 10^{10}$ strings were tested, and only 12 survived all 108 primes. Seven of them are solutions: the six known ones and $(1, 6, 46666)$. The other five have a root that is not an integer, or an integer root at most the largest digit L25.2.

The binomial families of Section 2.6 give infinitely many 3-digit solutions beyond this range. Their members are the strings $(\lambda c^2, 2\lambda ce, x)$ of case (i) of Theorem 5.46 below, and they can be listed in order.

**Proposition 5.4** (Members in order). PROVEN L25.2

Among the 3-digit members of the binomial families, a larger last digit $x$ gives a larger base.

*Proof sketch.* A member has $b \le \sqrt{x^x/\lambda} \le x^{x/2}$. For a member with last digit $x' \ge x + 1 \ge 7$, the other digits are at most $\sqrt{x'}$, so its base is at least $x'^{(x'-1)/2} - x'$, and Bernoulli's inequality shows that this exceeds $x^{x/2}$. □

By this ordering and a short case check, the next member after $(1, 2, 6)$ in base 215 is $(1, 6, 46666)$ in base $46666^{23333} - 3$ (Theorem 2.36).

**Computation 5.5** (The first members). COMPUTATION L25.2

The strings with $p, q \le 2000$ contain every 3-digit member with $x < 2001^{2001} \approx 10^{6605.8}$. There are 1,267 of them, all with $\lambda = 1$ and even $x$. Table 5.3 lists the first five.

Table 5.3. The first 3-digit members of the binomial families, in order of base L25.2.

| rank | $(p, q, x)$ | $(c, e)$ | base |
|---|---|---|---|
| 1 | $(1, 2, 6)$ | $(1, 1)$ | 215 |
| 2 | $(1, 6, 46666)$ | $(1, 3)$ | $46666^{23333} - 3 \approx 10^{108941.8}$ |
| 3 | $(4, 8, 16777476)$ | $(2, 2)$ | about $10^{10^{7.78}}$ |
| 4 | $(9, 6, 387467146)$ | $(3, 1)$ | about $10^{10^{9.22}}$ |
| 5 | $(1, 10, 10000000026)$ | $(1, 5)$ | about $10^{10^{10.70}}$ |

**Heuristic 5.6** (Nothing between). HEURISTIC L25.2

The strings that Computation 5.3 leaves open have a digit $M > 2000$ and a second digit above 200. By the root heuristic each is a solution with probability about $M^{-M/2}$, and there are fewer than $3M^2$ of them for each $M$. The expected number of such solutions is below $3 \cdot 2000^2 \cdot 2000^{-1000} \approx 10^{-3294}$.

So the claim "$(1, 6, 46666)$ is the next 3-digit solution after $(1, 7, 1)$" has three parts. It is proven that $(1, 6, 46666)$ is the next family member (Theorem 2.36), and that it is the only 3-digit solution in its base (Proposition 5.82). It is computed that no other 3-digit solution lies in bases from 905 to $10^{3301}$, nor below $46666^{23333} - 3$ with at most one digit above 200. It is heuristic that no string with two digits above 200 and one above 2000 fills the rest of the gap. An exhaustive run over all strings with digits up to 46,700 would need about $10^{14}$ tests and has not been done. Under an explicit form of the abc conjecture, the strings with two digits at most 200 are settled in every base (Corollary 5.55).

### 5.1.2 Which lengths occur

Under $0^0 = 0$ every length occurs, in infinitely many bases (Proposition 2.20). Under $0^0 = 1$ the searches show gaps.

**Computation 5.7** (Missing lengths). COMPUTATION L3.4 L11.4

Under $0^0 = 1$ no base up to 100 000 has a solution with 9 digits, and no base up to 10 000 has one with 18, 21, 22, 24, 26 or 28 digits. Among the lengths 2 to 30, these 7 are the only ones without a solution in any base up to 127.

For a large length the model of Chapter 3 expects about one solution of that length over all bases together, so a length is empty with probability about $e^{-1} \approx 37\%$. The data show 7 empty lengths out of 29, or 24%. These lengths are not empty for ever, though. The binomial families of Theorem 2.21 give solutions of every length $k \geq 3$ under $0^0 = 1$ (Corollary 2.22), in bases far beyond any search: the smallest 3-digit member after $(1, 2, 6)$ is $(1, 6, 46666)$ in base $46666^{23333} - 3$, about $10^{108942}$.

**Remark 5.8.** L3.5 L7.2 L22.3

The library first conjectured that under $0^0 = 1$ each length $k \geq 3$ has only finitely many solutions. The evidence was the lists above and the spacing heuristic of Section 5.5, which predicts a finite total for every $k \geq 3$. The binomial families refute the conjecture. They are instructive: a heuristic that treats the root of $Q_s$ as a random real number cannot see a family whose polynomial factors by an identity. What survives is a question (Q64): is every short solution beyond some size a member of an algebraic family? Conjecture 2.35 states the current conjecture.

## 5.2 Local obstructions

A polynomial with an integer root has a root modulo every integer. Turned around, this is a cheap test that rejects strings in every base at once.

**Proposition 5.9** (Local obstruction). PROVEN L11.3 L19.1 LEAN

If the polynomial $Q_s$ of a digit string $s$ has no root modulo some integer $q \geq 2$, then $s$ is not a solution in any base.

Lean: `not_isSolution_of_no_root_mod`

*Proof.* If $Q_s(b) = 0$, then $b \bmod q$ is a root of $Q_s$ modulo $q$, because $Q_s$ has integer coefficients and $Q_s(r) \bmod q$ depends only on $r \bmod q$. □

Two facts limit the useful moduli. By the Chinese remainder theorem, $Q_s$ has a root modulo $q_1 q_2$ with $\gcd(q_1, q_2) = 1$ exactly when it has a root modulo $q_1$ and modulo $q_2$. And a root modulo $p^e$ reduces to a root modulo $p^f$ for $f < e$. So only one power of each prime adds information, and a higher power makes the lower ones redundant L19.1.

How often does a random polynomial have no root? Over a large prime field, inclusion and exclusion over the roots shows that a random polynomial of degree $n$ has no root with probability

$$\sum_{j=0}^{n} \frac{(-1)^j}{j!} \quad \to \quad e^{-1} \approx 0.368,$$

which is 1/2, 1/3, 3/8 and 11/30 for $n = 2, 3, 4, 5$, the degrees $k - 1$ for $k = 3, \ldots, 6$ L19.1. The large primes of the sieve match these values to within 0.004 (Table 5.4). Section 5.4 explains why: the Galois group of $Q_s$ is almost always the full symmetric group, and the share of permutations without a fixed point is exactly this sum.

Table 5.4. Share of candidate strings that one modulus removes when it is tested first L19.1. $k$ digits, digits up to $M$.

| modulus | $k = 3$, $M = 80$ | $k = 4$, $M = 44$ | $k = 5$, $M = 28$ | $k = 6$, $M = 16$ |
|---|---|---|---|---|
| 2 | 0.012 | 0.030 | 0.055 | 0.092 |

| modulus | $k=3$, $M=80$ | $k=4$, $M=44$ | $k=5$, $M=28$ | $k=6$, $M=16$ |
|---|---|---|---|---|
| 32 | 0.286 | 0.218 | 0.298 | 0.336 |
| 3 | 0.204 | 0.229 | 0.253 | 0.281 |
| 27 | 0.386 | 0.350 | 0.391 | 0.417 |
| 5 | 0.306 | 0.318 | 0.326 | 0.328 |
| 25 | 0.416 | 0.362 | 0.403 | 0.402 |
| 127 | 0.496 | 0.333 | 0.373 | 0.366 |
| random, large $p$ | 1/2 | 1/3 | 3/8 | 11/30 |

The small primes are weak tests, and the reason is the digit-sum identity of Proposition 1.9. At $r \equiv 1$ the value $Q_s(1) = \sum_i (s_i - w(s_i))$ vanishes modulo $p$ for many strings. Modulo 2 it is worse.

**Proposition 5.10** (The test modulo 2). PROVEN L15.4 LEAN

$Q_s$ has no root modulo 2 exactly when both parity rules of Section 1.6 fail for $s$: $Q_s(1)$ odd excludes every odd base, and $Q_s(0)$ odd excludes every even base.

Lean: `exists_root_mod_two_iff`

*Proof.* $Q_s(0) = s_k - F(s)$ and $Q_s(1) = \sum_i s_i - F(s)$. For odd $b$, $Q_s(b) \equiv Q_s(1)$, and for even $b$, $Q_s(b) \equiv Q_s(0)$ modulo 2. □

Since $x^x \equiv x \pmod 2$, the last digit moves $Q_s$ only by even amounts modulo 2. Modulo $2^e$ the even digits $x \geq e$ have $x^x \equiv 0$, so the test sees the value of $x$ itself. That is why 32 removes 3 to 25 times as much as 2. The powers of 3 gain for a reason explained in Section 5.3: the values of $x^x - x$ modulo 27 fill only 10 residues.

### 5.2.1 The sieve

The test splits over positions. Fix the first $k-1$ digits. For a modulus $q$, the values of the prefix part of $Q_s(r)$ over all residues $r$ form a bit mask, and the last digit $x$ passes if the residue it needs lies in the mask. So the test costs a few bit operations per string and modulus. Only the strings that pass every modulus go on to the exact root.

**Computation 5.11** (Survivors of the sieve). COMPUTATION L11.5 L19.1

With the moduli 32, 27, 25, 49 and the primes 11 to 127, the sieve passes 36,309 of the $4.1 \cdot 10^{10}$ strings with 7 digits up to 32, and 133,240 of the $1.5 \cdot 10^{11}$ strings with 8 digits up to 24. With the 18 primes up to 61 alone the share was 3 to $6 \cdot 10^{-4}$, close to the $0.632^{18} = 2.6 \cdot 10^{-4}$ that independent tests would give.

The near-independence of the primes has a good explanation. For 3-digit strings with digits up to 80, the product of the measured survival shares predicts 28.9 survivors. Eleven further strings have a rational root and pass every test by definition. That makes about 40, against 46 observed L11.5. Section 4.8 describes the program and its speed.

## 5.3 Coverings and the residues of $x^x - x$

Local tests can do more than reject one string. Fix all digits but one and let that digit $x$ vary. Whether $Q_s$ has a root modulo $q$ then depends only on $x$ modulo a period, because $x^x \bmod q$ is periodic in $x$. If the residues excluded by several moduli cover all integers, the whole infinite family is excluded in every base. This is the idea of the covering congruences behind the Sierpinski numbers.

A varying last digit enters $Q_s$ as $x - x^x$. So the set

$$T(q) = \{x^x - x \bmod q : x \geq 1\}$$

decides which coverings exist. If $T(q)$ is all of $\mathbb{Z}/q$, the modulus $q$ can never exclude a last digit on its own.

**Lemma 5.12** (Small moduli). PROVEN L15.4 L20.2 LEAN

1. For $q = 2^e$, $T(q)$ is the set of even residues.
2. For a prime $p \geq 5$, $T(p) = \mathbb{Z}/p$. And $T(3) = \{0, 2\}$.
3. If $x \equiv 2 \pmod 6$, then $x^x - x \equiv 2 \pmod{27}$. Otherwise $x^x - x \equiv 0 \pmod 3$. So $T(9) = \{0, 2, 3, 6\}$ and $T(27) = 3\mathbb{Z}/27 \cup \{2\}$.

Lean: `T_prime_eq_univ, T_three, pow_self_modEq_27, multiple_mem_T (partial: part 1 modulo 2^e only for multiples of 2)`

*Proof.* (1) $x^x \equiv x \pmod 2$, and even $x \geq e$ give $x^x \equiv 0$, so $-x$ runs over all even residues. (2) Modulo $p$, $x^x$ depends on $u = x \bmod p$ and $j = x \bmod (p-1)$, which are independent by the Chinese remainder theorem. The residue 0 comes from $x \equiv 0 \pmod p$. For a target $v \neq 0$, pick a primitive root $u \neq -v$; then $u + v$ is a unit, so $u + v = u^j$ for some $j$, and the matching $x$ gives $x^x - x \equiv v$. A prime $p \geq 5$ has at least two primitive roots. (3) Only $x \equiv 2 \pmod 6$ gives $x^x \not\equiv x \pmod 3$. Put $u = -x = 1 + 3t$. Since $x$ is even, $x^x = u^x$, and the binomial theorem modulo 27 gives $x^x - x \equiv 2 + (x+1)^3(x-2)/2 \pmod{27}$, with $27 \mid (x+1)^3$. □

**Theorem 5.13** (Covering by 27). PROVEN L15.4 LEAN

Let $A(r) = \sum_{i=1}^{k-1} s_i r^{k-i}$ and $F_0 = \sum_{i=1}^{k-1} w(s_i)$ for a prefix $(s_1, \ldots, s_{k-1})$. If $A(r) - F_0 \not\equiv 0 \pmod 3$ and $A(r) - F_0 \not\equiv 2 \pmod{27}$ for every residue $r$ modulo 27, then no string $(s_1, \ldots, s_{k-1}, x)$ with $x \geq 1$ is a solution in any base, under either convention. Under $0^0 = 0$ this also holds for $x = 0$.

Lean: `covering_27`

*Proof.* A solution in base $b$ gives $A(b) - F_0 = x^x - x$, and by Lemma 5.12 the right side is $0 \pmod 3$ or $2 \pmod{27}$. □

**Example 5.14.** L15.4

For the prefix $(2, 1)$, $A(r) - F_0 = 2r^2 + r - 5$. Modulo 3 it takes the values 1, 1, 2 at $r = 0, 1, 2$. For $r \equiv 2 \pmod 3$, $2r^2 + r$ is 1 or 10 modulo 27, never 7. So $(2, 1, x)$ is not a solution for any $x \geq 1$, in any base. The digit $x$ can be as large as we like, so this excludes bases up to infinity in one stroke.

**Computation 5.15** (Coverings are common). COMPUTATION L15.4 L20.2

Among the 3-digit families with fixed digits up to 10, 27 of 341 are covered under $0^0 = 1$, and 22 of them need a modulus other than 2. For random prefixes with digits in $[3, 10^6]$, the last digit is excluded in every base for 11.2% of 3-digit prefixes and 7.2% to 8.6% of prefixes with 4 to 6 digits, using the moduli 16, 27, 25 and 7. The modulus 27 alone gives 11.2% and 7.2% to 8.4%, so almost all of this comes from Theorem 5.13. Adding the modulus 81 raises the shares by 0.2 to 0.4 percentage points.

So about one prefix in twelve makes its last digit impossible in every base. The minimal sets of moduli found are mostly $\{4\}$, $\{9\}$, $\{3\}$ and $\{3, 4\}$; one covering, of $(2, 0, 1, 0, x)$ under $0^0 = 1$, needs 8, 11 and 25 together L15.4.

### 5.3.1 The laws of $x^x - x$ modulo prime powers

The covering theorem rests on the gaps of $T(27)$. Which other prime powers have gaps? For $p \geq 7$ none do, and for 3 and 5 the gaps follow explicit laws. The tool is the $p$-adic exponential (Appendix C).

**Lemma 5.16** (Lifting). PROVEN L20.2 LEAN
Let $p$ be odd and $x_0$ prime to $p$ with $x_0^{x_0} \not\equiv 1 \pmod p$. As $x$ runs over the integers with $x \equiv x_0 \pmod{p(p-1)}$, the value $x^x - x$ modulo $p^e$ runs over the whole class $x_0^{x_0} - x_0 + p\mathbb{Z}$, for every $e \geq 1$. Multiples of $p$ also lie in $T(p^e)$ for every $e$.
Lean: `lemma20_7`

*Proof sketch.* Step from $x$ to $x' = x + p^{j+1}(p-1)s$. Then $x'^{x'} \equiv x^x(1 + p^{j+1}(p-1)s) \pmod{p^{j+2}}$, so $x^x - x$ moves by $p^{j+1}(p-1)(x^x-1)s$, and $x^x - 1$ is a unit. So the step $s$ runs through every value of the next $p$-adic digit. Induction on $j$. For the multiples of $p$, take $x \equiv 0 \pmod p$ with $x \geq e$: then $x^x - x \equiv -x$. □

**Theorem 5.17** (Primes from 7 on). PROVEN L20.2 LEAN
For every prime $p \geq 7$ and every $e \geq 1$, $T(p^e) = \mathbb{Z}/p^e$.
Lean: `theorem20_8`

*Proof sketch.* By Lemma 5.16 it suffices to reach every nonzero residue $v$ modulo $p$ with some $x$ whose $x^x \not\equiv 1$. Take a primitive root $u$ that avoids $-v$ and $1 - v$; then $u + v$ is a power $u^j \neq 1$. For $p \geq 11$ there are at least four primitive roots, and $p = 7$ is checked by hand. □

For $p = 3$ and $p = 5$ some classes have $x^x \equiv 1 \pmod p$, the step of Lemma 5.16 vanishes, and the class does not lift freely. On such a class write $x = \zeta e^a$ with $\zeta$ a root of unity in $\mathbb{Z}_p$ and $a \in p\mathbb{Z}_p$. Then $x^x = \exp(\zeta a e^a)$ when the order of $\zeta$ divides $x$, and $x^x - x$ becomes a power series in $a$ with explicit coefficients L20.2.

**Theorem 5.18** (The 3-adic law). PROVEN L20.2 LEAN
For $e \geq 4$,

$$T(3^e) = \{v : v \equiv 0 \pmod 3\} \cup \{v : v \equiv 2 \pmod{27},\ v \not\equiv 56 \pmod{81}\}.$$

So $T(3^e)$ has density $29/81$, and for $e \geq 5$ it is the full preimage of $T(81)$.
Lean: `pow_self_modEq_81 (partial: the inclusion modulo 81)`

*Proof sketch.* Only the class $x \equiv 2 \pmod 6$ matters. With $\zeta = -1$ and $a = 3w$, the values are $2 + 27G(w)$ with a power series $G$ whose coefficients lie in $\mathbb{Z}_3$ and tend to 0. Modulo 3, $G(w) \equiv -(w^3 + w^4)$, which takes the values 0, 1, 0 at $w \equiv 0, 1, 2$. So $G$ never takes the value 2 modulo 3, which excludes $v \equiv 56 \pmod{81}$. The derivative is a unit at $w \equiv 1, 2$, and Hensel's lemma shows that every other value is reached modulo every $3^e$. □

**Theorem 5.19** (The 5-adic law). PROVEN L20.2
Let $i$ be the square root of $-1$ in $\mathbb{Z}_5$ with $i \equiv 3 \pmod 5$, and $A = 1 - i$. For $e \geq 2$,

$$T(5^e) = \{v : v \not\equiv 3 \pmod 5\} \cup \{A(1 - 25s/2) \bmod 5^e : s \text{ a square in } \mathbb{Z}_5\}.$$

The class $3 \pmod 5$ holds 1, 3, 11, 53, 261, 1303 elements of $T(5^e)$ for $e = 2, \ldots, 7$, and the density of $T(5^e)$ tends to $49/60$.

The laws were checked against direct computation of $T(3^e)$ for $e \leq 10$ and $T(5^e)$ for $e \leq 7$ L20.2. They have two consequences for coverings.

**Corollary 5.20** (81 is the last useful power of 3). PROVEN L20.2 LEAN
If $A(r) - F_0 \notin T(81)$ for every residue $r$ modulo 81, then no string $(s_1, \ldots, s_{k-1}, x)$ with $x \geq 1$ is a solution in any base. For a single modulus $3^e$ with $e \geq 5$ this test excludes nothing new.
Lean: `covering_81`

For example, 81 excludes $(8, 7, x)$ for every $x$, and 27 does not: the value $8r^2 + 7r - 8^8 - 7^7$ is $2 \pmod{27}$ only for $r \equiv 8 \pmod 9$, and there it is $56 \pmod{81}$.

**Proposition 5.21** (Single powers of 5). PROVEN L20.2
If a single modulus $5^e$ excludes every last digit of a prefix, then every digit of the prefix is a multiple of 5, or the polynomial $A \bmod 5$ is a nonzero multiple of $(r^5 - r)^2$ and the prefix has at least 10 digits.

So the 5-adic gaps matter only together with other moduli. The prime powers $2^e$ and $3^e$ carry the coverings, because of the coupling between $x \bmod 3^e$ and $x \bmod 2 \cdot 3^{e-1}$. An open question (Q38) asks whether every covering needs a modulus on which $x^x - x$ misses residues, that is a power of 2 or 3, or a power of 5 from 25 on.

## 5.4 Galois groups and reducibility

How strong is the local test in the limit? A string that passes the test for every prime is either a solution or a string whose polynomial has a very special shape. This section makes that precise. Appendix B explains the tools: the Galois group of a polynomial, the Chebotarev density theorem and Jordan's theorem.

### 5.4.1 The Galois group is the full symmetric group

For a polynomial of degree $n$ with Galois group $G$, the Chebotarev density theorem says that the share of primes $p$ modulo which it has no root equals the share of elements of $G$ that fix no root. For $G = S_n$ this is the share of derangements, $\sum_{j \le n} (-1)^j / j!$.

**Computation 5.22** (Root-free primes). COMPUTATION L12.1
For 60 random strings of each length $k = 3, 4, 5, 6$ (digits up to 20, $0^0 = 1$), the share of primes from 200 to 20 000 modulo which $Q_s$ has no root has median 0.501, 0.335, 0.374 and 0.367. The derangement shares of $S_{k-1}$ are 0.500, 0.333, 0.375 and 0.367.

So $Q_s$ generically has the full symmetric Galois group, as Hilbert's irreducibility theorem leads one to expect: the constant term $F(s)$ is unrelated to the other coefficients. The one exception among the 240 strings is $(9, 18, 18, 9, 15)$, with a share of 0.631. Its value polynomial satisfies $P_s(r) - 15 = 9r(r+1)(r^2+r+1) = 9u(u+1)$ with $u = r^2 + r$. So $Q_s$ is a polynomial in $r^2 + r$, its group lies in the dihedral group $D_4$, and the derangement share of $D_4$ is 5/8. Decomposable value polynomials are the natural source of smaller groups.

### 5.4.2 The local test is complete for irreducible polynomials

**Proposition 5.23** (Irreducible polynomials fail locally). PROVEN L12.2
If $Q_s$ is irreducible of degree $n \ge 2$, then $Q_s$ has no root modulo $p$ for a set of primes $p$ of density at least $1/n$.

*Proof.* The Galois group acts transitively on the roots. By Jordan's theorem a transitive group contains an element that fixes no root, and Cameron and Cohen [23] proved that such elements make up at least a share $1/n$ of the group. The Chebotarev density theorem turns this share into a density of primes (Appendix B). □

So a string that passes the test for every prime has a reducible $Q_s$. A polynomial with a root modulo almost every prime but no rational root is called *intersective*. The question is whether some $Q_s$ is intersective.

**Proposition 5.24** (No intersective polynomial for $k \le 5$). PROVEN L16.3
If $k \le 5$ and $Q_s$ has a root modulo all but finitely many primes, then $Q_s$ has a rational root.

*Proof.* Without a rational root, $Q_s$ is irreducible, or for $k = 5$ a product of two irreducible quadratics with fields $\mathbb{Q}\big(\sqrt{D_1}\big)$ and $\mathbb{Q}\big(\sqrt{D_2}\big)$. The irreducible case is Proposition 5.23. If the two fields are equal, the primes inert in it (density $1/2$) give no root. If they differ, the primes inert in both (density $1/4$) give no root. □

**Proposition 5.25** (The criterion for $k = 6$). PROVEN L16.3
A quintic $Q_s$ has a root modulo almost every prime but no rational root exactly when $Q_s = gh$ with $g$ an irreducible quadratic, $h$ an irreducible cubic with group $S_3$, and $\mathbb{Q}(\sqrt{\operatorname{disc} g}) = \mathbb{Q}\big(\sqrt{\operatorname{disc} h}\big)$.

The classic example $(x^3 - 19)(x^2 + x + 1)$ has this shape, so the question for six digits is real. No string with digits up to 9 gives one L16.3.

### 5.4.3 Reducibility for five digits

For $k = 5$ a factorization into two quadratics can be analysed by comparing coefficients. Write $a, b', c, d$ for $s_2, s_3, s_4, s_5$ divided by $s_1$, and let $q = c - ab'/2 + a^3/8$. Then $q = 0$ exactly when $P_s(r)$ is a polynomial in $(r + a/4)^2$, that is, when $P_s$ is decomposable.

**Theorem 5.26** (Two quadratic factors). PROVEN L16.3
Let $k = 5$ and let the largest digit be at least 3. If $Q_s$ is a product of two quadratics over $\mathbb{Q}$, then $q = 0$, so $P_s$ is decomposable. Hence $Q_s$ is reducible exactly when it has a rational root, or when $P_s = s_1 G\big((r + a/4)^2\big)$ and the resulting quadratic in $(r + a/4)^2$ splits in a suitable way.

*Proof sketch.* Comparing coefficients expresses $F/s_1$ through the coefficients and one parameter $\tau$, with a term $q^2/(4\tau^2)$. Gauss's lemma makes $2s_1\tau$ an integer, so if $q \neq 0$ the formula bounds $F$ by a polynomial of degree 7 in the largest digit $M$. But $F \ge M^M$. For $M \ge 7$ this is a contradiction, and an exact check covers $3 \le M \le 12$. □

**Theorem 5.27** (Galois groups for four and five digits). PROVEN L16.4
1. For $k = 4$ and largest digit $M \ge 5$, every irreducible $Q_s$ has Galois group $S_3$.
2. For $k = 5$ and $M \ge 6$, every irreducible $Q_s$ has group $S_4$, unless $P_s$ is decomposable, and then the group lies in $D_4$.

*Proof sketch.* Once $F > (M + 1)^k$, $Q_s$ has exactly one real root for $k = 4$ and two for $k = 5$, because $P_s$ has no critical point outside $|r| \le M + 1$. For $k = 4$ the group $A_3$ needs three real roots. For $k = 5$ complex conjugation is a transposition, so the group is $S_4$ or $D_4$, and $D_4$ needs a rational root of the resolvent cubic. That root satisfies an equation that is linear in $F$ and bounds $F$ unless $q = 0$. □

**Computation 5.28** (Smaller groups). COMPUTATION L16.4
For $k = 4$ with digits up to 10, all 26,580 irreducible $Q_s$ have group $S_3$. For $k = 5$ with digits up to 8, 1,967 have group $D_4$, all with decomposable $P_s$, and 102,865 have $S_4$. For $k = 6$ with digits up to 5, 81 have the Frobenius group of order 20 and the rest $S_5$. The 81 are $s_1 r^5 + s_6$ and the strings $(1, 0, 5, 0, 5, x)$ and $(5, 0, 5, 0, 1, x)$, which are Dickson polynomials: $r^5 + 5r^3 + 5r = D_5(r, -1)$. These are the two exceptional families of Ritt's theory [24].

**Computation 5.29** (Six and seven digits). COMPUTATION L16.3
For $k = 6$ (digits up to 9) and $k = 7$ (digits up to 6), every factorization of $Q_s$ without a linear factor and without a decomposition of $P_s$ has largest digit at most 4. None of these polynomials has a root modulo every prime below 3000.

**Conjecture 5.30** (Reducibility for large digits). CONJECTURE L16.3
For every $k$ there are constants $C_k$ and $c_k$ such that, once $F(s) > C_k M^{c_k}$, every factorization of $Q_s$ without a linear factor comes from a decomposition $P_s = G(h(r))$. (Question Q33.)

If the conjecture holds, no $Q_s$ with large digits is intersective unless $P_s$ is decomposable. The sieve agrees: of 400 sampled survivors of the 18 primes with 4 digits, 394 are irreducible and would fall to more primes, and 6 have a rational root L12.2.

## 5.5 The root heuristic

The probabilistic model of Chapter 3 treats each base on its own. For a short string it is better to ask a different question: given $s$, is the real root of $Q_s$ an integer? This section builds a heuristic for that question, calibrates it, and explains what it cannot see.

### 5.5.1 Spacing

Let $s$ have $k$ digits and largest digit $x$. The real root $b^*$ of $Q_s$ is about $x^{x/(k-1)}$. As $s$ varies over strings with nearby roots, $Q_s$ changes by integers, so the root moves in steps of about $1/Q_s'(b^*)$. So the chance that $b^*$ is an integer is about $1/Q_s'(b^*)$, roughly $x^{-x(k-2)/(k-1)}$. There are about $kx^{k-1}$ strings with largest digit $x$.

**Heuristic 5.31** (Spacing heuristic). HEURISTIC L3.5 L11.4
The expected number of $k$-digit solutions is $E \approx \sum_s 1/Q_s'(b^*)$. It is finite for every $k \geq 3$. For $k = 2$ the chance is about $1/a$ for the leading digit $a$, and the sum diverges.

The divergence for two digits is right: there are infinitely many 2-digit solutions (Proposition 2.14). The finiteness for $k \geq 3$ is wrong, as Corollary 2.22 shows. We return to this below.

### 5.5.2 Local factors

A string whose polynomial has no root modulo $p$ cannot be a solution, and a string with two roots modulo $p$ is twice as likely to have a root at a random integer. So each prime contributes a factor, the number of roots.

**Heuristic 5.32** (Root heuristic with local factors). HEURISTIC L11.4 L19.3
The expected number of $k$-digit solutions is

$$E = \sum_s \frac{\prod_p N_s(p^e)}{Q_s'(b^*)},$$

capped at 1 per string, where $N_s(p^e)$ is the number of roots of $Q_s$ modulo a high power $p^e$ of each small prime $p$.

The first version of the heuristic used the number $\rho_s(p)$ of roots modulo $p$. The right factor is the density of the event $p^e \mid Q_s(b)$ for a random integer $b$, divided by $p^{-e}$, which is $N_s(p^e)$. When every root modulo $p$ is simple, Hensel's lemma gives $N_s(p^e) = \rho_s(p)$. A multiple root can lift to no root or to several.

**Proposition 5.33** (The local factor averages 1). PROVEN L19.3
Over a constant term that is uniform modulo $p^e$, the mean of $N_s(p^e)$ is exactly 1.

*Proof.* Each of the $p^e$ residues $r$ is a root for exactly one value of the constant term modulo $p^e$. □

So the local factors move individual strings but not the average. The factor stabilizes from $2^4$, $3^3$ and $5^2$ on.

Table 5.5. The root heuristic under $0^0 = 1$ against the observed counts L11.4, L19.3. The heuristic sums over strings with digits up to the stated bound.

| $k$ | digits up to | spacing only | roots mod $p$ | roots mod $p^e$ | observed |
|---|---|---|---|---|---|
| 3 | 60 | 2.31 | 5.09 | 5.59 | 6 |
| 4 | 22 | 1.38 | 1.39 | 0.75 | 2 |
| 5 | 11 | 1.13 | 2.42 | 3.32 | 6 |
| all | | 4.82 | 8.90 | 9.66 | 14 |

With local factors the 3-digit count matches. The 4-digit and 5-digit counts rest on 2 and 6 data points, and single strings near the cap decide them: 3435 has two roots modulo 2 but only one modulo 64, so its weight drops from 1 to 0.49 L19.3. Better calibration needs more data, which the offsets provide.

### 5.5.3 Calibration by offsets

A solution is a string with $P_s(b) - F(s) = 0$. A near-solution at offset $D$ has $P_s(b) - F(s) = D$. Near-solutions are further draws of the same kind of coincidence, so their counts calibrate the heuristic without trusting it. The program `kdigit` takes an option `--offset D` for this.

**Computation 5.34** (Offsets for four digits). COMPUTATION L19.2
Under $0^0 = 1$, digits up to 22, the 200 offsets $D = 1, ..., 200$ have 682 pairs $(s, b)$ in bases below 100. The heuristic predicts 731 with roots modulo $p$ and 677 with roots modulo $p^e$. The counts per offset have variance over mean 0.97, as for a Poisson law. Calibrated by the offsets, the expectation at $D = 0$ is 0.6 to 1.4, and 2 observed has probability 0.13 to 0.42.

Small offsets also catch neighbours of true solutions and algebraic families. 3435 has the neighbours $(3, 4, 1, 5)$ at $D = 6$ and $(3, 4, 2, 5)$ at $D = 13$. The string $(1, 0, 0, x)$ with $x^x$ a cube sits at $D = x - 3$ in base $x^{x/3}$, and $(1, 3, 3, x)$ at $D = x - 56$ in base $x^{x/3} - 1$. These identities hold with certainty where they hold, so the comparison above leaves out the bases of 100 and more, where 15 of 16 pairs belong to such families L19.2.

For five digits the offsets agree with the heuristic in bases below 10, 354 against 353.7. In bases 10 to 99 they show about twice the predicted count, mostly in clusters: groups of about four hits with a common prefix of three digits. Some clusters come from exact identities, such as $6 \cdot 11^4 + 6 \cdot 11^3 + 5 \cdot 11^2 = 6^6 + 6^6 + 5^5$, the self-paying block of Section 2.8. The cleanest calibration is base by base.

**Computation 5.35** (Base-by-base calibration). COMPUTATION L22.4
For each base $b$ and length $k$, count the strings with $P_s(b) - F(s) = D$ for all offsets $D$ in the class of 0 modulo $2(b-1)$ with $b \le |D| \le 0.05b^{k-1}$. The mean count per offset estimates the expected number of solutions in that base. For 5 digits in bases 13 to 99 this uses 1.24 million offsets with 257 hits and gives 0.33, with a range of 0.26 to 0.49 over the choices of window and class. For 6, 7 and 8 digits in bases 3 to 12 it gives 1.19, 0.99 and 1.13 against 1, 1 and 2 observed.

The 5-digit estimate agrees with the root heuristic, 0.28 with roots modulo $p$ and 0.33 with roots modulo $p^e$. But bases 13 to 99 hold three 5-digit solutions: $(3,3,6,6,1)$ in base 13, $(2,6,0,3,6)$ in base 14 and $(3,8,6,2,0)$ in base 48.

**Heuristic 5.36** (The 5-digit excess). HEURISTIC L19.2 L22.4

Three 5-digit solutions in bases 13 to 99 against 0.33 expected is an excess of about 9 times. Under a Poisson law the probability of 3 or more is 0.0047 (0.0024 to 0.014 over the choices of the calibration), and about 0.015 after allowing for the choice among the 10 cells that were examined.

The excess was once put at 17 times, from a calibration by far offsets $D = 75\,600j$ with only 10 hits. The base-by-base calibration replaces it L22.4. Could self-paying blocks explain the rest? The first solution is the exact block $(3,3,6,6)$ followed by a 1: $13 \cdot P_{(3,3,6,6)}(13) = 93\,366 = 3^3 + 3^3 + 6^6 + 6^6$. But the heuristic already counts this route.

**Lemma 5.37** (Blocks are inside the heuristic). PROVEN L22.4 LEAN

Under $0^0 = 1$, the polynomial of the string $(t,1)$ is $rP_t(r) - F(t)$, the polynomial of the block equation $bP_t(b) = F(t)$.

Lean: `P_sub_Fs_append_one, P_sub_Fs_append_zero`

*Proof.* $P_{(t,1)}(r) - F(t,1) = rP_t(r) + 1 - F(t) - 1$. □

The block route carries 0.039 of the expected 0.33, so blocks do not explain the excess. It stays an unexplained fluctuation at the level of $P \approx 0.015$ L22.4.

### 5.5.4 Local factors and the per-base model

The root heuristic gives each string a factor at every small prime. The per-base model of Chapter 3 has only the factor of the identity modulo $b-1$ (Section 3.3). The two look like different heuristics. They are one heuristic averaged in two ways, and the link is the way the arrangements of one multiset spread over the residues modulo $q$.

Fix a base $b$, a modulus $q$ and a multiset $c$ of $k \geq 3$ digits, and draw an arrangement of $c$ with a nonzero leading digit uniformly. Write $\sigma(c)$ for the digit sum. If $q \mid b-1$, every arrangement is congruent to $\sigma(c)$ modulo $q$ (Proposition 1.9). If $q \mid b$, the residue is fixed by the last digit. The remaining case is $\gcd(q, b(b-1)) = 1$. Since $b(b-1)$ is even, such a $q$ is odd.

**Theorem 5.38** (Arrangements spread over the residues). PROVEN L35.1

Let $\gcd(q, b(b-1)) = 1$. Fix the leading digit, let the other $k' = k-1$ digits be $x_1, \ldots, x_{k'}$, and put

$$\gamma = \max_{h \not\equiv 0 \pmod q} \left| \frac{1}{k'} \sum_{i=1}^{k'} e(hx_i/q) \right|.$$

Then for every residue $a$,

$$|qP(n \equiv a \pmod q) - 1| \leq (q-1)e^{1/8} \exp(-(1-\gamma)^2(k-1)/64).$$

The same holds without fixing the leading digit, with $\gamma$ the largest over the leading digits.

*Proof sketch.* Pair the positions $(2j, 2j+1)$. Exchanging the digits $u, v$ of a pair moves $n$ by $(v-u)b^{2j}(1-b)$, a unit times $v-u$ modulo $q$. Exchange each pair or not by a fair coin. The arrangement stays uniform, and the coins multiply $E\,e(hn/q)$ by factors $|\cos(\pi h\delta_j/q)|$. A pair is two distinct

digits drawn from those not yet used, so the mean of $\cos^2$ is at most $(1+\rho_j^2)/2$, with $\rho_j$ close to $\gamma$. Take about $(1-\gamma)k/8$ pairs and sum over $h$ L35.1. □

The rate depends on the length and on how spread out the digits are, not on the order of $b$ modulo $q$, as question Q91 expected. The proof uses nothing about $b$ except that $b$ and $b-1$ are units modulo $q$. So it applies to every residue $r$ of the base with $r$ and $r-1$ prime to $q$.

**Proposition 5.39** (The local factor is a base average). PROVEN L35.1
For a string $s$ let $N_s(q)$ be the number of residues $r$ modulo $q$ with $P_s(r) \equiv F(s) \pmod q$. For every multiset $c$, the mean of $N_s(q)$ over the arrangements $s$ of $c$ is

$$\sum_{r \pmod q} P(P_s(r) \equiv F(c) \pmod q).$$

The term $r=1$ is 1 if $q$ divides $F(c)-\sigma(c)$ and 0 otherwise. The term $r=0$ is the share of arrangements whose last digit is congruent to $F(c)$. For a prime $q$ each other term is $1/q$ up to the error of Theorem 5.38.

*Proof.* Exchange the sum over $r$ and the mean over the arrangements. Then $P_s(1)=\sigma(c)$ and $P_s(0)$ is the last digit. For a prime $q$ every other $r$ has $r$ and $r-1$ prime to $q$. □

So the local factor of a string, averaged over its arrangements, is the per-base factor averaged over the residue of the base. The refined model keeps the residue of the actual base: at bases $b \equiv 1$ it has the identity factor $q$ or 0, at the others the arrangement factor or the last-digit factor. The two heuristics agree on totals over many bases and differ base by base.

**Computation 5.40** (How fast the arrangement factors approach 1). COMPUTATION L35.1
Candidate multisets of $k$ digits in bases 9 to 30, 40 per base, under $0^0=1$, with the shares computed exactly. RMS of $qP(n \equiv F(c))-1$ over the bases with $\gcd(q, b(b-1))=1$:

| $q$ | $k=3$ | 4 | 6 | 8 | 10 | 12 |
|---|---|---|---|---|---|---|
| 3 | 0.76 | 0.59 | 0.28 | 0.17 | 0.087 | 0.043 |
| 5 | 0.93 | 0.73 | 0.30 | 0.18 | 0.082 | 0.036 |
| 7 | 1.14 | 0.71 | 0.30 | 0.21 | 0.075 | 0.040 |
| 9 | 1.48 | 1.03 | 0.51 | 0.25 | 0.12 | 0.084 |

The spread falls like $e^{-0.35k}$ and is below 0.1 from 10 digits on for $q=3,5,7$ and from 12 digits on for $q=9$. The mean of the factor is 0.98 to 1.04 in every cell from 5 digits on, and 0.87 to 1.05 at 3 and 4 digits.

The arrangement factors have mean 1. They add scatter to single strings, not an excess. So they cannot be why short solutions are more common than the basic model of Chapter 3 says, which was the lead of question Q91. Part of the excess has another source. At a short length a sum of a few values $x^x - x$ of small digits is concentrated modulo the small primes of $b-1$. For example $x^x \equiv x \pmod 3$ for 5 of every 6 digits $x \geq 1$. So the identity factor $(b-1)P((b-1) \mid F(c)-\sigma(c))$ is large at short lengths. The refined model contains it if it is computed at the short length itself.

**Computation 5.41** (Short solutions against the exact refined model). COMPUTATION L35.1
Expected and observed solutions, summed over the bases $\max(9,d)$ to 41:

| $d$ | basic | refined, $0^0=1$ | observed | refined, $0^0=0$ | observed |
|---|---|---|---|---|---|
| 3 | 0.46 | 0.72 | 1 | 0.97 | 3 |
| 4 | 0.39 | 0.56 | 2 | 0.83 | 2 |
| 5 | 0.40 | 0.45 | 2 | 0.61 | 4 |

| $d$ | basic | refined, $0^0 = 1$ | observed | refined, $0^0 = 0$ | observed |
|---|---|---|---|---|---|
| 6 | 0.38 | 0.47 | 1 | 0.66 | 2 |
| 8 | 1.13 | 1.23 | 2 | 1.74 | 3 |
| 12 | 1.12 | 1.20 | 3 | 1.79 | 2 |

The short exceptions of Section 3.9 (3 to 5 digits far below the base, bases 10 to 41 and 43) have the basic expectation 0.96 and the refined expectations 1.35 under $0^0 = 1$ and 1.86 under $0^0 = 0$, against 5 and 9 observed.

Under $0^0 = 0$ the nine are mostly structured: self-paying zeros, block pairs and a family member. Under $0^0 = 1$ one of the five is a family member. The other four are 3435, $(6, 5, 3, 4)$ in base 20, $(3, 3, 6, 6, 1)$ in base 13 and $(2, 6, 0, 3, 6)$ in base 14. Against 1.35 expected, four or more has probability 0.048, and two of the four are the 5-digit excess of Heuristic 5.36.

**Heuristic 5.42** (One heuristic for every length). HEURISTIC L35.1

The root heuristic and the per-base refined model are two averages of one local density. The root heuristic averages the per-base factors over the residue of the base; the refined model keeps the residue of the actual base. They agree on totals over many bases. Per base they differ by the arrangement factors, which have mean 1 and an RMS of about $e^{-0.35k}$, and by the last-digit factor. The excess of short solutions over the basic model has three parts: the identity factor modulo $b - 1$ at the short length (1.1 to 2.1 for $x^x$), identity families and self-paying blocks, and under $0^0 = 1$ a residual of about 3 times on four generic strings.

The medium lengths test this. For lengths 6 to 12 in bases 7 to 20 the library computed the root heuristic by sampling, the refined model exactly (with and without the forced top digit), and the base-by-base calibration of Computation 5.35 for lengths 6 to 10.

**Computation 5.43** (The two heuristics at 6 to 12 digits). COMPUTATION L35.2

Bases 7 to 20, all 83 cells of a length from 6 to 12:

| convention | observed | refined | root, roots mod $p$ | root, roots mod $p^e$ |
|---|---|---|---|---|
| $0^0 = 1$ | 10 | 8.85 | 9.51 | 9.74 |
| $0^0 = 0$ | 12 | 12.72 | 13.00 | 14.19 |

Lengths 6 to 10 under $0^0 = 0$, split by the parity of the base:

| bases | observed | calibration | refined, top digit forced | root, roots mod $p$ |
|---|---|---|---|---|
| even | 2 | 2.38 | 2.55 | 4.37 |
| odd | 8 | 5.77 | 5.56 | 5.27 |

Cell by cell the refined model with the top digit forced correlates with the calibration at 0.99 under $0^0 = 1$ and 0.94 under $0^0 = 0$.

So the two heuristics give the same totals, and both fit the census. Base by base only the refined model matches the calibration. The root heuristic averages the parity factor over even and odd bases, as Proposition 5.39 says it must. The forced top digit matters in single cells: in base 10 at 6 digits every candidate digit is at most 7, and one 7 makes $F \geq 7^7 = 823\,543$, a top digit 8 or 9 that no candidate contains. No missing factor shows at these lengths, so the 5-digit excess is not a factor that the medium lengths share.

### 5.5.5 What the heuristic cannot see

The completeness of the lists can be estimated with the same heuristic: sum the expected number of solutions with a digit above the list's bound.

**Heuristic 5.44** (Completeness of the lists). HEURISTIC L11.4 L19.1

For strings without an algebraic identity, the expected number of solutions missing from the lists is about $10^{-74}$ for 3 digits beyond the list with digits up to 80 (and below $10^{-3294}$ beyond Computation 5.3), $10^{-490}$ for 4, $10^{-109}$ for 5, $10^{-45}$ for 6, $1.7 \cdot 10^{-33}$ for 7 and $5.7 \cdot 10^{-21}$ for 8. For the missing lengths 9, 18, 21, 22, 24, 26 and 28 it is $10^{-29}$ or less.

These numbers say nothing about algebraic families. In the binomial family of Theorem 2.21 the digits are the coefficients of $(cr+e)^{k-1}$ without the constant term, followed by a digit $x$ with $x^x$ a $(k-1)$-th power. For these strings

$$Q_s(r) = (cr+e)^{k-1} - x^x$$

exactly, so the root is an integer by an identity. The heuristic gives such a string the weight $1/Q_s'(b^*)$, which is astronomically small. The same holds for the self-paying digits under $0^0 = 0$. So the refined heuristic of L22.4 adds the family members in a range, counted exactly, to the calibrated mean. In the searched ranges no family member under $0^0 = 1$ appears, and the tail estimates stand for generic strings.

The heuristic is blind to a second family as well. Under $0^0 = 1$ the strings $(c^j, 0, ..., 0, x)$ with $x = c^{jc^j} + j - 1$ are solutions for suitable $c$, because the last digit pays for the weight of $c^j$ and for the $j-1$ zeros (Theorem 2.30). For $c = 1$ it is the family $(1, 0, 0, x)$ seen above at the offset $D = x - 3$. These two linear families are the only identity families found.

Conjecture 2.35 states the resulting conjecture: under $0^0 = 1$, all but finitely many $k$-digit solutions are members of the two linear families, and every $k$-digit solution beyond the complete lists is one.

The evidence has four parts L25.1. The complete lists hold no other solution up to $10^{3301}$ (3 digits), $10^{186}$ (4), $10^{30.9}$ (5) and $10^{11}$ (6). The root heuristic puts the expected number beyond the lists at $10^{-74}$ or less. Every power identity $Q_s = \lambda A^j - W$ ends in a large digit, and every identity other than the two linear families needs one more coincidence, such as a large number being a perfect power. A census of all small prefixes, up to 7 digits, finds no member of any other mechanism. The conjecture cannot be strengthened to "finitely many solutions", since both families are infinite. It fails under $0^0 = 0$, where the self-paying digits and the swaps give further families. It replaces the old finiteness conjecture as the answer to question Q64.

## 5.6 The square class and near-integers

When one digit $x$ is large, $F(s)$ is dominated by $x^x$, and the question "is the root an integer?" becomes a question about how close a power comes to an integer. One case can be settled completely.

**Theorem 5.45** (Large leading digit). PROVEN L8.2 LEAN

Fix $k \geq 2$ and $k-1$ lower digits below a bound $B$, and let the leading digit $x$ vary with $x^{x-1}$ a perfect $(k-1)$-th power. For $x$ above an explicit bound that depends on $B$ and $k$, the only solutions are the self-paying families $(x, 0, ..., 0, 0)$ and $(x, 0, ..., 0, 1)$ under $0^0 = 0$, and $(x, 1)$ with $k = 2$ under $0^0 = 1$. For $k \geq 3$ there are none under $0^0 = 1$.

Lean: `theorem8_3, theorem8_3_one`

*Proof.* Write $x^{x-1} = Z^{k-1}$, let $Q(b)$ be the value of the lower $k-1$ digits and $C$ the sum of their weights. The equation is $xb^{k-1} + Q(b) - C = xZ^{k-1}$, with $0 \leq Q(b) < 2Bb^{k-2}$. If $b \geq Z+1$, the left side is at least $xZ^{k-1} + x(k-1)Z^{k-2} - C$, too large once $x(k-1)Z^{k-2} > C$. If $b \leq Z-1$, the left side is at most $xZ^{k-1} - (Z-1)^{k-2}(x(k-1) - 2B)$, too small once $x(k-1) > 2B$. So $b = Z$ and $Q(Z) = C$. A nonzero lower digit at a position $i \geq 1$ gives $Q(Z) \geq Z > C$ for large $x$. So the

only nonzero lower digit is the last one, $e$, and $e = (k-2)w(0) + w(e)$. Under $0^0 = 0$ this means $e^e = e$, so $e \in \{0, 1\}$. Under $0^0 = 1$ it means $e = k - 2 + w(e)$, and since $w(e) \geq e$ this forces $k = 2$, $e = 1$. □

### 5.6.1 Three digits: completing the square

For three digits $(p, q, r)$ the equation $pb^2 + qb + r = F$ becomes

$$(2pb + q)^2 = 4pF + q^2 - 4pr.$$

If one digit $x$ is large, write the right side as $4px^x + R$ when $x$ is not the leading digit (and similarly when it is). Call the string *in the square class* when the dominant term is a perfect square. Then the right side is a square plus a term of size about $x^2$, and two squares that close must be equal, which bounds $x$, unless $R = 0$.

The following theorem makes this precise and classifies every 3-digit string with one large digit.

**Theorem 5.46** (Three digits with one large digit). PROVEN L25.1 LEAN

Under $0^0 = 1$, let $s = (p, q, r)$ have two digits at most $B \geq 1$ and a third digit $x \geq 2B + 4$, and let $\ell$ be the leading digit. Write

$$(2pb + q)^2 = N + R, \quad N = 4\ell x^x, \quad R = 4p(F - r) + q^2 - N.$$

Then exactly one of the following holds.

1. $R = 0$. Then $x$ is the last digit, $p = \lambda c^2$ and $q = 2\lambda ce$ with $\lambda$ squarefree, $c \geq 1$, $e \geq 0$, and $x = w(p) + w(q) + \lambda e^2$. Here $Q_s(r) = \lambda(cr + e)^2 - x^x$, and $s$ is a solution exactly when $x^x = \lambda(cb + e)^2$. For even $x$ this needs $\lambda = 1$ and $c \mid x^{x/2} - e$, the binomial family with $j = 2$. For odd $x$ it needs $x/\lambda$ to be a square.
2. $R \neq 0$ and $N$ is a perfect square. Then $s$ is not a solution.
3. $R \neq 0$ and $N$ is not a perfect square. Then $Q_s$ has no power identity $\lambda A^2 - x^x$, and $s$ is a solution exactly when the Pell-type equation $U^2 - nV^2 = R$ holds with $U = 2pb + q$, where $N = nV^2$ with $n > 1$ squarefree.

Lean: `theorem25_7_ii_last, theorem25_7_ii_middle, theorem25_7_ii_leading, prop22_8 (partial: case (ii), and case (i) for even x)`

*Proof sketch.* If $x$ leads or sits in the middle, $R$ is a sum of positive terms, so $R = 0$ needs $x$ last, with $R = 4p(w(p) + w(q) - x) + q^2$. Then $4p \mid q^2$, and writing $p = \lambda c^2$ with $\lambda$ squarefree a count of prime exponents gives $q = 2\lambda ce$. The equation becomes $x^x = \lambda(cb + e)^2$. For (ii), let $N = V^2$. If $U > V$ then $R \geq 2V + 1$, and if $U < V$ then $|R| > V$. So a solution needs $|R| > \sqrt{N} \geq 2x^{x/2}$. But $|R| \leq 9x^2B^B$ in every position, and $x \geq 2B + 4$ makes $x^{x/2} \geq 6x^2B^B$. For (iii), an identity with one large digit puts it last, and $4p(P_s(r) - F(t))$ is a constant times a square exactly when $R = 0$. □

So among 3-digit strings with one large digit, the identities are exactly case (i), case (ii) has no solutions, and everything else is the generic case (iii). The six known 3-digit solutions all have $x \leq 7 < 2B + 4$, so the theorem does not apply to them. $(5, 1, 3)$ in base 25 sits on the boundary of case (ii): $N = 4 \cdot 5^6 = 250^2$, $U = 251$ and $R = 501 = 2V + 1$.

**Remark 5.47.** L8.2 L22.3 L25.1

An earlier corollary in the library claimed that the 3-digit solutions with two digits at most 20 and the third in its square class are the known ones. It assumed $R \neq 0$, and the binomial members with $R = 0$ refuted it: they add $(1, 6, x)$, $(1, 10, x)$, $(1, 14, x)$, $(1, 18, x)$, $(4, 8, x)$, $(4, 16, x)$, $(9, 6, x)$ and $(9, 18, x)$, in bases above $10^{108942}$. Theorem 5.46 is the final form of that corollary.

### 5.6.2 Outside the square class

When the dominant term is not a square, we are in case (iii) of Theorem 5.46: write $N = nV^2$ with $n$ squarefree. The equation asks whether $N + R$ is a square: a Pell-type equation $U^2 - nV^2 = R$ with the extra condition that $V$ equals a prescribed number such as $x^{(x-1)/2}$ (Appendix D). For each single $x$ this is decidable. For all $x$ at once it asks whether an exponentially growing sequence ever lands on the solution set of a Pell equation that itself changes with $x$. For longer strings it asks whether $x^{(x-1)/(k-1)}$ comes within about $B/x$ of an integer.

Such questions are beyond current methods. Even for the single sequence $(3/2)^n$ it is unknown whether the fractional parts are dense in $[0, 1]$. Mahler [25] proved that $(3/2)^k$ stays at distance more than $c^k$ from the nearest integer for every $c < 1$ and large $k$, without an effective bound, and Zudilin [26] gave the effective constant 0.5803. Until chapter 32 of the library, no result there handled equations where the base and the exponent of $x^x$ vary together L8.2, L9.10. Section 5.7 does it when the quadratic field stays fixed, with a $p$-adic form of Baker's theorem.

**Computation 5.48** (The near-integer law). COMPUTATION L12.5

Fix two digits of a 3-digit string and let the third, $x$, run over odd values up to 6,001, under $0^0 = 1$. For $(x, 1, 5)$, in the square class, the real root $b^*(x)$ sits just below an integer, with $x\|b^*(x)\| = 0.500$ for every $x$ from 19 to 6,001. For $(1, x, 1)$, $(6, 1, x)$ and $(2, x, 3)$ the fractional parts look uniform: 1.8% to 2.1%, 9.7% to 10.2% and 48.5% to 49.6% of them lie below 0.01, 0.05 and 0.25. The only exact hits are the known solutions.

So in the square class the root approaches the integers without reaching them, which is Theorem 5.45 seen numerically. Outside it the root behaves like a random number. That is what the spacing heuristic assumes, but a solution needs $b^*(x)$ to be an integer to within $x^{-x/2}$, far below what such a test can resolve. The family $(1, x, 1)$ leads to the equation $b(b + x) = x^x + 1$, the subject of Chapter 6.

## 5.7 Three regimes for three digits

Theorem 5.46 turns every 3-digit string with one large digit $x$ into a Pell-type equation $U^2 - nV^2 = R$, where $n$ is the squarefree part of $N = 4\ell x^x$. Whether this equation can be handled for all $x$ at once depends on one thing: how $n$ changes with $x$. The parity of $x$ decides most of it. If $x$ is even, $x^x = \left(x^{x/2}\right)^2$ is a square. If $x$ is odd, $x^x = x \cdot \left(x^{(x-1)/2}\right)^2$ is $x$ times a square. So $n$ is the squarefree part of $\ell$ (for even $x$ that does not lead), of $px$ (for odd $x$ that does not lead, with leading digit $p$), or of $x$ itself (for even $x$ that leads); for odd $x$ that leads, $N$ is a square. This gives three regimes and an open remainder L32.1.

Table 5.6. The regimes for 3-digit strings with one large digit $x$ L32.1.

| regime | when | method | result |
|---|---|---|---|
| $N$ a square | $x$ odd and leading; $x$ even behind a square leading digit | the gap of Theorem 5.46 (ii) | no solution outside the binomial family |
| $n$ fixed | $x$ even, not leading, leading digit not a square | 2-adic Baker (Theorem 5.49) | finitely many, explicit bound |
| $n$ fixed on a class | $x = \lambda t^2$ with $\lambda$ fixed: odd and not leading, or even and leading | real Baker (Proposition 5.52) | finitely many in each class |
| $n$ varies | the rest: odd $x$ not leading, even $x$ leading | none known; abc (Theorem 5.54) | open; finite under abc |

The first regime is elementary: two squares cannot be close unless they are equal. The other two use linear forms in logarithms, explained in Appendix E.

### 5.7.1 When the field does not change

Fix the digits $p \geq 1$ and $q \geq 0$, and let $x \geq 2$ be even. Consider the two strings

$$\text{last: } (p, q, x), \quad pb^2 + qb + x = w(p) + w(q) + x^x,$$

$$\text{middle: } (p, x, q), \quad pb^2 + xb + q = w(p) + x^x + w(q).$$

Write $p = nf^2$ with $n$ squarefree and assume $n > 1$, that is, $p$ is not a square. Multiply by $4p$ and complete the square. Since $x$ is even, both equations take the form

$$U^2 - nV^2 = R, \quad V = 2fx^{x/2},$$

with $U = 2pb + q$ and $R = q^2 + 4p(w(p) + w(q) - x)$ for the last position, and $U = 2pb + x$ and $R = x^2 + 4p(w(p) + w(q) - q)$ for the middle one. $R$ is not 0, since otherwise $n$ would be a square.

The point is that $n$ does not depend on $x$. All these equations, for every even $x$, live in one quadratic field $K = \mathbb{Q}(\sqrt{n})$, with one fundamental unit $\eta > 1$ (Appendix D).

**Theorem 5.49** (A 2-adic bound). PROVEN L32.1

Let $p$ be a positive integer that is not a square, $q \geq 0$, and $x$ even. If $(p, q, x)$ or $(p, x, q)$ is a solution in some base, under either convention, then $x \leq X_0(p, q)$, an explicit bound. It comes from an inequality whose left side is about $x/2$ and whose right side grows like $(\ln x)^3$.

*Proof sketch.* Put $\beta = U - V\sqrt{n}$ and $\beta' = U + V\sqrt{n}$, the two conjugates. Then $\beta\beta' = R$, and $\beta' - \beta = 2V\sqrt{n}$.

1. The difference is 2-adically tiny. $V = 2fx^{x/2}$ and $x$ is even, so $2^{x/2+1}$ divides $V$ and $v(\beta' - \beta) \geq x/2 + 2$, where $v$ is the 2-adic valuation, extended to $K$ (Section E.4). Since $\beta$ divides $R$, $v(\beta) \leq \log_2|R|$, and so $v(\beta'/\beta - 1) \geq x/2 + 2 - \log_2|R|$.
2. Make the height small. $\beta'$ is huge, so $\beta$ has a large height. Multiply by a power of the unit: $\delta = \beta\eta^{-m}$, with $m$ chosen so that both conjugates of $\delta$ have size about $\sqrt{|R|}$. Then $h(\delta) \leq (1/2)\ln(|R|\eta)$, and $m$ is about $\ln V / \ln \eta$, of order $x \ln x$.
3. A 2-adic linear form. Since $\eta\eta' = \pm 1$, the ratio $\beta'/\beta$ equals $\alpha_1\eta^{-2m}$ with $\alpha_1 = \pm\delta'/\delta$. So step 1 says that $\alpha_1 - \eta^{2m}$ has 2-adic valuation at least $x/2 + 2 - \log_2|R|$.
4. Bugeaud and Laurent (Theorem E.4) bound the same valuation from above by a constant times $(\ln m)^2 \ln(|R|\eta) \ln \eta$. With $m$ of order $x \ln x$ and $\ln|R|$ of order $\ln x$, the right side grows like $(\ln x)^3$, while the left side grows like $x/2$. So $x$ is bounded.

The cases where the theorem of Bugeaud and Laurent does not apply give smaller bounds: if $\alpha_1$ is not a 2-adic unit, then $x/2 + 2 \leq \log_2|R|$; if $\alpha_1 = \pm 1$, a direct computation of $v(\eta^{12m} - 1)$ works; and $m = 0$ forces $x$ to be tiny. □

The argument is the one Bennett and Bugeaud use for approximation by quadratic irrationals [27]. What is new is that the base of the power, $x$, moves together with the exponent. That costs nothing: only the 2-adic valuation of $V$ enters, and the height of $\delta$ grows only like $\ln x$.

**Computation 5.50** (No such string with small digits). COMPUTATION L32.1

For every non-square $p \leq 20$, every $q \leq 20$ and every even $x$, neither $(p, q, x)$ nor $(p, x, q)$ is a solution in any base, under either convention. The bounds $X_0(p, q)$ for the 704 cases (family, position, and convention when $q = 0$) reach $2.45 \cdot 10^9$, for $n = 19$, whose unit $170 + 39\sqrt{19}$ is large. A sieve tested all $1.98 \cdot 10^{11}$ values of $x$ below the bounds. The string is a solution exactly when $D = U^2$ is a perfect square, and $D$ must then be a square modulo each of 150 odd primes. The only values that passed every prime are $(3, 1, 2)$, $(3, 2, 2)$ and $(3, 2, 2)$ with $x$ in the middle, and their root is $b = 3$, too small for the digit 3.

**Corollary 5.51** (Even large digits). PROVEN L32.1

Let $s$ be a 3-digit solution in some base, under either convention, whose leading digit and one other digit are at most 20, and whose third digit is even and does not lead. Then $s$ is in the complete lists of Section 5.1, or it is a member $(c^2, 2ce, x)$ of the binomial family (Theorem 2.21 with $j = 2$).

*Proof.* If the leading digit is not a square, use Computation 5.50. If it is a square and $x < 44$, then $pb^2 \le F < 2 \cdot 20^{20} + 42^{42}$, so $b < 2 \cdot 10^{34}$, inside the lists. If $x \ge 44 = 2 \cdot 20 + 4$, Theorem 5.46 applies with $B = 20$: $N = 4px^x$ is a square, so case (iii) is empty, case (ii) has no solution, and case (i) is the family. That theorem is stated under $0^0 = 1$, but for $x$ in the middle or last position its proof works under $0^0 = 0$ as well. □

Of the known 3-digit solutions only $(1, 2, 6)$ in base 215 has this shape, and it is the first member of the family.

The bound can be made uniform in the small digits. If $p, q \le B$, the regulator $\ln \eta$ is at most $\sqrt{B}(\ln 4B + 2)$ by the class number formula, and the same inequality gives $x \le CB^{3/2}(\ln 3B)^4$ for an effective constant $C$ (Theorem 32.5 of L32.1). So the large digit of such a solution cannot be much larger than the small ones, unless the solution is a family member.

These families have no known covering (Section 5.3). Each odd prime $l$ leaves about $(l + 1)/(2l)$ of the residues of $x$, and the first five primes leave about 7% of the even $x$. Every single member fails modulo some prime, but a finite set of primes has a fixed period, while $x$ is unbounded. The bound from transcendence theory replaces the covering, and a finite sieve does the rest.

### 5.7.2 A fixed square class

When $x$ is odd and does not lead, $n$ is the squarefree part of $px$, so the field changes with $x$. It stays fixed if $x$ runs through one square class $x = \lambda t^2$ with $\lambda$ fixed. The same holds for an even $x$ that leads.

**Proposition 5.52** (One square class). PROVEN L32.1

Fix $p$, $q$, the position of $x$ and a squarefree $\lambda$. Then only finitely many $x$ with squarefree part $\lambda$ make the string a solution, and their number is effectively bounded.

*Proof sketch.* The field is fixed, $V$ is a constant times a power of $t$ and $\lambda$, and the associate $\delta$ of the previous proof has height $O(\ln x)$. Now the real absolute value does the work: the real linear form $\ln(2\sqrt{n}V) - \ln \delta - m \ln \eta$ is at most about $|R|/V^2$, roughly $x^{-x}$. It has four logarithms, and Matveev's theorem (Theorem E.3) bounds it below by $\exp(-c(\ln x)^3)$. For large $x$ this contradicts $x^{-x}$. There is no 2-adic shortcut here, because $V$ is odd. □

**Computation 5.53** (Odd squares). COMPUTATION L32.1

For every non-square $p \le 10$, every $q \le 10$ and every odd $t \ge 3$, neither $(p, q, t^2)$ nor $(p, t^2, q)$ is a solution in any base, under either convention. Matveev's bound gives $t \le 2.4 \cdot 10^9$ in these cases, and the sieve tested all $1.18 \cdot 10^{11}$ values of $t$. So Corollary 5.51 holds for digits at most 10 with "even" replaced by "even or an odd square".

### 5.7.3 Where the frontier is

What is left of the 3-digit problem with one large digit is the last row of Table 5.6: Pell equations in which the field $\mathbb{Q}(\sqrt{px})$ or $\mathbb{Q}(\sqrt{x})$ changes with $x$. The regulator of such a field can be as large as $\sqrt{x} \ln x$, and then every known form of Baker's theorem is too weak by a power of $\ln x$. A treatment

of a whole family of real quadratic fields at once, for example through results on $y^2 = ax^n + c$ that are uniform in $n$, would settle the case up to a finite computation (Q129, Q130).

The abc conjecture (Appendix F) closes every regime at once, in the completed-square form L33.4. In $U^2 = N + R$ the radical of the three terms is at most about $|R| \cdot 4\ell x \cdot U$, about $x^{x/2+3}B^B$, while $U^2$ is about $x^x$. The quality tends to 2. Written as $x^x = pb^2 + qb + (x - C)$ instead, the radical contains $b$, about $x^{x/2}$, and abc says nothing. The completed square is what makes it work.

**Theorem 5.54** (Three digits under abc). CONDITIONAL L33.4

Assume the abc conjecture. For every $\varepsilon > 0$ there is a constant $C_\varepsilon$ with the following property. Let $s$ be a 3-digit string, under either convention, with two digits at most $B$ and a third digit $x \geq 2B + 4$, in any position and of any parity, and write $U^2 = N + R$ as in Theorem 5.46. If $R \neq 0$ and $s$ is a solution, then $x \leq (2 + \varepsilon)B + C_\varepsilon$. The same holds for factorions.

*Proof sketch.* Let $g = \gcd(N, R)$. If $R > 0$, the triple $N/g + R/g = U^2/g$ is coprime; if $R < 0$, use $U^2/g + |R|/g = N/g$. Its radical is at most $\mathrm{rad}(N) \cdot U \cdot |R|/g$, and $\mathrm{rad}(N) \leq 2\ell x$. abc gives $N \leq K_\varepsilon(\mathrm{rad}(N)U|R|)^{1+\varepsilon}$. Since $|R| < \sqrt{N}$, $U^2 \leq 2N$. With $\ln N \geq x \ln x$ and $|R| \leq 9x^2B^B$ this becomes $x \ln x \leq (2 + O(\varepsilon))(B \ln B + 5 \ln x) + O_\varepsilon(1)$, so $x \leq (2 + O(\varepsilon))B + C_\varepsilon$. For factorials the radical of $N$ is about $e^x$ instead, and the same steps give a bound. □

So under abc every 3-digit solution with one dominant digit, a digit larger than about twice the others, is a member of the binomial family, case (i) of Theorem 5.46. Up to finitely many exceptions this proves Conjecture 2.35 for 3 digits in that range. With an explicit form of abc the constant is explicit (Section F.2).

Table 5.7. The largest $x$ that the explicit form $c < \mathrm{rad}(abc)^{7/4}$ allows when the other two digits are at most $B$ and $R \neq 0$ L33.4.

| $B$ | 1 | 10 | 100 | 200 | 500 | 1000 |
|---|---|---|---|---|---|---|
| $x^x$, last | 37 | 105 | 974 | 1,991 | 5,130 | 10,484 |
| $x^x$, middle | 45 | 105 | 974 | 1,991 | 5,130 | 10,484 |
| $x^x$, leading | 38 | 105 | 970 | 1,986 | 5,124 | 10,478 |
| $x!$, all | $4.10 \cdot 10^6$ | $4.10 \cdot 10^6$ | $4.11 \cdot 10^6$ | $4.12 \cdot 10^6$ | | $4.19 \cdot 10^6$ |

For $x^x$ the ratio $x/B$ is about 10 in this range and tends to 14 slowly. For $x!$ the bound hardly depends on $B$: the radical of $x!$ is about $e^x$, so abc beats $x!$ only once $\ln x$ exceeds about 15.

**Corollary 5.55** (The 3-digit list under the explicit abc). CONDITIONAL L33.4

Assume $c < \mathrm{rad}(abc)^{7/4}$ for all coprime $a + b = c$. Then every 3-digit solution under $0^0 = 1$ with two digits at most 200 is one of the six known ones or a binomial member, case (i) of Theorem 5.46.

*Proof.* By Table 5.7, $x \leq 1991$ when $R \neq 0$. Part 1 of Computation 5.3 decided every string with digits up to 2000. □

This removes the restriction to bases below $46666^{23333} - 3$ from part 2 of Computation 5.3, at the price of the hypothesis. What abc leaves is two large digits $u \geq v$. Moving the second large term into $R$ and applying abc again leaves only $v \ln v \geq (1/2 - \varepsilon)u \ln u$, about $v \geq u/2$ (with the exponent 7/4, about $v \geq u/14$). In that range the equation $U^2 = 4\ell(u^u + v^v) + R$ has four large terms. abc does not apply to four terms, its $n$-term generalization [28] is too weak here, and $u^u + v^v$ is not a perfect power, so Fermat-Catalan statements do not apply either (Q136).

For four or more digits the method needs the polynomial to be a perfect power up to a small remainder in a fixed field. That happens only for special prefixes, for example $q^2 = 3pr$ for

$(p, q, r, x)$, which gives a Thue equation in $\mathbb{Q}(p^{1/3})$ (Q131). Otherwise the equation defines a curve of genus at least 1 that changes with $x$.

## 5.8 Almost all bases have no short solution

The lists of Section 5.1 are complete only up to a bound. Counting arguments say something about all bases: short solutions exist only in a sparse set of bases. Recall that $X_k(b)$ is the largest digit $x$ with $x^x < b^k$. For fixed $k$, $X_k(b) \sim k \ln b / \ln(k \ln b)$.

**Proposition 5.56** (Few short solutions per base). PROVEN L4.1 LEAN
$N_{m,k}(b) \le 2(X_k(b)+1)^{k-1}$. For fixed $k$ this is a power of $\log b$.
Lean: `prop4_1`

*Proof.* Every digit is at most $X_k(b)$ (Proposition 1.5). Fix the last $k-1$ digits. The leading digit $a$ must satisfy $a^a - ab^{k-1} = R$ with $R$ fixed. The left side is strictly convex in $a$, so at most two values of $a$ work. □

**Proposition 5.57** (Few bases per length). PROVEN L10.5 LEAN
The number of bases up to $B$ with a $k$-digit solution is at most $X(X+1)^{k-1}$, where $X = X_k(B)$.
Lean: `prop10_4`

*Proof.* Every digit is at most $X$, and each digit string is a solution in at most one base (Proposition 1.7). □

Counting multisets instead of strings saves a factor of about $k!$. The price is that a multiset does not fix the base, but a window around a known value does.

**Lemma 5.58** (The base window). PROVEN L20.1
Let $n$ be a solution in base $b$ with $k \ge 2$ digits, leading digit $p$ and largest digit $M$. Then

$$b \le (n/p)^{1/(k-1)} < b + \frac{M+1}{p(k-1)}.$$

*Proof.* The second digit is at most $M$ and the rest add at most $b^{k-2}-1$, so $pb^{k-1} \le n < pb^{k-1} + (M+1)b^{k-2}$. Take $(k-1)$-th roots and use Bernoulli's inequality. □

All 3,135 known solutions satisfy the window, and it is close to sharp: the largest relative position of $(n/p)^{1/(k-1)}$ in it is 0.984 L20.1. A multiset $c$ determines $n = F(c)$, the leading digit is one of its digits, and the window then holds at most $M+2$ candidate bases.

**Proposition 5.59** (Counting multisets). PROVEN L20.1
Let $k \ge 2$ and $X = X_k(B)$. Under a fixed convention, the number of bases $b \le B$ with a $k$-digit solution is at most

$$\binom{X+k}{k}\left((X+1)\frac{H_k}{k-1} + k\right),$$

where $H_k = 1 + 1/2 + \ldots + 1/k$.

**Theorem 5.60** (Almost all bases). PROVEN L20.1 L11.6

For every $\varepsilon > 0$, almost all bases $b$ have no solution with $2 \le k \le (1-\varepsilon)\ln b/\ln\ln b$ digits, under either convention. More precisely, with $L = \ln B$ and fixed $\alpha < 1$, at most $B^{\alpha+o(1)}$ bases $b \le B$ have a solution with at most $\alpha L/\ln L$ digits. The same holds for $k$ up to

$$\frac{\ln b}{\ln\ln b}\left(1 + \frac{\ln\ln\ln b + \ln 2 - 1 - \delta}{\ln\ln b}\right)$$

for any fixed $\delta > 0$, and no argument that counts multisets reaches further.

*Proof sketch.* Put $K = \kappa L/\ln L$ and $X = X_K(B)$. From $X^X \approx B^K$ one gets $\ln(X/K) = \ln L - \ln\ln L - \ln 2 + o(1)$, so $X$ is about $\kappa L^2/(2(\ln L)^2)$ and $K^2/X$ is bounded. Stirling's formula then gives $\ln\binom{X+K}{K} = \kappa L(1 - a/\ln L + \ldots)$ with $a = \ln\ln L + \ln 2 - 1$. The other factors of Proposition 5.59 and the sum over $k \le K$ cost only $\exp(O(\ln L))$. With $\kappa = \alpha$ the count is $B^{\alpha+o(1)}$. The sharper range comes from taking $\kappa = 1 + (a-\delta)/\ln L$, and the same computation with $+\delta$ shows that the bound for $k = K$ alone then exceeds $B$. □

The first version of this theorem counted strings and reached only $(1/2-\varepsilon)\ln b/\ln\ln b$ L11.6. Counting multisets doubles the range.

**Computation 5.61** (The bounds at $B = 10^{96}$). COMPUTATION L20.1

At $B = 10^{96}$, where $L/\ln L = 40.95$, the string bound stays below $B$ up to 31 digits and the multiset bound up to 49 digits. At 16 digits the bounds are $B^{0.458}$ and $B^{0.343}$.

The theorem is weak in absolute terms. It covers lengths of order $\ln b/\ln\ln b$, while the model of Chapter 3 puts almost all solutions near $b$ digits. It is the first unconditional statement that solutions are absent from most bases in a whole range of lengths.

For a fixed length the multiset count alone is weaker than Proposition 5.57, because each window holds about $X/(pk)$ integers. The digit-sum identity repairs this: $b-1$ divides $F(c) - \sigma(c)$, where $\sigma(c)$ is the digit sum of $c$ (Proposition 1.9).

**Lemma 5.62** (Divisors in a short interval). PROVEN L20.1

Let $d_1 < \ldots < d_r$ be divisors of $m \ge 1$ in an interval $(Y-h, Y]$ with $Y - h \ge 1$. Then $m > (Y-h)^r/h^{r(r-1)/2}$.

*Proof.* $m$ is a multiple of $\operatorname{lcm}(d_1, \ldots, d_r)$, which is at least $\prod_i d_i / \prod_{i<j} \gcd(d_i, d_j)$ by a count of prime exponents. Each $\gcd(d_i, d_j)$ divides $d_j - d_i < h$. □

**Proposition 5.63** (Fixed length). PROVEN L20.1

For $k \ge 3$ the number of bases $b \le B$ with a $k$-digit solution is at most $X^k/(k-2)!(1+o(1))$ as $B \to \infty$, with $X = X_k(B)$. This improves Proposition 5.57 by the factor $(k-2)!$.

Beyond $\alpha = 1$ any argument that tests candidates one at a time stops, because the number of multisets itself exceeds $B$. Going further would need an equidistribution statement: that the integers $F(c) - \sigma(c) = \sum(d^d - d)$ spread evenly over residue classes modulo the integers $b-1$ of a short window that depends on $F(c)$ itself. No method for this is known L20.1.

Question Q101 asked whether averaging over the bases gets around this. Split a multiset into large digits and small digits, whose weights add up to less than $b-1$. Given the large part and the base, the congruence $(b-1) \mid F(c) - \sigma(c)$ fixes the small part of $F(c) - \sigma(c)$ exactly, and dissociation (Section 3.7.3) bounds the number of small multisets with a given value. The count becomes a sum

over the large parts and the bases of their windows. The gain is bounded, because most multisets have no small digit at all.

**Proposition 5.64** (Averaging gains a bounded factor). PROVEN L35.4
Let $K = \kappa L/\ln L$ with $\kappa$ bounded, $X = X_K(B)$, and let $z$ be the largest digit with $z^z \le B^A$ for a fixed $A > 0$. Among the multisets of $K$ digits from $0, \dots, X$, the share with no digit up to $z$ is $e^{-2A+o(1)}$. So the averaged count, with the window factor of Proposition 5.59, is at least $e^{-2A-o(1)}$ times the bound of Proposition 5.59. Small weights stay below $b \le B$, so $A < 1$, and averaging in this form gains at most a factor $e^{2+o(1)}$.

*Proof sketch.* The share is $\prod_{i=1}^{K}(1-(z+1)/(X+i))$. From $X\ln X \approx KL$ and $z\ln z \approx AL$ one gets $Kz/X \to 2A$. A multiset without small digits enters the averaged count exactly as in Proposition 5.59. The term $k = K$ carries almost all of the sum over $k \le K$ L35.4. □

**Computation 5.65** (The averaged bound at $B = 10^{96}$). COMPUTATION L35.4
With $z = 30$, the averaged bound is $B^{0.9698}$ at $K = 48$ against $B^{0.9725}$ for Proposition 5.59, a factor 1.8. Both stay below $B$ up to $K = 49$, and up to $K = 362$ at $B = 10^{1000}$.

The window does not see the small digits at all. That gives a range far beyond $\ln b/\ln\ln b$ for strings with few large digits.

**Theorem 5.66** (Few large digits). PROVEN L35.4
Fix $C \ge 1$ and $\varepsilon > 0$. Call a digit $x$ large when $x^x \ge \sqrt{b}$. Almost all bases $b$ have no solution of length at most $(\ln b)^C$ with at most $(1/C-\varepsilon)\ln b/\ln\ln b$ large digits, whatever the number of smaller digits. Up to $B$, at most $B^{\max(1/2,1-C\varepsilon)+o(1)}$ bases have one.

*Proof sketch.* The small digits weigh less than $k\sqrt{b}$ in all. That moves $(F/p)^{1/(k-1)}$ by less than 1, so the base lies in a window fixed by the large digits, the length and the leading digit. Count these data: the large part is one of $\binom{X+K'}{K'}$ multisets with $K' = (1/C-\varepsilon)L/\ln L$ and $\ln X \le (C+1)\ln L$, so its logarithm is at most $(1-C\varepsilon)L(1+o(1))$. The other factors are $\exp(O(\ln L))$ L35.4.□

For $C = 1$ this reaches lengths up to $\ln b$. Typical strings of that length have almost all digits large, so the theorem says nothing about them. It covers the structured strings: a few large digits padded with small ones, the shape of the self-paying and binomial families (Section 2.6). Those families live in sparse sets of bases, which agrees with the theorem.

## 5.9 Other roots: negative bases and rational roots

$Q_s$ has integer roots that are not ordinary bases. Some are Munchausen numbers in negative bases; others are too small to be bases. Rational roots pass every local test, which is why the local test can never prove alone that a string fails everywhere.

### 5.9.1 Negative bases

In base $-b$ with $b \ge 2$, every integer has exactly one representation with digits 0 to $b-1$ and no sign (Appendix I). So a string with digits below $b$ is a Munchausen number in base $-b$ when $Q_s(-b) = 0$: an integer root of the same polynomial at $r < -\max(s)$. Section 7.2 has the census and the model. Two facts belong here.

**Proposition 5.67** (Odd length). PROVEN L12.3 LEAN
A Munchausen number in a negative base has an odd number of digits.
Lean: `IsNegSolution.odd_length`

*Proof.* Its value $F(s)$ is positive. A string of length $k$ in base $-b$ with a nonzero leading digit has the sign of $(-b)^{k-1}$, because the leading term outweighs all others. So $k-1$ is even. □

**Proposition 5.68** (Partners). PROVEN L12.3 LEAN
Let $(s_1, s_2, s_3)$ be a solution in base $b$ with $s_1 \mid s_2$. Then it is also a solution in base $-(b + s_2/s_1)$.
Lean: `IsSolution.negPartner`

*Proof.* The two roots of the quadratic $Q_s$ add up to $-s_2/s_1$, so the other root is the integer $-b - s_2/s_1$. Its absolute value exceeds $b$, which exceeds every digit. □

The 3-digit solutions with leading digit 1, namely $(1, 3, 1)$ in base 4, $(1, 2, 6)$ in base 215 and $(1, 7, 1)$ in base 904, have the partners $-7$, $-217$ and $-911$. A fourth negative-base solution, $(1, 1, 1)$ in base $-2$, has the partner root 1, which is not a base. With five digits, $(1, 3, 4, 3, 3)$ is a solution in base $-5$: $625 - 375 + 100 - 15 + 3 = 338 = 1 + 27 + 256 + 27 + 27$.

**Computation 5.69** (Non-partners). COMPUTATION L16.6 L18.4
Among all 3-digit strings with digits up to 200, and those with two digits up to 1000 and one up to 30, the only negative-base solution under $0^0 = 1$ that is not a partner is $(1, 1, 1)$ in base $-2$.

For five digits partners need a symmetry of the value polynomial.

**Theorem 5.70** (Partners with five digits). PROVEN L16.6
Let $k$ be odd and let $s$ be a solution in base $b$ and in base $-c$. Then $c \geq b$ and $\delta = c - b < 4M/s_1$. If $P_s(r) \neq P_s(-r - \delta)$, then $b \leq 1 + 2kM(1 + \delta)^{k-1}$, which for $k = 5$ forces $M \leq 30$. If $P_s(r) = P_s(-r - \delta)$, then $P_s = G(r^2 + \delta r)$ and every solution in base $b$ is also one in base $-(b + \delta)$.

Under $0^0 = 1$ no 5-digit string with digits up to 300 is a solution in both a positive and a negative base L16.6.

### 5.9.2 Signs of the roots

Positive bases have three 3-digit solutions without a partner: $(3, 1, 3)$ in base 4, $(5, 1, 3)$ in base 25 and $(6, 1, 5)$ in base 91. Negative bases have none apart from $(1, 1, 1)$. Is something suppressing negative roots?

**Proposition 5.71** (The heuristic is symmetric). PROVEN L18.4
The root heuristic with local factors gives the two roots of every 3-digit string the same weight.

*Proof.* The local factors depend only on $Q_s$, and the derivative at the two roots has the same absolute value $\sqrt{\Delta}$, where $\Delta$ is the discriminant. □

Inside the square class the symmetry breaks. If the leading digit $p$ is odd and the largest digit, then $\sqrt{\Delta} = Y + t$ with $Y$ divisible by $2p$, and the positive root is an integer when $t \equiv q \pmod{2p}$, the negative root when $t \equiv -q$. For $0 < q < p$ the smallest admissible $t$ is $q$ for the positive root and $2p - q$ for the negative one L18.4.

**Heuristic 5.72** (Positive roots are favoured in the square class). HEURISTIC L18.4
For square-class strings with the largest digit in front, small values of $t$ make the positive root about 9 times as likely as the negative one: 0.0564 against 0.0064 expected solutions with digits up to 70. Outside the square class the weights are equal.

$(5,1,3)$ and $(3,1,3)$ follow this mechanism: for $(5,1,3)$, $Y = 250$ and $t = 1 = q$. The third non-partner, $(6,1,5)$, is an ordinary coin flip that came up positive.

### 5.9.3 Roots that are too small, and rational roots

An integer root $r$ with $|r|$ at most the largest digit is a valid identity $P_s(r) = F(s)$ in a representation whose digits are too large. Examples under $0^0 = 1$: $(3,1,2)$ at $r = 3$, where $27 + 3 + 2 = 32 = 27 + 1 + 4$, and $(4,1,3,1)$ at $r = 4$. With digits up to 28 there are 32 such 5-digit strings L12.4. They pass every local test. So the bound $r > \max(s)$ must come from outside the local theory.

**Proposition 5.73** (Rational roots of quadratics). PROVEN L16.5

If $Q_s$ with $k = 3$ has a rational root $p/q$ in lowest terms, its other root $p'/q'$ is rational with $qq' = s_1$, and the two roots have opposite signs.

*Proof.* By Gauss's lemma $Q_s = (qr - p)(q'r - p')$ over $\mathbb{Z}$. The product of the roots is $(s_3 - F)/s_1 < 0$. □

**Computation 5.74** (Rational roots). COMPUTATION L16.5

For $k = 3$ with digits up to 80 and $k = 4$ with digits up to 44, no $Q_s$ has a non-integer rational root above its largest digit. For $k = 3$ the non-integer rational roots that occur are the negative partners of the positive solutions with $s_1 \nmid s_2$, such as $-13/3$ for $(3,1,3)$, together with a few small ones.

So rational roots are rarer than integer roots. The meaningful version of a "rational base" is the number system of Akiyama, Frougny and Sakarovitch, treated in Section 7.4.

## 5.10 Bases with a 2-digit and a 3-digit solution

Two-digit solutions are plentiful and short ones are rare. Can one base have both?

**Lemma 5.75** (Two-digit bases are large). PROVEN L18.2 LEAN

If base $b$ has a 2-digit solution $(a,c)$ under $0^0 = 1$ and $m = \max(a,c)$, then $b \geq m^{m-1}$. So a base below $10^{460}$ with a 2-digit solution has digits at most 200 in it.

Lean: `lemma18_14`

*Proof.* By Proposition 2.14, $b = (a^a + g(c))/a$ with $g(c) = w(c) - c \geq 0$. If $a = m$, then $b \geq a^{a-1}$. If $c = m > a$, then $b \geq (c^c - c)/(c-1) \geq c^{c-1}$. Finally $201^{200} > 10^{460}$. □

**Computation 5.76** (Base 25 is alone). COMPUTATION L18.2

Under $0^0 = 1$, base 25 is the only base below $10^{3301}$ with both a 2-digit and a 3-digit solution, $(1,3)$ and $(5,1,3)$. Among the 8,502 bases of 2-digit solutions with digits up to 200, which include every such base below $10^{460}$, base 25 is the only one with a 3-digit solution. Under $0^0 = 0$ the same holds below $10^{51}$ and for the bases of 2-digit solutions with digits up to 120.

Base 25 works by a coincidence between powers. It is $3^3 - 3 + 1$, the base of $(1,3)$, and it is $5^2$, which makes 5 self-paying at position 2: $5 \cdot 25^2 = 5^5$. So $(5,1,3)$ is the self-paying 5 on top of the 2-digit solution $(1,3)$. The two descriptions meet because $3^3 - 2 = 5^2$, the same coincidence that gives $(1,0,3)$ in base 5 under $0^0 = 0$.

### 5.10.1 How many 3-digit solutions can one base have?

Base 4 has two 3-digit solutions, $(1,3,1)$ and $(3,1,3)$.

**Computation 5.77** (At most two). COMPUTATION L25.3
Under $0^0 = 1$, every base up to $10^{3301}$ has at most two 3-digit solutions, and base 4 is the only one with two.

Can a proof reach all bases? The first step is that the largest digit is almost fixed by the base.

**Lemma 5.78** (The largest digit). PROVEN L25.3 LEAN
Let $s$ be a 3-digit solution in base $b$ with leading digit $p$ and largest digit $M$. Then $b^2/3 \le w(M) < (M+1)b^2$ and $p = \lfloor F/b^2 \rfloor$.
Lean: `lemma25_13`

*Proof.* $F = pb^2 + qb + r$ with $qb + r < b^2$, so $p = \lfloor F/b^2 \rfloor$ and $b^2 \le F < (p+1)b^2$. Each weight is at most $w(M)$, so $F \le 3w(M)$, and $w(M) \le F < (p+1)b^2 \le (M+1)b^2$. □

**Corollary 5.79** (Two consecutive values). PROVEN L25.3 LEAN
The largest digits of the 3-digit solutions of one base take at most two values, and these are consecutive.
Lean: `corollary25_14`

*Proof.* If $M_1 \le M_2 - 2$ both occur, Lemma 5.78 gives $w(M_2) < 3(M_2+1)w(M_1) \le 3(M_2+1)w(M_2-2)$. But $w(M_2)/w(M_2-2) \ge 3M_2^2$. □

**Theorem 5.80** (At most 30 per base). PROVEN L25.3
Let $b \ge 5$. Under $0^0 = 1$, base $b$ has at most 30 three-digit solutions, unless $|b - M^{(M-1)/2}| \le 6$ for a value $M$ with $b^2/3 \le w(M) < (M+1)b^2$. Then it has at most $2M + 24$. There are at most 13 such exceptional bases for each $M$.

*Proof sketch.* Fix one of the at most two values $M$, and let $u, v$ be digits below $M$. Table 5.8 counts the solutions of each shape: in each, the equation fixes the value of an injective function of one digit, or of a function that takes each value at most twice. For the shapes whose leading digit is below $M$, $F/b^2$ lies in an interval of length below 1, so the leading digit takes at most two values. This leaves the shape $(M, u, v)$, with the equation $w(u) + w(v) - ub - v = Mb^2 - w(M)$. Two of its solutions must share their digits above the largest $T$ with $w(T) \le 2M(b+1)$, since a larger differing digit changes the weight sum by more than $M(b+1)$. With two or one digits above $T$ there are at most 1 or 3 solutions. With none, $|b^2 - M^{M-1}| \le 5(b+1)$, so $b$ is within 6 of $M^{(M-1)/2}$, and there are at most $M$ solutions. □

Table 5.8. The shapes of 3-digit solutions with largest digit $M$ in one base, with $u, v < M$ L25.3.

| shape | equation | solutions |
|---|---|---|
| $(M, M, M)$ | one equation in $b$ | at most 1 |
| $(M, M, u)$ | $g(u) = Mb^2 + Mb - 2w(M)$ | at most 1 |
| $(M, u, M)$ | $ub - w(u) = 2w(M) - M - Mb^2$ | at most 2 |
| $(u, M, M)$ | $ub^2 - w(u) = 2w(M) - Mb - M$ | at most 2 |
| $(u, M, v)$ | $u$ takes 2 values, then $g(v)$ is fixed | at most 2 |
| $(u, v, M)$ | $u$ takes 2 values, then $vb - w(v)$ is fixed | at most 4 |
| $(M, u, v)$ | $w(u) + w(v) - ub - v = Mb^2 - w(M)$ | at most 3, or $M$ |

This is a constant bound outside at most about $26 \ln X / \ln \ln X$ bases up to $X$, against the power of $\log b$ of Proposition 5.56. It does not reach 2. The weak point is the shape $(M, u, v)$, where each solution is an exact coincidence. In the bases of the binomial families more is known.

**Proposition 5.81** (One member per base). PROVEN L25.3 LEAN

A base has at most one 3-digit member of the binomial families.

Lean: `prop25_16_even (partial: members with even x)`

**Proposition 5.82** (Bases with a member). PROVEN L25.3

Let $(p, q, x)$ be a member in base $b$, with $x = w(p) + w(q) + \lambda e^2$. Every other 3-digit solution of base $b$ is $(p, x, r)$ with $w(r) - r = (x - q)b - w(p) - \lambda e^2$, or $(p, q', x)$ with $q' \neq q$ and $w(q') - w(q) = (q' - q)b$. Each equation has at most one solution, so base $b$ has at most 3 three-digit solutions.

For each of the 80 members with $x < 10^{300}$, among them base $46666^{23333} - 3$, the candidates for the two equations fail modulo some prime, so the member is the only 3-digit solution of its base L25.3.

**Problem 5.83** (At most two 3-digit solutions). OPEN L18.2 L25.3

Under $0^0 = 1$, does every base have at most two 3-digit solutions? (Question Q50.) It holds for every base up to $10^{3301}$ (Computation 5.77), and every base has at most 30 outside a sparse set (Theorem 5.80). A third solution needs three independent coincidences, so "at most two" is expected for every base. A proof needs control of the shape $(M, u, v)$, where a base near $M^{(M-1)/2}$ gives one equation in two free digits.

Chapter 6

# The equation $b(b+x) = x^n + c$

The 3-digit solution $(1, 7, 1)$ in base 904 looks like a pure coincidence. This chapter shows that it is not quite one. It belongs to a family of strings whose solutions are the solutions of a single exponential Diophantine equation, and that equation can be attacked with the tools of algebraic number theory: cyclotomic and Aurifeuillian factors, units and ideal classes of quadratic fields, and lower bounds for linear forms in logarithms. The family is far from solved, but the attack shows exactly where the difficulty lies. The appendices on quadratic fields (Appendix D), linear forms in logarithms (Appendix E), the abc conjecture (Appendix F) and curves (Appendix G) give the background.

## 6.1 Where the equation comes from

Take the 3-digit string $(1, x, r)$ in base $b$. Its value is $b^2 + xb + r$ and its weight is $1 + x^x + w(r)$. So it is a solution exactly when

$$b(b+x) = x^x + c, \qquad c = w(r) - r + 1.$$

For $r = 1$ under either convention, and for $r = 0$ under $0^0 = 0$, the constant is $c = 1$. The digits $r = 2$ and $r = 3$ give $c = 3$ and $c = 25$. The question becomes: for which $x$ does $x^x + c$ factor as $b(b+x)$, a product of two numbers that differ by exactly $x$? It is natural to replace the exponent $x$ by a free exponent $n$:

$$b(b+x) = x^n + c. \tag{6.1}$$

Completing the square, with $U = 2b + x$, turns it into $U^2 = 4x^n + x^2 + 4c$: a question about squares in a sequence that grows exponentially. For $n = 1$ and $c = 1$, $b = 1$ is a solution for every $x$.

**Computation 6.1** (Known solutions for $c = 1$, $n = x$). COMPUTATION L3.2 L16.2
The equation $b(b+x) = x^x + 1$ has the solutions $x = 3$, $b = 4$ and $x = 7$, $b = 904$, and no other with $x \le 4 \cdot 10^9$. They give $(1, 3, 1)$ in base 4 and $(1, 7, 1)$ in base 904 under $0^0 = 1$, and also $(1, 3, 0)$ and $(1, 7, 0)$ under $0^0 = 0$.

Every 3-digit solution $(p, q, r)$ in base $b$ gives a factorization $F - r = b(pb + q)$. Most of them come from an algebraic identity, completed by one small coincidence between powers (Table 6.1).

Table 6.1. The factor pair $F - r = b(pb + q)$ of each 3-digit solution L16.1.

| solution | $F - r$ | pair $(b, pb+q)$ | identity |
|---|---|---|---|
| $(1, 3, 1)$, base 4 | $3^3 + 1 = 28$ | $(4, 7)$ | cyclotomic and Aurifeuillian |
| $(1, 7, 1)$, base 904 | $7^7 + 1$ | $(8 \cdot 113, 911)$ | Aurifeuillian (Section 6.2) |
| $(1, 2, 6)$, base 215 | $6^6 - 1$ | $(6^3 - 1, 6^3 + 1)$ | difference of squares; needs $2^2 + 1 - 6 = -1$ |
| $(5, 1, 3)$, base 25 | $5^5 + 5^2$ | $(5^2, 5^3 + 1)$ | square class; needs $3^3 - 3 + 1 = 5^2$ |
| $(3, 1, 3)$, base 4 | $2(3^3 - 1) = 52$ | $(4, 13)$ | $4 = 2\Phi_1(3)$, $13 = \Phi_3(3)$ |
| $(6, 1, 5)$, base 91 | $7 \cdot 13 \cdot 547$ | $(91, 547)$ | none |

Only $(6, 1, 5)$ in base 91 has no algebraic factor pair: it needs $6 \cdot 5^5 \equiv 4 \pmod{91}$, a coincidence. The 4-digit solutions show no such structure. For 3435, $F - r = 3430 = 10 \cdot 7^3$ is a cube times 10 by chance.

## 6.2 Aurifeuillian factors and base 904

For odd $x$, $x^x + 1$ is the product of the cyclotomic values $\Phi_{2d}(x)$ over the divisors $d$ of $x$ (Appendix D). The two factors $b$ and $b + x$ of a solution are coprime: a common prime divides $x$ and $x^x + 1$, but $x^x + 1 \equiv 1 \pmod{x}$.

**Proposition 6.2** (The factor form for prime $x$). PROVEN L16.1

Let $x$ be an odd prime and $\Phi = \Phi_{2x}(x)$, so that $x^x + 1 = (x + 1)\Phi$. Every solution of $b(b + x) = x^x + 1$ has $b = (x + 1)A$, $b + x = B$ or $b = A$, $b + x = (x + 1)B$, with $AB = \Phi$ and $A \equiv B \equiv 1 \pmod{2x}$.

*Proof.* A prime $p \mid \Phi$ has $x^x \equiv -1 \pmod{p}$, so $x$ has order $2x$ modulo $p$ and $p \equiv 1 \pmod{2x}$. So every divisor of $\Phi$ is $1 \pmod{2x}$. Write $b = dA$, $b + x = eB$ with $de = x + 1$, $\gcd(d, e) = 1$ and $AB = \Phi$. Reducing $eB - dA = x$ modulo $x$ gives $e \equiv d \pmod{x}$. Since $|e - d| < x + 1$, either $e = d$, impossible for coprime $d, e$ with $de > 1$, or $|e - d| = x$, which forces $\{d, e\} = \{1, x + 1\}$. □

For $x = 7$ the pair is $A = 113$, $B = 911$, and $904 = 8 \cdot 113$. For $x = 3$ it is $A = 1$, $B = 7$. Both use the *Aurifeuillian* factorization of $\Phi_{2x}(x)$. For squarefree $x \equiv 3 \pmod{4}$, the polynomial $\Phi_{2x}(xt^2)$ splits in $\mathbb{Z}[t]$ into two factors of degree $\varphi(x)$. Their values at $t = 1$ are two integers $L < M$ with $LM = \Phi_{2x}(x)$. For $x = 7$, $\Phi_{14}(7) = 113 \cdot 911$. The same holds for squarefree $x \equiv 1 \pmod{4}$ with $\Phi_x(xt^2)$, a factor of $x^x - 1$.

### 6.2.1 The ratio of the two factors

For $x$ from 7 to 79 the ratio $M/L$ stays between 6.98 and 8.06, near $e^2 = 7.389$. This is a theorem.

**Theorem 6.3** (The ratio of Aurifeuillian factors). PROVEN L16.1

For squarefree odd $x > 1$, the Aurifeuillian factors $L < M$ of $x^x \pm 1$ satisfy

$$\ln\left(\frac{M}{L}\right) = 2 \sum_{j \text{ odd},\, \gcd(j,x)=1} \frac{\psi(j)}{j x^{(j-1)/2}},$$

with $\psi$ the quadratic character of $\mathbb{Q}(\sqrt{x})$. So $\ln(M/L) = 2 + (2/3)(x/3)/x + O(x^{-2})$ when $3 \nmid x$, where $(x/3)$ is the Legendre symbol, and always $|\ln(M/L) - 2| \le 2/(3(x - 1))$. In particular $M/L \to e^2$.

*Proof sketch.* Take $x \equiv 3 \pmod{4}$ and a primitive $4x$-th root of unity $\zeta$. The roots of $\Phi_{2x}(xt^2)$ are $\zeta^a/\sqrt{x}$ for $a$ prime to $4x$. The set $A = \{a : \psi(a) = 1\}$ is stable under the Galois group, so $L$ and $M$ are, up to order, $|G(1)|$ and $|G(-1)|$ with $G(t) = x^{\varphi(x)/2} \prod_{a \in A} (t - \zeta^a/\sqrt{x})$. Expanding $\ln|1 \pm \zeta^a/\sqrt{x}|$ in powers of $1/\sqrt{x}$ gives $\ln(M/L) = 2 \operatorname{Re} \sum_{j \text{ odd}} S_j / \left(j x^{j/2}\right)$ with $S_j = \sum_{a \in A} \zeta^{aj}$. Half of $S_j$ is a Ramanujan sum, which vanishes for odd $j$, and the other half is a Gauss sum $\psi(j)\tau(\psi)$ with $\tau(\psi) = 2\sqrt{x}$. So $S_j = \psi(j)\sqrt{x}$. □

The values agree with the series to within $10^{-30}$ for all 132 squarefree odd $x$ up to 300 L16.1. The term $x(\ln(M/L) - 2)$ tends to $\pm 2/3$ with the sign of $(x/3)$, and to 0 when $3 \mid x$. For $x = 7$ it is $+0.610$, so $M/L = 8.0619$.

### 6.2.2 Why $x = 7$ works and nothing larger does

**Proposition 6.4** (The pair is a solution modulo $x^2$). PROVEN L16.1
For a prime $x \equiv 3 \pmod 4$, $L \equiv 1 - x + x(x-1)/2$ and $M \equiv 1 + x + x(x-1)/2 \pmod{x^2}$. So $E = M - (x+1)L - x$ is divisible by $x^2$, and the pair $b = (x+1)L$, $b + x = M$ satisfies $b(b+x) \equiv x^x + 1$ modulo $x^2$.

The proof expands $L$ and $M$ in powers of $\sqrt{x}$; the coefficients are the elementary symmetric functions of the roots of unity in the set $A$ of Theorem 6.3.[6] So for every prime $x \equiv 3 \pmod 4$ the Aurifeuillian pair solves the equation to precision $x^2$. A true solution needs $E = 0$. By Theorem 6.3, $M - (x+1)L = L(M/L - x - 1)$, and $M/L < 8.1 < x + 1$ for $x \geq 11$, so $E < 0$. For $x = 7$, $M/L = 8.06$ lies within $x/L$ of $x + 1 = 8$, and $E = 0$. For $x = 3$, $L = 1$ and $E = 0$. So $(1, 7, 1)$ in base 904 is Aurifeuillian in a precise sense: the identity gives the factor pair and the congruence modulo $x^2$, and what remains is the numerical coincidence $M/L = e^{2+0.61/7} \approx 8.06$, close to $x + 1 = 8$.

**Theorem 6.5** (No other algebraic factor pair). PROVEN L16.1
Let $x \geq 3$ be odd. If $b(b+x) = x^x + 1$ and $b$ is a product of algebraic factors of $x^x + 1$, that is of cyclotomic values and Aurifeuillian halves, then $x = 3$ or $x = 7$.

*Proof sketch.* A solution has $|\ln((x^x+1)/b^2)| = \ln(1 + x/b)$, which is below $x^{2-x/2}$. On the other side, $\ln \Phi_{2d}(x)$ is $\varphi(d) \ln x$ up to $1/(x-1)$, and each Aurifeuillian pair splits its logarithm into halves that differ by $\ln(M_d/L_d)$. The difference $\ln((x^x+1)/b) - \ln b$ is then an odd multiple $D$ of $\ln x$, plus small errors, plus a sum of the ratios $\ln(M_d/L_d)$. By Theorem 6.3 each ratio is near 2 or small, and a case analysis shows that the total cannot be near 0 unless $x \in \{3, 7\}$. □

A computation over all odd $x$ up to 201 confirms this: the best algebraic split has $|\ln(x^x+1) - 2\ln b|$ equal to 0.0077 for $x = 7$ and at least 0.54 for every $x \geq 11$ L16.1. A separate exhaustive search of 3-digit strings with two digits up to 1000 and the third up to 30 finds no new 3-digit solution, in bases up to about $10^{1500}$.

### 6.2.3 The general series

The series of Theorem 6.3 is one case of a general law. Let $s > 1$ be squarefree, $D$ the discriminant of $\mathbb{Q}(\sqrt{s})$, $\chi$ its Kronecker symbol, $k \geq 1$ odd, and $m = k\,|D|\,/2$ for even $D$, $m = k\,|D|$ for odd $D$. Then $\Phi_m(st^2)$ splits into two Aurifeuillian factors, whose values at $t = y$ are $L \leq M$. Define the twisted Ramanujan sum $g_k(j) = \sum_{r \,|\, \gcd(j,k)} r\mu(k/r)\chi(k/r)\chi(j/r)$.

**Theorem 6.6** (The ratio in general). PROVEN L18.5
For $sy^2 > 1$,

$$\ln\left(\frac{M}{L}\right) = 2\left|\sum_{j \text{ odd}} \frac{g_k(j)}{j s^{(j-1)/2} y^j}\right|.$$

Since $g_k(1) = \mu(k)\chi(k)$, $M/L \to e^{2/y}$ as $s$ grows when $k$ is squarefree and prime to $D$, and $M/L \to 1$ otherwise.

The proof for even $D$ is the Gauss-sum computation above, with the standard formula for Gauss sums of induced characters. The odd case follows by the same computation with the character $\chi(a)(-1)^a$ on the exponents. Both cases were checked numerically: in 456 cases ($s \leq 40$, odd $k \leq 15$,

[6] The library states the congruences for $L$ and $M$ for squarefree $x \equiv 3 \pmod 4$, with $(x + \mu(x))/2$ in place of $(x-1)/2$. Only prime $x$ matters here: for composite $x$ the product $(x+1)LM$ is not $x^x + 1$, because $x^x + 1$ has further cyclotomic factors.

$y \le 4$) the series agrees with the factored polynomial to within $2 \cdot 10^{-62}$ L18.5. For the equation this says that an Aurifeuillian pair with $y \ge 2$ is balanced within a factor $e^{1+o(1)}$, so the $e^2$ coincidence of $x = 7$ needs $y = 1$.

## 6.3 Elementary exclusions and the limits of congruences

From now on write the equation as $U^2 = 4x^n + x^2 + 4$ with $U = 2b + x$, for $c = 1$.

**Proposition 6.7** (Elementary exclusions). PROVEN L16.2 L18.1 LEAN

The equation $b(b+x) = x^n + 1$ has no solution in each of these cases: (a) $n \ge 4$ even; (b) $n \ge 4$ and $x$ a square; (c) $n \ge 2$ and $x \equiv 2 \pmod 4$; (d) $n \ge 2$ and $x \equiv 4 \pmod 8$. For $n = x$ this excludes every even $x$ and every odd square.

Lean: `prop18_1_a, prop18_1_b, prop18_1_c, prop18_1_d, prop16_5`

*Proof.* (a), (b): then $x^n = Y^2$ for an integer $Y = x^{n/2}$, and $(2Y)^2 < U^2 < (2Y+1)^2$ because $0 < x^2 + 4 < 4Y + 1$. (c), (d): $x$ is even, so $U = 2V$ and $V^2 = x^n + (x/2)^2 + 1$. This is $2 \pmod 4$ if $x \equiv 2 \pmod 4$ and $5 \pmod 8$ if $x \equiv 4 \pmod 8$. Neither is a square. □

For odd $x$, $4x^x + x^2 + 4 \equiv 1 \pmod 8$ and it is $4 \pmod p$ for every prime $p \mid x$, so it is a square at 2 and at the primes of $x$. Could congruences at other primes finish the proof? A sieve gets very far.

**Computation 6.8** (The sieve to $4 \cdot 10^9$). COMPUTATION L16.2

For every $x$ from 2 to $4 \cdot 10^9$ other than 3 and 7, some odd prime $p \le 229$ makes $4x^x + x^2 + 4$ a quadratic non-residue modulo $p$. So $b(b+x) = x^x + 1$ has no further solution with $x \le 4 \cdot 10^9$, that is with $b$ up to about $10^{1.9 \cdot 10^{10}}$. The 31 primes $p$ with $p - 1 \mid 20160$ exclude all but a density $1.5 \cdot 10^{-4}$ of all integers $x$.

But no finite set of primes can finish the job.

**Proposition 6.9** (Congruences cannot finish the proof). PROVEN L16.2

For every finite set $S$ of primes, the integers $x \equiv 1 \pmod{\operatorname{lcm}(p-1 : p \in S)}$ pass the test at every $p \in S$. They have positive density.

*Proof.* If $x \equiv 1 \pmod{p-1}$ and $p \nmid x$, then $x^x \equiv x \pmod p$ by Fermat, so $4x^x + x^2 + 4 \equiv (x+2)^2$ is a square. If $p \mid x$, the value is 4, also a square. □

The reason is the trivial solution $b = 1$ of the equation with $n = 1$. At the primes where $x \equiv 1 \pmod{p-1}$, the power $x^x$ looks like $x^1$. Any proof must use infinitely many primes, or global information.

One more reformulation shows how thin the target is.

**Proposition 6.10** (The $x$-adic form). PROVEN L16.2

Let $x$ be an odd prime and $\beta$ the root of $t^2 + xt - 1 = 0$ in the $x$-adic integers $\mathbb{Z}_x$ with $\beta \equiv 1 \pmod x$. Let $R$ be the residue of $\beta$ modulo $x^x$ in $[0, x^x)$. Then $b(b+x) = x^x + 1$ has a solution exactly when $R < x^{x/2}$, and then $b = R$.

*Proof.* A solution has $b \equiv 1 \pmod x$ by Proposition 6.2, and $b$ is a root of $t^2 + xt - 1$ modulo $x^x$. Hensel's lemma (Appendix C) gives a unique such root that is 1 modulo $x$, so $b \equiv \beta$, and $b < x^{x/2}$. Conversely, if $R < x^{x/2}$, then $R(R+x) - 1$ is a positive multiple of $x^x$ below $2x^x$, so it equals $x^x$. □

So a solution means that the top $(x-1)/2$ base-$x$ digits of the $x$-adic number $\beta$, cut off at $x^x$, all vanish. For $x = 7$, $R = 904$ and $7^{3.5} = 907.5$. For the other primes up to 59, $R$ lies between

$x^{x-0.8}$ and $x^x$, as for a random residue. Heuristically the chance is $x^{-x/2}$. Statements that a block of consecutive digits of an algebraic $p$-adic number vanishes are open even for one fixed base.

## 6.4 The quadratic field

The equation is a norm equation in a real quadratic field (Appendix D). Let $x$ be odd, $D = x^2 + 4$, $K = \mathbb{Q}\big(\sqrt{D}\big)$, and

$$\varepsilon = \frac{x + \sqrt{D}}{2}, \qquad \varepsilon' = \frac{x - \sqrt{D}}{2} = -\frac{1}{\varepsilon}.$$

Then $\varepsilon$ is a unit of norm $-1$ and trace $x$, and for every integer $b$

$$N(b + \varepsilon) = (b + \varepsilon)(b + \varepsilon') = b^2 + bx - 1.$$

So the equation $b(b + x) = x^n + 1$ says exactly $N(b + \varepsilon) = x^n$.

**Lemma 6.11** (Splitting). PROVEN L18.1

Let $x$ be odd and $b$ a solution.

1. The ideals $(b + \varepsilon)$ and $(b + \varepsilon')$ are coprime.
2. Every prime $p \mid x$ splits in $K$.
3. $P = (\varepsilon + 1)$ has norm $x$, its conjugate is $P' = (\varepsilon' + 1)$, $PP' = (x)$, and $P$, $P'$ are coprime.

*Proof.* (1) A common prime ideal divides the difference $\sqrt{D}$ and the product $x^n$, but $\gcd(D, x) = 1$. (2) $D \equiv 4 \pmod p$ is a nonzero square. (3) $N(\varepsilon + 1) = \varepsilon\varepsilon' + \varepsilon + \varepsilon' + 1 = x$, and the difference of $\varepsilon + 1$ and $\varepsilon' + 1$ is $\sqrt{D}$, prime to $x$. □

**Lemma 6.12** (The ideal is an $n$-th power). PROVEN L18.1

For a solution with $x$ odd, $(b + \varepsilon) = B^n$, where $B$ is the product over the prime powers $p^a$ exactly dividing $x$ of $q_p^a$, with $q_p$ one of the two primes above $p$. Moreover $B \neq B'$.

*Proof.* $(b + \varepsilon)(b + \varepsilon') = (x)^n$ is the product of the ideals $\big(\pi_p \pi_p'\big)^{an}$ over the primes $p \mid x$, where $\pi_p$ and $\pi_p'$ are the two primes above $p$. By Lemma 6.11 each $\pi_p^{an}$ divides exactly one of the two coprime factors, and its conjugate divides the other. Collecting the factors of $(b + \varepsilon)$ gives $B^n$. If $B = B'$, then $B$ would contain both $\pi_p$ and $\pi_p'$ for some $p$, and both factors would share a prime. □

Call a solution *pure* if $B = P$ or $B = P'$, and *mixed* otherwise. If $x$ is a prime power, every solution is pure. A pure solution gives an explicit element: $b + \varepsilon = u(\varepsilon + 1)^n$ for a unit $u$.

**Proposition 6.13** (The unit form). PROVEN L16.2

Let $x \equiv 3 \pmod 4$ be squarefree with $x^2 + 4$ prime, and let $\eta$ be the fundamental unit of $\mathbb{Z}[\sqrt{x}]$. A solution of $b(b + x) = x^x + 1$ gives $(2b + x) + 2x^{(x-1)/2}\sqrt{x} = \pm(x + 2 \pm 2\sqrt{x})\eta^k$ for some integer $k$.

*Proof.* Both sides have norm $x^2 + 4 = p$, a prime that splits in $\mathbb{Q}(\sqrt{x})$. The two elements $x + 2 \pm 2\sqrt{x}$ generate the two primes above $p$, so the left side is one of them times a unit of norm 1. □

For $x = 3$ and $x = 7$, with $k = 2$ in both cases:

$$11 + 6\sqrt{3} = \big(5 - 2\sqrt{3}\big)\big(2 + \sqrt{3}\big)^2, \qquad 1815 + 686\sqrt{7} = \big(9 + 2\sqrt{7}\big)\big(8 + 3\sqrt{7}\big)^2.$$

So the equation asks whether one coordinate of a binary recurrence is ever $2x^{(x-1)/2}$. For fixed $x$ that is a classical finiteness problem [29]. The difficulty is that the field changes with $x$.

### 6.4.1 A lower bound for the exponent

Let $\eta_0 > 1$ be the fundamental unit of $K$ and $\mu_0 = \ln \eta_0$. Since $N(\varepsilon) = -1$, $\varepsilon = \eta_0^r$ with $r$ odd, and $\mu_0 \geq \operatorname{arcsinh}(1/2) = 0.4812$. Put $\Lambda_0 = \ln((b+\varepsilon)/(b+\varepsilon'))$. Since $b > x^{n/2} - x/2$,

$$0 < \Lambda_0 < \tau_n = \frac{\sqrt{D}}{x^{n/2} - x/2 - 1}, \tag{6.2}$$

which is tiny: for $x \geq 3$ and $n \geq 5$, $\tau_n < 0.276$. Let $z = (\varepsilon+1)/(\varepsilon-1)$. Then $\ln z = 2\operatorname{artanh}(1/\varepsilon) < 2.2/x$ for $x \geq 3$, and the exact series $x \ln z = 2 - \sum_{i \geq 1} 4\varepsilon^{-2i}/((2i-1)(2i+1))$ shows that $x \ln z$ lies just below 2.

**Proposition 6.14** (Pure solutions need a large exponent). PROVEN L18.1
Let $x$ be odd, $n \geq 5$ odd, and $b$ a pure solution. Then there is an odd integer $s \geq 1$ with $|n \ln z - s\mu_0| = \Lambda_0 < \tau_n$. In particular $n \ln z \geq \mu_0 - \tau_n$, and $x < 11n$.

*Proof.* Suppose $B = P$. Then $b + \varepsilon = u(\varepsilon+1)^n$ with $u = \pm\eta_0^k$, and $b + \varepsilon' = u'(\varepsilon'+1)^n$. The quotient of $\varepsilon + 1$ by $\varepsilon' + 1$ is $\varepsilon z$, and $|u/u'| = \eta_0^{2k}$. Taking logarithms gives $\Lambda_0 = 2k\mu_0 + n(r\mu_0 + \ln z) = n \ln z - s\mu_0$ with $s = -(2k + nr)$ odd. The case $B = P'$ is the same with the roles exchanged. If $s \leq -1$, the left side would be at least $\mu_0 > \tau_n$, against (6.2). Finally $n \geq (\mu_0 - \tau_n)/\ln z > (0.4812 - 0.276)x/2.2 > 0.093x$. □

For the diagonal $n = x$ this is already decisive, without any deep theorem.

**Proposition 6.15** (The diagonal has no pure solution beyond 7). PROVEN L18.1
For odd $x \geq 9$, $b(b+x) = x^x + 1$ has no pure solution. So for odd prime powers $x$ the only solutions are $x = 3$ and $x = 7$.

*Proof.* With $n = x$, $\tau_x < 5 \cdot 10^{-4}$ and $x \ln z$ is within 0.017 of 2. So $|2 - s\mu_0| < 0.018$ with $s \geq 1$ odd and $\mu_0 = \operatorname{arcsinh}(y/2)$ for a positive integer $y$. For $s = 1$ this needs $y \in (7.12, 7.39)$; for $s = 3$, $y \in (1.42, 1.45)$; for $s \geq 5$, $\mu_0 < 0.41$, which is impossible. No integer qualifies. The value $y = 7$, $s = 1$ is the solution $x = 7$, where $\tau_7 = 0.008$ is not small. $x = 5$ fails by a direct check. □

Here is the $e^2$ of Theorem 6.3 seen from the unit side: $n = x$ forces $s\mu_0$ close to 2, and $\mu_0 = \ln \varepsilon$ is about $\ln x$, which equals 2 near $x = e^2 = 7.39$.

## 6.5 Linear forms in logarithms

For general $n$ the relation $|n \ln z - s\mu_0| < \tau_n$ says that a linear form in two logarithms is exponentially small. Baker's theory gives lower bounds for such forms, and comparing the two bounds bounds $n$ (Appendix E).

**Theorem 6.16** (Laurent). CITED L18.1
Let $\alpha_1, \alpha_2 > 1$ be multiplicatively independent real algebraic numbers, $b_1, b_2$ positive integers, $\Lambda = b_2 \ln \alpha_2 - b_1 \ln \alpha_1$, $D_g = [\mathbb{Q}(\alpha_1, \alpha_2) : \mathbb{Q}]$, $\ln A_i \geq \max\bigl(h(\alpha_i), \ln(\alpha_i)/D_g, 1/D_g\bigr)$ with $h$ the absolute logarithmic height, and $b' = b_1/\bigl(D_g \ln A_2\bigr) + b_2/\bigl(D_g \ln A_1\bigr)$. Then

$$\ln|\Lambda| \geq -25.2 D_g^4 \max\bigl(\ln b' + 0.38, 10/D_g, 1\bigr)^2 \ln A_1 \ln A_2.$$

This is Corollary 2 of Laurent [22] with $m = 10$. The library quotes the constants from a secondary source and flags that they must be checked against the paper L18.1. The results below that use it inherit this caveat.

**Theorem 6.17** (All pure solutions). PROVEN L18.1
For odd $x$, the only pure solution of $b(b+x) = x^n + 1$ with $n \geq 4$ is $(x, n) = (7, 7)$.

*Proof sketch.* Apply Theorem 6.16 to $\alpha_1 = \eta_0$ and $\alpha_2 = z$, which are independent because $z$ is not an algebraic integer: it is a root of $xT^2 - 4T - x$. For $x \geq 10^5$, the lower bound for $\Lambda_0$ against (6.2) gives $n < 22\,000 \max(\mu_0, 1)$, while Proposition 6.14 gives $n > 0.49 x \mu_0$ roughly. Together they force $x < 10^5$, a contradiction. For odd $x < 10^5$ a computation finds $r$ and $\mu_0$ exactly, bounds $n$ by Laurent's theorem, checks $n \leq 15$ directly, and for larger $n$ uses that $s/n$ must be a convergent of $\ln z/\mu_0$ (Legendre's theorem, Appendix E). The only hit is $(7, 7)$. □

**Theorem 6.18** (Prime powers). PROVEN L18.1
Let $x$ be a prime power. The solutions of $b(b+x) = x^n + 1$ with $n \geq 2$ are exactly $(x, n) = (3, 2), (8, 2), (3, 3), (5, 3), (8, 3), (64, 3), (7, 7)$.

*Proof sketch.* Even $n \geq 4$ is excluded by Proposition 6.7. For $n = 2$ the solutions are the even-index Fibonacci numbers $x = F_{2k} = F_k L_k$; a prime power among them needs $F_k$ and $L_k$ to be powers of 2, which leaves $x = 3$ and $x = 8$. For $n = 3$ the list is complete by the elliptic curve of Section 6.6; $119 = 7 \cdot 17$ is not a prime power. For odd $n \geq 5$ and odd $x$ every solution is pure, and Theorem 6.17 applies. For $x = 2^a$, Proposition 6.7 forces $a \geq 3$, and with $V = U/2$ the equation reads $V^2 = 2^{an} + 2^{2a-2} + 1$. By Szalay's theorem on $2^A + 2^B + 1 = V^2$ [30, 31], the solutions with $A \geq B \geq 1$ are $(2t, t+1)$, $(5, 4)$ and $(9, 4)$, and none of them has $n \geq 5$. □

In the language of generalized Ramanujan-Nagell equations $U^2 - D = 4p^n$ [32], this gives the exact solution set for $D = p^2 + 4$ and every prime $p$: $n = 1$ always, $n = 2, 3$ for $p = 3$, $n = 3$ for $p = 5$, and $n = 7$ for $p = 7$.

### 6.5.1 Mixed solutions

A mixed solution has no explicit generator for $B$. Let $d$ be the order of the class of $B$ in the class group of $K$, so $d \mid n$. Then $B^d = (\eta)$ is principal, and the same argument with $\theta = \ln|\eta/\eta'|$ in place of $\ln z$ gives a linear form in $\mu_0$ and $\theta$.

**Theorem 6.19** (Mixed solutions). PROVEN L18.1
Let $x$ be odd and not a square, $n \geq 5$ odd, $b$ a solution, and $d$ the order of the class of $B$. If $d < n$, then $n < 6.4 \cdot 10^4 d \ln x$.

*Proof sketch.* Write $B^d = (\eta)$. Then $b + \varepsilon = u\eta^{n/d}$ with $u = \pm\eta_0^k$, and $\Lambda_0 = 2k\mu_0 + (n/d)\theta$ with $\theta = \ln|\eta/\eta'|$, reduced to $|\theta| \leq \mu_0$. The number $w = |\eta/\eta'|$ is a root of $x^d T^2 - \mathrm{Tr}(\eta^2)T + x^d$, so its height is at most $(d \ln x + |\theta|)/2$, and $w \neq 1$ because $B \neq B'$. This gives $|\theta| \geq 0.99 x^{-d/2}$, which forces $k\theta < 0$, and then $\Lambda_0$ is a linear form in the two logarithms $\mu_0$ and $|\theta|$. Laurent's theorem against (6.2) gives an inequality whose worst case, over a grid of $x$ from 3 to $10^{40}$ and $d$ up to 101, is $n/(d \ln x) \leq 63\,441$. □

There is no matching lower bound for $n$ here, and the $n = 3$ solution $x = 119 = 7 \cdot 17$ shows that mixed solutions exist: its ideal $B$ is not principal, and its class has order $d = 3 = n$.

**Corollary 6.20** (The diagonal $n = x$). PROVEN L18.1 L16.2
If $b(b+x) = x^x + 1$ has a solution with $x$ other than 3 and 7, then $x > 4 \cdot 10^9$ is odd, composite, not a square and not a prime power, the solution is mixed, the ideal $B$ is not principal, and the order $d$ of its class divides $x$ with $x/d < 3.3 \cdot 10^4 \ln x$. In particular the class number of $\mathbb{Q}\left(\sqrt{x^2+4}\right)$ is at least $x/(3.3 \cdot 10^4 \ln x)$.

*Proof.* Even $x$ and squares: Proposition 6.7. $x \le 4 \cdot 10^9$: Computation 6.8. Prime powers: Theorem 6.18. Pure solutions: Proposition 6.15. Principal $B$: the inequality behind Theorem 6.19, evaluated with $d = 1$ by the script `r2nt_principal.py`, allows only $x \le 747\,069$. Otherwise $3 \le d$ divides $x$. The same computation shows that for $x \ge 4 \cdot 10^9$ and $d \ge 3$ the inequality gives $n/(d \ln x) < 32\,500$; with $n = x$ this is the bound on $x/d$. □

So under $0^0 = 1$, the string $(1, x, 1)$ is a solution for no odd prime power $x$ other than 3 and 7, and any further $x$ must be composite, above $4 \cdot 10^9$, with a large non-principal ideal class.

For a fixed $x$ the whole argument is a finite procedure.

**Computation 6.21** (All exponents for odd $x \le 1001$). COMPUTATION L18.1

For odd $x \le 1001$ the only solution of $b(b+x) = x^n + 1$ with $n \ge 4$ is $(7, 7)$. The computation builds $K$ with a certified class group, runs over all $2^{\omega(x)}$ ideals $B$ (814 of the 1,818 are not principal), and tests the convergents of $\theta/(2\mu_0)$ up to the bound of Laurent's theorem, which allows exponents up to 7,816,158. With the threshold lowered to $n \ge 3$ it finds $(5, 3)$, $(7, 7)$ and $(119, 3)$.

### 6.5.2 How often the loophole is open

Corollary 6.20 leaves one loophole: a mixed ideal whose class order $d$ divides $x$ and is large. Does that happen at all?

**Computation 6.22** (Orders of mixed ideal classes). COMPUTATION L20.5

Assuming GRH for the class group computations: for all odd composite non-square $x$ up to 200 001 and five windows near 2.5, 4, 5.5, 7 and $8.5 \cdot 10^5$, the order $d$ of a mixed ideal class divides $x$ for some ideal in 6.5% of the $x$ below $10^4$ and 0.37% near $8.5 \cdot 10^5$. The share falls like $6.7x^{-0.54}$. The smallest case is $x = 15$, $d = 3$, and the largest $d$ found is 34 333 at $x = 514\,995 = 15 \cdot 34\,333$.

**Heuristic 6.23** (The class group cannot close the loophole). HEURISTIC L20.5

Extrapolating these statistics, the number of $x$ in $(4 \cdot 10^9, X]$ that pass every class-group condition of Corollary 6.20 grows like $10^4 \ln(X/(4 \cdot 10^9))$, without bound. Each such $x$ still needs the linear form to be smaller than $\tau_x$, a chance of order $x^{-x/2}$. So the expected number of further solutions is far below 1, and a proof must use the Diophantine condition.

## 6.6 Other exponents and the abc conjecture

For small fixed $n$ the equation $U^2 = 4x^n + x^2 + 4$ is a curve, and the theory of curves applies (Appendix G).

**Proposition 6.24** (Exponent $n = 2$). PROVEN L16.2

$b(b+x) = x^2 + 1$ has a solution exactly when $5x^2 + 4$ is a square, that is when $x$ is a Fibonacci number $F_{2k}$ of even index.

**Computation 6.25** (Exponent $n = 3$). COMPUTATION L16.2 L18.1

With $X = 4x$ and $Y = 4U$, $n = 3$ gives the elliptic curve $Y^2 = X^3 + X^2 + 64$, of conductor 433 and rank 2. It has exactly 10 integral points, and they give $x \in \{3, 5, 8, 64, 119\}$. This list is complete for all $x$.

**Computation 6.26** (Exponent $n=5$). COMPUTATION L18.1
$b(b+x)=x^5+1$ has no solution with $x \le 10^{10}$. The curve $b^2+xb=x^5+1$ has genus 2 and conductor 800 027, and its Jacobian has analytic rank 3. For $n=7,9,\dots,15$ the only solution with $x \le 10^9$ is $(7,7)$, and for odd $n$ from 17 to 201 there is none with $x \le 10^8$.

For each $n \ge 5$ the curve has genus at least 2, so Faltings's theorem [33] gives finitely many solutions. But for $n=5$ the known methods do not finish the job. Chabauty and Coleman's method needs a Mordell-Weil rank below the genus, and quadratic Chabauty needs rank equal to the genus. Here the rank is probably 3, larger than the genus 2. What remains is Baker's method for integral points on hyperelliptic curves with the Mordell-Weil sieve [34], which needs a saturated basis of the Mordell-Weil group. So the case $n=5$ is decidable but not decided.

Uniformly in $n$, the abc conjecture (Appendix F) gives finiteness.

**Proposition 6.27** (Solutions are abc triples of high quality). PROVEN L16.2 L18.1
For odd $x$, a solution of $b(b+x)=x^n+1$ gives the coprime triple $(x^2+4)+4x^n=(2b+x)^2$. For $n=x$ its quality is at least $\ln(4x^x)/\ln\bigl(2x(x^2+4)(2x^{x/2}+1)\bigr)$, which exceeds 1.6299 for $x \ge 29$ and tends to 2. With the sieve, any further solution with $n=x$ has quality above $2-10^{-8}$.

The best known abc triple, $2+3^{10}\cdot 109=23^5$ found by Reyssat, has quality 1.6299 [35].

**Theorem 6.28** (Finiteness under abc). CONDITIONAL L16.2 L18.1
Assume the abc conjecture, or even its weak form $c < K\,\mathrm{rad}\,(abc)^{2-\delta}$ for some $\delta > 0$. Then $b(b+x)=x^n+1$ has only finitely many solutions with $n \ge 4$. In particular $b(b+x)=x^x+1$ has only finitely many solutions.

*Proof sketch.* The radical of the triple is below $8.1x^{3+n/2}$. The weak abc inequality then bounds $x$ once $n > 12/\delta - 6$, and the remaining exponents are finitely many curves of genus at least 2, each with finitely many points by Faltings's theorem. For $x \equiv 0 \pmod 8$ divide the triple by 4 first. □

This is the same status as Brocard's problem $n!+1=m^2$, where abc gives finiteness [36], but here much less than abc suffices.

**Heuristic 6.29** (The unsearched region). HEURISTIC L18.1
Given $x$ and $n$, the number $4x^n+x^2+4$ is a square with chance about $1/\bigl(4x^{n/2}\bigr)$. Summed over the region not yet searched, the expected number of further solutions is below $10^{-14}$.

## 6.7 The even case

Proposition 6.7 leaves even $x$ open only when $x \equiv 0 \pmod 8$, and Theorem 6.18 handles $x=2^a$. So write $x=2^a m$ with $a \ge 3$ and $m > 1$ odd, and let $n \ge 5$ be odd. Put $V=b+x/2$ and $E=(x/2)^2+1$, so that $V^2-E=x^n$, and $\gamma=\bigl(V+\sqrt{E}\bigr)/2$ in the order $R=\mathbb{Z}\bigl[\bigl(1+\sqrt{E}\bigr)/2\bigr]$. Then $N(\gamma)=2^{an-2}m^n$.

**Lemma 6.30** (The ideal of $\gamma$). PROVEN L20.3
The prime 2 splits in $R$, and $(\gamma)=Q^{an-2}B^n$ with $Q$ one of the two primes above 2 and $B$ a product of primes above the primes of $m$.

Here the method of Section 6.4 breaks: the exponent $an-2$ of $Q$ is not a multiple of $n$. And $Q$ itself is far from principal.

**Lemma 6.31** (Small powers of $Q$ are not principal). PROVEN L20.3
$Q^j$ is not principal for $1 \le j \le \log_2 x - 3$.

*Proof.* If $Q^j = (\pi)$ with $\pi = \left(u + v\sqrt{E}\right)/2$, then $|u^2 - Ev^2|$ is $2^j$ or $2^{j+2}$ with $u$, $v$ coprime, and it is below $\sqrt{E}$. A coprime solution of $|u^2 - Ev^2| < \sqrt{E}$ comes from a convergent of $\sqrt{E}$ [37]. But $\sqrt{E} = [x/2; x, x, x, ...]$, and every convergent has norm $\pm 1$. □

So a solution gives a linear form in three logarithms: the fundamental unit, a generator of a power of the class of $Q^a B$, and a fixed number $\lambda$ that need not lie in the lattice of the other two. Laurent's two-logarithm theorem no longer applies, and the general theorem of Matveev [38] must be used. It has much larger constants.

**Theorem 6.32** (A bound for $n$). PROVEN L20.3
Let $x \equiv 0 \pmod 8$ have an odd prime factor, $n \ge 5$ odd, $b$ a solution, and $d$ the order of the class of $Q^a B$. Then $n < 4.1 \cdot 10^{12}(d+1)^2(\ln x)^2(1 + \ln n)$ for $x \ge 24$.

Only one pure configuration gives a bound in the other direction, the exponential $x < 2^{2n+1}$, an analogue of $x < 11n$ in Proposition 6.14 L20.3.

**Computation 6.33** (The even case for small $x$). COMPUTATION L20.3
Let $x \equiv 0 \pmod 8$ have an odd prime factor and not be a square. For $x \le 10^8$ there is no solution with $n \in \{5, 7, ..., 15\}$. For $x \le 10^4$ there is no solution with any odd $n \ge 5$: 1,219 values of $x$ and 6,794 configurations, each reduced with the lemma of Dujella and Petho [39].

## 6.8 The families $(1, x, 2)$ and $(1, x, 3)$

Under $0^0 = 1$ the strings $(1, x, 2)$ and $(1, x, 3)$ lead to $b(b+x) = x^x + 3$ and $b(b+x) = x^x + 25$. A neighbouring family is easier.

**Proposition 6.34** (The family $(1, 2, x)$). PROVEN L20.4 LEAN
$(1, 2, x)$ is a solution in some base exactly for $x = 6$, in base 215, under either convention.
Lean: `prop20_20`

*Proof.* The condition is $(b+1)^2 = x^x + 6 - x$. For odd $x$, $x^x \equiv x \pmod 8$, so the right side is $6 \pmod 8$, not a square. For even $x \ge 8$ it lies strictly between $\left(x^{x/2} - 1\right)^2$ and $\left(x^{x/2}\right)^2$. For $x = 6$ it is $6^6 = 216^2$, and the small cases fail by hand. □

For $c = 3$ and $c = 25$ put $D = x^2 + 4c$ and $\theta = \left(x + \sqrt{D}\right)/2$, of norm $-c$. Then $N(b + \theta) = b^2 + bx - c$, so the equation says $N(b + \theta) = x^n$ in the order $\mathbb{Z}[\theta]$. Now $\theta$ is not a unit. Does the method of Section 6.4 depend on the unit $\varepsilon$?

**Lemma 6.35** (Elementary cases). PROVEN L20.4
Let $c \in \{3, 25\}$ and $n \ge 2$. There is no solution with $n \ge 4$ even or with $n \ge 5$ and $x$ a square, except $(x, n, b) = (3, 6, 26)$ for $c = 25$. For $c = 3$, $x$ is prime to 3 and every odd prime factor of $x$ is $\pm 1 \pmod{12}$.

### 6.8.1 $c = 25$: an explicit generator

For $c = 25$ and $x$ odd and prime to 5, the prime 5 splits, $(\theta) = q^2$ for a prime $q$ above 5, and $(\theta - 5) = qP$ with $N(P) = x$. Then $P^2$ has the explicit generator

$$\pi = \sqrt{D} - 10 = \frac{(\theta - 5)^2}{\theta},$$

and $|\pi/\pi'| = e^{-2\ell}$ with $\ell = \operatorname{arcsinh}(10/x)$. This plays the role of $\varepsilon + 1$: a fixed power of the ideal of norm $x$ with a generator whose logarithm is about $C/x$ L20.4.

**Proposition 6.36** (Pure solutions for $c = 25$). PROVEN L20.4
Let $x$ be odd and prime to 5, $n \geq 5$ odd ($n \geq 7$ if $x = 3$), and $b$ a pure solution of $b(b + x) = x^n + 25$. Put $D = x^2 + 100$ and $\tau_n = \sqrt{D}/\left(x^{n/2} - x/2 - 25\right)$. Then $|n\ell - s\mu_0| < \tau_n$ for an integer $s \geq 1$, so $n > (\mu_0 - \tau_n)x/10$. For odd $x \geq 22\,027$ prime to 5 there is no pure solution with $n = x$.

The second statement is easier than its analogue for $c = 1$: every unit $e > 1$ of $\mathbb{Z}[\theta]$ satisfies $e > x$, so $\mu_0 > \ln x > 10$, while $x\ell < 10$.

**Theorem 6.37** (The family $(1, x, 3)$ for prime powers). PROVEN L20.4
Under $0^0 = 1$, $(1, x, 3)$ is a solution in no base for any prime power $x$. For an odd prime power $x$, the solutions of $b(b + x) = x^n + 25$ with $n \geq 3$ are exactly $(x, n, b) = (5, 3, 10), (7, 3, 16), (3, 6, 26)$.

*Proof sketch.* Even $x$ and squares are elementary. Powers of 5 are handled through the continued fraction of $\sqrt{y^2 + 4}$ with $y = x/5$. For other odd prime powers every solution is pure; Laurent's theorem bounds pure solutions for $x > 507\,211$, a computation with exact regulators covers smaller $x$, and sieves cover $n \leq 15$. □

### 6.8.2 $c = 3$: where the method stops

For $c = 3$ no element of $\mathbb{Z}[\theta]$ with a norm of the form $\pm x^j 3^i 2^h$ is known uniformly in $x$, because that would need a square root of 3 modulo $x$ given by a polynomial in $x$. So only the analogue of Theorem 6.19 remains. In Section 6.5 the fundamental unit was at most $\varepsilon$, so $\mu_0 \leq \ln \varepsilon \approx \ln x$. For $c = 3$ there is no unit of that size, and the regulator $\mu_0$ is typically of size $\sqrt{D} \approx x$.

**Proposition 6.38** (The regulator obstacle). PROVEN L20.4
Let $x > 10^6$ be prime and $b$ a solution of $b(b + x) = x^x + 3$, with $(b + \theta) = p^x$. Then either the class number of $\mathbb{Z}[\theta]$ is a multiple of $x$, or $p$ is principal and $\mu_0(\ln x + \mu_0) > x \ln x/(202M^2)$ with $M = \max(\ln(2x) + 0.38, 5)$.

The second case is the generic one: of the 1,121 primes $x \equiv \pm 1 \pmod{12}$ below $2 \cdot 10^4$, 621 have a principal prime above $x$, and all of them have a regulator above this bound (computed under GRH) L20.4.

**Computation 6.39** (The families for small $x$). COMPUTATION L20.4
$4x^x + x^2 + 12$ and $4x^x + x^2 + 100$ are not squares for $2 \leq x \leq 10^9$. For odd $x \leq 1001$, $b(b + x) = x^n + 3$ has no solution with $n \geq 2$, and $b(b + x) = x^n + 25$ with $x$ prime to 5 has only $(7, 3, 16)$ and $(3, 6, 26)$. For $n = 3$ the elliptic curves $Y^2 = X^3 + X^2 + 64c$ give no solution for $c = 3$ and the solutions $(x, b) = (5, 10), (7, 16), (656, 16477), (1600, 63205)$ for $c = 25$.

The lesson of this section: the method of Section 6.5 does not need a unit of norm $-1$. It needs a fixed power of the ideal of norm $x$ with an explicit generator whose logarithm is about $C/x$. For $c = 25$ that generator exists; for $c = 3$ it does not, and the regulator then blocks the argument even for prime $x$.

## 6.9 Cores in perfect-power bases

The abc conjecture also controls a related question from Chapter 2. Under $0^0 = 0$ every solution splits into a core, self-paying digits and an optional final 1 (Proposition 2.5). Can a nonzero core sit in a base with a self-paying digit, that is in a perfect power? The swaps $(a, a^a)$ of Section 2.3 give infinitely many such cores L15.3. The question is whether anything else does.

**Computation 6.40** (Two-digit cores in perfect-power bases). COMPUTATION L15.3
Among the 2-digit solutions with $a \le 200$ and $c \le 3000$ under $0^0 = 0$, other than $(a, 0)$ and $(a, 1)$, exactly four have a perfect-power base: $(1, 3)$ in $5^2$ and the swaps $(2, 4)$ in $2^7$, $(3, 27)$ in $3^{80}$ and $(4, 256)$ in $2^{2046}$.

**Theorem 6.41** (Cores under abc). CONDITIONAL L15.3
Assume the abc conjecture and fix $\delta > 0$. Only finitely many 2-digit solutions $(a, c)$ under $0^0 = 0$ with $c \ge 2$, $c \ne a^a$ and a perfect-power base satisfy $\min(a^a, c^c) \le \max(a^a, c^c)^{1/4-\delta}$.

*Proof sketch.* Say $c^c > a^a$, and write the base as $r^t$ with $t \ge 2$, so $r \le \sqrt{b}$. The equation $ar^t = c^c + (a^a - c)$ is an abc sum whose radical is at most about $c^{c/2}$ times a power of the small term. abc then bounds $c$. The other case is symmetric. □

The theorem leaves out the balanced case $a^a \approx c^c$, where four large terms appear (Q37). Part of one family is settled unconditionally: the case $t \mid c$ of $(1, c)$. It does not cover $(1, 3)$ in base $25 = 5^2$, where $t = 2$ does not divide $c = 3$.

**Proposition 6.42** (The family $(1, c)$). PROVEN L15.3
For $t \mid c$, the base $c^c - c + 1$ of $(1, c)$ is not a $t$-th power.

*Proof.* With $Y = c^{c/t}$ the base lies strictly between $(Y-1)^t$ and $Y^t$, because $Y^t - (Y-1)^t \ge Y^{t-1} \ge c^{c/2} > c - 1$. □

Heuristically the number of perfect powers among the 2-digit bases is $\sum_{(a,c)} b^{-1/2}$, which converges fast. So the swaps and $(1, 3)$ in base 25 are probably all 2-digit cores in perfect-power bases.

Chapter 7

# Other number systems and other weights

The definition of a Munchausen number has two ingredients: a way to write numbers with digits, and a weight for each digit. Chapters 1 to 6 keep both fixed: an integer base $b \geq 2$ and the weight $x^x$. This chapter changes them. It writes numbers in negative, imaginary, Gaussian, Eisenstein, rational and Pisot bases, in the polynomial ring $\mathbb{F}_q[T]$ and in the factorial number system, and it replaces $x^x$ by $x^{x+t}$, by $x!$ and by $x^d$.

The aim is to sort the facts of the earlier chapters into two kinds. Some hold in every world where the probabilistic model of Chapter 3 makes sense: about one solution per base and convention, a Poisson count, and a factor 2 whenever a certain modulus is even. Others belong to $x^x$ and the integers: the length shares 8, 55 and 23 percent, the twins, the self-paying digits. Two worlds break the model outright, and they show why it works elsewhere. Section 7.8 on function fields is the clearest case in this document of a coincidence model that fails, and of the exact reason.

## 7.1 A common frame

Every world in this chapter has a set of digits, a rule that gives each number exactly one digit string, and a weight $w(d)$ for each digit, with $w(0) = m$ for the convention $0^0 = m$. A solution is a number $n$ with at least two digits and $n = F(n)$, where $F(n)$ is the sum of the weights of its digits.

The model of Section 3.2 carries over almost word for word. Split the numbers into windows $W$, for example the numbers with exactly $k$ digits. Pretend that the sum $F$ of a random string from $W$ is a random number of its window.

**Heuristic 7.1** (The windows model). HEURISTIC L5.1 L21

The expected number of solutions is

$$E = \sum_{W} P(F(N) \in W), \quad N \text{ uniform in } W.$$

Each term is the chance that the sum of a random string of the window lands in the same window. In every world where the length of $F(N)$ barely depends on the length of $N$, these chances add up to about 1.

The second ingredient is an identity that every number obeys. In base $b$ every integer is congruent to its digit sum modulo $b - 1$ (Proposition 1.9). Most worlds have such a modulus.

**Definition 7.2** (Identity modulus). L21

An *identity modulus* of a number system is an integer $M \geq 2$ such that every number is congruent to its digit sum modulo $M$. A solution then satisfies $\sum_d (w(d) - d) \equiv 0 \pmod{M}$, the sum over its digits.

The refined model of Section 3.3 draws the sum at random among the numbers that obey this identity. A digit multiset that meets the condition becomes $M$ times as likely, and the others

become impossible. On average nothing changes, unless the condition holds more often than $1/M$ of the time. It does so in one situation.

**Heuristic 7.3** (Free residue factor). HEURISTIC L5.3 L14.6 L21

If $M$ is even and $w(d) \equiv d \pmod 2$ for every digit $d \geq 1$, then under $0^0 = 0$ the identity holds modulo 2 for free, and the expected number of solutions doubles. Under $0^0 = 1$ each zero adds 1, so the factor applies only to strings with an even number of zeros, and it is $1 + e^{-2} \approx 1.135$ when the number of zeros is a Poisson count with mean 1, as in a long string. When zeros are rare it tends to 2.

For $x^x$ the condition $w(d) \equiv d \pmod 2$ always holds, since $x^x$ and $x$ have the same parity. So the question in each world is whether its identity modulus is even. The function field $\mathbb{F}_q[T]$ has a stronger version of the same mechanism: there the weights agree with the digits at every prime factor of the modulus (Section 7.8).

Two kinds of structure sit outside the model, as in the integers. *Twins* are pairs of solutions that differ in the last digit (Proposition 2.1). *Self-paying digits* are digits whose weight equals their value at some position (Proposition 2.2). Both come from exact identities, not from chance.

The last tool is the near-solution count of Section 3.5. For an offset $D$ in the residue class of 0 modulo the identity modulus, count the $n$ with $n = F(n) + D$. Each such offset is another draw of the same coincidence with the true weights. The mean count per offset measures the real rate of solutions without any model. In this chapter "near" always means this mean.

Table 7.1 lists the worlds of this chapter.

Table 7.1. The number systems of this chapter.

| system | digits | identity modulus | section |
|---|---|---|---|
| base $b \geq 2$ | $0, \ldots, b-1$ | $b-1$ | Section 3.3 |
| base $-b$ | $0, \ldots, b-1$ | $b+1$ | Section 7.2 |
| base $ci$ | $0, \ldots, c^2-1$ | reduces to $-c^2$ | Section 7.3 |
| Gaussian base $-n+i$ | $0, \ldots, n^2$ | $(n+1)^2+1$ | Section 7.3 |
| Eisenstein base $-n+\omega$ | $0, \ldots, n^2+n$ | $n^2+3n+3$ | Section 7.3 |
| rational base $p/q$ | $0, \ldots, p-1$ | $p-q$ | Section 7.4 |
| Pisot base $\varepsilon_x$ | $0, \ldots, x$ | $x$ | Section 7.5 |
| $\mathbb{F}_q[T]$, base $B$ of degree $e$ | polynomials of degree $< e$ | $B-1$ | Section 7.8 |
| factorial system | $0 \leq d_i \leq i$ | none | Section 7.9 |

Appendix Appendix I explains how each of these number systems works and why its representations are unique.

### 7.1.1 Where the identity modulus comes from

The identity modulus of each world was found world by world. One statement covers all of them (Q108). Take a ring $R$ (the integers, $\mathbb{Z}[i]$, $\mathbb{Z}[\omega]$, $\mathbb{Z}[\varepsilon]$, $\mathbb{Z}[1/q]$ or $\mathbb{F}_q[T]$), place values $\beta_i$ in $R$, a digit set, and a set of admissible strings. A string has the value $v(s) = \sum s_i \beta_i$. An *exchange* swaps the digits at two positions of an admissible string and gives another admissible string.

**Proposition 7.4** (The invariant ideal). PROVEN L35.3

Let $I_0$ be the ideal generated by the differences $v(s) - v(s')$ of admissible strings with the same digit multiset. The multiset fixes the value modulo an ideal $I$ exactly when $I$ contains $I_0$.

1. $I_0$ contains every $(x-y)(\beta_i - \beta_j)$ for an exchange of the digits $x, y$ at the positions $i, j$, and is generated by them when the admissible strings of each multiset are connected by exchanges.

2. If $\beta_i = u\beta^i$ with a unit $u$, then $v(s) \equiv u \sum s_i \pmod{\beta - 1}$. If moreover two adjacent positions $i, i+1$ with $\beta^i$ a unit can exchange digits that differ by a unit, then $I_0 = (\beta - 1)$.
3. If $\beta$ is an algebraic integer with monic minimal polynomial $f$, then $(\beta - 1)$ meets $\mathbb{Z}$ in $|f(1)|\,\mathbb{Z}$. For $\beta = p/q$ or $-p/q$ in $\mathbb{Z}[1/q]$ it meets $\mathbb{Z}$ in $(p-q)\mathbb{Z}$ or $(p+q)\mathbb{Z}$.

*Proof.* An exchange moves the value by $(x-y)(\beta_i - \beta_j)$, and along a path of exchanges these steps add up. For (2), $\beta^i - 1$ is a multiple of $\beta - 1$, and an adjacent exchange with unit digit difference $e$ gives $eu\beta^i(\beta - 1)$. For (3), $\mathbb{Z}[t]/(f(t), t-1) = \mathbb{Z}/f(1)$, and $p/q - 1$ is a unit times $p - q$ in $\mathbb{Z}[1/q]$. □

So the identity modulus of a geometric system is $|f(1)|$: $b-1$ for $\beta = b$, $b+1$ for $-b$, $(n+1)^2 + 1$ for $-n+i$, $n^2+3n+3$ for $-n+\omega$, $x$ for $\varepsilon_x$ (where $f = t^2 - xt - 1$), $p - q$ for $p/q$. The digit differences matter: with only even digits the ideal would be $2(\beta - 1)$. The offsets of the swap lemma (Lemma 3.43) live in the same ideal.

**Proposition 7.5** (The residue factor is an index). PROVEN L35.3
Suppose $v(s) \equiv u \sum s_i$ modulo $I_0$ and $R/I_0$ is finite. Put $g(x) = w(x) - ux$, and let the digits of a string be independent with a fixed law that charges a digit with $g = 0$. If $H$ is the subgroup of $R/I_0$ generated by the $g(x)$ of the digits that occur, then $P(\sum g(x_i) \equiv 0) \to 1/\,|H|$, and the refined model gains the factor $[R/I_0 : H]$.

*Proof.* The partial sums are a random walk on $H$ that can stay put and whose steps generate $H$, so they become uniform on $H$. □

The parity doubling is the index 2 when every $g(x)$ is even. The factor $q^{k(B)}$ of $\mathbb{F}_q[T]$ is the index cut out by the trace conditions (Section 7.8). The factorial system has $I_0 = \mathbb{Z}$ and no factor.

**Computation 7.6** (The predicted moduli in 67 systems). COMPUTATION L35.3
Group the integers of a range by their digit multiset and take the gcd of all differences within groups. In integer bases 3 to 12, negative bases $-2$ to $-10$, bijective bases 2 to 10, balanced bases 3 to 9, ten rational bases $p/q$, seven bases $-p/q$, Gaussian bases $-n+i$ ($n \le 5$), Eisenstein bases $-n+\omega$ ($n \le 4$), Pisot bases $\varepsilon_x$ ($x \le 8$) and the factorial system, the gcd is the predicted modulus every time.

Two predictions follow for worlds not yet searched. Bijective base $b$, with digits 1 to $b$ and no zero, has the identity modulus $b-1$, and in odd bases the factor 2 with no convention to choose (Q95). Balanced ternary has the modulus 2 and the base $-3/2$ the modulus 5.

## 7.2 Negative bases

### 7.2.1 Representations and first examples

In base $-b$, with $b \ge 2$, every integer has exactly one representation $n = \sum_i s_i(-b)^i$ with digits $0 \le s_i \le b-1$ and no sign (Appendix I). A string $s$ is a Munchausen number in base $-b$ when $n = F(n)$, with the same weights as before. In the language of Section 1.3, these are the integer roots of the Munchausen polynomial $Q_s$ at negative arguments $r < -\max(s)$.

**Proposition 7.7** (Odd length). PROVEN L12.3 LEAN
A Munchausen number in a negative base has an odd number of digits.
Lean: `IsNegSolution.odd_length`

*Proof.* Its value $F(s)$ is positive. A string of length $k$ in base $-b$ with a nonzero leading digit has the sign of $(-b)^{k-1}$, because the leading term outweighs all the others. So $k-1$ is even. □

**Example 7.8.** L12.3

In base $-10$ the number 404201091 has the digits $(4,0,5,8,1,9,1,1,1)$. Indeed $4\cdot 10^8+5\cdot 10^6-8\cdot 10^5+10^4-9\cdot 10^3+100-10+1=404201091$, and under $0^0=1$

$$4^4+1+5^5+8^8+1+9^9+1+1+1=404201091.$$

A smaller example is $(1,3,4,3,3)$ in base $-5$: $625-375+100-15+3=338=1+27+256+27+27$.

The structure of the integer case carries over with $b+1$ in place of $b-1$. The key fact is that prefixes still give intervals.

**Proposition 7.9** (Prefix intervals). PROVEN L14.1

Let $\max_j=(b-1)(1+b^2+b^4+\ldots)$ and $\text{negmin}_j=(b-1)(b+b^3+\ldots)$, the sums over the even and the odd positions below $j$. The integers with at most $j$ digits in base $-b$ are exactly the integers in $\left[-\text{negmin}_j,\max_j\right]$. The integers whose digits at positions $\geq j$ are fixed form a translate of this interval by a multiple of $b^j$.

*Proof.* The $b^j$ strings of length $j$ have distinct values, and all lie in the interval. The interval has $\max_j+\text{negmin}_j+1=(b-1)(1+b+\ldots+b^{j-1})+1=b^j$ integers, so the strings fill it. Fixing the top part to a value $u$ gives the set $u(-b)^j+\left[-\text{negmin}_j,\max_j\right]$. □

So for odd $d$ the integers with exactly $d$ digits form the interval $\left(b^{d-1}-1\right)/(b+1)<n\leq\left(b^{d+1}-1\right)/(b+1)$, a window twice as wide in digits as in a positive base. The integers with an even number of digits are all negative. A second fact makes base $-b$ arithmetic as cheap as base $b$ arithmetic: $n+\text{negmin}_d$ has the base-$b$ digits $s_i$ at even positions and $b-1-s_i$ at odd positions (L14.1, Proposition 14.3). The search engine of Section 4.1 therefore runs unchanged on shifted digit vectors, and it grows about 1.9 times per base, as in positive bases (L14.2).

### 7.2.2 The identity modulus is b + 1

**Proposition 7.10** (The only invariant residue). PROVEN L14.1 LEAN

Let $M\geq 2$. The residue $n \bmod M$ depends only on the digit multiset of $n$, for all $n$ with at least 3 digits in base $-b$, exactly when $M$ divides $b+1$. Every solution satisfies $\sum_x(w(x)-x)\equiv 0\pmod{b+1}$.

Lean: `neg_residue_invariant_iff, IsNegSolution.dvd`

*Proof.* Since $-b\equiv 1\pmod{b+1}$, every $n$ is congruent to its digit sum modulo $b+1$. Conversely, swap the last two digits $x$ and $y$ of a number with at least 3 digits, with $x-y=1$. The leading digit does not move, the multiset stays the same, and the value changes by $(x-y)(-b)+(y-x)=-(b+1)$. So $M$ divides $b+1$. For a solution, $n=F(n)$ gives the identity. □

The consequences of Section 1.6 move with the modulus. For odd $b$ and $0^0=1$, a solution has an even number of zeros; for odd $b$ and $0^0=0$ the identity holds modulo 2 for free. So the parity doubling of Heuristic 7.3 happens in base $-b$ for odd $b$, where $b+1$ is even. Both $b-1$ and $b+1$ are even exactly when $b$ is odd, so the same $|b|$ doubles in both signs.

**Proposition 7.11** (Length). PROVEN L14.1

A solution in base $-b$ has an odd number $k\leq b+1$ of digits. So $k\leq b+1$ for even $b$ and $k\leq b$ for odd $b$.

*Proof sketch.* Odd length is Proposition 7.7. For odd $k \geq b+2$, $F(n) \leq k(b-1)^{b-1}$ while $n > (b^{k-1}-1)/(b+1)$, and a direct estimate shows $k(b+1)(b-1)^{b-1}+1 \leq b^{k-1}$ for all such $k$. The factor $(1-1/b)^{b-1} \leq 0.422$ for $b \geq 4$ does the work; bases 2 and 3 are checked by hand (L14.1). □

The bound is reached: base $-10$ has the 11-digit solution $(1,9,2,1,7,4,5,8,9,8,9)$ and base $-14$ a 15-digit one.

### 7.2.3 The length law

The large-base analysis of Section 3.2 adapts. The largest digit present fixes the size of $F$, and the windows of lengths now sit at even powers of $b$ apart.

**Heuristic 7.12** (Length shares in negative bases). HEURISTIC L14.1

In the large-base limit a negative base has about one solution per convention before the identity factor, with these shares by length:

| even $b$ | share | odd $b$ | share |
|---|---|---|---|
| $b+1$ digits (3 or more copies of $b-1$) | 8.0% | $b$ digits (at least one $b-1$) | 63.2% |
| $b-1$ digits | 78.4% | $b-2$ digits | 31.8% |
| $b-3$ digits | 11.7% | $b-4$ digits | 4.3% |
| $b-5$ digits | 1.6% | $b-6$ digits | 0.6% |
| fewer | 0.3% | fewer | 0.1% |

The refined factors are 1 and 2 under $0^0 = 0$, and 1 and $1+e^{-2}$ under $0^0 = 1$, for even and odd $b$.

For example, for even $b$ and $k = b+1$ the window starts near $b^{b-1}$, and $c$ copies of $b-1$ give $F \approx (c/e)b^{b-1}$. So $F$ lands in the window when $c \geq 3$, with probability $1-(5/2)e^{-1} \approx 0.080$, the same 8% as for $b$-digit solutions in positive bases.

### 7.2.4 The census

**Computation 7.13** (Negative bases $-2$ to $-38$). COMPUTATION L12.3 L14.3 L23.2

Table 7.2 lists the number of solutions and their lengths in every base from $-2$ to $-38$, under both conventions. The search is exhaustive in each base.

Table 7.2. Solutions in negative bases. "lengths" lists the length of each solution. $E_m$ is the refined model. Data: L14.3, L23.2, `data/negative-bases.csv`.

| base | $0^0=1$ | lengths | $E_1$ | $0^0=0$ | lengths | $E_0$ |
|---|---|---|---|---|---|---|
| $-2$ | 1 | 3 | 0.75 | 2 | 3, 3 | 2.25 |
| $-3$ | 0 | | 1.11 | 0 | | 1.33 |
| $-4$ | 0 | | 1.05 | 1 | 5 | 0.63 |
| $-5$ | 1 | 5 | 1.16 | 2 | 3, 5 | 2.54 |
| $-6$ | 0 | | 1.20 | 0 | | 1.17 |
| $-7$ | 3 | 3, 7, 7 | 1.48 | 4 | 3, 3, 7, 7 | 2.55 |
| $-8$ | 1 | 7 | 1.36 | 2 | 3, 3 | 1.58 |
| $-9$ | 0 | | 1.40 | 0 | | 2.31 |
| $-10$ | 3 | 9, 9, 11 | 1.30 | 4 | 9, 9, 9, 11 | 1.24 |
| $-11$ | 2 | 9, 11 | 1.61 | 2 | 11, 11 | 2.80 |
| $-12$ | 1 | 11 | 1.23 | 0 | | 1.27 |
| $-13$ | 1 | 13 | 1.44 | 4 | 7, 13, 13, 13 | 2.48 |
| $-14$ | 2 | 13, 15 | 1.20 | 1 | 15 | 1.12 |
| $-15$ | 0 | | 1.38 | 0 | | 2.37 |
| $-16$ | 0 | | 1.18 | 1 | 15 | 1.13 |
| $-17$ | 1 | 17 | 1.34 | 2 | 17, 17 | 2.29 |
| $-18$ | 0 | | 1.12 | 0 | | 1.12 |

| base | $0^0=1$ | lengths | $E_1$ | $0^0=0$ | lengths | $E_0$ |
|---|---|---|---|---|---|---|
| −19 | 1 | 19 | 1.30 | 1 | 19 | 2.27 |
| −20 | 0 | | 1.12 | 0 | | 1.12 |
| −21 | 2 | 21, 21 | 1.31 | 1 | 21 | 2.25 |
| −22 | 0 | | 1.11 | 1 | 21 | 1.11 |
| −23 | 1 | 23 | 1.30 | 2 | 21, 23 | 2.27 |
| −24 | 0 | | 1.10 | 3 | 23, 23, 23 | 1.09 |
| −25 | 0 | | 1.27 | 2 | 3, 3 | 2.18 |
| −26 | 1 | 27 | 1.08 | 0 | | 1.07 |
| −27 | 1 | 25 | 1.24 | 2 | 25, 27 | 2.15 |
| −28 | 2 | 23, 27 | 1.06 | 1 | 23 | 1.05 |
| −29 | 0 | | 1.21 | 1 | 27 | 2.13 |
| −30 | 0 | | 1.05 | 0 | | 1.05 |
| −31 | 4 | 31, 31, 31, 31 | 1.19 | 5 | 29, 29, 31, 31, 31 | 2.10 |
| −32 | 2 | 31, 31 | 1.05 | 4 | 13, 13, 31, 31 | 1.06 |
| −33 | 1 | 33 | 1.21 | 3 | 33, 33, 33 | 2.09 |
| −34 | 0 | | 1.05 | 0 | | 1.04 |
| −35 | 0 | | 1.23 | 1 | 33 | 2.14 |
| −36 | 3 | 33, 35, 35 | 1.06 | 1 | 35 | 1.06 |
| −37 | 1 | 37 | 1.18 | 1 | 37 | 2.08 |
| −38 | 0 | | 1.04 | 0 | | 1.04 |

The small bases show the same algebraic structure as positive bases. Under $0^0=0$ twins appear in bases $-2$, $-7$, $-8$, $-10$, $-25$ and $-32$. For $x$ odd or a square, $(x,0,0)$ and $(x,0,1)$ are solutions in base $-x^{(x-1)/2}$ as well as in base $x^{(x-1)/2}$, which gives $(4,0,0)$ in base $-8$ and $(5,0,0)$ in base $-25$. In base $-32$ the digit 16 pays for itself at position 12, because $16\cdot 32^{12}=2^{64}=16^{16}$ and $(-32)^{12}>0$. The solutions with $b+1$ digits, in bases $-10$, $-14$ and $-26$, all start with the digit 1 followed by $b-1$, as the length analysis predicts.

**Proposition 7.14** (Partners between the conventions). PROVEN L23.2
If $n$ is a solution under $0^0=0$ that ends in the digit 2 and has exactly two zero digits, then $n-1$ is a solution under $0^0=1$. This holds in every base $b$ and $-b$.

*Proof.* The number $n-1$ ends in 1 and has the same other digits. So $F_1(n-1)=F_0(n)+2-w(2)+w(1)=n+2-4+1=n-1$, where the 2 counts the two zeros under $0^0=1$. □

Base $-36$ shows it: a 35-digit string ending in $(\ldots,7,2)$ under $0^0=0$ and the same string ending in $(\ldots,7,1)$ under $0^0=1$.

### 7.2.5 Counts against the model

**Computation 7.15** (Counts against the model). COMPUTATION L14.3
Over bases $-10$ to $-35$:

| convention | bases | observed | refined model | basic model |
|---|---|---|---|---|
| $0^0=0$ | even $b$ | 15 | 14.5 | 14.6 |
| $0^0=0$ | odd $b$ | 26 | 29.5 | 14.5 |
| $0^0=1$ | even $b$ | 11 | 14.6 | 14.6 |
| $0^0=1$ | odd $b$ | 14 | 17.0 | 14.5 |

Every group is within one standard deviation of the refined model. The parity doubling is plain: under $0^0=0$ the odd bases have 26 solutions against 15 in the even ones, while the basic model predicts the same for both. The length shares follow Heuristic 7.12: in even bases most solutions have $b-1$ digits, in odd bases most have $b$ digits.

### 7.2.6 Empty bases

Nine bases from $-2$ to $-38$ have no solution under either convention: $-3$, $-6$, $-9$, $-15$, $-18$, $-20$, $-30$, $-34$ and $-38$. The model gives each base a chance $\exp(-E)$ to be empty, with $E = E_0 + E_1 - E_s$ and $E_s$ the expected number of zero-free solutions, which count in both conventions. The expected number of empty bases is 4.63, and nine or more has probability 0.033.

Six of the nine are multiples of 3, which looked like a pattern when the census stopped at $-35$. The bases $-3$, $-6$, $-9$ were noticed first, so a fair test leaves them out. Among the multiples of 3 from $-12$ to $-36$, three of nine are empty against 1.15 expected ($P = 0.095$). The other bases from $-10$ have three of 20 against 2.44 ($P = 0.45$).

**Heuristic 7.16** (The empty multiples of 3 are chance). HEURISTIC L14.3 L23.2

Nothing in the arithmetic of the multiples of 3 lowers their chance of a solution:

- The identity modulus $b + 1 \equiv 1 \pmod 3$ gives no factor 3.
- For $3 \mid b$ the last digit is fixed modulo 3 by $F$. The exact effect of this on the model, measured by sampling, is a factor between 0.9985 and 1.0006 for $b = 12$ to 39.
- The near-solution counts of the multiples of 3 match the model, also at offsets that keep $n - F(n)$ modulo $b$ and hence modulo 3.

The excess of empty negative bases is mild and shared by all bases.

One side result is worth noting. Offsets that keep the last digit of $n$ give 6% to 23% fewer near-solutions in negative bases than offsets that only keep the identity. This is the better calibration of the count at $D = 0$, and against it the solutions of bases $-10$ to $-27$ number 42 against 52. Why the last digit matters here is open (Section 8.2).

**Remark 7.17** (The verdicts with clustered near-solutions). L23.2 L26.4

Near-solutions at different offsets are not independent: rearranging the digits of a number keeps $F$ and moves $n - F(n)$ by a multiple of $b + 1$, so one coincidence can be counted at several offsets (Chapter 3). Recomputed with this clustering, the P values of this section become 0.11 for the 42 solutions against 52 (it was 0.09) and 0.056 for the multiples of 3 against their offsets (it was 0.044). The lower near-solution rate at offsets that keep the last digit drops from about 4 to about 2.9 standard deviations. The verdicts do not change.

## 7.3 Imaginary, Gaussian and Eisenstein bases

### 7.3.1 Imaginary bases reduce to negative bases

**Proposition 7.18** (Imaginary bases). PROVEN L14.4

Let $c \geq 2$. A string $S$ is a Munchausen number in base $ci$, with digits $0, \ldots, c^2 - 1$, exactly when its digits at odd positions are all 0 and its digits at even positions form a string $t$ of odd length $d$ with

$$\text{value of } t \text{ in base } -c^2 = F_m(t) + m(d-1).$$

So $S$ has $2d - 1 \equiv 1 \pmod 4$ digits, and under $0^0 = 0$ the Munchausen numbers of base $ci$ and base $-c^2$ correspond one to one.

*Proof.* $(ci)^{2k} = (-c^2)^k$ and $(ci)^{2k+1} = ci(-c^2)^k$. So the value of $S$ is $A + ciB$, where $A$ is the value of the even-position digits in base $-c^2$ and $B$ that of the odd-position digits. $F(S)$ is a positive integer, so $B = 0$, and by uniqueness in base $-c^2$ every odd-position digit is 0. The top digit sits at an even position, so $S$ has $2d - 1$ digits, and $d$ is odd by Proposition 7.7. The $d - 1$ zeros at odd positions add $m(d-1)$ to $F$. □

Under $0^0 = 1$ the zeros cost something, so the solutions differ from those of base $-c^2$. Base $2i$ has $(1, 0, 2, 0, 0)$: its value is $16 - 8 = 8$, and $1 + 1 + 4 + 1 + 1 = 8$. Base $3i$ has $(1, 0, 3, 0, 3)$ and two longer ones, base $4i$ has $(1, 0, 0, 0, 4)$ and a 29-digit one, and base $5i$ has none (L14.4, computed).

### 7.3.2 Gaussian bases

**Theorem 7.19** (Kátai and Szabó). CITED L14.5

For $n \geq 1$, every Gaussian integer has exactly one finite expansion $\sum_i a_i \beta^i$ in base $\beta = -n + i$ with digits $a_i \in \{0, 1, ..., n^2\}$ [40].

The last digit of $a + bi$ is $(a + nb) \bmod (n^2 + 1)$, because $i \equiv n \pmod{\beta}$. A Munchausen number is a rational integer equal to the sum of the weights of its digits.

**Proposition 7.20** (Gaussian identity). PROVEN L14.5

Every Gaussian integer satisfies $z \equiv \text{digit sum} \pmod{\beta - 1}$. A rational integer is divisible by $\beta - 1 = -(n + 1) + i$ exactly when it is divisible by its norm $M = (n + 1)^2 + 1$. So a solution satisfies $\sum(w(x) - x) \equiv 0 \pmod{M}$, and no other Gaussian modulus is a function of the multiset.

$M$ is even exactly when $n$ is even. So Heuristic 7.3 predicts doubling under $0^0 = 0$ in the bases $-2 + i$ and $-4 + i$, and not in $-1 + i$ and $-3 + i$.

A length bound needs a little more work than in base $b$, because the digits of a Gaussian integer are produced by repeated division and the remainders shrink only geometrically. The result is a bound on the length of a solution: 18, 16, 24 and 38 digits for $n = 1, 2, 3, 4$ (L14.5, Proposition 14.10, proven with a finite computation).

**Computation 7.21** (Gaussian bases). COMPUTATION L14.5

The solutions in bases $-1 + i$, $-2 + i$ and $-3 + i$ are complete up to the length bound; base $-4 + i$ is searched up to 22 digits only.

| base | $0^0 = 1$ | $E_1$ | $0^0 = 0$ | $E_0$ |
|---|---|---|---|---|
| $-1 + i$ | $(1, 1, 1, 0, 0, 0, 0, 0, 1) = 9$ | | six: 2, 3, and four with 9 digits | |
| $-2 + i$ | none | 1.05 | $(4, 2, 4, 0, 4, 1, 0, 4) = 1029$ | 2.22 |
| $-3 + i$ | three with 19 digits, one with 21 | 1.11 | the 21-digit one | 1.10 |
| $-4 + i$ | none up to 22 digits | 1.06 | not searched | 2.08 |

The 21-digit solution of base $-3 + i$ is $(1, 5, 9, 2, 5, 4, 9, 7, 1, 3, 9, 1, 9, 2, 9, 1, 7, 8, 2, 8, 8) = 1989087728$. It has no zero, so it counts under both conventions.

The counts match the model, with one exception: base $-3 + i$ has four solutions under $0^0 = 1$ against 1.11 expected ($P = 0.026$). Base $-4 + i$ is out of reach: the model puts 83% of its mass at 34 digits, and the prefix test of the search tree does not work, because the numbers with fixed top digits are not an interval but points of a fractal tile.

Positive integers do not use every length in a Gaussian base. In base $-2 + i$ no positive integer has 2, 3, 5, 7, 10 or 12 digits, for example. The lengths come in runs of one parity, and the parity switches about every $\pi / \arctan(1/n)$ digits (L14.5, computed and heuristic). For base $-1 + i$ the pattern has a clean proof.

**Theorem 7.22** (Base $-1 + i$ is base $-4$ in blocks). PROVEN L23.4 LEAN

In base $\beta = -1 + i$, with digits 0 and 1, the representation of a rational integer $n$ is its base $-4$ representation with each base $-4$ digit $B$ written as the block $(B \bmod 2, 0, [B \geq 2], [B \geq$

2]) at positions $4j$ to $4j+3$, least significant first. So a positive integer has $8m+1$ digits if its leading base $-4$ digit is 1, and $8m+4$ digits if it is 2 or 3.
Lean: `gaussian_blocks (partial: the block form; existence and uniqueness cited)`

*Proof.* $\beta^4 = -4$. Cut a representation of $z$ into blocks of four digits, so that $z = \sum_j C_j(-4)^j$ with $C_j = a_0 + a_1\beta + a_2\beta^2 + a_3\beta^3$ and $a_k \in \{0,1\}$. Since $\beta^2 = -2i$ and $\beta^3 = 2+2i$,

$$C = (a_0 - a_1 + 2a_3) + i(a_1 - 2a_2 + 2a_3).$$

The imaginary part $y_j$ of $C_j$ lies in $\{-2,...,3\}$. If $z$ is real, $\sum_j y_j(-4)^j = 0$. Reducing modulo 4 gives $y_0 \equiv 0$, so $y_0 = 0$, the only multiple of 4 in that range. Divide by $-4$ and repeat: every $y_j$ is 0. Now $a_1 - 2a_2 + 2a_3 = 0$ forces $a_1 = 0$ and $a_2 = a_3$, and $C_j = a_0 + 2a_2$ is a base $-4$ digit. By uniqueness of base $-4$ representations this is the base $-4$ representation of $z$. A positive integer has an odd number $2m+1$ of base $-4$ digits (Proposition 7.7), and its top block ends at position $8m$ or $8m+3$. □

**Corollary 7.23** (The solutions of base $-1+i$). PROVEN L23.4
The Munchausen numbers of base $-1+i$ with at least two digits are 2, 3, 4, 5, 6, 7 under $0^0 = 0$, and 9 under $0^0 = 1$.

*Proof sketch.* Under $0^0 = 0$, $F(n)$ is the number of ones, which is the base $-4$ digit sum $s(n)$. Under $0^0 = 1$, $F(n)$ is the number of digits. With $2m+1$ base $-4$ digits, $n > (4^{2m}-1)/5$, while $s(n) \le 3(2m+1)$ and the length is at most $8m+4$. So $m \le 1$ and $n \le 19$, and a short check finishes the proof. □

So the census of base $-1+i$ is complete. The argument needs a power of $\beta$ that is a rational integer. For $-n+i$ with $n \ge 2$ no power of $\beta$ is real, and the length runs of those bases remain unproven (Section 8.2).

### 7.3.3 Eisenstein bases

Let $\omega^2 + \omega + 1 = 0$ and $\beta = -n + \omega$ with $n \ge 1$. Then $\beta$ is a root of $t^2 + (2n+1)t + (n^2+n+1)$. A theorem of Kátai and Kovács and of Gilbert says that such a quadratic is the base of a canonical number system exactly when $-1 \le 2n+1 \le n^2+n+1$ [41, 42]. So every element of $\mathbb{Z}[\omega]$ has exactly one expansion with digits $0,...,n^2+n$.

**Proposition 7.24** (Eisenstein identity). PROVEN L21.3
$\beta - 1 = -(n+1) + \omega$ is primitive of norm $M = n^2+3n+3$. A rational integer solution satisfies $\sum(w(x) - x) \equiv 0 \pmod M$. Since $M = n(n+3)+3$, $M$ is odd for every $n$.

So Heuristic 7.3 predicts no doubling in any Eisenstein base. This makes them the sharpest test of the claim that the factor 2 follows the parity of the modulus and not the shape of the base.

**Computation 7.25** (Eisenstein bases). COMPUTATION L21.3
"near" is the mean number of near-solutions per offset $jM$, $j = 1,...,30$.

| base | $M$ | $0^0$ | solutions (lengths) | model | near |
|---|---|---|---|---|---|
| $-1+\omega$ | 7 | 0 | none (complete) | 0.79 | |
| $-1+\omega$ | 7 | 1 | none (complete) | 1.49 | |
| $-2+\omega$ | 13 | 0 | 1 (15), complete | 1.22 | 1.03 |
| $-2+\omega$ | 13 | 1 | 1 (15), complete | 1.19 | 1.07 |
| $-3+\omega$ | 21 | 0 | 2 (12, 25), to 25 of 30 digits | 1.07 | 1.27 |
| $-3+\omega$ | 21 | 1 | 2 (23, 25), to 25 of 30 digits | 1.06 | 1.13 |

| base | $M$ | $0^0$ | solutions (lengths) | model | near |
|---|---|---|---|---|---|
| $-2+i$ (control) | 10 | 0 | 1 (8), complete | 2.22 | 2.57 |
| $-2+i$ (control) | 10 | 1 | none (complete) | 1.05 | 1.07 |

The answer is clean. The Eisenstein bases show no doubling: their near-solution rates under $0^0 = 0$ are 1.03 and 1.27 per offset, against 1.07 and 1.13 under $0^0 = 1$. The Gaussian control with an even modulus doubles in the same test, 2.57 against 1.07.

## 7.4 Rational bases

In the rational base $p/q$ of Akiyama, Frougny and Sakarovitch, with $\gcd(p, q) = 1$ and $1 \le q < p$, every positive integer $N$ has exactly one expansion

$$N = \sum_i \frac{a_i}{q}\left(\frac{p}{q}\right)^i, \quad 0 \le a_i < p,$$

found by the recursion $qN_i = pN_{i+1} + a_i$ with $N_0 = N$ [43]. The base $p/1$ is the integer base $p$. A Munchausen number is an $N \ge 1$ with $N = \sum w(a_i)$. The length of an expansion grows like $\log_{p/q} N$, while $F$ grows at most linearly in the length, so each base has finitely many.

**Example 7.26.** L16.5

In base $3/2$, $8 = (2, 1, 0, 1, 1)$:

$$\frac{1}{2} + \frac{1}{2}\cdot\frac{3}{2} + \frac{1}{2}\left(\frac{3}{2}\right)^3 + \left(\frac{3}{2}\right)^4 = 8 = 4 + 1 + 1 + 1 + 1.$$

**Proposition 7.27** (Rational identity). PROVEN L16.5 L21.2

Every $N$ satisfies $qN \equiv \sum_i a_i \pmod{p-q}$. A solution satisfies $\sum(qw(a) - a) \equiv 0 \pmod{p-q}$. For $q > 1$ there are no twins.

*Proof.* Modulo $p - q$ we have $p \equiv q$, so $qN_i \equiv qN_{i+1} + a_i$. Summing over $i$ gives $qN_0 \equiv qN_L + \sum a_i$ with $N_L = 0$. Since $\gcd(q, p-q) = 1$, the equation $N = F$ turns into the second form. Changing the last digit by 1 changes the value by $1/q$, which is not an integer, so twins cannot occur. □

If $p$ and $q$ are both odd, $p - q$ is even and $qw(a) - a \equiv w(a) - a \equiv 0 \pmod 2$ under $0^0 = 0$. The identity holds modulo 2 for free, and the model doubles. If $p$ or $q$ is even, $p - q$ is odd and nothing doubles. The identity modulus has nothing to do with the digit range $p$, so this is a clean test.

The integers form a tree under the parent map $N \mapsto (qN - a_0)/p$, and a search walks it from the root. A subtree can be cut when its numbers grow faster than any sum of weights can follow (L21.2, Proposition 21.9, proven). The tree has about $2(p-1)^{p-1}/\ln(p/q)$ nodes, so the cost is worst for $q$ near $p$.

**Computation 7.28** (Rational bases with $p \le 11$). COMPUTATION L16.5 L21.2

All bases $p/q$ with $q > 1$ and $p \le 11$ except 11/8 and 11/10 were searched exhaustively under both conventions. Totals, with "near" the summed mean count of $N = F(N) + j(p - q)$ for $j = 1, \ldots, 30$:

| group | bases | $0^0 = 0$ | refined | basic | near | $0^0 = 1$ / refined |
|---|---|---|---|---|---|---|
| $p$, $q$ odd | 9 | 14 | 19.8 | 10.0 | 19.9 | 11 / 9.9 |
| others | 20 | 20 | 23.4 | 23.3 | 21.9 | 22 / 24.0 |

Under $0^0 = 1$ the bases with $p \le 8$ have the solutions 8 and 15 (base 3/2), 60 and 546 (5/2), 583 (5/3), 15227 and 33827 (6/5), 96983, 146841 and 103858 (7/2, 7/3, 7/4), 299637 and 346791 (7/5), 741782 and 744115 (7/6), 2665246 and 6011678 (8/5), and 11673299 (8/7).

The near-solutions show the doubling best: 2.2 per offset and base in the odd/odd bases against 1.1 in the others, under $0^0 = 0$ only, exactly as the refined model predicts. The solutions themselves fall a little short in the odd/odd bases, 14 against 19.8 ($P(X \le 14) = 0.11$), while their near-solutions do not.

## 7.5 A Pisot base

Let $x \ge 1$ and $\varepsilon = \varepsilon_x = \left(x + \sqrt{x^2+4}\right)/2$, the larger root of $t^2 - xt - 1$. It is a Pisot unit: its conjugate $-1/\varepsilon$ has absolute value below 1. This $\varepsilon$ is also the unit that controls the family $(1, x, 1)$ in Section 6.4. Since $1 = x/\varepsilon + 1/\varepsilon^2$, the greedy expansions in base $\varepsilon$ are the strings over $0, ..., x$ in which every digit $x$ is followed by a 0, except at the end [44]. Every positive integer has a finite greedy expansion [45], possibly with digits after the point. A Munchausen number is an integer equal to the sum of the weights of all its digits.

**Proposition 7.29** (Length). PROVEN L21.4

A positive integer $N$ has at most $2\log_\varepsilon N + 2$ digits in base $\varepsilon$. So every Pisot base $\varepsilon_x$ has finitely many solutions.

*Proof sketch.* Let the expansion run from position $k$ down to $-m$. Then $N \ge \varepsilon^k$. If $m \ge 1$, multiply by $\varepsilon^m$ and look at the conjugate: it equals $N(-1/\varepsilon)^m$, and also a digit expansion whose last digit is at least 1. The digit rule bounds that expansion below by $1/\varepsilon$. So $N\varepsilon^{-m} \ge 1/\varepsilon$, which bounds $m$. A solution has $N \le Tx^x$ for $T$ digits, and the two bounds together bound $T$ (L21.4). □

**Proposition 7.30** (Identity and twins). PROVEN L21.4

$\varepsilon - 1$ has norm $-x$, so $\mathbb{Z}[\varepsilon]/(\varepsilon - 1) = \mathbb{Z}/x$, and $\varepsilon \equiv \varepsilon^{-1} \equiv 1$ modulo $\varepsilon - 1$. Hence $N \equiv$ digit sum $(\mathrm{mod}\, x)$. Under $0^0 = 0$, if $N$ is a solution whose digit at position 0 is 0 and whose digit at position 1 is not $x$, then $N + 1$ is a solution. For $x = 1$ the digit at position $-1$ must also be 0.

There are no self-paying digits: a digit at a position other than 0 adds an irrational amount to the value. The identity modulus $x$ is even for even $x$, so Heuristic 7.3 predicts doubling there.

**Computation 7.31** (Pisot bases $x = 1$ to 12). COMPUTATION L21.4

All solutions, found by enumerating digit multisets up to the length bound and checked by direct expansion for $x \le 7$:

| $x$ | $0^0 = 0$ | $0^0 = 1$ |
|---|---|---|
| 1 | 2 | 9, 12 |
| 2 | 3, 4, 5 | 4, 8, 11, 14 |
| 3 | none | 7, 37 |
| 4 | 9, 10, 54, 55, 84, 138, 312 | 10, 138 |
| 5 | 40, 370 | 40, 370 |
| 6 | 877, 3423, 3475, 6615, 6616, 7031, 50349 | 877, 3423, 3475, 6616, 7031 |
| 7 | 3933, 106077, 917942, 917943 | 3933, 870775, 917943 |
| 8 | 96959, 96960, 917915, 923417, 1787625, 2520454, 2564464, 17697718 | none |
| 9 | 405944671 | none |

| $x$ | $0^0 = 0$ | $0^0 = 1$ |
|---|---|---|
| 10 | 310, 421798810, 877154544, 1600837696, 1956350341, 1956350342 | 1956350342 |
| 11 | 10405164675, 10405164721 | 1971580245, 10405164675, 10405164721 |
| 12 | none | none |

Summed over $x = 2$ to 12 under $0^0 = 0$: the six even $x$ have 31 solutions (7 twin pairs) against 17.0 in the refined model and 8.0 in the basic one; the five odd $x$ have 9 against 5.4. Near-solutions at the offsets $jx$, $j = 1, ..., 30$, number 2.65 per base and offset for even $x$ and 1.43 for odd $x$.

For the golden ratio ($x = 1$), the first entry reads $2 = \varepsilon + \varepsilon^{-2} = w(1) + w(1)$. The pairs $(N, N+1)$ in the table are the twins of Proposition 7.30. The doubling for even $x$ is plain in the near-solutions. The solutions of even $x$ exceed even the doubled model, and the twins do not explain this: near-solutions come in twin pairs at least as often as solutions do.

The calibration explains part of the excess. For even $x$ under $0^0 = 0$ the offsets $jx$ are not all in the class of 0: offsets with even $j$ carry 1.5 to 2.4 times as many near-solutions as offsets with odd $j$, much as the class of 0 is taken modulo $2(b-1)$ in odd integer bases. So for even $x$ the class of 0 is, at least in part, the class modulo $2x$, and the library's offsets $jx$ mixed two classes.

**Proposition 7.32** (A congruence modulo $2x$). PROVEN L26.6
For even $x$, every positive integer $N$ with digits $d_i$ in base $\varepsilon_x$ satisfies

$$\frac{N - \sum_i d_i}{x} \equiv A_2 + A_3 \pmod 2,$$

where $A_r$ is the sum of the digits at the positions $i \equiv r \pmod 4$.

The proof expands $\varepsilon^2 = 1 + x\varepsilon$ and uses $\varepsilon^4 \equiv 1$ modulo $2x$. The congruence depends on the arrangement of the digits, not only on their multiset, so it does not by itself give the factor between the two classes. Its cause is open. It is not an invariant of the multiset either: for $x = 4, 6, 8$ and $N < 8000$ the gcd of the differences within multiset groups is $2x$ only at length 4, and $x$ at every other length L35.3. So the class effect is a bias of the greedy expansions.

**Computation 7.33** (The Pisot excess, recalibrated). COMPUTATION L21.4 L26.6
For $x = 2$ to 8 under $0^0 = 0$, the 25 solutions of even $x$ against the near-solutions per offset in the class of 0 modulo $2x$:

| calibration | mean per offset | clustered P value |
|---|---|---|
| all offsets $jx > 0$ | 10.60 | 0.0016 |
| even $j$ only | 12.53 | 0.0095 |
| local: $D = \pm 2jx$, $j = 1, 2, 3$ | 14.83 | 0.049 |
| far: $D = \pm 2jx$, $j = 11, ..., 20$ | 9.70 | |

The defect spectrum peaks near $D = 0$ at the short lengths that hold most solutions, so far offsets miss the peak. Under $0^0 = 1$ and for odd $x$ the excesses that remain (11 against 6.8 and 7 against 2.2 with the local calibration, $P = 0.14$ and 0.03) rest on the tiny solutions of $x \le 5$.

So the Pisot bases follow the parity law, and with the right class and a local calibration their excess is a 5% event. The class factor modulo $2x$ is a new structure without an explanation (Section 8.2).

## 7.6 Other weights

The search engine accepts other weight tables. Two of the resulting experiments, forced residues and the convention $0^0 = m$ for other $m$, test the refined model directly and are described in Section 3.4. This section treats three families of weights that occur naturally: $x^{x+t}$, factorials, and the powers $x^d$ of narcissistic numbers.

### 7.6.1 Two facts behind the length law

The length shares 8, 55 and 23 percent of Heuristic 3.3 rest on two facts about $w(x) = x^x$:

- the top weight sits at a fixed height between powers of $b$: $(b-1)^{b-1} \approx b^{b-1}/e$, so $c$ copies of the top digit cross a power of $b$ at $c = e$;
- one digit value lower costs a factor of about $eb$: $(b-1)^{b-1}/(b-2)^{b-2} \approx e(b-1)$.

For $x^{x+t}$ with fixed $t$ both facts hold, with every length moved up by $t$, and $x^{x+t} \equiv x \pmod 2$. So everything carries over.

**Computation 7.34** (The weights $x^{x+t}$). COMPUTATION L14.6 L17.4

For $t = -1, 1, 2$ in bases 10 to 32, the totals per convention and base parity, the parity factors (about 2 under $0^0 = 0$, 1.2 under $0^0 = 1$), the shifted length law, the Poisson dispersion and the number of empty bases all match the refined model. For example, for $x^{x+1}$ under $0^0 = 0$ the even bases have 18 solutions against 14.3 and the odd bases 34 against 25.4.

### 7.6.2 Factorions

A *factorion* is a number equal to the sum of the factorials of its digits, with $0! = 1$. In base 10 the factorions are 1, 2, 145 and 40585 [4]. For $x!$ both facts above fail. One digit value lower costs only a factor $b-1$, and the height of $(b-1)!$ between powers of $b$, $\theta(b) = \mathrm{frac}(\log_b (b-1)!)$, moves with $b$.

**Heuristic 7.35** (Length law of factorions). HEURISTIC L14.6 L17.4

Let $L(b)$ be the number of digits of $(b-1)!$. For almost all large $b$, a factorion has $L(b) - k$ digits with probability about $\left(1 - e^{-\lambda}\right)e^{-k\lambda}$, $k = 0, 1, 2, \ldots$, where $\lambda \approx L(b)/b = 1 - 1/\ln b + o(1/\ln b)$. The shares tend to 63.2%, 23.3%, 8.6%, 3.2%, a geometric law with ratio $e^{-1}$. The expected total is still about 1 per base and convention. There is no parity factor.

The reason: the weights $(b-1-k)!$ fall by a factor of about $b$ per step, so the largest digit present fixes the power of $b$, and the number of its copies only fixes the leading digit. The parity factor disappears because $x!$ is even for $x \geq 2$, so $x! - x$ is odd for odd $x$, and the identity modulo 2 is a genuine condition. Convergence to the limit is slow, because $\lambda - 1 \approx -1/\ln b$.

**Computation 7.36** (Factorions in bases 10 to 41). COMPUTATION L14.6 L17.4

The census agrees with OEIS A193163 for bases 2 to 20 [5]. In bases 10 to 41 there are 37 factorions against 37.1 in the model, and 11 bases without one against 10.1. The long ones follow the geometric length law loosely (9 against 15.8 at the top length $L$); short ones, mostly with two digits, are four times too common. Short Munchausen numbers are about 3 times too common on generic strings against the exact refined model (Section 5.5.4). The odd bases from 11 to 41 have 15 factorions and the even bases from 10 to 40 have 22: no parity effect.

For a fixed number of digits, short factorions obey exact theorems (Section 7.7). For example, every $x \geq 3$ gives the 2-digit factorion $(1, x)$ in base $x! - x + 1$.

### 7.6.3 Narcissistic numbers

A $d$-digit *narcissistic number* equals the sum of the $d$-th powers of its digits; 153 is an example in base 10 [3]. The weights $x^d$ change with the length, and so do their residues modulo $b-1$. This makes the published lists of narcissistic numbers an outside test of the refined model, on data that nobody tuned for it.

Two cases are exact. Since $x^d \equiv x \pmod 2$, odd bases get the factor 2 at every length. And if $x^d \equiv x \pmod{b-1}$ for every digit, for example when $b-1$ is squarefree and $d \equiv 1 \pmod{\lambda(b-1)}$, the identity holds for every multiset and the factor is the full $b-1$.

**Computation 7.37** (Narcissistic numbers in bases 3 to 14). COMPUTATION L19.4

`fast search --power d` finds every narcissistic number with every length in bases 3 to 10, up to 62 digits in bases 11 to 13 and up to 50 digits in base 14. The counts agree with the OEIS lists. Against the models:

| cells | observed | basic model | refined model |
|---|---|---|---|
| all bases 3 to 14 | 760 | 252.5 | 671.3 |
| $d \geq 6$, $x^d \equiv x \pmod{b-1}$ for all $x$ | 207 | 24.3 | 208.5 |
| $d \geq 6$, odd $b$ | 354 | 100.4 | 349.7 |
| $d \geq 6$, even $b$ | 280 | 118.0 | 238.3 |
| $d \leq 5$ | 126 | 34.2 | 83.3 |

The refined model, which adds only the residues, follows the bases from 3 to 189 solutions, where the basic model is off by a factor of 3. In the forced cells the residue factor reaches $b-1=13$, and the model is right to 1%: base 14 with $d=49$ has 13 narcissistic numbers, the model expects 9.0 and the basic model 0.69. The deviations are the short lengths, as for $x^x$, and an excess of about 18% in even bases, mostly from bases 8 and 10.

Length 2 can be solved exactly, and the answer comes from the theory of sums of two squares (Section D.8). The 2-digit narcissistic numbers are also the 2-digit fixed points of the map "sum of the squares of the digits", known from happy numbers.

**Theorem 7.38** (Two-digit narcissistic numbers). PROVEN L30.3

For every $b \geq 2$, base $b$ has exactly $\tau(b^2+1)-2$ two-digit narcissistic numbers, where $\tau$ counts divisors.

*Proof.* A 2-digit narcissistic number satisfies $s_1 b + s_0 = s_1^2 + s_0^2$ with digits $1 \leq s_1 \leq b-1$ and $0 \leq s_0 \leq b-1$. Multiply by 4 and complete the squares:

$$(2s_1 - b)^2 + (2s_0 - 1)^2 = b^2 + 1.$$

So a solution gives a representation $b^2+1 = A^2 + B^2$ with $B = 2s_0 - 1$ odd. Conversely, take a representation with $B$ odd and positive. Then $A^2 \equiv b^2 \pmod 2$, so $s_1 = (A+b)/2$ is an integer. If $B=1$, then $A = \pm b$, and $s_1$ is 0 or $b$, not a digit. If $B \geq 3$, then $|A| \leq b-2$, so $s_1$ is a digit, and $B \leq b$ makes $s_0 = (B+1)/2$ a digit. So the solutions correspond to the representations with $B$ odd and positive, minus the two with $B=1$.

Count them with Theorem D.10. If $b$ is even, $b^2+1$ is odd, $r_2 = 4\tau(b^2+1)$, and exactly one of $A$, $B$ is odd. Swapping $A$ and $B$, and changing the sign of $B$, shows that a quarter of the representations have $B$ odd and positive: $\tau(b^2+1)$ of them. If $b$ is odd, $b^2+1 = 2m$ with $m$ odd, $r_2 = 4\tau(m)$, both $A$ and $B$ are odd, and half of the representations have $B > 0$: that is $2\tau(m) = \tau(b^2+1)$. In both cases subtract 2. □

Base 10 has none, because 101 is prime. That is why the base-10 list of narcissistic numbers jumps from 9 to 153 [3].

**Corollary 7.39** (Empty bases and Landau's problem). PROVEN L30.3

1. Every odd base $b \geq 3$ has at least two 2-digit narcissistic numbers.
2. An even base has none exactly when $b^2 + 1$ is prime. So "infinitely many bases have no 2-digit narcissistic number" is equivalent to Landau's open problem whether $n^2 + 1$ is prime for infinitely many $n$.
3. Infinitely many bases have at most two.
4. The mean number over the bases $b \leq X$ is $(3/\pi) \ln X + O(1)$.

*Proof.* (1) $b^2 + 1 = 2m$ with $m > 1$ odd, so $\tau(b^2 + 1) \geq 4$. (2) The count is 0 exactly when $\tau(b^2 + 1) = 2$, that is, when $b^2 + 1$ is prime. For odd $b > 1$ the number $b^2 + 1$ is even and larger than 2, so only even $b$ can give a prime. (3) By Iwaniec's theorem, infinitely many $b^2 + 1$ have at most two prime factors, and then $\tau(b^2 + 1) \leq 4$. (4) Hooley's theorem. Both theorems are in Section D.8. □

The parity effect of Heuristic 3.17 is a theorem here.

**Proposition 7.40** (The parity doubling is exact). PROVEN L30.3

The mean number of 2-digit narcissistic numbers is $(2/\pi) \ln X + O(1)$ over the even bases $b \leq X$, and $(4/\pi) \ln X + O(1)$ over the odd ones.

*Proof sketch.* Write $\tau(n^2 + 1)$ as twice the number of divisors $d \leq n$, and let $\nu(d)$ be the number of square roots of $-1$ modulo $d$. Summing over $n \leq X$ of one parity and exchanging the sums gives sums of $\nu(d)/d$ over $d$, up to $O(X)$. An odd $d$ divides $n^2 + 1$ equally often for even and odd $n$. An even $d$ is $2d'$ with $d'$ odd, because 4 never divides $n^2 + 1$, and it forces $n$ odd. The even $d$ add $X \sum_{d' \leq X/2} \nu(d')/d'$ to the odd $n$, which equals the share of the odd $d$ up to $O(X)$. So the odd bases get twice the sum of the even bases, up to $O(X)$, and Hooley's total fixes the constants. □

So the factor 2 of odd bases is the prime 2 dividing $b^2 + 1$, which happens exactly when $b$ is odd. In the refined model the same factor comes from $x^2 \equiv x \pmod 2$.

**Computation 7.41** (Two-digit narcissistic numbers up to $10^6$). COMPUTATION L30.3

Means of $\tau(b^2 + 1) - 2$ over the bases $b \leq X$:

| $X$ | even $b$ | $(2/\pi) \ln X$ | odd $b$ | $(4/\pi) \ln X$ | even $b$ with none |
|---|---|---|---|---|---|
| $10^2$ | 1.720 | 2.932 | 5.102 | 5.863 | 36.0% |
| $10^3$ | 3.136 | 4.398 | 7.896 | 8.795 | 22.2% |
| $10^4$ | 4.630 | 5.863 | 10.797 | 11.727 | 16.8% |
| $10^5$ | 6.088 | 7.329 | 13.735 | 14.659 | 13.3% |
| $10^6$ | 7.555 | 8.795 | 16.670 | 17.590 | 10.8% |

A brute force over all digit pairs agrees with Theorem 7.38 for every $b \leq 400$.

Both means grow at the predicted rates, with constant terms of about $-1.2$ and $-0.9$. The coincidence model misses this growth. It gives a 2-digit string a chance that does not depend on $b^2 + 1$, so its mean stays bounded, while the truth grows like $\ln b$. The missing factor is local. Modulo an odd prime $\ell$, the conic $A^2 + B^2 \equiv b^2 + 1$ has $\ell - \chi(\ell)$ points if $\ell \nmid b^2 + 1$. If $\ell \mid b^2 + 1$, it splits into the two lines $A \equiv \pm iB$ and has $2\ell - 1$ points. So each prime factor of $b^2 + 1$ roughly doubles the expected count. These are the local factors of the root heuristic (Section 5.5) in their simplest exact case. In bases 3 to 14 the 2-digit counts sum to 22, against 14.4 in the refined

model. That is part of the excess of short narcissistic numbers in Computation 7.37. No analogue of Theorem 7.38 is known for 3 digits, where the equation is a cubic surface (Q122).

### 7.6.4 The factorion shortfall in odd bases

One deviation in the factorion data resisted every explanation the library tried. In the odd bases 14 to 36, long factorions (at least $L(b) - 5$ digits) number 3 under $w(0) = 0$ and 4 under $w(0) = 1$, against about 11.9 and 11.8 in the model.

**Computation 7.42** (The shortfall against its near-solutions). COMPUTATION L17.4 L19.5 L23.1

Over the odd bases 21 to 35, the mean number of long near-factorions per offset, against $D = 0$:

| offsets | keep $n - F(n) \equiv 0$ modulo | $w(0) = 0$ | $w(0) = 1$ |
|---|---|---|---|
| $D = 0$ (the factorions) | everything | 1 | 2 |
| model | | 8.52 | 8.37 |
| $j(b-1)$ | $b-1$ | 8.33 | 8.37 |
| $jb(b-1)$ | $b-1$, $b$ | 8.17 | 8.13 |
| $jb^3(b-1)$ | $b-1$, $b^3$ | 7.80 | 8.20 |
| $jb^2$ lcm$(1, \ldots, b+1)$ | all moduli up to $b+1$, and $b^3$ | 8.23 | 7.83 |

Two independent engines and a brute force agree on the census. The fresh odd bases 37 to 47, searched after the shortfall was noticed, have 3 long factorions per convention against 6.2 expected ($P = 0.13$). Base 47 took 6292 seconds and has one 34-digit factorion under each convention.

**Heuristic 7.43** (An unexplained fluctuation). HEURISTIC L23.1

A mechanism would have to separate $D = 0$ from every small offset that keeps all residues up to $b + 1$ and the last three digits. By Proposition 1.12 the only residue that the digit multiset fixes is $n \bmod (b-1)$, and the offsets rule out the low digits and the forced top digits. Per offset family and convention the shortfall has $P = 0.006$ to $0.025$, counted with the clustering of near-solutions (Chapter 3). After the choice among about 16 groups it has $P \approx 0.13$, and with the fresh bases combined $P \approx 0.09$; an earlier, naive calibration gave about 0.05. The library counts it as an unexplained fluctuation.

The local arguments of Section 7.7 do not reach this question. They need one digit much larger than the others, and a long factorion has many digits of similar size.

## 7.7 Short factorions and Brocard's problem

For a fixed number of digits, a factorion is a root of one polynomial equation in the base, as in Chapter 5. For the weight $x!$ this equation has a feature that $x^x$ lacks: $x!$ is divisible by every prime up to $x$. This gives congruences that get stronger as the digit grows, and with them theorems about short factorions in all bases at once L33.1, L33.2. Where the congruences stop, a famous open problem takes over: Brocard's equation $n! + 1 = m^2$. In this section $w(x) = x!$ for $x \geq 1$, and $w(0) = 1$ (the usual $0! = 1$) or $w(0) = 0$.

### 7.7.1 Two digits

**Proposition 7.44** (Infinitely many 2-digit factorions). PROVEN L33.2

For every $x \geq 3$ the string $(1, x)$ is a factorion in base $b = x! - x + 1$, under either convention.

*Proof.* The value is $b + x = x! + 1 = w(1) + w(x)$, and $b > x$ for $x \geq 3$. □

So 2-digit factorions are infinitely many, as 2-digit Munchausen numbers are (Section 2.4). The interesting lengths start at 3.

### 7.7.2 Three digits: a square modulo every small prime

Take a 3-digit string with a large digit $x$ and two digits at most $B$. Multiply the equation by 4 times the leading digit and complete the square, as in Section 5.6. For a large last digit, $(p, q, x)$ in base $b$,

$$(2pb + q)^2 = 4p \cdot x! + K(x), \quad K(x) = q^2 + 4p(w(p) + w(q) - x).$$

The other positions give the same shape with another $K$:

- middle, $(p, x, r)$: $(2pb + x)^2 = 4p \cdot x! + K(x)$ with $K(x) = x^2 + 4pA$ and $A = w(p) + w(r) - r$;
- leading, $(x, q, r)$: $(2xb + q)^2 = 4x \cdot x! + K(x)$ with $K(x) = q^2 + 4xA$ and $A = w(q) + w(r) - r$.

In all three cases $K(x)$ is small: of size about $x$ or $x^2$. The dominant term is a multiple of $x!$.

**Lemma 7.45** (A square modulo every small prime). PROVEN L33.1

If $U^2 = c \cdot x! + K$ for integers $U$, $c$ and $K$, then $K$ is a square modulo every prime $\ell \le x$. In particular, if some odd prime $\ell \le x$ does not divide $K$ and $K$ is not a square modulo $\ell$, then there is no solution.

*Proof.* A prime $\ell \le x$ divides $x!$, so $U^2 \equiv K \pmod{\ell}$. □

The library uses a sharper form with $\ell$-adic numbers (Appendix C). It also covers primes that divide $K$ to a low power, and the prime 2 (Lemmas 33.1 and 33.2 of L33.1). For the weight $x^x$ nothing of the kind holds. The term $4px^x$ is divisible only by the primes of $p$ and $x$, so modulo most primes it runs through many residues as $x$ varies, and no finite set of congruences bounds the large digit (Section 5.2).

So everything depends on one question: if $K(x)$ is not a perfect square, is there a prime $\ell \le x$ modulo which it is not a square? Write $K = f^2 D$ with $D$ squarefree. A prime that is inert in the quadratic field $\mathbb{Q}\left(\sqrt{D}\right)$ and does not divide $f$ does the job. The least inert prime of a quadratic field is small: below 80 for every discriminant up to $4 \cdot 10^6$ in absolute value, and at most a small power of the discriminant in general (Section D.4). Since $|K(x)|$ grows only like $x$ or $x^2$, the least inert prime is far below $x$ once $x$ is large.

**Theorem 7.46** (Three-digit factorions with one large digit). PROVEN L33.1

Fix the convention, the position of the large digit $x$, and the two other digits.

1. If $x$ is the last or the middle digit, only finitely many $x$ give a factorion, and there is an explicit bound.
2. If $x$ is the leading digit, the same holds for the $x$ with $K(x)$ not a perfect square. If $K(x) = s^2$, the string is a factorion exactly when $4x \cdot x! + s^2 = U^2$ for an integer $U \equiv q \pmod{2x}$ with $(U - q)/(2x) > x$. There are infinitely many such $x$, about $\sqrt{X}$ of them up to $X$.

When the two other digits are at most $B$, the bounds hold uniformly in $B$. They grow like a power of $B!$, and like a polynomial in $B$ under the generalized Riemann hypothesis (Section B.4).

*Proof sketch.* First the $x$ with $K(x)$ a perfect square. In the last position $K(x) < 0$ once $x > w(p) + w(q) + q^2/(4p)$, so these $x$ are finitely many. In the middle, $x^2 + 4pA = t^2$ means $(t - x)(t + x) = 4pA$ with $A \ge 1$, which has finitely many solutions. In front, $q^2 + 4Ax = s^2$ holds for every $x = (s^2 - q^2)/(4A)$ that is an integer, infinitely often. (If $A = 0$, which happens only under $w(0) = 0$, then $4x \cdot x!$ would be a square. But by Bertrand's postulate a prime in $(x/2, x)$ divides it exactly once.)

For the other $x$, apply Lemma 7.45 with the least inert prime $\ell_0$ of $\mathbb{Q}\big(\sqrt{K(x)}\big)$. The bounds of Section D.4 give $\ell_0$ of size at most about $x^{0.9}$, below $x$ for large $x$. If $\ell_0$ divides $f$, the field has a smaller discriminant, and the $\ell$-adic form of the lemma excludes $x$ once $x$ is a little larger than $\ell_0 \log_{\ell_0}|K|$. A computer finds the point $X_0$ beyond which all of this holds. □

Table 7.3. The bound $X_0(B)$ beyond which a 3-digit factorion with one large digit and the other two at most $B$ needs $K(x)$ to be a perfect square, unconditionally and under GRH L33.1.

| $B$ | last | last, GRH | middle | middle, GRH | leading | leading, GRH |
|---|---|---|---|---|---|---|
| 5 | 60,640 | 241 | 1,340 | 1,340 | 137 | 137 |
| 10 | $1.28 \cdot 10^6$ | 2,158 | 17,560 | 2,158 | $4.42 \cdot 10^6$ | 4,626 |
| 20 | $3.0 \cdot 10^{12}$ | 23,380 | $3.8 \cdot 10^9$ | 23,380 | $2.1 \cdot 10^{16}$ | 37,650 |
| 100 | $3.9 \cdot 10^{83}$ | $6.5 \cdot 10^6$ | $4.6 \cdot 10^{72}$ | $6.5 \cdot 10^6$ | $3.4 \cdot 10^{130}$ | $7.1 \cdot 10^6$ |

Below the bound, the remaining $x$ can be checked one by one. This gives the complete list for small digits, up to the leading case.

**Computation 7.47** (Three-digit factorions with two digits at most 10). COMPUTATION L33.1

Let $s$ be a 3-digit factorion, under either convention, whose digits other than one digit $x$ are at most 10. Then $s$ is one of

- $(1,4,4)$ in base 5, $(1,4,5)$ in base 10, $(6,6,1)$ and $(6,6,2)$ in base 15, $(7,2,5)$ in base 27, $(1,6,6)$ in base 35, $(2,4,10)$ in base 1346,
- $(6,6,0)$ in base 15 under $w(0)=0$,

or $s=(x,q,r)$ with $x > 4{,}103{,}641$, $K(x)=s'^2$ a perfect square and $4x \cdot x! + s'^2$ a perfect square.

The check covers 679 cases (convention, position, two digits). Every $x \le 400$ is tested exactly. Every $x$ up to $X_0$ with $K(x)$ not a square is excluded by a prime below 300 or, for a few $x$, by a larger prime. The 211,668 values of $x$ with $K(x)$ a square and $x > 3000$ are sieved: $c \cdot x! + s^2$ must be a square modulo 40 primes above $x$, where $x!$ is a unit. None survives.

The solution $(2,4,10)$ in base 1346 is the square case of the last position: $K(10)=144=12^2$, and the equation reads $10!/2+9=1347^2$. Its digits are small, so earlier searches over all strings with digits up to 150 already found it. What is new is that no large third digit, in any base, adds a factorion, except possibly in the leading case.

### 7.7.3 Brocard's problem

In 1876 Brocard asked for all solutions of $n!+1=m^2$. The known ones are $n=4,5,7$, and Berndt and Galway found no other with $n \le 10^9$ [46]. Whether there are finitely many is open. The leading case of Theorem 7.46 is a problem of the same kind: $4x \cdot x! + s^2 = U^2$ asks for two divisors $U-s$ and $U+s$ of $4x \cdot x!$ that differ by only $2s$, about $4\sqrt{Ax}$. Brocard's equation asks for two divisors $m-1$ and $m+1$ of $n!$ that differ by 2.

The argument of Lemma 7.45 is due, for a fixed constant, to Dabrowski [47]: $x!+A=y^2$ has finitely many solutions when $A$ is not a square, by one fixed prime $\ell$ modulo which $A$ is not a square. Here $K(x)$ moves with $x$, so one needs a bound for the least inert prime instead of one fixed prime. When $A$ is a square, as in Brocard's equation, no prime helps: every $U^2 = c \cdot x! + s^2$ is solvable modulo every prime. This is where congruences stop.

Under the abc conjecture (Appendix F), Brocard's equation has only finitely many solutions (Overholt [36]), and so does the leading case.

**Corollary 7.48** (The Brocard case under abc). CONDITIONAL L33.1

Assume the abc conjecture. Then for fixed $q$ and $r$ only finitely many leading digits $x$ give a factorion $(x, q, r)$. Assume the explicit form $c < \mathrm{rad}(abc)^{7/4}$ (Section F.2). Then the leading case of Computation 7.47 does not occur, and the list there is complete.

*Proof sketch.* Put $N = 4x \cdot x!$ and $R = s^2$, and divide $N + R = U^2$ by $g = \gcd(N, R)$. The radical of the triple is at most $\mathrm{rad}(N) \cdot U \cdot R$. Here $\mathrm{rad}(N)$ is the product of the primes up to $x$, about $e^x$, while $\ln N$ is about $x \ln x - x$, and $U$ is about $\sqrt{N}$. So abc gives $(x \ln x - x)/2 \lesssim (1 + \varepsilon)x + O(\ln x)$, which bounds $x$. With the exponent $7/4$ the inequality reads $\ln N < 7 \ln 2 + 14 \ln(\mathrm{rad}(N)R)$. Since $\ln \mathrm{rad}(N) < 1.01624x$ [48], this fails for $x > 4{,}103{,}641$, where $x \ln x - x$ overtakes $14.23x$. □

So for three digits the factorion problem is solved up to a question of Brocard type, and that question is finite under abc. This is as far as current methods go.

Two large digits $u \le v$ and one small digit lead to the same kind of equation. Under GRH the congruence argument works with $M = 4d(u! + v!)$ when $v$ is at most about $u^3$, and with $M = 4d \cdot v!$ above that, where $d$ is the leading digit. What it leaves are equations $U^2 - t^2 = 4d(u! + v!)$ and $U^2 - t^2 = 4d \cdot v!$, again of Brocard type (Proposition 33.8 of L33.1). Without GRH a range between $v \approx u^{1.1}$ and $v \approx (u!)^{0.45}$ is out of reach.

### 7.7.4 Longer factorions

For $k \ge 4$ digits the completed square is no longer available, but the congruence argument still works with the polynomial of the string. Fix a prefix $t = (t_0, ..., t_{k-2})$ of digits at most $B$, with $t_0 \ge 1$, and let the last digit $x$ be large. With $P_t$ the value polynomial of the prefix (Section 1.3) and $F(t)$ the sum of its weights, the string $(t, x)$ is a factorion in base $b$ exactly when

$$G_x(b) = x!, \quad G_x(r) = rP_t(r) + x - F(t).$$

If $G_x$ has no root modulo some prime $\ell \le x$, there is no solution. By the Chebotarev density theorem (Appendix B), $G_x$ has no root modulo a positive share of the primes, unless it has a root modulo every prime. The question is whether such a prime lies below $x$.

**Theorem 7.49** (A large last digit under GRH). CONDITIONAL L33.2

Assume the generalized Riemann hypothesis. Let $k \ge 3$ and let the prefix $t$ have digits at most $B$. There is an effective constant $c_k$ with the following property. If $x \ge c_k(\ln x + B \ln B)^2(\ln \ln x + \ln B)^2$ and $G_x$ has no root modulo some prime, then $(t, x)$ is a factorion in no base. For $k \le 5$, $G_x$ has a root modulo every prime only when it has a rational root. Up to $X$ this happens for $O\big(X^{1/(k-1)}\big)$ values of $x$.

*Proof sketch.* The discriminant of $G_x$ has logarithm of order $\ln x + B \ln B$. Under GRH, Lagarias and Odlyzko [49] count the primes up to $X$ whose Frobenius element fixes no root, with an error of order $\sqrt{X}$ times logarithms. For $X$ of the stated size this count exceeds the number of primes that divide the discriminant. So some prime $\ell \le x$ has no root of $G_x$ and divides $x!$, a contradiction. For degree at most 4, Jordan's theorem (Theorem B.6) and a short check of products of two quadratics show that a fixed-point-free element exists unless $G_x$ has a linear factor. A rational root $-u/v$ has $v \mid t_0$, and then $x$ is a polynomial of degree $k - 1$ in $u$. □

For $k = 3$ the rational-root values of $x$ are finitely many, and Theorem 7.46 is the unconditional form. For $k = 4$ and 5 they are infinitely many. For such $x$ the polynomial factors as $G_x(r) = (vr + u)H(r)$, and the factorion equation becomes $(vb + u)H(b) = v \cdot x!$: two factors of a factorial in a fixed proportion, a problem of Brocard type again. So GRH alone does not prove that 4-digit factorions are finitely many. The abc conjecture does, for each fixed prefix (Theorem 2.40 with Proposition 2.38).

**Computation 7.50** (Four-digit factorions with a large last digit). COMPUTATION L33.2
For all prefixes $(p, q, s)$ with digits up to 6, both conventions and $2 \le x \le 1500$ (881,412 strings), there is no 4-digit factorion. Where a prime $\ell \le x$ without a root of $G_x$ exists, the least one is at most 101, and 2, 3 and 5 settle 65% of these strings. Every $x \ge 76$ without such a prime has $G_x$ with a rational root, as Theorem 7.49 says.

The GRH bound is far from tight here. The least prime without a root behaves like a random variable with a geometric tail, not like $(\ln x)^2$.

**Conjecture 7.51** (Finitely many factorions of each length). CONJECTURE L33.2
For each $k \ge 3$ there are only finitely many $k$-digit factorions.

The root heuristic (Section 5.5) gives a string with largest digit $M$ a chance of about $(M!)^{-(k-2)/(k-1)}$, and there are about $M^{k-1}$ such strings. The sum over $M$ converges very fast. Unlike for $x^x$, no identity family can make it diverge, because $x!$ is almost never a power (Proposition 2.38).

## 7.8 Function fields, and why the model misses there

The polynomial ring $\mathbb{F}_q[T]$ over a finite field is the classic analogue of the integers (Appendix J). It has bases, digits and a natural analogue of $x^x$. This world is special because its weights are additive. The model of Chapter 3 gets the count roughly right, but for the wrong reasons, and the way it fails shows exactly which assumption the model makes.

### 7.8.1 The setting

Let $q$ be a prime and $B \in \mathbb{F}_q[T]$ a polynomial of degree $e \ge 2$. The digits are the $q^e$ polynomials of degree below $e$, and every polynomial $n$ has exactly one expansion $n = \sum_i d_i B^i$, by repeated division with remainder. For a digit $d$ of degree $j$ put $|d| = q^j$, the analogue of the absolute value, and give $d$ the weight

$$w(d) = d^{|d|} = d^{q^{\deg d}},$$

with $w(0) = m$. A nonzero constant $c$ has $|c| = 1$ and $w(c) = c$, so the $q-1$ constants are the one-digit solutions.

**Proposition 7.52** (Linearity). PROVEN L21.1 LEAN
For a digit $d$ of degree $j$, $w(d) = d\big(T^{q^j}\big)$. Hence

$$F(n) = \sum_{j=0}^{e-1} D_j\big(T^{q^j}\big) + mz,$$

where $D_j$ is the sum of the digits of $n$ of degree exactly $j$ and $z$ is the number of zero digits.
Lean: `ffWeight_eq_expand, ffWeight_sum_of_degree`

*Proof.* In characteristic $p$ the map $y \mapsto y^q$ is additive, and it fixes the coefficients, which lie in $\mathbb{F}_q$. So $d^{q^j} = d\big(T^{q^j}\big)$, which is linear in $d$. □

So $n$ is a solution exactly when its class sums $(D_0, \dots, D_{e-1}, z)$ are a fixed point of an explicit map on a finite group, and there are at most $q^{e(e+1)/2}$ solutions, times $q$ if $m = 1$. The problem is finite and, for a fixed pattern of digit degrees, linear. Two kinds of structure follow at once.

**Proposition 7.53** (Families and self-paying digits). PROVEN L21.1 LEAN
Under $0^0 = 0$, if $n$ is a solution, so is $n + c$ for every constant $c$; solutions come in families of exactly $q$, which differ only in the last digit. A digit $d$ of degree $j \geq 1$ pays for itself at position $k$ when $d^{q^j-1} = B^k$.

Lean: `ffWeight_add_C (partial: the weight identity)`

*Proof.* $w(d + c) = w(d) + c$ for every digit $d$: if $\deg d = j \geq 1$, then $(d + c)^{q^j} = d^{q^j} + c$, and for constants and 0 both sides equal $d + c$. Adding $c$ to $n$ changes only the last digit, so $F$ grows by $c$ as well. □

The families replace the twins of Proposition 2.1. For example, in base $T^3$ over $\mathbb{F}_2$ the digit $T^2$ pays for itself at position 2, since $(T^2)^4 = T^8 = T^2(T^3)^2$.

### 7.8.2 Free residues at every prime

Every $n$ satisfies $n \equiv \sum_i d_i \pmod{B-1}$, so $B - 1$ is the identity modulus, and a solution satisfies $\sum_j \left(D_j\left(T^{q^j}\right) - D_j\right) + mz \equiv 0 \pmod{B-1}$ (L21.1, Proposition 21.5). In the integers the identity holds for free only modulo 2. Here it holds for free at every prime factor of $B - 1$.

**Theorem 7.54** (Free residues). PROVEN L21.1
Let $P$ be an irreducible factor of $B - 1$, and let Tr be the trace from $\mathbb{F}_q[T]/(P)$ to $\mathbb{F}_q$. Then for all class sums $D$,

$$\mathrm{Tr}\left(\sum_j \left(D_j\left(T^{q^j}\right) - D_j\right) \bmod P\right) = 0.$$

So under $0^0 = 0$ the identity holds for free on a subspace of codimension at least $\omega(B - 1)$, the number of distinct irreducible factors of $B - 1$.

*Proof.* Let $\theta$ be a root of $P$ and $\sigma$ the Frobenius of $\mathbb{F}_q(\theta)$. Then $D_j\left(\theta^{q^j}\right) = \sigma^j\left(D_j(\theta)\right)$, because $D_j$ has coefficients in $\mathbb{F}_q$, and $\mathrm{Tr}(\sigma^j y) = \mathrm{Tr}(y)$. Distinct factors give independent conditions by the Chinese remainder theorem. □

For a linear factor $T - a$ the statement reads $D\left(a^{q^j}\right) = D(a)$: every weight agrees with its digit at $a$, just as $x^x \equiv x \pmod 2$. Let $k(B)$ be the codimension of the image of the identity map $D \mapsto \sum_j \left(D_j\left(T^{q^j}\right) - D_j\right) \bmod (B-1)$. Then $k(B) \geq \omega(B - 1)$, and the refined model predicts about $q^{k(B)}$ times the basic count, the analogue of the factor 2.

### 7.8.3 The free residue dimension

Is $k(B)$ exactly the number of factors? In most cases yes, and the proof is linear algebra with Frobenius on $R = \mathbb{F}_q[T]/(B-1)$ (Section J.6). Let $t$ be the class of $T$ in $R$ and $\varphi(x) = x^q$. A class sum $D_j$ of degree at most $j$ is a polynomial in $t$ of degree at most $j$, and $D_j\left(T^{q^j}\right)$ reduces to $\varphi^j(D_j)$. So the image of the identity map is

$$J = \sum_{j=1}^{e-1} (\varphi^j - 1) W_j,$$

where $W_j$ is the span of $1, t, \ldots, t^j$, and $k(B) = e - \dim J$. Call an irreducible factor $P$ of $B - 1$ *good* if $\gcd(\deg P, e - 1) = 1$.

**Theorem 7.55** ($k(B)$ is the number of factors). PROVEN L29.2
1. If $B - 1$ has a good factor, then $k(B) = \omega(B - 1)$.

2. If $e-1$ is a prime power, then $k(B)=\omega(B-1)$ for every base $B$ of degree $e$. This covers $e=2,...,6,8,9,10,12,14,...$.
3. In every case $k(B)\le\omega(B-1)+1$, and $k(B)=\omega(B-1)$ when $B-1$ is not squarefree.

*Proof sketch.* (1) Since $\varphi^j-1=(\varphi-1)(1+\varphi+...+\varphi^{j-1})$, the space $J$ lies in the image $I$ of $\varphi-1$, which has codimension $\omega(B-1)$ by fact 1 of Section J.6. The term $j=e-1$ has $W_{e-1}=R$, and on a good factor its image is already the image of $\varphi-1$ (fact 4 with $S=\{e-1\}$). The bad factors have total degree at most $e-1$, so the spaces $W_j$ with $j\ge e-2$ reach every class modulo their product. There the two consecutive exponents $e-2$ and $e-1$ give the image of $\varphi-1$ again (fact 4). So $J$ contains the good part of $I$ and projects onto the bad part of $I$, and therefore $J=I$. (2) If every factor were bad, a prime $r$ dividing $e-1$ would divide every degree, hence $e=\sum a_i\deg P_i$, and also $e-1$. (3) Pass to linear functionals. A functional vanishes on $J$ when it is invariant under $\varphi^j$ on $W_j$ for each $j$. The difference $\lambda\circ\varphi-\lambda$ of such a functional is forced onto one line, which gives at most one dimension beyond the $\varphi$-invariant functionals. When $R$ has a nilradical, the conditions for that extra dimension make a nondegenerate pairing vanish on it, which is impossible. □

The library also excludes the extra dimension for squarefree $B-1$ when $q\le e-2$, and by hand for the remaining factor types of degree 7. So $k(B)=\omega(B-1)$ holds for every $e\le10$ and for every base of every census, by proof (L29.2).

**Computation 7.56** ($k(B)$ by computation). COMPUTATION L29.2
$k(B)=\omega(B-1)$ in all 110,654 monic bases of degree $e\ge3$ with $q=2$ and $e\le13$, $q=3$ and $e\le8$, $q=5$ and $e\le6$, $q=7$ and $e\le5$, $q=11$ and $e=4$, $q=13$ and $e\le4$; in 379,700 polynomials of the factor types with no good factor, up to $e=22$ for $q=2$; and in random samples of the first open type.

**Conjecture 7.57** ($k(B)=\omega(B-1)$). CONJECTURE L29.2
$k(B)=\omega(B-1)$ for every base $B$.

An exception needs a squarefree $B-1$ with no good factor, no factor of degree 2, and $q\ge e-1$. The first such case is $e=11$, with factors of degrees 5 and 6 and $q\ge11$ (Q111).

In degree 2 the count is a formula.

**Theorem 7.58** (Degree 2). PROVEN L21.1 L24.4
Let $e=2$. Under $0^0=0$, base $B$ has exactly $q(q-1)$ solutions if $B-1$ has two distinct roots in $\mathbb{F}_q$, and none otherwise. They are the numbers $c+\beta T^q$ with $c\in\mathbb{F}_q$ and $\beta\neq0$. Under $0^0=1$, for $q\ge5$, a base with two roots has no solution if $B=\lambda(T-c)^2$ (proven), and $q(q-1)$ solutions otherwise (computed for $q\le19$).

*Proof sketch.* Write $D_1=\alpha+\beta T$. Then $F=c+\beta T^q$ for a constant $c$. If $\beta=0$ the sum has one digit. Otherwise the identity gives $(B-1)\mid\beta(T^q-T)$, and $T^q-T$ is the product of all $T-a$, so $B-1$ needs two distinct roots. Conversely, writing $(T^q-T)/(B-1)$ in base $B$ gives the base-$B$ digits of $F$ explicitly, and each $\beta$ turns out to give exactly $q$ values of $c$. Under $0^0=1$ the square bases follow from Proposition 7.65 below: in base $\lambda(T-c)^2$ every solution of degree $q$ has $(q-3)/2$ or $(q-1)/2$ zero digits, which is not divisible by $p=q$. The count $q(q-1)$ in the other bases is a computation for $q\le19$; its proof is open. □

**Remark 7.59.** L21.1 L24.4

The library first stated that under $0^0 = 1$ and for $q \geq 5$ each base with two roots has $q(q-2)$ solutions. That is the average: the $q(q-1)/2$ square bases $\lambda(T-c)^2$ have none, and the other bases with two roots have $q(q-1)$. For $q \geq 5$ the census shows 0 or $q(q-1)$ in every such base. For $q = 3$ the average is 5 per base with two roots, and the per-base pattern is not established.

### 7.8.4 The census and the gap

**Computation 7.60** (Census for $e \geq 3$). COMPUTATION L21.1 L24.3

Mean number of solutions per base over all bases of degree $e$, against the refined model, which is an exact sum here. The last column is the share of $0^0 = 0$ solutions whose number of zero digits is divisible by $p$.

| $q$ | $e$ | bases | $0^0 = 0$ | refined | $0^0 = 1$ | refined | ratio | share $p \mid z$ |
|---|---|---|---|---|---|---|---|---|
| 2 | 5 | 32 | 3.750 | 4.071 | 2.125 | 3.183 | 0.67 | 0.57 |
| 2 | 7 | 128 | 5.031 | 5.026 | 3.000 | 3.768 | 0.80 | 0.60 |
| 3 | 3 | 54 | 8.222 | 6.479 | 4.111 | 5.477 | 0.75 | 0.49 |
| 3 | 4 | 162 | 14.444 | 9.858 | 3.778 | 8.025 | 0.47 | 0.26 |
| 3 | 5 | 486 | 14.296 | 12.924 | 5.111 | 10.315 | 0.50 | 0.36 |
| 5 | 3 | 500 | 34.400 | 26.230 | 11.840 | 23.338 | 0.51 | 0.34 |
| 5 | 4 | 2500 | 60.480 | 45.414 | 19.664 | 39.674 | 0.50 | 0.33 |
| 7 | 3 | 2058 | 76.286 | 64.810 | 2.286 | 59.438 | 0.038 | 0.030 |
| 11 | 3 | 13310 | 240.000 | 233.004 | 17.273 | 220.000 | 0.079 | 0.072 |
| 13 | 3 | 26364 | 407.077 | 379.118 | 8.923 | 360.982 | 0.025 | 0.022 |
| 17 | 3 | 78608 | 857.412 | 835.518 | 30.118 | 804.398 | 0.037 | 0.035 |

The column "ratio" is census over refined model under $0^0 = 1$.

Two facts stand out. Under $0^0 = 0$ the model is 3% to 47% too low for every odd $q$, about 5 standard deviations in all. Under $0^0 = 1$ it is far too high, and the gap grows with $q$: for $q = 13$ the true count is 2.5% of the prediction. The model also gets the dependence on $B$ right in one respect: each factor of $B-1$ multiplies the count by about $q$ (L21.1, computed). So the free residue factor is real. Something else fails.

### 7.8.5 Frobenius builds the digits of F in blocks

The model assumes that the digits of $F$ look like independent uniform digits. Here they do not, and the reason is that $F$ is a Frobenius image. Write $\Phi(n) = n(T^q) = n^q$.

**Lemma 7.61** (Frobenius blocks). PROVEN L24.2 LEAN

For a digit $d$ write $d(T^q) = \sum_r \beta_r(d) B^r$ in base $B$. Then for every $n = \sum_i d_i B^i$,

$$\Phi(n) = \sum_i \sum_r \beta_r(d_i) B^{qi+r}.$$

So $\Phi$ replaces each digit by its own block of $q$ digits, without carries. The digits $\beta_r$ with $r > R = \lfloor (e-1)q/e \rfloor$ are zero.

Lean: `expand_ringVal (partial: the identity before uniqueness of expansions)`

*Proof.* The coefficients of $n$, $d_i$ and $B$ lie in $\mathbb{F}_q$, so $n^q = \sum_i d_i^q (B^q)^i = \sum_i d_i (T^q) B^{qi}$. Substituting the expansion of $d_i(T^q)$ gives distinct exponents $qi + r$ with digits as coefficients, so this is the base-$B$ expansion by uniqueness. Since $\deg d(T^q) \leq (e-1)q$, the digits above position $R$ vanish. □

Since $F(x) = D_0 + mz + \Phi(D_1 + \Phi(D_2 + \ldots + \Phi(D_{e-1})))$, a Horner form, the whole digit string of $F$ is assembled from $e$ single digits by repeated blocks (L24.2, Proposition 24.2 and Corollary 24.3). Two consequences destroy the independence that the model assumes.

**Corollary 7.62** (Forced zeros and rank one). PROVEN L24.2

1. The digit of $F$ at a position $i \geq 1$ is zero whenever some base-$q$ digit of $i$ exceeds $R$. For $e = 3$ and $q > e$ this forces about a third of all positions to be zero.
2. For $\lfloor (e-2)q/e \rfloor < r \leq R$, all digits of $F$ at the positions $i \geq 1$ with $i \equiv r \pmod q$ are multiples of one fixed digit $\tau_r$ of the base, and they add up to $c(x)\tau_r$ for a single number $c(x) \in \mathbb{F}_q$.

In the model the digits at the positions of part 2 are about $L/3$ independent uniform digits, where $L$ is the length. In truth their total contribution to the class sums is one number times a fixed vector.

**Computation 7.63** (The digit law of $F$). COMPUTATION L24.4

For $q = 5$, $e = 3$, all bases and all full-length class sums, the share of each digit degree by position:

| position | zero | degree 0 | degree 1 | degree 2 |
|---|---|---|---|---|
| model | 0.008 | 0.032 | 0.160 | 0.800 |
| 0 | 0.020 | 0.078 | 0.102 | 0.800 |
| 1 | 0.066 | 0 | 0.134 | 0.800 |
| 2 | 0.200 | 0 | 0.128 | 0.672 |
| 3, 8, 13 | 0.200 | 0 | 0.800 | 0 |
| 4, 9, 14 | 1 | 0 | 0 | 0 |
| 16 (top) | 0 | 0 | 0 | 1 |

Over the cases $q \leq 13$, between 15% and 45% of the digits of $F$ are zero, where the model expects $(L-1)/q^e$, 4 to 700 times fewer.

### 7.8.6 The control that was not a control

The library first judged the gap a fluctuation, because near-solutions $n = F(n) + (B-1)g$ with constant or low-degree $g$ matched the census. That check was not independent.

**Lemma 7.64** (Near-solutions that are solutions in disguise). PROVEN L24.4

Let $n$ be a solution under $0^0 = 0$ whose last two digits $d_0$, $d_1$ have the same degree $j \geq 1$, and let $g$ be a digit with $\deg g < j$. Then $n + (B-1)g$ has the digits $d_0 - g$ and $d_1 + g$ in place of $d_0$, $d_1$, the same class sums, and so it is the near-solution $F(x) + (B-1)g$. This is a bijection between such solutions and such near-solutions. Constant offsets give a bijection whenever both digits have positive degree.

*Proof.* $(B-1)g$ has the digits $-g$ and $g$ at positions 0 and 1. The digits $d_0 - g$ and $d_1 + g$ keep degree $j$, and the class sum of degree $j$ changes by $-g + g = 0$. □

In the integers nothing like this happens. Adding $j(b-1)$ changes the two lowest digits by $-j$ and $+j$, and it changes $F$ by $w(d_0 - j) - w(d_0) + w(d_1 + j) - w(d_1) \neq 0$, because $x^x$ is not additive. So a small offset does not carry a solution to a near-solution one for one. Integer near-solutions do cluster in a weaker way: a rearrangement of the digits keeps $F$ and moves $n - F(n)$ by a multiple of $b-1$, and about two thirds of the near-solutions in bases 12 to 26 share their multiset with a near-solution at another offset (Chapter 3). That inflates the variance of the offset means by a factor 1.5 to 3.5 but not the means themselves: offsets with no transposition between them give the same mean within 4%. In $\mathbb{F}_q[T]$, by contrast, the mean itself was the census. Offsets that randomize more and more low digits bring the near-solution count down to the model: for $q = 5$, $e = 3$ it falls from the census value 34.4 per base to 20.1 when 13 low digits can change, against the model's 26.2.

### 7.8.7 The exact channel under $0^0 = 1$

**Proposition 7.65** (The conventions). PROVEN L24.4
Let $p$ be the characteristic. If $B-1$ has an irreducible factor whose degree is prime to $p$, then the solutions in base $B$ under $0^0 = 1$ are exactly the solutions under $0^0 = 0$ whose number of zero digits is divisible by $p$.

*Proof.* $F_1(n) = F_0(n) + z(n)$, and $z(n)$ is a constant of $\mathbb{F}_q$. Let $n$ be a solution under $0^0 = 1$ and $P$ such a factor. The trace argument of Theorem 7.54, applied to the identity under $0^0 = 1$, gives $z \deg P = 0$ in $\mathbb{F}_q$, so $p \mid z(n)$, and then $F_0(n) = F_1(n) = n$. The converse is immediate. □

The census confirms this in 124,257 bases. The refined model takes zeros to be rare, so it puts the share of solutions with $p \mid z$ at 0.72 to 0.95. The true share is the last column of the census table, 0.02 to 0.67, because most zeros of $F$ are forced (Corollary 7.62) and the rest come in batches of $q$. With $r_m$ the ratio of census to model under $0^0 = m$, the proposition gives the exact identity

$$r_1 = r_0 \cdot (\text{true share of } p \mid z)/(M_1/M_0),$$

where $M_1/M_0$ is close to the model's share. For $q = 5$, $e = 3$: $1.311 \cdot 0.344/0.890 = 0.507$, the observed ratio. So the whole gap under $0^0 = 1$ is the zero count.

### 7.8.8 The channel under $0^0 = 0$

Under $0^0 = 0$ zeros do not matter, and the totals are close, but the solutions are a different population from the one the model imagines. For $q = 13$, $e = 3$, 43% of the solutions have $D_1 = 0$, against 2.5% in the model, and only 1.6% have $D_0 \neq 0$, against 44%. Constants never occur above position 0, and the block structure forces the lower class sums.

**Proposition 7.66** (The generic cell). PROVEN L24.4
Let $0^0 = 0$, $q \geq e$ and $q \equiv 1 \pmod{e}$. Call a class sum $x$ generic if every digit of $F(x)$ that is neither a forced zero nor at a position $\equiv R \pmod{q}$ has degree $e-1$. Then a generic $x$ is a solution if and only if $D_j = 0$ for all $j < e-1$ and $\sigma'^{e-1}(D_{e-1}) = D_{e-1}$, where $\sigma'$ is the Frobenius map modulo $B-1$.

So on this sector the fixed-point system collapses to the kernel of $\sigma'^{e-1} - 1$ on $\mathbb{F}_q[T]/(B-1)$, of dimension $\kappa(B) = \sum \gcd(e-1, \deg P)$ over the irreducible factors $P$ of $B-1$ (for squarefree $B-1$). The model uses the free factor $q^{k(B)}$ with $k(B)$ the number of factors. The two differ exactly when some factor of $B-1$ has a degree that shares a divisor with $e-1$.

**Heuristic 7.67** (The source of the excess). HEURISTIC L24.4
The $0^0 = 0$ excess comes from the sector of class sums that the Frobenius blocks force. The solutions concentrate there, and where the identity restricted to that sector has a larger kernel, $q^\kappa$ instead of $q^k$, the model is low. Where $B-1$ splits completely, $\kappa = k$ and the model is right within 2%: for $q = 13$, $e = 3$ the bases with three roots have 2148.6 solutions against 2188.0, while the bases with a root and a quadratic factor have 242.0 against 169.7.

### 7.8.9 A model that keeps the blocks

The heuristic suggests a better model: keep the Frobenius blocks exactly and randomize only what they do not fix. By Lemma 7.61, $F(x)$ is assembled from $e$ generating digits $p_0, \ldots, p_{e-1}$, with $p_{e-1} = D_{e-1}$ and $p_j = \sigma(p_{j+1}) + D_j$, where $\sigma$ is a fixed map on digits (L24.2, Corollary 24.3). The digit $p_j$ fills the positions from $q^{j-1}$ to $q^j - 1$ through its block. So the class sums of $F(x)$ are the class of $p_0$ plus one fixed function $K_j(p_j)$ for each $j \geq 1$, and $x \mapsto (p_0, \ldots, p_{e-1})$ is a bijection.

**Definition 7.68** (Block model). L29.4
Keep the maps $K_j$, and replace each generating digit $p_j$ by a uniform digit of the same degree (zero stays zero). In the refined block model the random digits must also give the same value of $F$ modulo $B-1$ as the true ones.

The refined block model needs no $\kappa(B)$: the larger kernel of Proposition 7.66 is built in, because the generic sector keeps its degrees. A model solution also satisfies the identity modulo $B-1$, as a true solution does. The expected count is an exact finite sum, which takes seconds per base orbit.

**Computation 7.69** (The block model against the census). COMPUTATION L29.4
Mean number of solutions per base under $0^0 = 0$:

| $(q, e)$ | census | refined model | refined block model | Pearson, per orbit |
|---|---|---|---|---|
| $(5, 3)$ | 34.4 | 26.2 | 31.6 | 26.5 to 6.6 (25 orbits) |
| $(7, 3)$ | 76.3 | 64.8 | 73.5 | 48.5 to 13.8 (63) |
| $(5, 4)$ | 60.5 | 45.4 | 58.7 | 749 to 187 (160) |

The block model comes within 17% of the census in 16 of the 21 factor types of $B-1$, against 5 of 21 for the refined model. The irreducible quartic of $(5, 4)$ gets 8.8 against a census of 10.0; the refined model gives 4.9.

**Heuristic 7.70** (Keep the forcing in full). HEURISTIC L29.4
Keeping all $e$ generating digits and their degrees, and randomizing the rest, explains most of the $\mathbb{F}_q[T]$ gap: the total error drops from 18% to 33% down to 3% to 9%, and the misfit per orbit by a factor of 4. Keeping only the top digit is not enough. The types with a repeated factor of $B-1$ still miss: two double roots in $(5, 4)$ have 2.09 times the block model, and two types without solutions get 5 and 6 from the model.

For a repeated factor the ring $R$ has a nilradical, where the identity behaves differently, as in part 3 of Theorem 7.55 (Q115). The recipe "keep the forced digits, randomize the rest" is the same one that the forced top digits apply in the integers (Section 3.5). The next section applies it to the factorial system.

### 7.8.10 Short solutions for every degree

Call a solution short if $D_j$ is constant for every $j \geq 2$. Then $F = C + bT^q$. The degree-2 count of Theorem 7.58 extends to every degree.

**Theorem 7.71** (Short solutions). PROVEN L24.5
Let $0^0 = 0$, $e \geq 2$ and $q \geq e$. If base $B$ has a short solution, then $B-1$ has $e$ distinct roots in $\mathbb{F}_q$. If it does, base $B$ has either exactly $q(q-1)$ short solutions, the numbers $C + bT^q$ with $C \in \mathbb{F}_q$ and $b \neq 0$, or none, according to an explicit test on the base-$B$ digits of $(T^q - T)/(B-1)$. If $e \mid q-1$ and $B = \lambda T^e + \mu$ with $(1-\mu)/\lambda$ a nonzero $e$-th power, the test passes.

### 7.8.11 What the function field teaches

**Remark 7.72** (Which assumption fails). L24.4
The model of Chapter 3 assumes that the digits of the sum behave like independent uniform digits, subject only to the identity. In $\mathbb{F}_q[T]$:

- independence fails: the $L$ digits of $F$ come from $e$ single digits, and about $1/e$ of the positions carry multiples of one fixed digit;
- uniformity fails: whole positions are always zero or always of one degree;

- the identity modulo $B-1$ is exact, and the model uses it correctly; the error lies in which class sums the solutions have.

The root cause is that $F$ is additive.

In the integers $x \mapsto x^x$ is not additive, and nothing like the blocks exists. What remains is the forced top digits of Section 3.5: the largest digits of a multiset fix the leading two or three digits of its sum.

**Computation 7.73** (The integer digits of the sum). COMPUTATION L24.6
For 100,000 random $(b-1)$-digit strings per base, with a $(b-1)$-digit sum, in bases 11, 15, 21, 25, 31 and 41, $0^0 = 1$: the top two or three positions of the sum are almost never zero or zero far more often than $1/b$. Below them the rate of zero digits is $1/b$ within 3%. The forced top digits move the integer predictions by a few percent: the parity factor under $0^0 = 1$ by 2% to 4%, and the total by about 1%.

One place in the integer data leans the way the toy world does: the long solutions under $0^0 = 0$ in bases 12 to 41 have 29 zero digits against 42.5 expected, 2.1 standard deviations low. The forced top digits explain 2 to 4 of the missing 13.5. Whether the rest is chance is open (Section 8.2).

## 7.9 The factorial number system

Every $n \geq 1$ has exactly one expansion $n = \sum_{i=1}^{L} d_i i!$ with $0 \leq d_i \leq i$ and $d_L \geq 1$, so $L! \leq n < (L+1)!$. A solution is an $n = \sum_i w(d_i)$. This world has neither a length bound nor an identity.

**Proposition 7.74** (No invariant residue). PROVEN L21.5 LEAN
No modulus $M \geq 2$ makes $n \bmod M$ a function of the digit multiset on the numbers with 3 or more digits.
Lean: `factorial_no_invariant_residue`

*Proof.* $(1,0,1) = 3! + 1 = 7$ and $(1,1,0) = 3! + 2! = 8$ have the same multiset and differ by 1. □

**Proposition 7.75** (Self-paying digits). PROVEN L21.5 LEAN
A digit $a$ at position $i$ pays for itself, $a \cdot i! = a^a$, only for $(a,i) = (1,1)$ and $(2,2)$.
Lean: `factorial_selfPaying`

*Proof.* We need $a^{a-1} = i!$. For $i \geq 3$ take a prime $p \geq 3$ with $i/2 < p \leq i$: for $i = 3$ take $p = 3$, and for $i \geq 4$ Bertrand's postulate gives a prime in that range, which is at least 3. It divides $i!$ exactly once. Then $p \mid a$, so $p$ divides $a^{a-1}$ at least $a - 1 \geq p - 1 \geq 2$ times, a contradiction. For $i \leq 2$, $i!$ is 1 or 2. □

Under $0^0 = 0$ the solutions $1 = (1)$, $4 = (2,0)$ and $5 = (2,1)$ are the combinations of these two digits, and 4, 5 are twins. Under $0^0 = 1$ the solutions are 1, $2 = (1,0)$ and $5 = (2,1)$.

**Computation 7.76** (Census to 34!). COMPUTATION L21.5
For $n < 34!$, about $3 \cdot 10^{38}$, the only solutions are 1, 4, 5 under $0^0 = 0$ and 1, 2, 5 under $0^0 = 1$. The search fixes the counts from the largest digit value down, as in Section 4.1, and uses that a digit $u$ can only sit at a position $\geq u$.

The windows model predicts $\sum_{L=3}^{33} E_L = 4.5$ solutions per convention in this range, and about 5.7 more between 34! and 301!. Zero against 4.5 has probability 0.011. Near-solutions show that the model itself is wrong here: with offsets $D = 1, \ldots, 100$ there are none for $7 \leq L \leq 29$ except one

cluster at $L = 26$, and with offsets $D = 1000j$ the hits sit in 5 of 17 lengths, in clusters of 10 to 50 offsets.

**Heuristic 7.77** (Why the model fails). HEURISTIC L21.5

A digit $v$ can only sit at a position $\geq v$, and the large digits have $v$ close to $L$. So they sit in the few top positions. But the top digits of $F$ are fixed by the counts of the large digits, $F \approx c_v v^v + c_{v-1}(v-1)^{v-1}$, which take only a few values. At a given length either these values can place the digit $v$ in a top position or they cannot, for all multisets at once. The coincidences come all or nothing per length. In integer bases a large digit can sit at any position, mostly among the random low digits, and this effect is absent.

The model's mean may still be about right: the near-solutions at $D = 1000j$ total 218 and 263 against its 231. But their variance is far larger than Poisson.

### 7.9.1 A model with the top digits kept

The fix that works in $\mathbb{F}_q[T]$ also works here: keep the true top digits of $F$ and randomize the rest. For a multiset $c$ whose sum $F(c)$ lies in the window $[L!, (L+1)!)$, let $T$ be the top $K$ factorial digits of $F(c)$. Treat only the digits at positions 1 to $L-K$ as random. The chance that the digits of $F(c)$ form $c$ is then

$$\mathbb{1}(T \subset c) \cdot \frac{A(c-T, L-K)}{\prod_{i \leq L-K}(i+1)},$$

where $A(c, N)$ counts the ways to place a multiset of $N$ digits at the positions 1 to $N$ with each value $u$ only at a position $\geq u$. These placements form a Ferrers board, so $A$ is a product of binomial coefficients: place the values from the largest down, each into the free positions at or above it.

**Proposition 7.78** (The top-digit model). PROVEN L29.5

Let $n$ be uniform in the window, $W = [L!, (L+1)!)$. Summed over all multisets, the top-$K$ model expects

$$E_L(K) = E_n[\mathbb{1}(F(n) \in W) \cdot r(n)]$$

solutions of length $L$. Here $c(n)$ is the multiset of $n$, $T$ is the top $K$ digits of $F(n)$, and $r(n)$ is the chance that a random arrangement of $c(n)$ has the top digits $T$, divided by the chance that a uniform member of $W$ has them. $E_L(0)$ is the plain model and $E_L(L)$ is the exact number of solutions in $W$.

*Proof.* The plain model is the sum over multisets $c$ with $F(c) \in W$ of the number of arrangements of $c$ in $W$, divided by $|W|$. So a sum over multisets is an expectation over $n$, with weight $|W|$ over the number of arrangements of $c(n)$. The ratio of $A(c-T, L-K)$ to the number of arrangements is the chance that a random arrangement has top digits $T$, and $|W|$ over the product is the inverse of the second chance. For $K = L$ the indicator says that the digits of $F(c)$ are $c$. □

The adaptive version keeps the positions where the largest digit $v$ of $n$ can sit, $K = L - v + 1$: the forcing of Heuristic 7.77. Call a length $L$ open if this model expects more than 0.01 solutions.

**Computation 7.79** (Open lengths). COMPUTATION L29.5

By sampling, $3 \cdot 10^6$ strings per length under $0^0 = 0$ and $10^6$ under $0^0 = 1$, for $3 \leq L \leq 100$:

- for $L \leq 33$, under both conventions, the open lengths are 8, 10, 13, 14, 15, 18, 19, 20, 24, 26, 28, 29, 30 and 32: 14 of 31;
- for $34 \leq L \leq 100$, 11 to 14 lengths are open, among them 35 and 40;

- among the lengths 10 to 26, all 8 closed lengths have no near-solution at the offsets $D = 1000j$, where the plain model expects 81; all 5 lengths with near-solutions are open;
- the model expects 3.85 solutions below 34! under $0^0 = 0$ and 3.90 under $0^0 = 1$, against 4.6 and 4.8 for the plain model.

So about half the lengths are open, not 5 of 17. The model moves the expected solutions between lengths and hardly changes their total. Under a Poisson law, no solution below 34! then has probability $e^{-3.85} = 0.021$. But coincidences come in clusters: the offsets that carry at least one near-solution are only 0.24 to 0.39 times the model mean, the rest being repeats of one multiset. With this calibration the empty census has probability 0.22 to 0.39, an ordinary outcome (heuristic).

**Heuristic 7.80** (Beyond 34!). HEURISTIC L29.5

For $34 \le L \le 40$ the top-digit model expects about 0.29 solutions, so at least one with probability 0.07 to 0.10 after the cluster calibration. The open lengths there are 35 and 40. Up to $L = 100$ it expects about 2.5, at least one with probability 0.45 to 0.6. Beyond $L = 50$ these numbers are uncertain by a factor of 2.

A census to $L = 40$ needs integers wider than 128 bits and a few $10^8$ tree nodes per length; it has not been run. Like the plain model, the top-digit model expects finitely many solutions, but it gives nothing a proof could use. Whether the factorial system has finitely many solutions is open; no size argument and no residue are available (Section 8.2).

## 7.10 What is universal

Table 7.4 collects the features of all the worlds of this chapter.

Table 7.4. Features of the number systems and weights. "one per base" means about one coincidence per base and convention before the residue factor.

| world | one per base | identity modulus | free residue factor | twins | self-paying digits |
|---|---|---|---|---|---|
| integers, $x^x$ | yes | $b-1$ | 2 if $b$ odd | yes, $0^0 = 0$ | perfect-power bases |
| integers, $x^{x+t}$ | yes | $b-1$ | 2 if $b$ odd | yes | yes |
| factorions $x!$ | yes | $b-1$ | none | triples, $0^0 = 0$ | not studied |
| narcissistic $x^d$ | yes | $b-1$ | up to $b-1$, by cell | | |
| negative base $-b$ | yes | $b+1$ | 2 if $b$ odd | yes | yes |
| Gaussian $-n+i$ | yes | $(n+1)^2+1$ | 2 if $n$ even | | |
| Eisenstein $-n+\omega$ | yes | $n^2+3n+3$ | none (odd) | not studied | no |
| rational $p/q$ | yes | $p-q$ | 2 if $p$, $q$ odd | none for $q > 1$ | no |
| Pisot $\varepsilon_x$ | yes | $x$ | 2 if $x$ even | yes | no |
| $\mathbb{F}_q[T]$ | yes, 0.6 to 1.0 | $B-1$ | $q^{k(B)}$ | families of $q$ | yes |
| factorial | no: clustered | none | none | yes, at position 1 | only $(1,1)$, $(2,2)$ |

Three statements summarize the chapter.

**Heuristic 7.81** (Universal laws). HEURISTIC L14.6 L21

1. In every world where a digit value can appear at many positions, a base has about one coincidence per convention, the count is close to Poisson, and a base is empty with probability about $\exp(-E)$. The reason is that the length of the sum barely depends on the length of the string, so the expectations over all lengths add to about 1.
2. The free residue factor follows the identity modulus and the residues of the weights (Proposition 7.5). It is 2 exactly when the modulus is even and the weights agree with the digits modulo 2. In $\mathbb{F}_q[T]$ it is $q$ per prime of $B-1$.

3. The length shares are not universal. They depend on the height of the top weight between powers of the base and on the cost of one digit value less: $x^x$ and $x^{x+t}$ give 8/55/23%, even negative bases 8/78/12%, odd negative bases and factorions geometric laws with ratios $e^{-2}$ and $e^{-1}$.

The coincidence model itself is not universal. It needs the digits of the sum below the top to behave like random digits. That fails in the factorial system, where large digits are tied to the top positions, and in $\mathbb{F}_q[T]$, where additive weights build the sum from a few blocks. In the integers the only structure of this kind is the forced top digits, a correction of a few percent.

A pooled test puts the worlds side by side. Compare, in each world, the number of solutions with the mean count of near-solutions at offsets in the class of 0, which measures the real rate of the coincidence without the model.

**Computation 7.82** (A pooled test of the worlds). COMPUTATION L26.5
Over eight worlds (integer bases 12 to 32, factorions in odd and in even bases, negative bases $-10$ to $-27$, rational bases, Pisot bases, Eisenstein bases and 2-cycles), 284 solutions stand against 297 to 304 expected from the near-solutions. The worlds disagree more than chance allows: the heterogeneity statistic is 17.2 on 7 degrees of freedom ($P = 0.016$) with the Pisot offsets in the class modulo $2x$, and 23.3 ($P = 0.002$) with the original offsets. Two worlds drive it, the odd-base factorions (ratio 0.33) and the Pisot bases. Without these two, the other six fit one common ratio, 0.91, with $P = 0.60$ for heterogeneity.

So outside the two named worlds nothing suggests that the exact equation $n = F(n)$ differs from its near misses. Both exceptions were chosen after looking at the data: the factorion shortfall among about 16 groups, the Pisot excess as the largest positive deviation.

A last test asked whether one statistic could predict where the model fails. In every world, compare the low digits of $F(c)$, below its top three, with the digits that $c$ still needs, and divide by the same agreement for random low digits. The ratio is within 1.6% of 1 in the integers, base $-20$, the factorions and the factorial system, within 3.7% in the Pisot base $\varepsilon_6$, and 2.2% to 3.4% above 1 per digit value in $\mathbb{F}_5[T]$ (L26.8, computed). It does not track the model's error. It is near 1 in the factorial system, where the model fails completely, because that failure sits in the top digits, and it has the same sign in $\mathbb{F}_q[T]$ under both conventions, where the model's error changes sign. The failures of the model live in the top digits or in a single channel such as the zero count, and an average over the low digits dilutes them.

Chapter 8

# Open problems

This chapter collects what is not known. The first section gives the status of the main questions in one table. The second states the open problems, by theme, with the evidence for each and with what a solution would need. The questions are numbered as in the library's list, `docs/library/questions.md`, so that Q64 means question 64 there.

Several open problems run into the same barrier. A proof that some coincidence between powers never happens, for infinitely many cases at once, is beyond current methods. Section 5.6 explains the barrier for short solutions, and Section 4.7 for algorithms. It is worth keeping in mind when judging which problems are realistic.

## 8.1 Status

Table 8.1. The main questions and their status.

| question | status | answer | where |
|---|---|---|---|
| Does every base have finitely many solutions? | PROVEN | yes: at most $b$ digits | Proposition 1.3 |
| Finitely many $k$-digit solutions under $0^0 = 1$, $k \geq 3$? | PROVEN | no: binomial families give infinitely many | Corollary 2.22 |
| Are large short solutions family members? | CONJECTURE | positions and linear families classified; the rest needs one more coincidence; true under abc for one dominant digit | Conjecture 8.1 |
| 3-digit strings with an even large digit behind a non-square leading digit | PROVEN | finitely many for each pair of small digits; none with both at most 20 | Section 5.7 |
| 3-digit strings under abc | CONDITIONAL | one large digit: at most about twice the others unless a family member; two large digits $u \geq v$ open for $v$ above about $u/2$ | Theorem 5.54 |
| Which weights have identity families? | PROVEN | a sufficient condition for every weight; factorials never meet it; exact under abc for one dominant digit | Section 2.7.1 |
| 3-digit factorions with one large digit | PROVEN | finitely many, except a leading case of Brocard type; complete apart from it when the other digits are at most 10 | Theorem 7.46 |
| Finitely many $k$-digit factorions, $k \geq 3$? | CONJECTURE | 2 digits: infinitely many; GRH reduces a large last digit to Brocard-type equations; abc gives finiteness with one dominant digit | Conjecture 7.51 |
| Is $(1, 6, 46666)$ the next 3-digit solution? | OPEN | next family member (proven); none in bases 905 to $10^{3301}$ | Problem 8.3 |
| At most two 3-digit solutions per base? | OPEN | at most 30 outside a sparse set of bases (proven); at most two up to $10^{3301}$ | Problem 8.6 |
| Does every length occur? | PROVEN | yes, under both conventions | Section 2.6 |
| Is $N_0(b)$ bounded? | PROVEN | no | Proposition 2.45 |

| question | status | answer | where |
|---|---|---|---|
| Is $N_1(b)$ bounded? | OPEN | unknown; constructions give at most 2 in the bases searched | Problem 8.4 |
| Infinitely many bases without solutions? | OPEN | model: about 1 base in 8 | Problem 8.8 |
| A $b$-digit solution under $0^0 = 1$? | OPEN | none in bases 3 to 45; model expects some | Problem 8.9 |
| Bases with two 2-digit solutions under $0^0 = 1$ | PROVEN | only base 3 | Theorem 2.19 |
| $b(b+x) = x^x + 1$ only for $x = 3, 7$? | OPEN | proven for prime powers and $x \le 4 \cdot 10^9$; abc implies finitely many | Corollary 6.20 |
| $b(b+x) = x^n + 1$ with $n \ge 4$ | OPEN | only $(7, 7)$ known; finite under abc | Section 6.6 |
| Count of solutions per base | HEURISTIC | Poisson with mean about 1 per convention, doubled in odd bases under $0^0 = 0$ | Section 3.7 |
| Length shares in the basic model | PROVEN | 8.0%, 55.2%, 23.3%, … with error $O(1/b)$ | Theorem 3.7 |
| Is the expected count a trace? | PROVEN | yes: the count over lengths 2 to $b$ is $\operatorname{tr} M_b - E_1 = 1 + C/b$, $C = 1.1046$, one eigenvalue; the uniform sum over all lengths is computed, not proved | Section 3.2.2 |
| Do the parity factors follow from the characters? | PROVEN | the order-2 character gives 2 and $1 + e^{-2}$; the other characters are open | Section 3.3.1 |
| The 12% excess under $0^0 = 1$ | HEURISTIC | not a third correction: zero-free surplus in odd bases, one cell in even bases, about 2 standard deviations | Heuristic 3.35 |
| Are the P values of the library uniform? | COMPUTATION | no, through selection; the 12 groups named in advance look uniform | Section 3.6.3 |
| Are neighbouring bases independent? | COMPUTATION | no dependence found once the top digits are forced; the two conventions of one base correlate at 0.78 | Section 3.6.4 |
| Do the arrangements of a multiset spread over the residues? | PROVEN | yes for $q$ prime to $b(b-1)$, at a rate set by the length and the spread of the digits, not by the order of $b$ | Theorem 5.38 |
| Are the root heuristic and the per-base model one heuristic? | HEURISTIC | yes on totals: the root heuristic is the refined model averaged over the residue of the base; base by base only the refined model fits | Heuristic 5.42 |
| One invariant ideal for every number system? | PROVEN | yes: the ideal of the exchanges, $(\beta - 1)$ for geometric place values; the predicted modulus in all 67 systems tested | Proposition 7.4 |
| Past $\ln b / \ln \ln b$ by averaging over bases? | PROVEN | not for multiset counts: the gain is bounded; strings with few large digits are excluded up to length $(\ln b)^C$ | Proposition 5.64 |
| Near-solutions in every base? | PROVEN | yes: $\lvert n - F(n)\rvert < nb^{-t}$ with $t = \sqrt{b/e} - O(1)$ for every $b \ge 27$; $t$ of order $b$ open | Theorem 3.40 |
| Assumption R | OPEN | false for short lengths as first stated; proven by computation for bases 12 to 6000; not needed for the Poisson law in the limit | Problem 8.10 |
| Infinitely many bases without 2-digit narcissistic numbers? | OPEN | equivalent to Landau's problem on primes $n^2 + 1$ | Corollary 7.39 |
| $k(B) = \omega(B-1)$ in $\mathbb{F}_q[T]$? | OPEN | proven with a good factor, for $e \le 10$ and every census base; at most one more always | Problem 8.35 |
| Excess of $(b-1)$-digit solutions | HEURISTIC | resolved: chance; the totals fit | Section 3.11 |
| Empty bases $\equiv 3 \pmod 8$ | COMPUTATION | resolved: base 43 has a solution | Section 3.11 |

| question | status | answer | where |
|---|---|---|---|
| Excess of 5-digit solutions in bases 13 to 99 | HEURISTIC | unexplained fluctuation, $P \approx 0.015$ | Section 5.5 |
| Factorion shortfall in odd bases | HEURISTIC | unexplained fluctuation; $P \approx 0.13$ after the choice among 16 groups | Section 7.6 |
| Excess of solutions in Pisot bases | HEURISTIC | mostly calibration: a class modulo $2x$; $P \approx 0.05$ after recalibration | Problem 8.16 |
| Are near-solutions at different offsets independent? | PROVEN | no: rearrangements link them; means stay unbiased, variances grow 1.5 to 3.5 times | Section 3.5 |
| Do all worlds share one rate of solutions to near-solutions? | COMPUTATION | six of eight do (ratio 0.91); odd-base factorions and Pisot bases differ | Section 7.10 |
| A search faster than $1.9^b$ | HEURISTIC | exact: open, with constant factors such as the residue rule (1.1 to 1.3); with a 1 to 3% miss rate about $1.75^b$; tree searches need $1.638^b$ in the model | Section 4.4 |
| Can another order of the tree beat $1.638^b$? | PROVEN | no: every order, fixed or adaptive, in the model | Problem 8.18 |
| Certificates of emptiness | PROVEN | in an interval oracle model, at least the widest level of the exact tree, up to a small factor | Problem 8.19 |
| Constraint solvers | COMPUTATION | exact but grow 1.95 to 2.3 per base, worse than the tree | Problem 8.18 |
| Polynomial-time search | OPEN | as hard to settle as P versus NP | Section 4.7 |
| Arbitrary weights | PROVEN | NP-complete, counting #P-complete | Theorem 4.24 |
| Galois groups of $Q_s$ | PROVEN | $S_{k-1}$ generically; classified for $k = 4, 5$ | Section 5.4 |
| Integers with three bases under $0^0 = 1$ | OPEN | none known | Problem 8.5 |
| Lengths in base $-1 + i$ | PROVEN | exactly $\equiv 1, 4 \pmod 8$ | Theorem 7.22 |
| Finitely many solutions in the factorial system? | OPEN | only 1, 2, 4, 5 below 34!; a top-digit model leaves 14 of 31 lengths open and finds the empty census ordinary | Problem 8.34 |
| Why the model misses in $\mathbb{F}_q[T]$ | HEURISTIC | Frobenius blocks and the $0^0 = 1$ channel proven; a model that keeps the blocks closes most of the $0^0 = 0$ gap | Section 7.8 |

## 8.2 Problems

### 8.2.1 Structure under $0^0 = 1$

Under $0^0 = 0$ the zero digits cost nothing, and constructions are easy: self-paying digits combine freely, and every length and every count occurs. Under $0^0 = 1$ every zero costs 1, and the known structure is thin: the 2-digit families, two linear families of power identities, and a few chains of power identities. Section 2.6 describes the families.

**Conjecture 8.1** (Large short solutions are family members). CONJECTURE L25.1 L22.3
Under $0^0 = 1$ and for each $k \geq 3$, all but finitely many $k$-digit solutions are members of the two linear families: the binomial families of Theorem 2.21, and the family $(c^j, 0, ..., 0, x)$ with $x = c^{jc^j} + j - 1$. Moreover every $k$-digit solution in a base above the largest base of the complete list (904, 20, 48, 9 and 9 for $k = 3, ..., 7$) is such a member. (Q64)

Much of Q64 is now settled (L25.1). In every power identity $Q_{s(r)} = \lambda A(r)^j - W$, where $W$ is the weight of the large digits, the last digit is one of the large digits (Theorem 25.2, proven). With $A$ linear the members are the binomial families and a second family with $e = 0$, for example

$(8,0,0,16777218)$ in base $16777218^{5592406}/2$; it gives infinitely many solutions of every length $k \geq 4$ and the smallest known ones with 4, 5 and 6 digits (Theorems 25.4 and 25.5, proven). Nonlinear $A$, decomposable $P_s$, products, Aurifeuillian factors, several large digits and self-paying digits with a paying tail each need one more Diophantine coincidence (Proposition 25.6), and a census of all small prefixes finds no member of any of them. For 3 digits with two digits at most $B$ and the third at least $2B+4$ the classification is complete (Theorem 25.7). With an even large digit that does not lead, two digits at most 20 and either convention, the conjecture holds in every base (Corollary 5.51), by a 2-adic form of Baker's theorem and a sieve (Section 5.7). The evidence for the conjecture is the complete lists of Section 5.1, which hold no solution above these bases up to $10^{51}$, $10^{186}$, $10^{30.9}$ and $10^{11}$ for 3 to 6 digits, extended to $10^{3301}$ for 3 digits, and the root heuristic of Section 5.5, which leaves less than $10^{-74}$ expected generic 3-digit solutions beyond the list. Under the abc conjecture the first part holds for every length in the range of one dominant digit, a digit larger than a constant times the others (Theorem 2.40), and for 3 digits the constant is about 2 (Theorem 5.54). Under $0^0 = 0$ the conjecture fails: self-paying digits and swaps give further families.

**Problem 8.2** (Nonlinear identity families). OPEN L25.1
Is there an identity family with nonlinear $A$, that is, a parametrized $A$ for which $A(b) = x^{x/j}$ is solvable by structure?

Proposition 25.6 reduces this to values of polynomials at perfect powers. For each fixed $A$ the Schinzel-Tijdeman theorem bounds the exponent, so a family would need $A$ to vary.

**Problem 8.3** (The next 3-digit solution). OPEN L25.2 L22.3
Prove that $(1,6,46666)$ in base $46666^{23333}-3$, about $10^{108942}$, is the next 3-digit solution under $0^0 = 1$ after $(1,7,1)$ in base 904. (Q65)

What is known (L25.2):

- proven: $(1,6,46666)$ is the next member of the families after $(1,2,6)$, and the only 3-digit solution in its base (Theorem 25.10);
- computed: no 3-digit solution lies in a base from 905 to $10^{3301}$, and below $46666^{23333}-3$ every other 3-digit solution has two digits above 200 and one above 2000 (Theorem 25.11, a sieve with 108 primes over $1.36 \cdot 10^{10}$ strings);
- heuristic: the strings left open add an expected $10^{-3294}$ solutions;
- proven and computed, in every base: a solution with two digits at most 20 and an even large digit that does not lead is in the lists or a family member (Corollary 5.51).

The open part is the strings with two digits in $[201,46666]$ and one above 2000. They are generic Pell-type equations with two large digits (Section 5.6). A sieve over all of them has about $10^{14}$ strings; a sieve over pairs of large digits, with the third digit fixed by size, could cut this to about $10^{10}$ tests and close the gap.

**Problem 8.4** (Is $N_1(b)$ bounded?). OPEN L8.4 L15.5
Is there a constant $C$ such that every base has at most $C$ solutions under $0^0 = 1$?

Under $0^0 = 0$ the answer is no (Proposition 2.45). Under $0^0 = 1$ a self-paying digit replaced by 0 leaves a near-solution with $F(n) = n+1$, so a construction like Proposition 2.45 needs near-solutions of every small defect in one base. The chain-and-cycle construction of power identities (L15.5) gives at most 2 solutions in every base $r^t$ with $r \leq 40$ not a perfect power and $t \leq 300$, the range searched. The model says that the count is a Poisson variable with mean about 1, which has no bound, but a large count would be a chance event in a single base. A chain whose top lands

within the number of available self-paying digits of the zero count it needs would give three or more solutions; bases $r^t$ with $r$ in $\{2,3,5,6\}$ and $t$ up to $10^4$ with many divisors are the natural place to look (Q88). The same construction gives no infinite family of long solutions: chains of one step give only base 64 (Proposition 2.12), and a search up to $t = 10^5$ finds nothing new. Whether swaps, the 2-cycles of the construction, allow a family with a one-step chain is open (Q138).

**Problem 8.5** (Three bases). OPEN L15.6 L18.3
Is there an integer that is a solution under $0^0 = 1$ in three different bases? (Q39)

Under $0^0 = 0$ the number of bases is unbounded: $2^{2048} + 1$ has 26. Under $0^0 = 1$ an integer has at most two 2-digit representations, and every other representation of an integer $a^a + c^c$ with two 2-digit bases has a rigid shape (L18.3). None exists among the 2,117 integers with a 2-digit representation with digits up to 90.

**Problem 8.6** (Two 3-digit solutions per base). OPEN L25.3 L18.2
Does every base have at most two 3-digit solutions under $0^0 = 1$? (Q50)

Base 4 has two, $(1,3,1)$ and $(3,1,3)$, and every other base up to $10^{3301}$ has at most one (computed, Theorem 25.11). Partial results (L25.3, proven): the largest digits of the 3-digit solutions of one base take at most two consecutive values; every base has at most 30 three-digit solutions, except at most 13 bases near $M^{(M-1)/2}$ for each $M$, which have at most $2M + 24$ (Theorem 25.15); a base has at most one family member, and a base with a member has at most three 3-digit solutions (Propositions 25.16 and 25.17). The method of Theorem 2.19 fails at the shape $(M, u, v)$, where a base near $M^{(M-1)/2}$ gives an equation in two free digits. A proof that this shape has at most two solutions per base would give an absolute bound near 26, and with more care perhaps 2. See also Section 5.10.

**Problem 8.7** (Cores in perfect-power bases). OPEN L15.3
Under $0^0 = 0$, is $(1,3)$ in base 25 the only 2-digit core in a perfect-power base that is not a swap $(a, a^a)$? (Q37)

The abc conjecture settles the case where one of $a^a$, $c^c$ is much smaller than the other (Section 6.9). The case $a^a \approx c^c$ has four large terms, and abc does not apply.

### 8.2.2 The model

**Problem 8.8** (Infinitely many empty bases). OPEN L8.5 L9.7 L11.2
Are there infinitely many bases without a solution under either convention?

The model says yes: about 17% of even and 8% of odd bases near base 40 are empty, and 20% and 9% in the large-base limit. The data agree: bases 19, 27 and 35 are the empty ones up to 41 (base 43 has solutions), and each base's own near-solution spectrum predicts its chance of being empty. A proof would have to exclude a coincidence in infinitely many bases at once. Congruences remove only a fixed share of the candidates, and constructions only show that bases have solutions. The opposite statement, that every large base has a solution, is equally out of reach and false in the model. Counting cannot help beyond Theorem 5.60: past that length there are more digit strings than bases. The only residue the multiset fixes is the one modulo $b - 1$ (Proposition 1.12), which removes a share $1/(b-1)$ at best. So a proof would have to use the size and the digits together, as Theorem 5.46 does for 3 digits L32.2. The analogue for 2-digit narcissistic numbers shows how hard such questions are: there it is equivalent to Landau's problem whether $n^2 + 1$ is prime infinitely often (Corollary 7.39).

**Problem 8.9** ($b$-digit solutions under $0^0 = 1$). OPEN L7.1 L8.6 L11.2
Does some base $b$ have a solution with $b$ digits under $0^0 = 1$?

Under $0^0 = 0$ bases 15 and 16 have one. Under $0^0 = 1$ none exists in bases 3 to 45. The model expects a $b$-digit solution in about 8% of large bases, but it overestimates this length at small $b$ because the top digits of a $b$-digit sum are forced. The near-solution calibration expects 1.3 in bases 10 to 30, where 0 is observed ($P = 0.27$). So the evidence does not support the original conjecture that none exists, and a search with `--min-digits b` in bases near 50 is the natural test.

**Problem 8.10** (Assumption R). OPEN L13.2 L29.3
Prove that $\rho(b) < 1 - 3\ln b/b$ for every base $b > 6000$, where $\rho(b)$ is the largest normalized exponential sum of $x^x - x$ modulo $b-1$ over the digit ranges $[0, L)$ with $L \geq b/2$. (Q54, Q80, Q112, Q113, Q118, Q119, Q120)

This is the only gap in the refined version of Theorem 3.54, and it is the factor 1 for odd primes in Section 3.3. Theorem 3.56 reduces Assumption R to this one number per base, and a computation proves it for $12 \leq b \leq 6000$ (Corollary 3.58). The computed $\rho(b)$ tends to 0.7638, the 3-adic limit of Proposition 3.59, and Conjecture 3.60 predicts $\rho(b) \leq 0.77$ from base 100 on. Small prime factors of $b-1$ are handled by Gauss sums (Proposition 3.61). The hard case is a prime $p = b - 1$, where the digits see each residue of $x$ once. There the sums $\sum_{x<p} e(h(x^x - x)/p)$ show square-root cancellation in the data, but no bound below $p$ for a single $h$ is known. Over every odd prime power dividing $b-1$ for $b \leq 20,000$, the Poisson rate of the residues is at least 0.2143, and at least 0.2308 from base 10 on.

The Poisson law in the limit no longer needs Assumption R. Anti-concentration of the small digits gives it for every base, at the slow rate $\exp(-c \ln b/\ln\ln b)$ (Theorem 3.64). So what remains open is the rate $O(1/b)$. Two reformulations may help. For prime $b-1$ it suffices to count the collisions of $x^x - x$ modulo $p$, which the data put at the value of a random map (Conjecture 3.67, Q118). An inverse Littlewood-Offord theorem for steps drawn from a fixed set of residues would reduce Assumption R to a statement about $x^x - x$ alone (Q119). The dissociation argument might also use the digits near $b$, whose weights are known modulo a prime $b-1$ (Q120). The same argument should give the Poisson limit of a coupled model for factorions, which is not yet written out (Q123).

The length shares of the basic model, once listed here, are now a theorem (Theorem 3.7), and so is the trace form of the count (Section 3.2.2). The first-order correction $c_j/b$ is known in the basic model, and its sum is $C = 1.104636$ in closed form (Proposition 3.13). In the refined model it is not known (Q114).

**Problem 8.11** (Is the refined model complete?). OPEN L34

1. What carries the extra variance between perturbed worlds, 1.3 to 1.5 times the clustered sampling error? Short lengths, the fourth and later top digits, and the low digits of $F$ are candidates. (Q139)
2. Do fresh odd bases 27 to 41 repeat the surplus of zero-free near-solutions, 1.26 times the model? (Q140)
3. Prove $\operatorname{tr} M_b - E_1 = 1 + C/b + o(1/b)$ with a bound uniform in the length. (Q141)
4. Which property of $b-1$ makes the characters other than 0 and $q/2$ large, and do they vanish in the limit? (Q142, Conjecture 3.21)
5. Pre-register the prediction of every group of the ledger that extends to fresh bases. (Q143)

The answers decide how much of the remaining scatter is a property of $x^x$. So far the data need no correction beyond the residues and the top digits (Section 3.5.2).

**Problem 8.12** (Existence and averages). OPEN L35

1. Can near-solutions with $|n - F(n)| < nb^{-cb}$ be proved in every base? Theorem 3.40 stops at $t \approx \sqrt{b/e}$ because of the zone that the matched digits must avoid. (Q144)
2. Is there a closed form for the identity factor modulo $b-1$ at short lengths, in the small primes of $b-1$, that gives the short-solution rates of all weights at once? (Q145)
3. Prove that sums of $K$ weights $x^x$ with no small digit avoid the residue 0 modulo $b-1$ for all but a share $B^{-\theta}$ of the pairs of a multiset and a base of its window. That would extend Theorem 5.60 to $\alpha = 1 + \theta$. (Q146)
4. Sharpen Theorem 5.38 for small digits and large $q$, where $\gamma$ is close to 1 but the arrangements still spread. (Q147)
5. Combine Theorem 5.66 with the congruence modulo $b-1$ for strings with a long, dissociated small part. (Q148)

The first question asks for existence in the range where the model predicts it. The others ask whether the residues can push the counting arguments past their present limits.

**Problem 8.13** (The length tail). OPEN L10.5

Does the tail of the length law beyond $J = \ln b + 1$, which supports Conjecture 3.72, appear in many perturbed worlds? (Q56)

**Problem 8.14** (Second-order effects in the model). OPEN L22.4 L23.5 L24.6

Several small deviations from the refined model are unexplained:

1. The base-by-base calibration of Section 5.5 has not been applied to the 4-digit list or to the long solutions of bases 12 to 40. (Q66)
2. Factorions: what could separate $D = 0$ from every small offset that keeps all residues up to $b+1$ and modulo $b^3$? (Q67)
3. Why is base 19 low in its near-solutions under $0^0 = 0$, not only in its solutions? (Q68) With clustered near-solutions it is 2.9 standard errors below the model with forced top digits, one case of 42, so this is no longer a separate anomaly.
4. Why is the model about 12% low under $0^0 = 1$ in both parities of the base? (Q70) The excess survives the corrected variance at 3.0 standard deviations. It is not tied to the zeros, and the scatter between perturbed worlds brings it to about 2 (Heuristic 3.35).
5. Do the long solutions under $0^0 = 0$ really have fewer zero digits than the model, 29 against 43.1 in bases 12 to 43? (Q71) The near-solutions show no such law (Section 3.5.3).
6. In negative bases, offsets that keep the last digit give 6% to 23% fewer near-solutions, about 2.9 standard deviations with clustering. Which digits of $F \bmod b$ cause this?
7. Narcissistic numbers with $d \geq 6$ exceed the refined model by 18% in even bases, mostly bases 8 and 10. Do bases 16, 18 and 20 show the same? (Q87)

Each of these is a deviation of one to three standard deviations. The per-base scatter of about 20% that the library first reported is half an artefact: near-solutions at different offsets share multisets, and the sample variance of the offset means overstated their precision (L26.4). With the corrected variance and the top three or four digits of the sum forced, the scatter is noise under $0^0 = 0$ ($P = 0.09$ to $0.14$) and marginal under $0^0 = 1$ ($P = 0.03$). A model that also fixes the bottom digits and adds local factors for short lengths is the natural next step.

**Problem 8.15** (Near-solutions as a calibration). OPEN L26.1

Find a closed form of the design effect, the factor by which clusters of rearranged near-solutions inflate the variance of an offset mean, for factorions, negative bases and Pisot bases. Rerun

the rational and Eisenstein offsets with the strings kept, so that their design effect is measured rather than assumed.

For integer bases a heuristic based on swaps of the bottom four or five digits predicts the design effect within a few percent (L26.1). Offsets with no transposition between them, or with steps divisible by many small numbers, avoid the clusters and keep the mean.

**Problem 8.16** (The class modulo $2x$ in Pisot bases). OPEN L26.6
In the Pisot bases $\varepsilon_x$ with even $x$, under $0^0 = 0$, offsets $jx$ with even $j$ carry 1.5 to 2.4 times as many near-solutions as offsets with odd $j$. Which identity causes this? (Q74)

The congruence of Proposition 7.32 ties the parity of $(N - \sum d_i)/x$ to the arrangement of the digits, not to their multiset, so it does not give the factor by itself. With the offsets in the class modulo $2x$ and a local calibration, the Pisot excess has $P \approx 0.05$ (Section 7.5). The class is not an invariant of the multiset: the gcd of the differences within multiset groups is $2x$ only at length 4 (Section 7.5). So a bias of the greedy expansions must cause it.

**Problem 8.17** (Cycles of $n \mapsto F(n)$). OPEN L15.1 L23.3
The full census of cycles stops at base 17; base 18 was searched only for 2-cycles. Do the cycle counts keep following the class-lumped random-mapping model, and are the zero-free cycles, which count under both conventions, more common than the model's share?

The image of the map is far from that of a random mapping: 14% to 21% of the multisets of bases 10 to 16 are hit, against 31% to 38% for the weighted random mapping. Exact coincidences between sums and the forced top digits explain most of the gap (L26.7, computed). A model of the image that forces the top digits and counts the coincidences exactly is open.

### 8.2.3 Search and complexity

**Problem 8.18** (A fast exact search). OPEN L6.4 L8.1 L10.4 L27.2
Is there an exact search whose cost grows more slowly than about $1.9^b$? Is there one with polynomial cost?

The set of bases with a solution is a tally language in NP (Proposition 4.21). A proof that it is not in P would prove that NE differs from E, which is stronger than P different from NP; a proof that it is NP-hard would prove P = NP (Section 4.7). So neither direction is provable with current methods. Within the model, every tree search that walks the digits from the top needs about $1.638^b$ nodes (Theorem 4.11), and a quantum search could reach about $1.41^b$. A large classical gain needs a new source of information about the low digits of $n$.

Several routes are now closed (L27, L28):

- the entropy bound holds for every order of revealing digit counts, fixed or chosen node by node, so the top-down order is optimal among trees (Theorem 27.2, proven in the model); bottom-up and random orders peak near $3.6^b$;
- in bases $2^k$ the bottom digits of the sum are fixed only once the counts of all odd digits are known, so a check of bottom digits prunes nothing in a top-down tree, and orders that reveal the odd counts early lose (Proposition 27.8, Corollary 27.9);
- the constraint solvers CP-SAT and CaDiCaL are exact but grow 1.95 to 2.3 per base, as fast as the tree or faster, at $10^4$ to $10^5$ times its cost (computed, bases 8 to 20);
- the exact search gains constant factors only: the forcing loop, run to its fixed point, saves 1.46 times the nodes, and the interval it reaches is the exact hull of the possible sums (Lemma 4.3);

the residue rule saves 1.1 to 1.3 times more (Theorem 4.4); the other count and residue results of the library are implied by the interval test or cost more than they save (L31.4);
- the inexact search gains a constant factor: a corrected law for the z-score rule visits 1.1 to 1.3 times fewer nodes at the same miss rate, and a GPU runs it 2.6 to 2.9 times faster than 8 CPU threads.

**Problem 8.19** (Certificates of emptiness). OPEN L27.3
In an oracle model where a query reveals the interval of the sum for a partial assignment of digit counts, every certificate that a base has no solution needs at least $N_k$ queries for every level $k$, a number at least 0.86 times the widest level of the exact tree in bases 5 to 21 (Theorem 27.5, proven). Prove that this share stays bounded below as $b$ grows, which would give a lower bound $2^{b-O(\log b)}$ in this model. What can an oracle that also reveals low digits do?

**Problem 8.20** (Hardness with long solutions). OPEN L13.1
Is the problem with arbitrary weights still NP-complete on weight tables whose solutions must have at least $b - C$ digits, or at least $b/2$? (Q55)

The hard instances of Theorem 4.24 have solutions with at most $b/2$ digits, while real solutions have about $b - 1$ digits.

**Problem 8.21** (Powers as weights). OPEN L13.1 L18.6
With weights $x^{e(x)}$ and exponents in unary, is the problem NP-hard? (Q40)

The reduction of Theorem 4.24 needs digits whose weights have a prescribed base-$b$ expansion. With pure powers this forces exponents of size $\exp\left(c\sqrt{k/\log k}\right)$ for $k$ items, by linear forms in logarithms (L18.6). A reduction would have to use long expansions, for example the explicit digits of $(b-c)^e$.

**Problem 8.22** (Two search projects). OPEN L6 L19.1
1. Compute the near-solution spectrum in one pass: widen the interval of the search tree by $W$ and check every offset $|D| \le W$ at the leaves. (Q25)
2. Find where the tree search and the string sieve cross over for lengths 8 to 12, and push those lengths to bases $10^4$ to $10^5$. (Q29) A first step is done: the 8-digit list is complete for bases up to 23,364 and the 7-digit list up to $10^{7.2}$ (L19.1).

**Problem 8.23** (Constant factors in the exact search). OPEN L31
Is there a cheap exact test for nodes with two free slots, which must combine the residue with the interval (a 2-sum problem over the digits below $v$)? Can a mask of the digits still available make the last-digit check pay? Is the law of the free slots over the exact tree a fixed geometric law? (Q124, Q127, Q128)

The residue rule removes a share of the nodes with one free slot, 43% of the tree. With two free slots the residue alone reaches almost every class, but the two digits must also put the sum in the interval. A test that uses both could act on another quarter of the nodes. The answer changes only the constant, not the growth rate.

### 8.2.4 The Munchausen polynomial

**Problem 8.24** (Reducibility comes from decompositions). OPEN L16.3
For every $k$, is there a bound $F(s) \le C_k M^{c_k}$, with $M$ the largest digit, beyond which every factorization of $Q_s$ without a linear factor comes from a decomposition $P_s = G(h(r))$? (Q33)

The statement is a theorem for $k = 5$ (Section 5.4). For $k = 6$ the same elimination gives a curve of genus 1, and the integrality step becomes a question of approximating $Z\cos(2\pi/5)$ by rationals with bounded denominator.

**Problem 8.25** (Which moduli coverings need). OPEN L15.4 L20.2
Does every covering congruence that excludes a family of digit strings in every base use a modulus $q$ on which $x^x - x$ misses residues, that is a power of 2, 3 or 5? (Q38)

Every covering found so far uses such a modulus, sometimes together with others: the covering of $(2, 0, 1, 0, x)$ under $0^0 = 1$ uses 8, 11 and 25.

### 8.2.5 Exponential Diophantine equations

**Problem 8.26** ($b(b + x) = x^x + 1$ for composite $x$). OPEN L16.2 L18.1 L20.5
Prove that $b(b + x) = x^x + 1$ has no solution with $x > 7$.

What is left is described in Corollary 6.20: $x > 4 \cdot 10^9$ odd, composite, not a prime power, with a mixed ideal whose class has order $d \mid x$ and $x/d < 3.3 \cdot 10^4 \ln x$. Class orders with $d \mid x$ do occur, from $x = 15$ on, in computations that assume the generalized Riemann hypothesis, so the class group alone cannot close the case (L20.5). The weak abc inequality $c < K\mathrm{rad}(abc)^{2-\delta}$ implies that there are finitely many solutions; an effective form of it, together with the sieve, would finish the proof (Appendix F).

**Problem 8.27** (Three digits when the field changes). OPEN L32.1
Find a bound for the solutions of $U^2 - nV^2 = R$, with $V$ a power such as $x^{x/2}$, that is uniform over the fields $\mathbb{Q}(\sqrt{px})$ with $x$ odd. Close the other square classes $x = \lambda t^2$ with $\lambda > 1$, and carry the method to 4 digits with a binomial prefix. (Q129, Q130, Q131)

This is the last row of Table 5.6. In a fixed field, Baker's method bounds the large digit (Theorem 5.49, Proposition 5.52). When the field changes with $x$, its regulator can be as large as $\sqrt{x}\ln x$, and the known bounds are too weak by a power of $\ln x$. A result on $y^2 = ax^n + c$ that is uniform in $n$ would be the natural tool. Under abc every regime is closed (Theorem 5.54).

**Problem 8.28** (Two comparable large digits). OPEN L33.4
Under abc, a 3-digit string with two large digits $u \ge v$ and $R \ne 0$ is bounded unless $v$ is at least about $u/2$. Is there a conditional statement that covers the comparable range, for the equation $U^2 = 4\ell(u^u + v^v) + R$ with four large terms? (Q136)

The $n$-term generalization of abc has exponent 3 for four terms [28], too weak here, and $u^u + v^v$ is not a perfect power. This is the 3-digit form of the case $a^a \approx c^c$ that Section 6.9 leaves open for 2 digits.

**Problem 8.29** (The leading Brocard case of 3-digit factorions). OPEN L33.1 L33.2
Is $4x \cdot x! + s^2 = U^2$, with $s^2 = q^2 + 4Ax$ as in Theorem 7.46, solvable for any $x > 10$? Does the reduction for two large digits leave any solution? For $k \geq 6$, are there prefixes for which $G_x$ has a root modulo every prime without a rational root, for infinitely many $x$? (Q133, Q135, Q137)

The sieve of Computation 7.47 reaches $x = 4{,}103{,}641$, the bound that the explicit abc inequality gives for digits up to 10. Pushing it to $10^9$, as Berndt and Galway did for Brocard's equation [46], would cover the explicit bound for digits up to about $10^6$. An explicit Burgess bound for imaginary quadratic characters would cut the unconditional bound $X_0(B)$ of Table 7.3 from a power of $B!$ to about $(B!)^{1/4}$ (Q134).

**Problem 8.30** (The exponent $n = 5$). OPEN L18.1
Find all integer points on the genus 2 curve $b^2 + xb = x^5 + 1$.

No solution has $x \leq 10^{10}$. The Jacobian has analytic rank 3, so its Mordell-Weil rank is probably 3, larger than the genus. If so, Chabauty's method does not apply (Appendix G). The problem is decidable with Baker's method for hyperelliptic curves and the Mordell-Weil sieve, which needs a 2-descent and a saturated basis of the Mordell-Weil group. More generally, is the set of $(x, n)$ with $n \geq 4$ finite? Only $(7, 7)$ is known.

**Problem 8.31** (The even case). OPEN L18.1 L20.3
Solve $b(b + x) = x^n + 1$ for $x \equiv 0 \pmod 8$ with an odd prime factor and odd $n \geq 5$.

Here the 2-part of the ideal has exponent $an - 2$, not a multiple of $n$, and the linear form in logarithms has three terms. Matveev's theorem bounds $n$ in terms of $x$ and a class order, but nothing bounds $x$ in terms of $n$ except in one pure configuration. No solution has $x \leq 10^4$ (all $n$) or $x \leq 10^8$ ($n \leq 15$).

**Problem 8.32** (The families $(1, x, 2)$ and $(1, x, 3)$). OPEN L20.4
Prove that $(1, x, 2)$ is never a solution under $0^0 = 1$, that is, that $b(b + x) = x^x + 3$ has no solution; and finish $(1, x, 3)$ for the $x$ that are not prime powers.

No solution exists for $x \leq 10^9$ in either family. For $(1, x, 3)$ the method of Section 6.5 works for prime powers, because the ideal of norm $x$ has an explicit generator. For $(1, x, 2)$ no such generator is known, and the regulator of the quadratic order is not bounded by $\log x$, so even prime $x$ remains open.

**Problem 8.33** (Short solutions as mean-value problems). OPEN L30.2 L30.3
What is the asymptotic of the mean density of the fixed points of $x \mapsto x^x$ modulo $a$, averaged over $a \leq Y$? It governs the share of 2-digit strings that are solutions. Is there an exact count, or an average order, for 3-digit narcissistic numbers, $s_2 b^2 + s_1 b + s_0 = s_2^3 + s_1^3 + s_0^3$? (Q121, Q122)

The data suggest that the mean density falls like $Y^{-0.44}$ for $Y \leq 300$ (Computation 2.18). The density is not multiplicative in $a$, because the exponent ties the primes of $a$ together. For 2-digit narcissistic numbers the count is $\tau(b^2 + 1) - 2$ (Theorem 7.38). For 3 digits the equation is a cubic surface for each $b$, and the model predicts a bounded mean, but the local factors that give $\tau(b^2 + 1)$ at length 2 suggest that they matter again.

### 8.2.6 Other number systems

**Problem 8.34** (The factorial number system). OPEN L21.5
Does the factorial number system have finitely many solutions?

Only 1, 4, 5 under $0^0 = 0$ and 1, 2, 5 under $0^0 = 1$ exist below 34!. No length bound and no residue are available, and the windows model fails there (Section 7.9). A model that keeps the true top digits of the sum explains the failure: it leaves 14 of the 31 lengths up to 33 open, closes every length without near-solutions, and makes the empty census an ordinary outcome (Computation 7.79). It expects about 0.3 solutions for $34 \le L \le 40$ and 2.5 up to $L = 100$ (Heuristic 7.80). A census to $L = 40$ with integers wider than 128 bits is the next test. Why $L = 13$ has the largest model mean but few near-solutions (Q116), and how to sample the model beyond $L = 50$ (Q117), are open.

**Problem 8.35** (A formula for $\mathbb{F}_q[T]$). OPEN L21.1 L24.5
Find a formula, or a good heuristic, for the average number of solutions over all bases of degree $e \ge 3$ in $\mathbb{F}_q[T]$. Is $k(B) = \omega(B-1)$ always (Conjecture 7.57)? Which values of the leading coefficient pass the test of Theorem 7.71 for $e = 3$ and $q \equiv 2 \pmod 3$? Prove that $T^q$ has no zero digit in a non-square base of degree 2 with two roots of $B - 1$. (Q111, Q115)

Theorem 7.55 proves $k(B) = \omega(B-1)$ whenever $B-1$ has a factor of degree prime to $e-1$, and for every $e \le 10$; the first open case is $e = 11$ with factors of degrees 5 and 6. The block model of Computation 7.69, which keeps the Frobenius blocks and randomizes the rest, predicts the census within 3% to 9% in total. It still misses the bases where $B-1$ has a repeated factor, and a formula would have to explain those.

**Problem 8.36** (Length runs in Gaussian bases). OPEN L14.5 L23.4
Prove the runs of lengths of positive integers in base $-n+i$ for $n \ge 2$. (Q69)

For $n = 1$ the proof uses $\beta^4 = -4$ (Theorem 7.22). For $n \ge 2$ no power of $\beta$ is real.

**Problem 8.37** (Out-of-reach censuses). OPEN L14.5 L21.2 L21.3
Complete the census of base $-4+i$ past 22 digits, of base $-3+\omega$ at 27 and 28 digits and of $-4+\omega$, and of the rational bases 11/8, 11/10 and $p \ge 12$. Each needs a better pruning rule: in Gaussian and Eisenstein bases the numbers with fixed top digits form a fractal tile, not an interval. (Q89)

For base $-4+i$ a disk around each tile could serve as the prefix test. The model puts 83% of its expectation at 34 digits, so such a test would complete the Gaussian census one base further.

**Problem 8.38** (Base $-3+i$). OPEN L14.5
Base $-3+i$ has four solutions under $0^0 = 1$ against 1.11 expected, three of them with 19 digits ($P = 0.026$). Is this chance?

### 8.2.7 Lengths without solutions

The binomial families give solutions of every length, so no length is missing in all bases. The question that survives concerns bases near the length.

**Conjecture 8.39** (Missing lengths near the base). CONJECTURE L3.4 L11.4
Under $0^0 = 1$, every solution with 9, 18, 21, 22, 24, 26 or 28 digits belongs to an algebraic family such as the binomial families of Theorem 2.21, and so lies in an astronomically large base.

No base up to 100,000 has a 9-digit solution, and no base up to 10,000 has one with the other lengths (computed). The root heuristic puts the expected number of 9-digit solutions in larger bases, outside algebraic families, at $10^{-29}$. In bases up to 127, lengths 31 to 45 have not been surveyed against the model's prediction that about 37% of long lengths are missing (Q18).

## 8.3 Connections to other fields

Library chapter 30 reads the problem against other fields L30.4. Three transfers have proofs and appear in the earlier chapters: anti-concentration gives the Poisson law of the refined model in the limit (Section 3.7.3), the self-power map $x \mapsto x^x$ modulo a prime describes 2-digit solutions (Proposition 2.16), and sums of two squares count 2-digit narcissistic numbers (Theorem 7.38). The other connections are analogies so far. Each could become a route to one of the problems above.

- Random mappings. The map $x \mapsto x^x$ modulo $p$ has more fixed points and shorter cycles than a random map [8], and $n \mapsto F(n)$ has a smaller image than a random map (Section 3.10). In both cases the map factors through a coarser object, the order of $x$ or the digit multiset, which a random-mapping model must keep. After the shift to $x^x - x$ the collisions are those of a random map (Computation 3.66).
- Covering systems. The coverings of Section 5.3 exclude digit strings in every base by congruences modulo powers of 2 and 3, as Sierpinski excluded primes of the form $k2^n + 1$. The modern theory of coverings, such as Hough's solution of the minimum modulus problem [50], is about distinct moduli and does not apply.
- Knapsacks and lattices. A $k$-digit solution is a choice of one digit per position with balances that sum to 0 (Section 1.5), a multiple-choice subset sum of density near 1. Lattice attacks need density well below 1 [14], and generic knapsack algorithms cost about $b^{0.3b}$. The tree beats them because it works with multisets and the order of the weights.
- Communication complexity. The lower bound for certificates of emptiness (Section 4.4) is a fooling-set argument in the sense of Kushilevitz and Nisan [51]. Nothing new follows yet.
- Singular series. The local factors of the root heuristic (Section 5.5) play the role of the singular series of Hardy and Littlewood. The 2-digit narcissistic numbers are the simplest case where the product of local factors gives the exact count, through $\tau(b^2 + 1)$.
- Function fields. The free residues of $\mathbb{F}_q[T]$ (Section 7.8) are additive Hilbert 90 for the Frobenius of $\mathbb{F}_q[T]/(B - 1)$, and the weights there are additive polynomials in the digits.
- Cryptography. Some signature schemes use the map $x \mapsto x^x$ modulo $p$, so its collisions matter there [10]. The library adds data on the shifted map but nothing about those schemes.
- Perfect powers and S-units. Self-paying digits solve $a^{a-1} = b^k$, and the identity families use that $x^x$ is a $j$-th power on a progression, like the families of Ko for $x^x y^y = z^z$ [52]. S-unit theorems give finiteness for equations with a fixed number of terms. The Munchausen equation has as many terms as digits, and no transfer is known.

Appendix A

# Probability: Poisson approximation, Borel-Cantelli and entropy

The probabilistic model of Chapter 3 treats a Munchausen number as a rare coincidence. Rare coincidences are counted with the Poisson law. This appendix collects the facts from probability that the main text uses: the Poisson limit and how good it is, the two Borel-Cantelli lemmas, the digit counts of a random number and their Poisson coupling, anti-concentration of sums, the statistics that compare counts with a model, and the entropy argument behind the search bound of Section 4.4. A good general reference is Durrett's textbook [53]. For Poisson approximation the standard book is [54], and for entropy [55].

## A.1 Rare events and the Poisson law

**Definition A.1** (Poisson law).
A random variable $Z$ with values in $\{0, 1, 2, ...\}$ is Poisson with mean $\lambda \geq 0$ if $P(Z = k) = e^{-\lambda}\lambda^k/k!$. Its mean and its variance are both $\lambda$. In particular $P(Z = 0) = e^{-\lambda}$.

The Poisson law appears whenever many unlikely and nearly independent events are counted. Let $X_1, ..., X_N$ be indicators with $P(X_i = 1) = p_i$, and let $W = X_1 + ... + X_N$ and $\lambda = p_1 + ... + p_N$. If every $p_i$ is small and the $X_i$ are independent, then $W$ is close to Poisson with mean $\lambda$. To say how close, we need a distance between laws.

**Definition A.2** (Total variation distance).
For two probability laws $P$ and $Q$ on the same countable set, the total variation distance is

$$d_{\mathrm{TV}}(P, Q) = \sup_A |P(A) - Q(A)| = \frac{1}{2}\sum_x |P(x) - Q(x)|,$$

the supremum over all events $A$. For random variables we write $d_{\mathrm{TV}}(W, Z)$ for the distance between their laws.

So $d_{\mathrm{TV}}(W, Z) \leq \varepsilon$ means that every statement about $W$, such as "$W = 0$" or "$W \geq 3$", has the same probability for $Z$ up to $\varepsilon$. Two facts make the distance easy to use. It does not grow when one applies a function: $d_{\mathrm{TV}}(f(W), f(Z)) \leq d_{\mathrm{TV}}(W, Z)$. And for vectors with independent components it is subadditive: the distance between $(W_1, W_2)$ and $(Z_1, Z_2)$ is at most $d_{\mathrm{TV}}(W_1, Z_1) + d_{\mathrm{TV}}(W_2, Z_2)$.

**Theorem A.3** (Le Cam's inequality). CITED
Let $X_1, ..., X_N$ be independent indicators with means $p_i$, $W = \sum X_i$ and $\lambda = \sum p_i$. Then

$$d_{\mathrm{TV}}(W, \mathrm{Poisson}(\lambda)) \leq \sum_i p_i^2.$$

The proof ([56]) couples each $X_i$ with a Poisson variable of mean $p_i$ that differs from it with probability at most $p_i^2$. For Munchausen numbers in one base, the indicators are "this digit multiset is a solution", there are about $4^b$ of them, and each has a tiny probability. So the squared sum is negligible, if the events are independent. They are not quite: twins under $0^0 = 0$ (Proposition 2.1) come in pairs, and a zero-free solution counts under both conventions (Proposition 1.8). The Chen-Stein method handles such dependence.

## A.2 The Chen-Stein method

Chen [57] adapted Stein's method for the normal law to the Poisson law. The form below is due to Arratia, Goldstein and Gordon [58]. Give each index $\alpha$ a neighbourhood $B_\alpha$ that contains $\alpha$: the indices whose events may depend strongly on $X_\alpha$.

**Theorem A.4** (Chen-Stein bound). CITED
Let $W = \sum_\alpha X_\alpha$ be a sum of indicators with $p_\alpha = EX_\alpha$ and $\lambda = \sum p_\alpha$. Put

$$b_1 = \sum_\alpha \sum_{\beta \in B_\alpha} p_\alpha p_\beta, \quad b_2 = \sum_\alpha \sum_{\beta \in B_\alpha, \beta \neq \alpha} E\big[X_\alpha X_\beta\big],$$

$$b_3 = \sum_\alpha E\big|E\big[X_\alpha - p_\alpha \mid X_\beta : \beta \notin B_\alpha\big]\big|.$$

Then $d_{\mathrm{TV}}(W, \mathrm{Poisson}(\lambda)) \le b_1 + b_2 + b_3$.

The three terms have a plain meaning. $b_1$ is small when every neighbourhood has a small total probability. $b_2$ is small when two events in one neighbourhood rarely happen together. $b_3$ measures how much $X_\alpha$ depends on the events outside its neighbourhood; if $X_\alpha$ is independent of them, $b_3 = 0$. The theorem has sharper constants, $\big(1 - e^{-\lambda}\big)/\lambda$ in front of $b_1 + b_2$, which matter when $\lambda$ is large.

**Example A.5** (How the Poisson theorem of chapter 3 uses it).
The coupled model of Section 3.7 replaces the large-digit part $V(u)$ of the sum of a multiset by a random number $Y_u$, one for each upper part $u$. All multisets with the same upper part form a cluster, and different clusters are independent. So $b_3 = 0$ when the neighbourhoods are the clusters. The library proves that inside a cluster two events can both happen only as a twin pair (Lemma 13.7 of L13.2). So $b_2$ is twice the expected number of twin pairs, which is $O(1/b)$, and $b_1$ is a sum of squared cluster masses, which decays faster than any power of $b$. The library proves a cluster version directly, without the general theorem: it compares the indicator vector of each cluster with independent Bernoulli variables, and then uses Le Cam's inequality. The result is Theorem 3.54: the pair $(N_0, N_1)$ of counts under the two conventions is within $O(1/b)$ of a bivariate Poisson law.

A bivariate Poisson pair has the form $(Z_s + Z_0, Z_s + Z_1)$ with independent Poisson variables $Z_s, Z_0, Z_1$. The shared part $Z_s$ counts the zero-free solutions. Its correlation is $\lambda_s / \sqrt{(\lambda_s + \lambda_0)(\lambda_s + \lambda_1)}$, which is how the model predicts the correlation of $N_0$ and $N_1$ across bases (L13.2).

## A.3 Chebyshev and the Borel-Cantelli lemmas

**Theorem A.6** (Chebyshev's inequality). CITED
For a random variable $Y$ with mean $\mu$ and variance $\sigma^2$, and every $t > 0$, $P(|Y - \mu| \ge t) \le \sigma^2 / t^2$.

The two Borel-Cantelli lemmas decide whether infinitely many of a sequence of events $A_1, A_2, \ldots$ happen.

**Theorem A.7** (Borel-Cantelli lemmas). CITED

1. If $\sum_n P(A_n) < \infty$, then with probability 1 only finitely many $A_n$ happen.
2. If the $A_n$ are independent and $\sum_n P(A_n) = \infty$, then with probability 1 infinitely many $A_n$ happen.

*Proof.* (1) The number of events that happen has finite expectation $\sum P(A_n)$, so it is finite with probability 1. (2) The chance that none of $A_n, \ldots, A_N$ happens is $\prod(1 - P(A_i)) \le \exp\bigl(-\sum_{i=n}^{N} P(A_i)\bigr)$, which tends to 0 as $N \to \infty$. So after every $n$ some event happens. □

The model of Chapter 3 treats the bases as independent trials. A statement such as "only finitely many solutions lie far from their length" then follows from the first lemma, and its converse from the second. This is how the library derives the length window, Conjecture 3.72 (L9.9, L10.5). The heuristic step is the model itself. The lemmas are exact.

**Example A.8** (The longest run of heads).
Flip a fair coin forever, and let $A_n$ be the event that a run of at least $r_n$ heads starts at flip $n$. Then $P(A_n) = 2^{-r_n}$. Take $r_n = \log_2 n + a \log_2 \log_2 n$. Then $P(A_n) = 1/(n(\log_2 n)^a)$, and $\sum_n P(A_n)$ is finite exactly when $a > 1$. The events are not independent, but they are nearly so for far apart $n$, and the conclusion holds: runs longer than $\log_2 n + (1+\varepsilon)\log_2 \log_2 n$ occur only finitely often, while runs longer than $\log_2 n + (1-\varepsilon)\log_2 \log_2 n$ occur infinitely often. In base $b$, the model gives a solution with $J = b - k > \ln b + a \ln \ln b$ digits missing a probability of about $C/(b(\ln b)^a)$, the same sum. So Conjecture 3.72 has the same shape as the law of the longest run.

## A.4 Digit counts of a random number

Draw a $d$-digit string uniformly in base $b$. The number of copies of one fixed digit value is binomial with $d$ trials and success probability $1/b$, so its mean is $d/b$. If $b$ grows and $d = b - j$ for a fixed $j$, the mean tends to 1, and the binomial law tends to the Poisson law with mean 1 (by Le Cam's inequality, the error is at most $d/b^2$). The counts of finitely many fixed values, such as $b-1, b-2, b-3$, become independent Poisson variables in the limit.

This is the whole content of Heuristic 3.3. A digit $b - y$ weighs $(b-y)^{b-y} \approx e^{-y} b^{b-y}$, so the sum $F$ is decided by the largest digit present and its number $c$ of copies. With $c$ copies of $b-1$ the sum is about $(c/e)b^{b-1}$:

| length of $F$ | event | limit probability |
|---|---|---|
| $b$ | $c \ge 3$ copies of $b-1$ | $1 - (5/2)e^{-1} = 0.080$ |
| $b-1$ | $c = 1$ or 2 | $(3/2)e^{-1} = 0.552$ |
| $b-2$ | no $b-1$, some $b-2$ | $e^{-1}(1-e^{-1}) = 0.233$ |
| $b-3$ | largest digit $b-3$ | $e^{-2}(1-e^{-1}) = 0.086$ |

The probabilities add to 1, because the events cover almost every string. That is why the model expects about one solution per base and convention, before the parity effect of Section 3.3. The same computation with other weights gives other length laws (Section 7.6).

### A.4.1 Poisson coupling of several counts

The table treats the counts of the top values as independent Poisson(1) variables. In a finite base they are not: they come from one string of fixed length, so they are slightly dependent, and each is binomial. A *coupling* makes the step exact. It bounds the total variation distance between the true law of the vector of counts and the law of independent Poisson variables. Every event that

depends only on these counts then has the same probability under both laws, up to that distance. Two facts give the bound.

**Theorem A.9** (Binomial against Poisson). CITED
For every $n$ and $p$, $d_{\mathrm{TV}}(\mathrm{Bin}(n,p), \mathrm{Poisson}(np)) \le p$.

This is the bound of Barbour and Hall [59]; Le Cam's inequality gives the weaker $np^2$. The second fact is a conditioning trick. Let $N_1, ..., N_K$ be the counts of $K$ fixed values in $d$ independent uniform digits. Given their total $M$, the counts are multinomial with $K$ equal cells. Independent Poisson variables with a common mean have the same property: given their total, they are multinomial with equal cells. So the distance between the two vectors equals the distance between their totals. The total $M$ is binomial with $d$ trials and success probability $K/b$, and the Poisson total has mean $dK/b$, so the distance is at most $K/b$. The same trick compares Poisson$(d/b)$ counts with Poisson(1) counts. With $d = b - j$ and the leading digit handled separately, the vector of the $K = j + 1$ top counts is within $((j+1)^2 + 1)/b$ of independent Poisson(1) variables (Lemma 29.3 of L29.1). Theorem 3.7 uses this to turn the table above into a theorem about the model.

## A.5 Stochastic matrices, trace and eigenvalues

A process that moves between finitely many states $1, ..., n$ at random is described by its transition matrix $M$: the entry $M_{ij}$ is the chance to move from state $i$ to state $j$.

**Definition A.10** (Stochastic matrix).
A square matrix $M$ is *stochastic* if its entries are nonnegative and each row sums to 1. Row $i$ is then a probability law on the states. The *trace* $\mathrm{tr}\, M = \sum_i M_{ii}$ is the sum of the chances to stay put.

Three facts from linear algebra are all that Section 3.2.2 needs.

- The vector $\mathbf{1} = (1, ..., 1)$ satisfies $M\mathbf{1} = \mathbf{1}$, because the rows sum to 1. So 1 is an eigenvalue.
- Every eigenvalue $\lambda$ has $|\lambda| \le 1$. If $Mv = \lambda v$ and $|v_i|$ is the largest coordinate, then $|\lambda|\ |v_i| = |\sum_j M_{ij} v_j| \le \sum_j M_{ij}\ |v_i| = |v_i|$.
- The trace is the sum of all eigenvalues, counted with multiplicity. This holds for every square matrix: both are the coefficient of $t^{n-1}$ in the characteristic polynomial $\det(tI - M)$, up to sign.

So $\mathrm{tr}\, M = 1 + (\lambda_2 + ... + \lambda_n)$: one plus the other eigenvalues. When these are small, the trace is close to 1.

**Example A.11** (Two states).
Let

$$M = \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix}.$$

The eigenvalues are 1 and $1 - p - q$, and $\mathrm{tr}\, M = 2 - p - q$, which is indeed $1 + (1 - p - q)$. The second eigenvalue measures how much the next state depends on the present one. If the two rows are equal, so $q = 1 - p$, the next state does not depend on the present one at all. Then the second eigenvalue is 0 and the trace is exactly 1.

The same holds in any size. If all rows equal one probability law $s$, then $M = \mathbf{1}s^T$ has rank one. Its eigenvalues are $s \cdot \mathbf{1} = 1$ and 0, and its trace is $\sum_i s_i = 1$. A small change of the rows moves the eigenvalues a little, and the trace follows. For example, let $M = \mathbf{1}u^T + \varepsilon v\sigma^T$ with $u \cdot \mathbf{1} = 1$ and $\sigma \cdot \mathbf{1} = 0$. Then $M\mathbf{1} = \mathbf{1}$ and $Mv = (u \cdot v)\mathbf{1} + \varepsilon(\sigma \cdot v)v$. A vector orthogonal to both $u$ and $\sigma$ is sent

to 0. So $M$ maps the plane spanned by $\mathbf{1}$ and $v$ into itself, and in the basis $(\mathbf{1}, v)$ it acts by the matrix

$$\begin{pmatrix} 1 & u \cdot v \\ 0 & \varepsilon\sigma \cdot v \end{pmatrix},$$

so the eigenvalues are 1 and $\varepsilon(\sigma \cdot v)$. Every other eigenvalue is 0. This is the computation behind Heuristic 3.14.

A last fact concerns blocks. If the states split into groups and the process never leaves a group, $M$ is block diagonal, and each block is a stochastic matrix with its own eigenvalue 1. So the eigenvalue 1 has one copy per closed group. In Section 3.3 the parities of odd bases under $0^0 = 0$ are two such groups, and this is the doubling of the expected count (Corollary 3.19).

## A.6 Anti-concentration: the Littlewood-Offord problem

Concentration inequalities, such as Chebyshev's, say that a sum of independent steps stays near its mean. Anti-concentration says the opposite: the sum is spread out, and no single value has a large probability. Littlewood and Offord met this question in 1943, when they counted the real roots of random polynomials. Erdos gave the sharp answer.

**Theorem A.12** (The Littlewood-Offord bound). CITED
Let $a_1, ..., a_n$ be real numbers with $|a_i| \geq 1$, and let $\varepsilon_1, ..., \varepsilon_n$ be independent signs, each $\pm 1$ with probability $1/2$. Then every open interval of length 2 contains $\sum_i \varepsilon_i a_i$ with probability at most $\binom{n}{\lfloor n/2 \rfloor}/2^n$, which is about $\sqrt{2/(\pi n)}$.

The proof [60] is a counting argument with Sperner's theorem on antichains of subsets. The bound is attained when every $a_i = 1$: then the sum is a simple random walk, which sits at its most likely value with probability of order $n^{-1/2}$. Steps with more varied sizes spread the sum further. Halasz [61] gave bounds in terms of the arithmetic structure of the steps. The inverse theorems of Tao and Vu [62] and of Nguyen and Vu [63] go the other way: if the sum hits one value with large probability, then most steps lie in a short generalized arithmetic progression.

**Example A.13** (Dissociated steps).
Take $a_i = 2^i$. Different choices of signs give different sums, because the binary expansion of $\sum_i (\varepsilon_i + 1)2^{i-1}$ shows the signs. So each value has probability $2^{-n}$, the extreme case of anti-concentration. Steps with this property are called *dissociated*. The weights $x^x - x$ behave the same way on a window of small digits: a sum of a few of them shows which digits were used (Lemma 3.62). Theorem 3.63 turns this into a bound that holds modulo any integer $Q$.

Anti-concentration is weaker than equidistribution. Equidistribution modulo $q$ says that each residue class has probability close to $1/q$. Anti-concentration says only that no class has probability above some small $\rho$. In Section 3.7, $\rho \to 0$ is enough for the Poisson law in the limit, while the rate $O(1/b)$ needs $\rho \leq \kappa/q$, which is Assumption R.

The proof of Theorem 3.63 also needs a tail bound for the number of steps that land in a window. That number is binomial.

**Theorem A.14** (Chernoff bounds). CITED
Let $N$ be a sum of independent indicators with mean $\mu$, and let $h(t) = t \ln t - t + 1$. Then $P(N \leq a) \leq \exp(-\mu h(a/\mu))$ for $a \leq \mu$, and $P(N \geq a) \leq \exp(-\mu h(a/\mu))$ for $a \geq \mu$.

See [64]. The proof applies Markov's inequality to $e^{tN}$ and optimizes over $t$.

## A.7 Comparing counts with a model

The library tests the model with simple statistics. Three of them recur.

- Poisson tail probabilities. If a model predicts a mean $\mu$ and the data show $k$, the library reports $P(X \geq k)$ or $P(X \leq k)$ for a Poisson variable $X$ with mean $\mu$. For example 3 observed against 0.33 expected gives $P(X \geq 3) = 0.0047$ (L22.4).
- The dispersion statistic. For counts $N_1, ..., N_r$ with predicted means $\lambda_1, ..., \lambda_r$, the sum $\sum_i (N_i - \lambda_i)^2/\lambda_i$ is about $r$ when every count is Poisson. A much larger value shows extra scatter, for example from families of solutions that come in groups (L17.2). This is Pearson's chi-square statistic.
- The look-elsewhere effect. If one picks the most extreme of $N$ groups after looking at the data, a tail probability $p$ for that group is worth about $1 - (1-p)^N \approx Np$. The library uses this to discount several striking patterns. For example a deviation with $p = 0.019$, chosen among about 14 groups, has a chance of about 23% to appear somewhere (L5.5).

Many apparent anomalies of the data turned out to be chance after one of these corrections, or after fresh data. Section 3.11 tells these stories.

## A.8 Clustered samples and the design effect

The near-solution test of Section 3.5 estimates the expected number of solutions of a base by the mean count of near-solutions $n = F(n) + D$ over $J$ offsets $D$. If the offset counts were independent Poisson draws, that mean would have variance $\lambda/J$. They are not independent. A rearrangement of the digits of $n$ keeps $F(n)$ and moves $n$ by a multiple of $b - 1$, so one multiset whose sum lies close to its own arrangements is a near-solution at several offsets at once (Lemma 26.1 of L26.1). The counts come in clusters. Survey statistics has a standard language for this.

**Definition A.15** (Intraclass correlation and design effect).
Suppose that $n$ observations fall into clusters of $m$ each, all with variance $\sigma^2$, and that two observations in one cluster have correlation $\rho$ while observations in different clusters are independent. The number $\rho$ is the intraclass correlation. The mean of all $nm$ observations has variance

$$\frac{\sigma^2}{nm}(1 + (m-1)\rho).$$

The factor $\text{deff} = 1 + (m-1)\rho$ is the design effect, and $nm/\text{ deff}$ is the effective sample size: the number of independent observations that would give the same precision.

With $\rho = 0$ nothing changes. With $\rho = 1$ every cluster repeats one value, the design effect is $m$, and only the $n$ clusters count. Ignoring a positive $\rho$ makes an estimate look more precise than it is.

For rare events the clusters have random sizes, and a cleaner form applies. Write the total over the offsets as $T = \sum_c K_c$, where $K_c$ is the number of offsets at which the multiset $c$ is a near-solution. Different multisets are independent in the model, and each one is a near-solution only rarely, so

$$\text{Var } T \approx \sum_c E[K_c^2] = \varphi E[T], \quad \varphi = \frac{E[K^2]}{E[K]},$$

with the moments taken over the multisets that appear at least once. The design effect $\varphi$ is the mean cluster size seen from a random near-solution. A Poisson variance, $E[T]$, understates the scatter of $T$ by the factor $\varphi$, and the mean over $J$ offsets is worth only $J_{\text{eff}} = J/\varphi$ independent offsets. If several near-solutions can also clump at one offset, such as twins under $0^0 = 0$ (Proposition 2.1), each single count has variance $\psi\lambda$ with $\psi = E[\text{clump}^2]/E[\text{clump}]$, a compound Poisson law, and $J_{\text{eff}} = J\psi/\varphi$ (Proposition 26.5 of L26.1).

**Example A.16** (The integer offsets).
In bases 12 to 26, with the 60 offsets $D = 2j(b-1)$, about two thirds of the near-solutions share their multiset with another offset. The design effect is 2.1 to 2.5, so the 60 offsets are worth about 27 independent ones (L26.2). A swap of the last two digits explains the largest share of the partner pairs, and a simple count of partners predicts $\varphi$ within a few percent. Offsets chosen so that no two differ by a single transposition cut the shared share to about 15% and nearly double $J_{\text{eff}}$, with the same mean.

How the P values change depends on the question. A single offset count is still Poisson, because each multiset is a near-solution at a given offset at most once. So when one compares the count $N_0$ of true solutions with the offset mean $\hat{\lambda}$, the variance of the difference is $\psi\lambda + \varphi\lambda/J$, and the first term dominates for $J \geq 20$. Most "chance" verdicts of the library therefore stand (Table 26.4.1 of L26.4). Comparisons of the offset means themselves change a lot, because there the second term is everything. The chi-square of $\lambda/$ model over 21 bases falls from about 97 to about 42 once the clustered variance is used, and the low near-solution rate of base 19 goes from 4.6 to 2.9 standard errors. The library also uses a rank P value, $(1 + \#\{D : N_D \leq N_0\})/(J+1)$, which is exact when the counts at $D = 0$ and at the offsets are exchangeable, whatever the clusters, but cannot go below $1/(J+1)$ (L26.3).

Two tools combine evidence across groups.

- Fisher's method. If $p_1, ..., p_k$ are independent P values, then under the null hypothesis $-2\sum_i \ln p_i$ has the chi-square law with $2k$ degrees of freedom. The library uses it to combine an in-sample P value, after the look-elsewhere correction, with the P value of fresh bases; for the factorion shortfall the combined value moved from about 0.05 to about 0.09 with the clustered calibration (L23.1, L26.4).
- Pooled ratios and heterogeneity. Each group $i$ gives an observed count $O_i$ and an offset estimate $E_i$, with a variance $v_i$ that includes the design effect. A common ratio $\theta = O/E$ is estimated by weighting each group by $1/v_i$. Cochran's statistic $Q = \sum_i \left(O_i - \hat{\theta}E_i\right)^2/v_i$ has about the chi-square law with $k-1$ degrees of freedom if all groups share one ratio. Over eight number systems the library finds $Q = 23.3$ on 7 degrees of freedom (P about 0.002). Without the odd-base factorions and the Pisot bases, the other six worlds fit one ratio, 0.91, with $Q = 3.6$ on 5 degrees of freedom (P = 0.60) (L26.5). So outside two named worlds the solutions behave like their near-solutions, which is the claim of Chapter 3.

## A.9 Many P values at once

A P value is the chance, computed under the null hypothesis, of a result at least as extreme as the one observed. If the null hypothesis is true and the statistic is continuous, the P value is a uniform random number in $[0, 1]$: $P(p \leq a) = a$ for every $a$. For counts the law is a little lumpy, but close to uniform. So a list of P values from tests of a correct model should look like a list of uniform random numbers. About 5% fall below 0.05, about 1% below 0.01, and the median is near $1/2$. The Kolmogorov-Smirnov test compares the whole list with the uniform law: it measures the largest gap between the share of P values below $a$ and $a$ itself.

Fisher's method combines independent P values into one. Under the null hypothesis $-\ln p$ is exponential with mean 1, so $-2\ln p$ has the chi-square law with 2 degrees of freedom. A sum of $k$ independent ones has the chi-square law with $2k$ degrees of freedom, and its upper tail is the combined P value.

**Example A.17** (Fisher's method).
Three independent tests give $p = 0.10, 0.20, 0.30$. Then $-2(\ln 0.1 + \ln 0.2 + \ln 0.3) = 10.23$. For $2k$ degrees of freedom the tail is $P(\chi^2_{2k} \geq x) = e^{-x/2} \sum_{i<k} (x/2)^i / i!$, which gives 0.115 here. No single test is striking, and neither is the combination.

Two things break this picture. The first is dependence. If two tests reuse the same data, their P values move together, and Fisher's sum treats one piece of evidence as two. The combined P value is then too small. The second is selection. If a reported group is the most extreme of $k$ groups that were looked at, its P value is the smallest of $k$ uniform numbers, with $P(\min \leq a) = 1 - (1-a)^k$. The Sidak correction reports $1 - (1-p)^k$ in place of $p$. A list of such minima is far from uniform. With $k = 3$ the median is $1 - 2^{-1/3} = 0.21$. With $k = 6$ the share below 0.05 is $1 - 0.95^6 = 0.26$. Tests whose hypothesis was written down before the data were seen avoid selection, which is why fresh bases and pre-registered predictions count for more. Section 3.6.3 applies all of this to the P values of the library.

## A.10 Entropy and the asymptotic equipartition property

**Definition A.18** (Entropy and varentropy).
Let $X$ take values in a countable set with probabilities $P(x)$. The entropy of $X$ is $H = E[-\ln P(X)] = -\sum_x P(x) \ln P(x)$, measured in nats. The varentropy is $V = \mathrm{Var}(-\ln P(X))$, the variance of the same quantity.

Entropy measures how many values a random variable really takes. A uniform variable on $M$ values has $H = \ln M$. For a vector of independent components, entropies and varentropies add. The key fact is that a long vector of independent components almost always takes a value with probability close to $e^{-H}$.

**Theorem A.19** (Asymptotic equipartition property). CITED
Let $X_1, X_2, \ldots$ be independent and identically distributed with entropy $H$. Then $-\frac{1}{n} \ln P(X_1, \ldots, X_n) \to H$ in probability. So for large $n$ the vector lies, with probability close to 1, in a typical set of about $e^{nH}$ values, each with probability about $e^{-nH}$. Every set that holds the vector with probability at least $p > 0$ has at least about $pe^{nH}$ elements.

See [55]. The last sentence is the useful one for searching. If a search must find a hidden object that is drawn from a law with entropy $H$, and the search tries candidates one by one, it must try about $e^H$ of them to succeed with a fixed probability. A one-shot version needs only Chebyshev's inequality.

**Proposition A.20** (A one-shot entropy bound). PROVEN L10.2
Let $X$ have entropy $H$ and varentropy $V$. If a set $S$ contains $X$ with probability $p$, then $|S| \geq (p - V/t^2)e^{H-t}$ for every $t > 0$.

*Proof.* By Chebyshev, $P(-\ln P(X) < H - t) \leq V/t^2$. Every other value has $P(x) \leq e^{-H+t}$, so the values of $S$ outside the rare set hold at most $|S|\, e^{-H+t}$. Hence $p \leq V/t^2 + |S|\, e^{-H+t}$. □

**Example A.21** (Where 1.638 comes from).
The search of Section 4.1 fixes the counts of the large digits first. At the level where $\kappa b$ digit positions are still unknown, and $(1-\kappa)b$ digit values are decided, the model says that each decided value appears in the unknown part a Poisson($\kappa$) number of times, independently. So the node above a hidden solution has entropy $H = (1-\kappa)b \cdot H(\mathrm{Poisson}(\kappa))$. As a function of $\kappa$

this is largest at $\kappa \approx 0.368$, where it equals $0.4933b$. The varentropy grows only linearly in $b$, so the proposition gives a bound of $e^{0.4933b - O(\sqrt{b})}$ nodes at that level: about $1.638^b$, since $e^{0.4933} = 1.638$. This is Theorem 4.11. By contrast, the exact tree counts every count vector that fits, about $\binom{b}{\kappa b}$ of them, which is largest at $\kappa = 1/2$ and grows like $2^b$ (L10.1).

Appendix B

# Galois groups, Chebotarev and Jordan

A digit string $s = (s_1, ..., s_k)$ is a solution in base $b$ exactly when $b$ is a root of the Munchausen polynomial

$$Q_s(r) = s_1 r^{k-1} + s_2 r^{k-2} + ... + s_k - F(s)$$

(Section 1.3). Chapter 5 studies how this polynomial behaves modulo primes. The tools are the Galois group of a polynomial, the way it controls roots modulo primes, and a little group theory. This appendix states them. Good references are the survey of Stevenhagen and Lenstra [65], Serre's article on Jordan's theorem [66], and Serre's book [67] for Hilbert irreducibility.

## B.1 The Galois group of a polynomial

Let $f$ be a polynomial with rational coefficients, of degree $n$, without repeated roots. Its splitting field $K$ is the smallest field that contains $\mathbb{Q}$ and all $n$ complex roots of $f$.

**Definition B.1** (Galois group).
The Galois group $G$ of $f$ is the group of field automorphisms of $K$. Each automorphism permutes the roots of $f$, and it is determined by this permutation. So $G$ is a subgroup of the symmetric group $S_n$ on the roots, well defined up to a renumbering of the roots.

$G$ acts transitively on the roots exactly when $f$ is irreducible over $\mathbb{Q}$. For a "random" polynomial of degree $n$, the group is all of $S_n$. Hilbert's irreducibility theorem makes this precise: if a polynomial $f(t, r)$ in two variables has group $G$ over the field $\mathbb{Q}(t)$, then for all integers $t$ outside a thin set, which has density 0, the specialized polynomial has group $G$ again ([67]). For $Q_s$ the free parameter is the constant term $F(s)$, which has nothing to do with the other coefficients. So one expects the group $S_{k-1}$ for almost all strings. Section 5.4 confirms this by the statistics below, and proves it for $k = 4$ and $k = 5$ once the digits are not too small (L16.4).

Smaller groups come from structure. A polynomial is decomposable if $P = G(h(r))$ with $G$ and $h$ of degree at least 2. Then the roots fall into blocks, one for each root of $G$, and the Galois group preserves the blocks. For example $Q_s$ for the string $(9, 18, 18, 9, 15)$ is a polynomial in $u = r^2 + r$, so its group lies in the dihedral group $D_4$ of order 8 (L12.1). Ritt's theory [24] describes all ways to decompose a polynomial. Two families of indecomposable polynomials have small groups for another reason: the power maps $r^n$ and the Dickson polynomials

$$D_n(r, a) = \sum_{i=0}^{\lfloor n/2 \rfloor} \frac{n}{n-i} \binom{n-i}{i} (-a)^i r^{n-2i},$$

which satisfy $D_n(y + a/y, a) = y^n + (a/y)^n$. For example $D_5(r, -1) = r^5 + 5r^3 + 5r$. The strings $(1, 0, 5, 0, 5, x)$ have exactly this digit polynomial, and their group is the group $F_{20}$ of order 20 instead of $S_5$ (L16.4).

## B.2 Reduction modulo a prime

The roots of $f$ modulo a prime $p$ are visible in the Galois group. Let $f$ have integer coefficients, and let $p$ be a prime that does not divide the leading coefficient or the discriminant of $f$. Then $f \bmod p$ has no repeated factor, and it factors over $\mathbb{F}_p$ into irreducible factors of degrees $n_1, ..., n_r$ with $n_1 + ... + n_r = n$.

**Theorem B.2** (Dedekind, Frobenius). CITED
For such a prime $p$ there is an element $\sigma_p$ of $G$, the Frobenius element, well defined up to conjugation, whose cycle type on the roots is $(n_1, ..., n_r)$. In particular $f$ has a root modulo $p$ exactly when $\sigma_p$ fixes at least one root.

So the question "how often does $f$ have no root modulo $p$" becomes the question "how often is the Frobenius element a fixed-point-free permutation". The answer is the Chebotarev density theorem.

**Theorem B.3** (Chebotarev density theorem). CITED
For every conjugacy class $C$ of $G$, the set of primes $p$ with $\sigma_p$ in $C$ has natural density $|C| \,/\, |G|$ among all primes.

**Corollary B.4** (Primes without a root). CITED
The primes modulo which $f$ has no root have density equal to the share of elements of $G$ that fix no root. For $G = S_n$ this share is the share of derangements,

$$\frac{D_n}{n!} = \sum_{j=0}^{n} \frac{(-1)^j}{j!},$$

which is $1/2, 1/3, 3/8, 11/30$ for $n = 2, 3, 4, 5$ and tends to $1/e \approx 0.368$.

The formula for $D_n/n!$ is inclusion and exclusion over the set of fixed points. The same formula gives the share of all polynomials of degree $n$ over $\mathbb{F}_p$ without a root, in the limit of large $p$: inclusion and exclusion over the roots gives $\sum_{j \le n} (-1)^j \binom{p}{j} p^{-j} \to \sum_{j \le n} (-1)^j / j!$. So for large primes a polynomial with group $S_n$ behaves like a random polynomial.

**Example B.5** (The sieve as a Galois statistic).
Section 5.4 reports, for 60 random strings of each length with digits up to 20, the share of primes between 200 and 20,000 modulo which $Q_s$ has no root. For $k = 3, 4, 5, 6$ the medians are 0.501, 0.335, 0.374 and 0.367, against $\frac{1}{2}, \frac{1}{3}, \frac{3}{8}, \frac{11}{30}$. The local test of Proposition 5.9 therefore removes about $1/e$ of the strings per large prime, independently from prime to prime, which explains the speed of the sieve of Section 4.8. Small primes are weaker, for a reason that has nothing to do with Galois theory: at $r \equiv 1$ the polynomial reduces to the digit-sum identity of Proposition 1.9 (L11.5).

## B.3 Jordan's theorem and intersective polynomials

Could a string pass the local test for every prime without being a solution anywhere? Jordan's theorem rules this out for irreducible polynomials.

**Theorem B.6** (Jordan). CITED
A transitive group of permutations of $n \ge 2$ points contains an element with no fixed point.

*Proof.* By Burnside's lemma, the average number of fixed points of an element of a transitive group is 1. The identity fixes $n \geq 2$ points, more than the average. So some element fixes fewer than one point, that is none. □

Cameron and Cohen [23] sharpened the count: at least a share $1/n$ of the elements of a transitive group fix no point. With Chebotarev this gives Proposition 12.1 of the library: if $Q_s$ is irreducible of degree $n$, it has no root modulo a set of primes of density at least $1/n$. Serre's survey [66] gives the history and many applications.

For reducible polynomials the theorem can fail. A polynomial that has a root modulo every prime (or every integer), but no rational root, is called intersective.

> **Example B.7** (An intersective polynomial).
> $f(r) = (r^3 - 19)(r^2 + r + 1)$ has a root modulo every prime. For $p \equiv 1 \pmod 3$ the factor $r^2 + r + 1$ has a root, since $\mathbb{F}_p^*$ contains an element of order 3. For $p \equiv 2 \pmod 3$, cubing is a bijection of $\mathbb{F}_p$, so $r^3 = 19$ has a root. For $p = 3$ the root is $r = 1$, and for $p = 19$ it is $r = 0$. But $f$ has no rational root.

In Galois terms, the group of a product $gh$ acts on the roots of $g$ and of $h$ together, and an element can move every root of both factors even when each factor alone always keeps a root. The library proves that no Munchausen polynomial with $k \leq 5$ is intersective without a rational root, and gives an exact criterion for $k = 6$: a quadratic and a cubic factor with the same quadratic field, the shape of the example above (L16.3). No such string is known. So in practice a string that passes every local test has a rational root, and the rational roots that are not bases are the "roots too small to be bases" of Section 5.9.

## B.4 Effective Chebotarev and the generalized Riemann hypothesis

Chebotarev's theorem says how often a Frobenius class occurs among all primes. Some arguments need more: a prime with a given class below an explicit bound. The bound depends on the zeros of the Dedekind zeta function of the splitting field.

The Riemann hypothesis says that the nontrivial zeros of the Riemann zeta function have real part $1/2$. The generalized Riemann hypothesis (GRH) says the same for the $L$-functions of Dirichlet characters and the zeta functions of number fields. It is open, and widely believed. Under GRH, Lagarias and Odlyzko [49] count the primes up to $X$ in a Frobenius class with an error of order $\sqrt{X}\ln(d_L X^n)$, where $d_L$ is the discriminant and $n$ the degree of the splitting field. As a consequence, the least prime in a given class is at most a constant times $(\ln d_L)^2$. Bach [68] and Bach and Sorenson [69] give explicit constants. Without GRH the known bounds are a power of $d_L$, far weaker.

Theorem 7.49 uses this for the polynomial $G_x$ of a factorion with a large last digit: under GRH a prime without a root of $G_x$ comes below a power of $\ln x$, so below $x$, and it divides $x!$. For quadratic fields the same idea bounds the least inert prime (Section D.4).

## B.5 Further reading

The Chebotarev theorem and its history are in [65]. Jordan's original paper is [70], and [66] surveys its descendants. Hilbert irreducibility is in [67]. Ritt's decomposition theory starts with [24].

Appendix C

# $p$-adic numbers and Hensel lifting

Congruences modulo a prime power $p^e$ appear throughout the main text: in the sieve for short solutions (Section 4.8), in the local factors of the root heuristic (Section 5.5), in the residues of $x^x - x$ that make coverings (Section 5.3), and in the $x$-adic form of the equation $b(b+x) = x^x + 1$ (Chapter 6). The $p$-adic numbers organize all congruences modulo powers of one prime into a single object. Gouvea's book [71] is a friendly introduction.

## C.1 Valuations and the $p$-adic integers

**Definition C.1** ($p$-adic valuation).
Fix a prime $p$. For a nonzero integer $n$, the valuation $v_p(n)$ is the exponent of $p$ in $n$, so $n = p^{v_p(n)}m$ with $p \nmid m$. Put $v_p(0) = \infty$, extend to fractions by $v_p(a/c) = v_p(a) - v_p(c)$, and define the $p$-adic absolute value $|x|_p = p^{-v_p(x)}$.

A number is $p$-adically small when it is divisible by a high power of $p$. So $3^{10}$ is tiny in the 3-adic sense, and $1 + 3 + 9 + 27 + \ldots$ converges 3-adically (to $-1/2$, by the geometric series). The absolute value satisfies the strong triangle inequality $|x + y|_p \le \max(|x|_p, |y|_p)$.

**Definition C.2** ($p$-adic integers).
A $p$-adic integer is a sequence $(a_1, a_2, \ldots)$ with $a_e$ a residue modulo $p^e$ and $a_{e+1} \equiv a_e \pmod{p^e}$ for every $e$. They form a ring $\mathbb{Z}_p$ that contains $\mathbb{Z}$ (an integer $n$ is the sequence of its residues). Equivalently, $\mathbb{Z}_p$ is the set of formal series $c_0 + c_1 p + c_2 p^2 + \ldots$ with digits $0 \le c_i < p$, added and multiplied with carries.

An element of $\mathbb{Z}_p$ is a unit exactly when $p$ does not divide its first residue $a_1$. A statement that holds modulo $p^e$ for every $e$ is a statement about $\mathbb{Z}_p$. This is the language of the "$p$-adic laws" of $x^x - x$ in the library (L20.2).

## C.2 Hensel's lemma

How do roots modulo $p$ extend to roots modulo $p^2, p^3, \ldots$? Newton's method answers this, with the $p$-adic absolute value in place of the real one.

**Theorem C.3** (Hensel's lemma). CITED
Let $f$ be a polynomial with coefficients in $\mathbb{Z}_p$ and let $a$ be an integer with $f(a) \equiv 0 \pmod{p}$ and $f'(a) \not\equiv 0 \pmod{p}$. Then for every $e \ge 1$ there is exactly one residue $a_e$ modulo $p^e$ with $a_e \equiv a \pmod{p}$ and $f(a_e) \equiv 0 \pmod{p^e}$. Together they form the unique root of $f$ in $\mathbb{Z}_p$ that reduces to $a$.

*Proof.* Suppose $f(a_e) \equiv 0 \pmod{p^e}$. Try $a_{e+1} = a_e + tp^e$. By Taylor expansion, $f(a_e + tp^e) \equiv f(a_e) + tp^e f'(a_e) \pmod{p^{e+1}}$. Since $f'(a_e) \equiv f'(a)$ is a unit modulo $p$, there is exactly one $t$ modulo $p$ that makes the right side vanish. □

A simple root modulo $p$ therefore lifts uniquely. A multiple root, with $f'(a) \equiv 0$, can lift to no root or to many.

**Example C.4** (Multiple roots).
$f(r) = r^2 - p$ has the double root 0 modulo $p$, but no root modulo $p^2$, since $r^2 \equiv p$ forces $p \mid r$ and then $p^2 \mid r^2$. On the other hand $g(r) = r^2$ has the double root 0 modulo $p$, and modulo $p^2$ it has the $p$ roots $0, p, 2p, \ldots$. In the root heuristic of Section 5.5 this matters: $3435 = (3, 4, 3, 5)$ in base 10 has two roots of $Q_s$ modulo 2, but only one modulo 64, so its 2-adic weight halves (L19.3).

## C.3 Counting roots modulo $p^e$

Let $N(p^e)$ be the number of roots of a polynomial modulo $p^e$. The root heuristic of Section 5.5 weighs each digit string by such counts. The right local factor at $p$ is $N(p^e)$ for large $e$, not the count modulo $p$ alone, because the event "$p^e$ divides $Q_s(b)$" for a random integer $b$ has probability $N(p^e)/p^e$, against $p^{-e}$ for a random value. If all roots modulo $p$ are simple, $N(p^e) = N(p)$ for every $e$ by Hensel's lemma.

**Proposition C.5** (The average is one). PROVEN L19.3
Fix a polynomial $g$ with integer coefficients and let the constant term vary: $f_c(r) = g(r) - c$. Averaged over the $p^e$ residues $c$ modulo $p^e$, the number of roots of $f_c$ modulo $p^e$ is exactly 1.

*Proof.* Each of the $p^e$ residues $r$ is a root of $f_c$ for exactly one $c$, namely $c = g(r)$. So the total number of roots over all $c$ is $p^e$. □

For a Munchausen polynomial the constant term $s_k - F(s)$ is the part that the weights $x^x$ control, and in the model it behaves like a random residue. So the local factors are unbiased: they average to 1 and only redistribute the weight among strings. This is why the completeness estimates of Section 5.5 do not depend on which version of the factor one uses.

## C.4 The exponential and the logarithm

The $p$-adic exponential and logarithm are defined by the usual series,

$$\exp(z) = \sum_{n \geq 0} \frac{z^n}{n!}, \quad \log(1+z) = \sum_{n \geq 1} (-1)^{n+1} \frac{z^n}{n}.$$

The logarithm converges for $|z|_p < 1$, that is for $z \in p\mathbb{Z}_p$. The exponential converges only for smaller $z$, because $n!$ is divisible by a high power of $p$: one needs $v_p(z) > 1/(p-1)$. For odd $p$ that means $z \in p\mathbb{Z}_p$, and for $p = 2$ it means $z \in 4\mathbb{Z}_2$.

**Theorem C.6** (Exp and log). CITED
For odd $p$, exp and log are inverse bijections between $p\mathbb{Z}_p$ and $1 + p\mathbb{Z}_p$, and they turn sums into products. For $y \in 1 + p\mathbb{Z}_p$ and an integer $x$, $y^x = \exp(x \log y)$.

See [71]. With this, a power $x^x$ with $x$ in a fixed residue class becomes a power series in a $p$-adic variable. The library uses it to find the exact set $T(p^e)$ of residues of $x^x - x$ modulo $p^e$ (L20.2).

## C.5 The 3-adic law of $x^x - x$

The covering theorem Theorem 5.13 rests on one congruence. Here is its elementary proof, as a model for the $p$-adic arguments of Section 5.3.

**Proposition C.7** (Residues modulo 27). PROVEN L15.4
For every integer $x \geq 1$: if $x \equiv 2 \pmod 6$ then $x^x - x \equiv 2 \pmod{27}$; otherwise $x^x - x \equiv 0 \pmod 3$.

*Proof.* If $3 \mid x$, both terms vanish modulo 3. If $x \equiv 1 \pmod 3$, then $x^x \equiv 1 \equiv x$. If $x \equiv 5 \pmod 6$, then $x$ is odd and $x \equiv -1 \pmod 3$, so $x^x \equiv (-1)^x = -1 \equiv x$.

Now let $x \equiv 2 \pmod 6$. Then $x$ is even and $3 \mid x+1$. Put $u = -x = 1 + 3t$ with $3t = -(x+1)$. Since $x$ is even, $x^x = u^x$. By the binomial theorem,

$$u^x = \sum_{j} \binom{x}{j} (3t)^j \equiv 1 + x(3t) + \binom{x}{2}(3t)^2 \pmod{27},$$

because $(3t)^j$ is divisible by 27 for $j \geq 3$. Substituting $3t = -(x+1)$ and simplifying gives

$$x^x - x \equiv 2 + \frac{(x+1)^3(x-2)}{2} \pmod{27},$$

and $27 \mid (x+1)^3$. □

So modulo 27 the values of $x^x - x$ prime to 3 are all equal to 2. The full law (Theorem 20.11 of L20.2) uses the exponential: on the class $x \equiv 2 \pmod 6$ one writes $x = -\exp(3w)$ and expands $x^x - x$ as $2 + 27G(w)$ with a power series $G$. Modulo 3, $G(w) \equiv -(w^3 + w^4)$, which never takes the value 2. Hensel's lemma applied to $G$ then gives the exact set for every power of 3: the multiples of 3, together with the residues $\equiv 2 \pmod{27}$ that are not $\equiv 56 \pmod{81}$. Beyond 81 a single power of 3 adds no new constraint. For primes $p \geq 7$, every residue modulo $p^e$ is a value of $x^x - x$, so those primes alone can never cover a family (Theorem 20.8).

Appendix D

# Quadratic fields, Pell equations and cyclotomic factors

Chapter 6 studies the equation $b(b+x)=x^n+c$. Completing the square turns it into $U^2=4x^n+x^2+4c$ with $U=2b+x$, and the natural place to factor $U^2-(x^2+4c)$ is a quadratic field. The same chapter explains the solution $(1,7,1)$ in base 904 by a special factorization of $7^7+1$. This appendix collects the algebraic number theory behind both, and the facts on sums of two squares that count 2-digit narcissistic numbers in Section 7.6. Marcus [72] and Ireland and Rosen [73] are good textbooks; Niven, Zuckerman and Montgomery [37] cover continued fractions; Cohen [74] covers the computations with PARI [75].

## D.1 Quadratic fields and their integers

**Definition D.1** (Quadratic field).
For a squarefree integer $D\neq 0,1$, the quadratic field $K=\mathbb{Q}\big(\sqrt{D}\big)$ is the set of numbers $\alpha=a+c\sqrt{D}$ with rational $a,c$. The conjugate of $\alpha$ is $\alpha'=a-c\sqrt{D}$. The norm and the trace are

$$N(\alpha)=\alpha\alpha'=a^2-Dc^2,\quad \operatorname{Tr}(\alpha)=\alpha+\alpha'=2a.$$

The norm is multiplicative: $N(\alpha\beta)=N(\alpha)N(\beta)$.

The ring of integers $\mathcal{O}_K$ is the set of elements of $K$ that are roots of a monic polynomial with integer coefficients. It is $\mathbb{Z}\big[\sqrt{D}\big]$ if $D\equiv 2,3 \pmod 4$ and $\mathbb{Z}\Big[\big(1+\sqrt{D}\big)/2\Big]$ if $D\equiv 1 \pmod 4$. Its discriminant is $4D$ or $D$. A subring of finite index, such as $\mathbb{Z}\big[\sqrt{D}\big]$ when $D\equiv 1 \pmod 4$, is called an order. The library works in orders like $\mathbb{Z}\Big[\big(x+\sqrt{x^2+4c}\big)/2\Big]$, whose discriminant $x^2+4c$ need not be squarefree (L20.4).

## D.2 Units and the Pell equation

A unit is an element of $\mathcal{O}_K$ whose inverse is also in $\mathcal{O}_K$; equivalently, its norm is $\pm 1$. For a real quadratic field ($D>0$) the units are exactly the numbers $\pm\varepsilon^k$, $k\in\mathbb{Z}$, for one fundamental unit $\varepsilon>1$. Its logarithm $\mu_0=\ln\varepsilon$ is the regulator.

Units are solutions of the Pell equation $u^2-Dv^2=\pm 1$ (or $\pm 4$ in the other ring). They can be found with continued fractions. Write $\sqrt{D}=[a_0;a_1,a_2,\ldots]$, a periodic continued fraction, and let $p_j/q_j$ be its convergents.

**Theorem D.2** (Legendre). CITED
If $\alpha$ is irrational and $|\alpha-p/q|<1/(2q^2)$ with $q\geq 1$, then $p/q$ is a convergent of $\alpha$. In particular, if $u,v>0$ are coprime and $\left|u^2-Dv^2\right|<\sqrt{D}$, then $u/v$ is a convergent of $\sqrt{D}$.

See [37]. The second statement turns questions about small norms into a finite check along one period of the continued fraction.

**Example D.3** (The unit of the family $(1, x, 1)$).
For odd $x$, the number $\varepsilon = \left(x + \sqrt{x^2+4}\right)/2$ is a unit of norm $(x^2 - (x^2+4))/4 = -1$ and trace $x$. For every integer $b$, $N(b+\varepsilon) = b^2 + bx - 1$. So the equation $b(b+x) = x^n + 1$ says exactly $N(b+\varepsilon) = x^n$: an element of prescribed norm, in a field that changes with $x$ (Section 6.4). When $E = (x/2)^2 + 1$ for even $x$, the continued fraction is $\sqrt{E} = [x/2; x, x, x, ...]$, every convergent has norm $\pm 1$, and Legendre's theorem shows that no element has a small norm such as $\pm 2^j$ (Lemma 20.16 of L20.3).

Fields of Richaud-Degert type, $D = a^2 + r$ with $r \mid 4a$, have a fundamental unit of size about $2a$, as in the example. For most real quadratic fields the fundamental unit is much larger, of size up to $e^{\sqrt{D}}$. This difference decides whether the methods of Section 6.5 work (L20.4).

## D.3 Ideals and the class group

Unique factorization of elements can fail in $\mathcal{O}_K$. The classic example is $6 = 2 \cdot 3 = \left(1 + \sqrt{-5}\right)\left(1 - \sqrt{-5}\right)$ in $\mathbb{Z}\left[\sqrt{-5}\right]$. Ideals repair this.

**Theorem D.4** (Unique factorization of ideals). CITED
Every nonzero ideal of $\mathcal{O}_K$ is a product of prime ideals in exactly one way. An odd prime $p$ splits as $p\mathcal{O}_K = PP'$ with $P \neq P'$ when the discriminant is a nonzero square modulo $p$, stays prime when it is a nonsquare, and ramifies as $P^2$ when $p$ divides the discriminant.

The norm of an ideal $I$ is the size of $\mathcal{O}_K/I$, and $N(\alpha\mathcal{O}_K) = |N(\alpha)|$. Two ideals are equivalent if one is a nonzero element times the other. The classes form a finite abelian group, the class group, whose order $h$ is the class number. An ideal is principal, that is generated by one element, exactly when its class is trivial. If $B^n$ is principal, the order $d$ of the class of $B$ divides $n$.

**Example D.5** (Why ideals enter the equation).
For odd $x$ every prime $p \mid x$ splits in $\mathbb{Q}\left(\sqrt{x^2+4}\right)$, since $x^2 + 4 \equiv 4$ is a square modulo $p$. A solution of $N(b+\varepsilon) = x^n$ gives $(b+\varepsilon) = B^n$ for an ideal $B$ of norm $x$ that takes one prime above each $p \mid x$. If $x$ is a prime power, $B$ is determined up to conjugation by the explicit element $\varepsilon + 1$, which has norm $x$ ("pure" solutions). Otherwise $B$ can mix the primes, and its class need not be trivial ("mixed" solutions). The case that remains open for $b(b+x) = x^x + 1$ is a mixed $B$ whose class has a large order dividing $x$ (Corollary 6.20).

For real quadratic fields the class number and the regulator are tied by the class number formula,

$$h \cdot \mu_0 = \frac{\sqrt{\Delta}}{2} \cdot L(1, \chi),$$

where $\Delta$ is the discriminant and $L(1, \chi)$ is the value at 1 of the $L$-function of the quadratic character $\chi$. The value $L(1, \chi)$ is of moderate size, roughly between $\Delta^{-\varepsilon}$ and $\ln \Delta$. So a small regulator forces a large class number, and the reverse. In $\mathbb{Q}\left(\sqrt{x^2+4}\right)$ the regulator is at most $\ln x$ or so, and the class number can be as large as about $x/\ln x$, which is why the class group conditions of Corollary 6.20 cannot be excluded by size alone (L20.5).

## D.4 The least inert prime

The splitting law above has a compact form. For a quadratic field with discriminant $\Delta$, the Kronecker symbol $\chi(\ell) = (\Delta/\ell)$ is 1 if the prime $\ell$ splits, $-1$ if it stays prime (is *inert*), and 0 if it ramifies. For odd $\ell$ that does not divide $\Delta$, $\ell$ is inert exactly when $\Delta$ is not a square modulo $\ell$. Half of all primes are inert, by the Chebotarev density theorem (Appendix B). The question is how soon the first one comes.

**Definition D.6** (Least inert prime).
The least inert prime $\ell_0(\Delta)$ is the smallest prime $\ell$ with $\chi(\ell) = -1$.

Section 7.7 needs a bound for $\ell_0$ that is much smaller than $|\Delta|$. The truth is far smaller than what can be proven.

**Theorem D.7** (Bounds for the least inert prime). CITED

1. For every fundamental discriminant with $1 < |\Delta| \le 4 \cdot 10^6$, $\ell_0(\Delta) \le 79$ (computed, L33.1).
2. Trevino [76]: for $\Delta > 1596$, $\ell_0(\Delta) \le \Delta^{0.45}$. This improves the bound $\sqrt{\Delta}/2$ of Granville, Mollin and Williams [77].
3. For $\Delta < 0$, the Polya-Vinogradov inequality in the explicit form of Pomerance [78] gives $\ell_0(\Delta) \le C\sqrt{|\Delta|}(\ln|\Delta|)^2$ with an explicit $C$.
4. Under the generalized Riemann hypothesis, Bach [68]: for a character modulo $m \ge 10^8$ that is not trivial, the least prime $\ell$ with $\chi(\ell) \ne 1$ is below $(1.17 \ln m - 6.36)^2$.

The idea behind (iii) is short. If no prime up to $y$ were inert, then $\chi(n) = 1$ for every $n \le y$ prime to $\Delta$, since $\chi$ is multiplicative. So the sum $\sum_{n \le y} \chi(n)$ would be about $y$. The Polya-Vinogradov inequality says that this sum is at most about $\sqrt{|\Delta|}\ln|\Delta|$, so $y$ cannot be larger. Under GRH the same argument works with sums over primes, and the bound drops to a power of $\ln|\Delta|$.

**Example D.8** (How the bound is used).
For the 3-digit factorion $(p, q, x)$ the quantity $K(x)$ of Section 7.7 is of size about $x$. If $K(x)$ is not a square, the least inert prime of $\mathbb{Q}\big(\sqrt{K(x)}\big)$ is at most about $x^{0.45}$ by (ii), or a power of $\ln x$ under GRH. Either way it is below $x$ for large $x$, so it divides $x!$, and Lemma 7.45 excludes the string.

## D.5 Cyclotomic polynomials, Gauss sums and Ramanujan sums

The cyclotomic polynomial $\Phi_m(t)$ is the monic polynomial whose roots are the primitive $m$-th roots of unity. It has integer coefficients and degree $\varphi(m)$, and $t^n - 1 = \prod_{d \mid n} \Phi_d(t)$. For odd $x$,

$$x^x + 1 = \prod_{d \mid x} \Phi_{2d}(x),$$

so $x^x + 1 = (x+1)\Phi_{2x}(x)$ when $x$ is prime.

Two exponential sums appear in the proofs of Section 6.2. Let $\zeta$ be a primitive $m$-th root of unity.

- The Ramanujan sum $c_m(j) = \sum_a \zeta^{aj}$, over the residues $a$ prime to $m$. It is an integer, $c_m(j) = \sum_{d \mid \gcd(j,m)} \mu(m/d)d$ with the Moebius function $\mu$, and $|c_m(j)| \le \gcd(j, m)$.
- The Gauss sum of a character $\chi$ modulo $m$, $\tau(\chi) = \sum_a \chi(a)\zeta^a$. For the primitive quadratic character of a real quadratic field with discriminant $\Delta$, $\tau(\chi) = \sqrt{\Delta}$. The twisted sums $\sum_a \chi(a)\zeta^{aj}$ are $\chi(j)\tau(\chi)$ when $j$ is prime to $m$ and 0 otherwise.

See [73] and [79].

## D.6 Aurifeuillian factorizations

Some values of cyclotomic polynomials split for an algebraic reason. The oldest example is

$$2^{4k+2} + 1 = \big(2^{2k+1} - 2^{k+1} + 1\big)\big(2^{2k+1} + 2^{k+1} + 1\big).$$

In general, for squarefree $x \equiv 3 \pmod 4$ the polynomial $\Phi_{2x}(xt^2)$ factors over $\mathbb{Z}$ into two polynomials of degree $\varphi(x)$, and setting $t = 1$ splits $\Phi_{2x}(x)$ into two Aurifeuillian factors $L < M$. The reason is

that $\sqrt{x}$ lies in the field of $4x$-th roots of unity, so the roots split into two halves that the Galois group of $\mathbb{Q}(\zeta)$ over $\mathbb{Q}(\sqrt{x})$ preserves. Brent [80] and Stevenhagen [81] give formulas for the factors.

**Example D.9** (The factors behind base 904).
For $x = 3$: $\Phi_6(3t^2) = 9t^4 - 3t^2 + 1 = (3t^2 + 3t + 1)(3t^2 - 3t + 1)$, which gives $\Phi_6(3) = 7 = 7 \cdot 1$. Then $3^3 + 1 = 4 \cdot 1 \cdot 7$, and $b = 4 \cdot 1 = 4$, $b + 3 = 7$ solve $b(b + 3) = 3^3 + 1$: this is $(1, 3, 1)$ in base 4.

For $x = 7$: $\Phi_{14}(7) = 102943 = 113 \cdot 911$, an Aurifeuillian split with $L = 113$ and $M = 911$. Then $7^7 + 1 = 823544 = 8 \cdot 113 \cdot 911$, and $b = 8 \cdot 113 = 904$, $b + 7 = 911$ solve $b(b + 7) = 7^7 + 1$: this is $(1, 7, 1)$ in base 904. It works because $M/L = 8.06$ is close to $x + 1 = 8$. The library proves that $M/L \to e^2 \approx 7.39$ as $x$ grows (Theorem 6.3), so the coincidence cannot repeat for larger $x$.

The proof of Theorem 6.3 writes $\ln(M/L)$ as a sum over the roots in one half, expands $\ln(1 \pm \zeta^a/\sqrt{x})$ in powers of $x^{-1/2}$, and evaluates the resulting sums of roots of unity with the Ramanujan and Gauss sums above.

## D.7 Two reformulations of $b(b + x) = x^x + 1$

These illustrate how the tools above combine. Both are proven in L16.2.

- The $x$-adic form. Let $x$ be an odd prime. The polynomial $t^2 + xt - 1$ has the simple root $t \equiv 1$ modulo $x$, so by Hensel's lemma (Appendix C) it has a root $\beta \in \mathbb{Z}_x$ with $\beta \equiv 1$. Let $R$ be the residue of $\beta$ modulo $x^x$, in $[0, x^x)$. Then the equation has a solution exactly when $R < x^{x/2}$, and then $b = R$. For $x = 7$, $R = 904$ and $7^{3.5} \approx 907.5$. For a random residue this happens with probability about $x^{-x/2}$.
- The unit form. Let $x \equiv 3 \pmod 4$ be squarefree with $x^2 + 4$ prime, and $\varepsilon$ the fundamental unit of $\mathbb{Z}[\sqrt{x}]$. A solution gives $(2b + x) + 2x^{(x-1)/2}\sqrt{x} = \pm(x + 2 \pm 2\sqrt{x})\varepsilon^k$. For $x = 7$: $1815 + 686\sqrt{7} = \left(9 + 2\sqrt{7}\right)\left(8 + 3\sqrt{7}\right)^2$. So the equation asks whether one coordinate of the sequence $\gamma\varepsilon^k$ ever equals $2x^{(x-1)/2}$. For fixed $x$ this is a problem for linear forms in logarithms (Appendix E). The difficulty is that the field changes with $x$.

## D.8 Sums of two squares and the divisors of $n^2 + 1$

Theorem 7.38 counts the 2-digit narcissistic numbers of base $b$ by the divisors of $b^2 + 1$. This section collects the facts behind it. The first one is classical [73]. The others are research results.

**Theorem D.10** (Sums of two squares). CITED
Let $r_2(N)$ be the number of pairs of integers $(A, B)$ with $A^2 + B^2 = N$, counting signs and order. Then

$$r_2(N) = 4 \sum_{d \mid N} \chi(d),$$

where $\chi(d)$ is 0 for even $d$, 1 for $d \equiv 1 \pmod 4$ and $-1$ for $d \equiv 3 \pmod 4$.

The proof works in the Gaussian integers $\mathbb{Z}[i]$, where $A^2 + B^2 = (A + Bi)(A - Bi)$ is a norm. The ring $\mathbb{Z}[i]$ has unique factorization. A prime $p \equiv 1 \pmod 4$ splits into two conjugate Gaussian primes, a prime $p \equiv 3 \pmod 4$ stays prime, and $2 = -i(1 + i)^2$. Counting the elements of norm $N$ prime by prime gives the formula.

For $N = n^2 + 1$ the formula is simple. If an odd prime $p$ divides $n^2 + 1$, then $-1$ is a square modulo $p$, so $p \equiv 1 \pmod 4$. And 4 never divides $n^2 + 1$. So every odd divisor has $\chi = 1$, and $r_2(n^2 + 1) = 4\tau(m)$, where $m$ is the odd part of $n^2 + 1$ and $\tau$ counts divisors.

The size of $\tau(n^2+1)$ is a classical topic. Its divisors pair up as $d$ and $(n^2+1)/d$, with exactly one of each pair at most $n$, so $\tau(n^2+1)$ is twice the number of divisors $d \le n$. A given $d$ divides $n^2+1$ for $\nu(d)$ residues of $n$ modulo $d$, where $\nu(d)$ is the number of square roots of $-1$ modulo $d$. The values $\nu(d)$ are bounded on average, so summing over $n \le X$ gives about $X \sum_{d \le X} \nu(d)/d$, which grows like $X \ln X$. Three results make this precise or show its limits.

- Hooley [82] proved $\sum_{n \le X} \tau(n^2+1) = (3/\pi) X \ln X + O(X)$. So a typical $n^2+1$ has about $(3/\pi) \ln n$ divisors.
- Landau's problem. Is $n^2+1$ prime for infinitely many $n$? Landau listed it in 1912 among four problems on primes that he considered out of reach, and it is still open. The heuristic of Hardy and Littlewood predicts about $0.69X/\ln X$ such $n \le X$.
- Iwaniec [83] proved the best result in this direction: $n^2+1$ has at most two prime factors, counted with multiplicity, for infinitely many $n$.

So questions about $\tau(n^2+1)$ on average are solved, while questions about single values, such as how often $\tau(n^2+1) = 2$, are among the hardest open problems in number theory. Corollary 7.39 shows that the question "are there infinitely many bases without a 2-digit narcissistic number" is exactly of the second kind.

Appendix E

# Linear forms in logarithms

The strongest results of Chapter 6 bound the exponent $n$ in $b(b+x) = x^n + 1$, and Section 5.7 uses the same tool to exclude whole families of 3-digit strings. They follow one pattern, due to Baker. The equation forces a linear combination of logarithms of algebraic numbers to be extremely small. A deep theorem says that such a combination, if it is not zero, cannot be too small. Comparing the two bounds bounds the unknown. A computer then checks the remaining range, with continued fractions or a sieve. This appendix explains the pattern, first for the ordinary absolute value and then for the $p$-adic one. The book of Baker and Wustholz [84] is the standard reference, and Smart [85] explains the computational side.

## E.1 Two powers that come close

Start with a question that needs no algebraic numbers. How close can a power of 2 come to a power of 3? If $2^a$ and $3^c$ are close, then their ratio is near 1, so

$$\Lambda = a \ln 2 - c \ln 3 = \ln(2^a / 3^c)$$

is near 0. This $\Lambda$ is a *linear form in logarithms*: integers times logarithms of fixed numbers. It is never 0, because $2^a \neq 3^c$ for $a, c \geq 1$. The question is how small it can be.

A trivial bound is easy. $|2^a - 3^c| \geq 1$, so $|\Lambda|$ is at least about $3^{-c}$, which is exponentially small in the exponents. That bound is useless for equations, because the equations we meet also make $\Lambda$ exponentially small. Baker's theorem replaces it with a bound that is only polynomially small in the exponents. So if an equation forces $\Lambda$ to be exponentially small, the exponents cannot be large.

For 2 and 3 the answer is known in a strong form: $|2^a - 3^c|$ tends to infinity, and the solutions of $|2^a - 3^c| = 1$ are $(a, c) = (1, 1), (2, 1), (3, 2)$. The proofs of such statements for other numbers all go through a lower bound for $\Lambda$. To state it for algebraic numbers we need a way to say how complicated a number is.

## E.2 Heights

Lower bounds for linear forms depend on how complicated the algebraic numbers are. The measure is the height.

**Definition E.1** (Logarithmic height).
Let $\alpha$ be algebraic of degree $d$ with minimal polynomial $a_0(t - \alpha_1)...(t - \alpha_d)$ over $\mathbb{Z}$, with coprime integer coefficients. Its absolute logarithmic height is

$$h(\alpha) = \frac{1}{d}\left( \ln|a_0| + \sum_{i=1}^{d} \ln \max(1, |\alpha_i|) \right).$$

Think of the height as the number of digits needed to write the number down, measured with natural logarithms. For a rational number $p/q$ in lowest terms, $h(p/q) = \ln \max(|p|, |q|)$. So $h(2) =$

$\ln 2$ and $h(1000/7) = \ln 1000$: the height grows with the size of the numerator and the denominator, not with the size of the number. For a unit $\varepsilon > 1$ of a real quadratic field, $h(\varepsilon) = (\ln \varepsilon)/2$.

Three rules make heights easy to bound:

- $h(\alpha\beta) \le h(\alpha) + h(\beta)$ and $h(\alpha + \beta) \le h(\alpha) + h(\beta) + \ln 2$;
- $h(\alpha^n) = |n| h(\alpha)$, so in particular $h(1/\alpha) = h(\alpha)$;
- for an algebraic integer, $a_0 = 1$, so the height is the average of $\ln \max(1, |\alpha_i|)$ over the conjugates. If every conjugate of an integer of a quadratic field has absolute value at most $Y$, its height is at most $\ln Y$.

The last rule is how the 3-digit argument of Section 5.7 keeps heights small: it replaces a large number by one of the same norm whose two conjugates are both of moderate size. For the number $z$ of Section 6.5, a root of $xT^2 - 4T - x$, the height is $(\ln(xz))/2$ (L18.1).

## E.3 Baker's theorem and its explicit forms

Let $\alpha_1, ..., \alpha_r$ be positive real algebraic numbers and $b_1, ..., b_r$ integers, and put

$$\Lambda = b_1 \ln \alpha_1 + ... + b_r \ln \alpha_r, \quad B = \max |b_i|.$$

If the $\alpha_i$ are multiplicatively dependent, $\Lambda$ can vanish. Otherwise $\Lambda \neq 0$, and the question is how small it can be. Trivial estimates give nothing better than $|\Lambda| > e^{-cB}$, which is useless. Baker proved a lower bound that is only polynomial in $B$.

**Theorem E.2** (Baker). CITED
If $\Lambda \neq 0$, then $|\Lambda| > B^{-C}$, where $C$ is an effectively computable constant that depends only on $r$, on the degree of the field generated by the $\alpha_i$, and on their heights.

See [86] and [84]. For applications one needs explicit constants. Two versions appear in the library.

**Theorem E.3** (Matveev). CITED L20.3
Let the $\alpha_i$ lie in a number field of degree $D$, let $\Lambda \neq 0$, and let $A_i \ge \max(Dh(\alpha_i), |\ln \alpha_i|, 0.16)$. Then

$$\ln |\Lambda| > -1.4 \cdot 30^{r+3} r^{4.5} D^2 (1 + \ln D)(1 + \ln B) A_1 ... A_r.$$

This is from [38], in the form of [87]. For three logarithms in a quadratic field the constant is about $9.7 \cdot 10^{11}$.

For two logarithms Laurent's bound [22] is far sharper. Its shape is

$$\ln |\Lambda| \ge -C D^4 (\max(\ln b' + 0.38, 10/D, 1))^2 \ln A_1 \ln A_2, \quad b' = \frac{b_1}{D \ln A_2} + \frac{b_2}{D \ln A_1},$$

with $\ln A_i \ge \max(h(\alpha_i), (\ln \alpha_i)/D, 1/D)$, for $\Lambda = b_2 \ln \alpha_2 - b_1 \ln \alpha_1$ with positive $b_i$. The library uses $C = 25.2$, Corollary 2 of [22] with the parameter $m = 10$. The library took these constants from a secondary source and notes that they must be checked against the paper (L18.1). The structure of the argument does not depend on the exact constants, but the numerical ranges do.

The important feature of both bounds: the dependence on $B$ is through $\ln B$ (Matveev) or $(\ln B)^2$ (Laurent). An equation that makes $\Lambda$ exponentially small in $B$ therefore bounds $B$.

## E.4 The $p$-adic version

The $p$-adic absolute value of Appendix C measures a different kind of closeness: two integers are 2-adically close when their difference is divisible by a high power of 2. For example, $3^{2^k} - 1$ is divisible by $2^{k+2}$, so the powers $3^{2^k}$ come 2-adically close to 1. The analogue of Baker's question

asks how large the valuation $v_p\left(\alpha_1^{b_1} - \alpha_2^{b_2}\right)$ can be. The trivial answer is again useless: nothing obvious stops it from being as large as the size of the numbers. Kunrui Yu proved $p$-adic analogues of Baker's theorem, and for two numbers Bugeaud and Laurent [88] gave a sharp explicit form.

To use valuations on algebraic numbers, extend $v_p$ from $\mathbb{Q}$ to a number field $K$. For a quadratic field this is concrete: choose a root of the defining polynomial in an extension of the $p$-adic numbers, and measure each element of $K$ there. The extension keeps the rules $v(\alpha\beta) = v(\alpha) + v(\beta)$ and $v(\alpha + \beta) \geq \min(v(\alpha), v(\beta))$, with $v(p) = 1$. An algebraic integer has $v \geq 0$. A unit has $v = 0$.

**Theorem E.4** (Bugeaud and Laurent). CITED L32.1

Let $p$ be a prime and let $\alpha_1, \alpha_2$ be multiplicatively independent algebraic numbers with $v_p(\alpha_1) = v_p(\alpha_2) = 0$. Let $D = [\mathbb{Q}(\alpha_1, \alpha_2) : \mathbb{Q}]$, and let $g$ be the least positive integer with $v_p(\alpha_i^g - 1) > 0$ for both $i$. Let $\ln A_i \geq \max(h(\alpha_i), (\ln p)/D)$ and $b' = b_1/(D \ln A_2) + b_2/(D \ln A_1)$. Then for positive integers $b_1, b_2$,

$$v_p\left(\alpha_1^{b_1} - \alpha_2^{b_2}\right) \leq \frac{24pg}{(p-1)(\ln p)^4} D^4 (\max(\ln b' + \ln\ln p + 0.4, 10(\ln p)/D, 10))^2 \ln A_1 \ln A_2.$$

This is Theoreme 3 of [88], in the simpler form of Bennett and Bugeaud [27]. The shape is the same as in Laurent's real bound: the exponents enter only through $(\ln b')^2$, and the heights through the product $\ln A_1 \ln A_2$. So if an equation makes $\alpha_1^{b_1} - \alpha_2^{b_2}$ divisible by a power of $p$ that grows linearly in the exponents, the exponents are bounded.

The number $g$ is small. In a quadratic field the residues modulo the prime above $p$ form a field with at most $p^2$ elements, so $\alpha^{p^2-1} \equiv 1$ for every $\alpha$ with $v(\alpha) = 0$, and $g \leq p^2 - 1$. For $p = 2$ this gives $g \leq 3$, and the constant in front becomes $24 \cdot 2 \cdot 3 \cdot 2^4/(\ln 2)^4 \approx 9991$.

Why use the 2-adic form at all? In the 3-digit families of Section 5.7 the equation says that two conjugate numbers differ by $2V\sqrt{n}$, where $V$ contains the factor $x^{x/2}$. If $x$ is even, this difference is divisible by $2^{x/2}$, a power of 2 that grows linearly in $x$. The real size of the difference is huge, so the real form of Baker's theorem has nothing to say about it; its 2-adic size is tiny, and that is what the 2-adic theorem measures.

## E.5 The method on an example

Here is the argument for the pure solutions of $b(b + x) = x^n + 1$ with odd $x$ (Theorem 18.7 of L18.1), stripped of constants.

1. The small linear form. In the quadratic field $\mathbb{Q}\left(\sqrt{x^2+4}\right)$ with fundamental unit $\eta_0$, $\mu_0 = \ln \eta_0$, a pure solution gives an odd integer $s$ with
$$|n \ln z - s\mu_0| = \Lambda_0 < \tau_n \approx \sqrt{x^2+4} \cdot x^{-n/2},$$
where $z = (\varepsilon + 1)/(\varepsilon - 1)$ and $\ln z \approx 2/x$ (Proposition 18.4). The left side is a linear form in the two logarithms $\ln z$ and $\ln \eta_0$, with coefficients $n$ and $s$. The right side is exponentially small in $n$.
2. The lower bound. Laurent's theorem gives $\ln \Lambda_0 \geq -C'(\ln n)^2 \ln x \cdot \max(\mu_0, 1)$ for a constant $C'$.
3. Comparison. So $(n/2 - 1)\ln x \lessapprox C'(\ln n)^2 \ln x \max(\mu_0, 1)$. This bounds $n/\max(\mu_0, 1)$: the library gets $n < 22000 \max(\mu_0, 1)$ once $x \geq 10^5$.
4. A second inequality. Since $s \geq 1$, the same linear form gives $n \ln z \geq \mu_0 - \tau_n$, so $n$ is at least about $x\mu_0/2$. With step 3 this forces $x < 10^5$.
5. The finite range. For each odd $x < 10^5$ the computer finds $\mu_0$, the largest $n$ that step 3 allows, and the candidates for $n$ by continued fractions (below). Only $(x, n) = (7, 7)$ survives.

The mixed solutions follow the same plan with an ideal class of order $d$ (Theorem 18.10). The even case $x \equiv 0 \pmod 8$ leaves a linear form in three logarithms that is inhomogeneous: an extra fixed logarithm $\lambda$ appears. Laurent's theorem no longer applies, Matveev's bound is much weaker, and there is no analogue of step 4, so $n$ is bounded in terms of $x$ but not $x$ in terms of $n$ (L20.3).

## E.6 Reduction with continued fractions

The bounds from Baker's method are huge, for example $n <$ several million for each $x$. Continued fractions cut them down. For a homogeneous form, step 1 says

$$\left| \ln \frac{z}{\mu_0} - \frac{s}{n} \right| < \frac{\tau_n}{n\mu_0}.$$

Once $n\tau_n < \mu_0/2$, the right side is below $1/(2n^2)$, and Legendre's theorem (Section D.2) says that $s/n$ is a convergent of $\ln z/\mu_0$. So the candidates are the convergents with denominator up to the bound, a list of a few dozen, each tested exactly.

For an inhomogeneous form $|j\theta - k + \lambda| < AB_0^{-j}$, with known reals $\theta, \lambda$ and constants $A > 0$, $B_0 > 1$, the tool is the lemma of Dujella and Petho [39].

**Lemma E.5** (Dujella and Petho). CITED

Let $M$ be an upper bound for $j$, let $p/q$ be a convergent of $\theta$ with $q > 6M$, and let $\varepsilon = \|q\lambda\| - M\|q\theta\|$, where $\|\cdot\|$ is the distance to the nearest integer. If $\varepsilon > 0$, there is no solution with $\ln(Aq/\varepsilon)/\ln B_0 \le j \le M$.

One application of the lemma usually reduces a bound of $10^{19}$ to a few hundred. The library uses it for the even case (Theorem 20.19 of L20.3).

## E.7 Where the method stops

Baker's method needs a fixed number field, or a family of fields with uniformly controlled units. The same limit appears for 3 digits: the method settles the families in which the field $\mathbb{Q}(\texttt{sqrt}\ n)$ does not change with the large digit, and stops where it does (Section 5.7). In Chapter 6 the field $\mathbb{Q}\left(\sqrt{x^2+4}\right)$ changes with $x$. The method still works for pure solutions, because the generator $\varepsilon + 1$ is explicit for every $x$ and $\mu_0 \le \ln \varepsilon \approx \ln x$. It stops at mixed solutions with a non-principal ideal (Corollary 6.20), and for $b(b+x) = x^n + 3$, where the regulator is not bounded by $\ln x$ (L20.4).

Related questions about powers are open even in the simplest case. Mahler [25] proved that the distance from $(3/2)^k$ to the nearest integer exceeds $c^k$ for every $c < 1$ and large $k$, without an effective bound; Zudilin [26] gave the effective constant 0.5803. Whether the fractional parts of $(3/2)^k$ are dense in $[0,1]$ is unknown. The finiteness problems for short solutions (Section 5.6) ask how close $x^{(x-1)/(k-1)}$ comes to an integer, a question of the same kind with base and exponent moving together.

The method also appears in complexity theory. To build the hard instances of Section 4.7 from pure powers $x^e$, one would need many digits $x$ with $x^{e-1}$ very close to a power of $b$. Laurent's theorem with $D = 1$ shows that this needs exponents of size $\exp\left(c\sqrt{k/\ln k}\right)$ for $k$ items, so the reduction has no polynomial-size analogue (L18.6).

Appendix F

# The abc conjecture

Several finiteness statements in Chapter 5, Chapter 6 and Chapter 7 are proven only on the assumption of the abc conjecture. This appendix states the conjecture, explains how it is measured, and shows how it applies to the equations of the main text. Granville and Tucker [35] give a readable survey.

## F.1 The radical and the conjecture

**Definition F.1** (Radical and quality).
The radical $\operatorname{rad}(n)$ of a positive integer $n$ is the product of the distinct primes that divide $n$. For coprime positive integers $a + b = c$, the quality of the triple is

$$q(a, b, c) = \frac{\ln c}{\ln \operatorname{rad}(abc)}.$$

For most triples $\operatorname{rad}(abc)$ is much larger than $c$, and the quality is below 1. A high quality needs $a$, $b$ and $c$ to be divisible by high powers of small primes. For example $1 + 8 = 9$ has radical 6 and quality $\ln 9 / \ln 6 \approx 1.226$.

**Conjecture F.2** (The abc conjecture of Masser and Oesterle). CONJECTURE
For every $\delta > 0$ there is a constant $K_\delta$ such that every coprime triple $a + b = c$ of positive integers satisfies $c < K_\delta \operatorname{rad}(abc)^{1+\delta}$. Equivalently, for every $\delta > 0$ only finitely many triples have quality above $1 + \delta$.

Infinitely many triples have quality above 1, so the exponent $1 + \delta$ cannot be replaced by 1. The best known triple is due to Reyssat:

$$2 + 3^{10} \cdot 109 = 23^5, \quad q = \frac{\ln 23^5}{\ln(2 \cdot 3 \cdot 109 \cdot 23)} \approx 1.6299.$$

The conjecture is widely believed and widely used as a heuristic guide. A proof has been claimed (Mochizuki, 2012), but that proof is not accepted by most experts, and this document treats the conjecture as open. Much weaker forms are also unproven: no one knows how to show that $c < K \operatorname{rad}(abc)^{2-\delta}$ for some fixed $\delta > 0$. The best unconditional bounds, from linear forms in logarithms (Appendix E), are exponential in a power of the radical.

## F.2 An explicit form

The constant $K_\delta$ in the conjecture is not known, so the conjecture alone gives finiteness but no explicit bound. A. Baker proposed an explicit version. Laishram and Shorey [89] showed that Baker's explicit conjecture implies a simple statement.

**Conjecture F.3** (An explicit abc inequality). CONJECTURE
Every coprime triple $a + b = c$ of positive integers satisfies $c < \mathrm{rad}(abc)^{7/4}$.

This is much weaker than the full conjecture for large triples, but it has no unknown constant. It turns a finiteness statement into a bound that a computer can reach. The exposition uses it in Corollary 5.55 and Corollary 7.48. No triple with quality above 1.63 is known, so the exponent $7/4$ leaves a wide margin.

The conjecture is about three terms. Browkin and Brzezinski [28] proposed a version for $n$ terms, with exponent $2n - 5 + \delta$. For four terms this is 3, too weak for the equations of the library with two large digits.

## F.3 Consequences

The conjecture implies many finiteness theorems at once. Two examples near the main text:

- Brocard's equation $n! + 1 = m^2$ has only finitely many solutions (Overholt [36]). The known ones are $n = 4, 5, 7$.
- Fermat's equation $x^p + y^p = z^p$ has no solution with $xyz \neq 0$ for all large $p$, because a solution would be a triple of quality at least $p/3$.

The pattern in both: an equation produces a triple $a + b = c$ in which two terms are high powers or have only small prime factors, so the radical is much smaller than $c$.

## F.4 Use in the main text

**Example F.4** (The equation $b(b + x) = x^n + 1$).
Completing the square gives $(x^2 + 4) + 4x^n = (2b + x)^2$ for odd $x$. This is a coprime triple, and its radical is at most $2x(x^2 + 4)(2b + x)$, which is about $4x^{3+n/2}$, while $c$ is about $4x^n$. The quality is therefore about $n/(3 + n/2)$, which tends to 2. So the full abc conjecture implies that the equation has only finitely many solutions with $n \geq 7$, and even the weak form with exponent $2 - \delta$ is enough (Section 6.6). For the diagonal $n = x$ the library sieves every $x \leq 4 \cdot 10^9$, so any further solution of $b(b + x) = x^x + 1$ would be a triple of quality above $2 - 10^{-8}$ (L16.2). The record is 1.6299.

**Example F.5** (Cores in perfect-power bases).
Under $0^0 = 0$, a 2-digit solution $(a, c)$ in a base $b = r^t$ with $t \geq 2$ gives $ar^t = a^a + c^c - c$. When one of $a^a$ and $c^c$ is much larger than the other, this is an abc triple with two large terms that are powers, and the conjecture leaves only finitely many such solutions (Section 6.9, Theorem 15.10 of L15.3). The case $a^a \approx c^c$ has four large terms and is out of reach of abc.

**Example F.6** (Three-digit strings and factorions).
A 3-digit string with one large digit $x$ becomes $U^2 = N + R$ after completing the square, with $N = 4\ell x^x$ and $R$ small (Theorem 5.46). The radical of $N$ is at most $2\ell x$, tiny compared with $N$, and $U$ is about $\sqrt{N}$. So the triple has quality close to 2, and abc bounds $x$ by about twice the other digits (Theorem 5.54). For factorions, $N = 4\ell \cdot x!$, and the radical of $x!$ is the product of all primes up to $x$, about $e^x$. The quality is now only about $\ln(x!)/(x + \ln(x!)/2)$, which still tends to 2 but slowly. With the explicit exponent $7/4$ the bound is $x \leq 4{,}103{,}641$, almost independent of the other digits (Corollary 7.48).

So abc would settle several of the open problems of Chapter 8 in one stroke, but none of them needs the full strength of abc. The weak form with exponent $2 - \delta$, which is also open, would already finish the family $(1, x, 1)$.

Appendix G

# Elliptic and genus 2 curves

For a fixed exponent $n$, the equation $b(b+x) = x^n + 1$ of Chapter 6 is the question which integer points lie on the plane curve

$$U^2 = 4x^n + x^2 + 4, \quad U = 2b + x.$$

For $n = 3$ this is an elliptic curve, and the question is settled. For $n = 5$ it is a curve of genus 2, and the question is decidable in principle but not decided. This appendix explains the difference. Silverman [90] is the standard text on elliptic curves. McCallum and Poonen [91] survey the method of Chabauty and Coleman.

## G.1 Curves and their genus

A curve $y^2 = f(x)$, with $f$ a polynomial with rational coefficients and no repeated roots, has genus $g = \lfloor (\deg f - 1)/2 \rfloor$. Genus 0 curves (degree 1 or 2) have either no rational point or infinitely many, parametrized by rational functions. Genus 1 curves with a rational point are elliptic curves. Curves of genus at least 2 are different in kind.

**Theorem G.1** (Faltings). CITED
A curve of genus at least 2 over $\mathbb{Q}$ has only finitely many rational points.

See [33]. The proof does not say how to find the points. For the equation above, $U^2 = 4x^n + x^2 + 4$ has genus $\lfloor (n-1)/2 \rfloor$, which is at least 2 for $n \geq 5$. So for each fixed $n \geq 5$ there are only finitely many solutions (Proposition 18.13 of L18.1). This is the "each $n$ separately" half of the finiteness question; abc would give the uniform half (Appendix F).

## G.2 Elliptic curves

An elliptic curve over $\mathbb{Q}$ can be written $y^2 = x^3 + ax^2 + cx + d$ with a cubic without repeated roots, together with a point at infinity. Its rational points form an abelian group under the chord and tangent construction: three points add to zero when they lie on a line.

**Theorem G.2** (Mordell-Weil). CITED
The group $E(\mathbb{Q})$ of rational points of an elliptic curve is finitely generated: $E(\mathbb{Q}) \cong \mathbb{Z}^r \times T$ with $T$ finite. The number $r$ is the rank.

Integral points, with $x$ and $y$ integers, are much rarer than rational points.

**Theorem G.3** (Siegel). CITED
An elliptic curve given by an equation with integer coefficients has only finitely many integral points.

Siegel's proof is not effective. Baker's method (Appendix E), applied to elliptic logarithms, makes it effective: given generators of $E(\mathbb{Q})$, one can bound and then list all integral points. PARI and Sage implement this.

**Example G.4** (The exponent $n = 3$).
For $n = 3$, multiply $U^2 = 4x^3 + x^2 + 4$ by 16 and put $X = 4x$, $Y = 4U$:

$$Y^2 = X^3 + X^2 + 64.$$

This elliptic curve has conductor 433 and rank 2, and exactly 10 integral points (computed with Sage, L16.2, L18.1). They give the solutions $x = 3, 5, 8, 64, 119$ of $b(b + x) = x^3 + 1$. For instance $x = 3$ gives $U^2 = 108 + 9 + 4 = 121$, $U = 11$ and $b = 4$: the string $(1, 3, 1)$ satisfies $b(b + 3) = 3^3 + 1$ in base 4. The point with $x = 119 = 7 \cdot 17$ is the mixed solution that shows the limits of the method of Section 6.5. The same computation with $x^n + 25$ in place of $x^n + 1$ gives the curve $Y^2 = X^3 + X^2 + 1600$ (L20.4).

The rank is hard to compute in general. Its analytic counterpart is the order of vanishing at $s = 1$ of the $L$-function of the curve, the analytic rank. The conjecture of Birch and Swinnerton-Dyer says that the two ranks agree. It is proven when the analytic rank is 0 or 1, and open otherwise.

## G.3 Genus 2 curves and Chabauty's method

For a curve $C$ of genus $g \geq 2$, the rational points embed into the Jacobian $J$, an abelian variety of dimension $g$ whose rational points again form a finitely generated group $J(\mathbb{Q}) \cong \mathbb{Z}^r \times T$.

**Theorem G.5** (Chabauty, Coleman). CITED
If the rank $r$ of $J(\mathbb{Q})$ is smaller than the genus $g$, then $p$-adic integration on $J$ produces a nonzero function that vanishes on all rational points of $C$. This gives an explicit finite set that contains all of them, and often the exact set.

See [92] and [93]. The condition $r < g$ is essential: the closure of $J(\mathbb{Q})$ inside the $p$-adic points of $J$ has dimension at most $r$, and when $r < g$ it meets the curve in finitely many points. A more recent method, quadratic Chabauty, handles some curves with $r = g$. For $r > g$ neither applies.

**Example G.6** (The exponent $n = 5$).
For $n = 5$ the curve is $b^2 + xb = x^5 + 1$, of genus 2, with conductor $800027 = 331 \cdot 2417$. Its known rational points have $x \in \{-1, 0, 1, 1/64\}$, plus the points at infinity. PARI computes the analytic rank of the Jacobian as 3 (L18.1). If the rank is 3, it exceeds the genus, so neither Chabauty-Coleman nor quadratic Chabauty applies. What remains is Baker's method for integral points on hyperelliptic curves, combined with the Mordell-Weil sieve (Bugeaud, Mignotte, Siksek, Stoll and Tengely [34]). It needs a 2-descent on the Jacobian and a basis of $J(\mathbb{Q})$, which the available software did not provide here. So $n = 5$ is decidable but not decided. A sieve shows that there is no solution with $x \leq 10^{10}$.

For larger odd $n$ the genus grows, and the same tools apply in principle, one curve at a time. They cannot treat all $n$ at once, and the diagonal $n = x$, the case of Munchausen numbers, is an infinite family of curves of growing genus. This is why Chapter 6 uses the quadratic field and linear forms in logarithms for the uniform results, and curves only for small $n$.

Appendix H

# Complexity: NP, #P and sparse sets

Section 4.7 asks whether Munchausen numbers can be found fast, and why the answer is out of reach. The tools are the classes P and NP, reductions, counting classes, and the special status of problems with very few instances. This appendix explains them. Garey and Johnson [94] and Arora and Barak [95] are standard references; [21] is a short introduction.

## H.1 P, NP and certificates

A decision problem is a set $L$ of strings over a finite alphabet: the instances whose answer is "yes". The size of an instance is its length as a string. So the encoding matters. The base $b$ written in binary has about $\log_2 b$ symbols; written in unary, as $1^b$, it has $b$ symbols.

**Definition H.1** (P and NP).
$L$ is in P if a deterministic algorithm decides membership in time polynomial in the size $s$ of the instance. $L$ is in NP if there is a polynomial $p$ and a polynomial-time verifier $V$ such that an instance $u$ is in $L$ exactly when some certificate $c$ of length at most $p(s)$ makes $V(u, c)$ accept.

A Munchausen number is its own certificate. Let $L_m$ be the set of strings $1^b$ such that base $b$ has a nontrivial solution under $0^0 = m$. A solution has at most $b$ digits (Proposition 1.3), so it can be written with $O(b \log b)$ bits, and checking $F(n) = n$ takes time polynomial in $b$. So $L_m$ is in NP (Proposition 4.21). Note the unary encoding: in binary the same problem would be in NEXP, a much larger class, because a solution is exponentially long in $\log b$.

## H.2 Reductions and NP-completeness

A polynomial-time reduction from $L$ to $L'$ is a polynomial-time map $f$ with $u \in L$ exactly when $f(u) \in L'$. A problem is NP-hard if every problem in NP reduces to it, and NP-complete if it is also in NP. The first NP-complete problem was SAT, the satisfiability of Boolean formulas (Cook and Levin). Thousands followed. The one used in the library is SUBSET-SUM.

**Definition H.2** (SUBSET-SUM).
Given positive integers $a_1, ..., a_k$ and a target $T$, all in binary, is there a set $S$ of indices with $\sum_{i \in S} a_i = T$?

SUBSET-SUM is NP-complete [94]. It is hard only because the numbers are written in binary: dynamic programming solves it in time polynomial in $k$ and $T$. The Munchausen equation looks like a subset sum, $\sum_i (s_i b^i - w(s_i)) = 0$, but with the digits as unknowns and the target equal to the number itself.

The library proves that the problem with arbitrary weights is NP-complete (Theorem 4.24). The instance is a base $b$ in unary and a table of weights $w(0), ..., w(b-1)$ in binary, and the question is whether some $n \geq b$ equals the sum of the weights of its digits. The reduction from SUBSET-SUM

builds a weight table whose solutions correspond one to one to the subsets with sum $T$. Each item $a_i$ becomes a digit $x_i$ at a position $P_i$, with weight $x_i b^{P_i} + p a_i$ for a prime $p$ dividing $b-1$; the target becomes a mandatory digit with weight $z b^{P_z} - pT$; every other digit gets the weight $y+1$, and the digit-sum identity modulo $p$ (Proposition 1.9) forbids it. The details are in L13.1.

## H.3 Counting: #P

**Definition H.3** (#P).
A function $f$ from strings to nonnegative integers is in #P if there is a polynomial-time verifier $V$ and a polynomial $p$ such that $f(u)$ is the number of certificates $c$ of length at most $p(s)$ that make $V(u, c)$ accept.

A reduction is parsimonious if it preserves the number of solutions, not only their existence. Valiant [96] introduced #P and showed that counting can be hard even when deciding is easy. #SAT is #P-complete under parsimonious reductions, and so is #SUBSET-SUM. The reduction above is parsimonious, so counting the solutions of a general weight table is #P-complete. The counting function $b \mapsto N_m(b)$ of the real problem is in #P.

## H.4 Sparse and tally sets

$L_m$ has a special shape: for each length it contains at most one string, $1^b$. Such a set is a tally set. A set is sparse if it contains at most polynomially many strings of each length. Sparse sets carry very little information, and this limits what one can prove about them.

**Theorem H.4** (Berman, Mahaney). CITED
If a sparse set is NP-hard under polynomial-time many-one reductions, then P = NP.

Berman [18] proved this for tally sets, and Mahaney [19] for all sparse sets. So, unless P = NP, $L_m$ is not NP-complete, and no reduction can give evidence that Munchausen numbers are hard to find. On the other side, $L_m$ is in NP, so a proof that $L_m$ is not in P would prove P $\neq$ NP. Book's theorem sharpens this.

**Theorem H.5** (Book). CITED
NP contains a tally set that is not in P if and only if NE $\neq$ E. Here E is the class of problems solvable in time $2^{O(s)}$, and NE its nondeterministic version.

See [20]. So a proof that no polynomial algorithm finds Munchausen numbers would prove NE $\neq$ E, a statement stronger than P $\neq$ NP. Both directions are therefore closed to current methods: a lower bound would separate complexity classes, and hardness would collapse them. This is the barrier of Section 4.7. What remains possible is an algorithm that uses the arithmetic of $x^x$. Every algorithm that treats the weights as an arbitrary table is subject to the NP-completeness of the general problem.

## H.5 Lattices and low-density subset sums

Some subset-sum instances are easy. If the numbers $a_i$ are very large compared with their count $k$, the density $k/\log_2 \max a_i$ is small, and the solution is usually the shortest vector of a lattice built from the $a_i$. Lattice reduction (the LLL algorithm) then finds it; Lagarias and Odlyzko [14] analyse this. The Munchausen equation, seen as a subset sum over digit placements, has density about $b/\log_2 b$, far above 1. At that density the relaxed lattice problem has many short vectors, and the method fails (L6.3). The best generic algorithms for random subset sums (Howgrave-Graham and

Joux [15], Becker, Coron and Joux [16]) also do not apply, because here the target is the unknown itself (Section 4.5).

Appendix I

# Numeration systems

Chapter 7 asks the Munchausen question in number systems other than the positive integer bases: negative bases, complex bases, rational bases, a Pisot base, the factorial system, and polynomial bases over a finite field (Appendix J). Each system needs three things: a digit set, a proof that every number has exactly one expansion, and a description of which residues the digit multiset fixes. This appendix supplies them. Knuth [97] covers negative and complex bases with many historical notes.

## I.1 The common pattern: the identity modulus

In base $b$ every integer satisfies $n \equiv$ (digit sum) $\pmod{b-1}$, because $b \equiv 1 \pmod{b-1}$. The same argument works for any base $\beta$ in a ring: modulo $\beta - 1$ every power of $\beta$ is 1, so

$$n = \sum_i s_i \beta^i \equiv \sum_i s_i \pmod{\beta - 1}.$$

When $\beta$ is an algebraic integer and $\beta - 1$ is primitive (not divisible by a rational integer greater than 1), a rational integer is divisible by $\beta - 1$ exactly when it is divisible by the norm $N(\beta - 1)$. So the identity becomes a congruence modulo the integer $M = |N(\beta - 1)|$, the identity modulus. For a Munchausen number $n = F(n)$ it reads $\sum(w(x) - x) \equiv 0 \pmod{M}$, as in Proposition 1.9.

No other modulus is a function of the digit multiset. Swapping two adjacent digits $x$ and $y = x - 1$ at the bottom changes the value by $(x - y)(\beta - 1) = \beta - 1$ and keeps the multiset. So any modulus that the multiset determines divides $\beta - 1$ (Proposition 1.12). Section 7.1 shows that this one modulus governs the parity effect in every system: the expected count doubles when $M$ is even and $w(x) \equiv x \pmod 2$.

| system | digits | identity modulus $M$ | parity doubling under $0^0 = 0$ |
|---|---|---|---|
| base $b$ | $0, \ldots, b-1$ | $b - 1$ | $b$ odd |
| base $-b$ | $0, \ldots, b-1$ | $b + 1$ | $b$ odd |
| base $-n + i$ | $0, \ldots, n^2$ | $(n+1)^2 + 1$ | $n$ even |
| base $-n + \omega$ | $0, \ldots, n^2 + n$ | $n^2 + 3n + 3$ | never |
| base $p/q$ | $0, \ldots, p-1$ | $p - q$ | $p, q$ odd |
| base $\varepsilon_x$ | $0, \ldots, x$ | $x$ | $x$ even |
| factorial | $0, \ldots, i$ at position $i$ | none | never |

## I.2 Negative bases

**Theorem I.1** (Negative bases). CITED
Let $b \geq 2$. Every integer $n$, positive, negative or zero, has exactly one expansion $n = \sum_i s_i(-b)^i$ with digits $0 \leq s_i \leq b - 1$.

The proof is division with remainder: $s_0$ is the residue of $n$ modulo $b$ in $\{0, \ldots, b-1\}$, and the rest is the expansion of $(n - s_0)/(-b)$. Negative bases go back to Grunwald in 1885 (see [97]).

Since $-b \equiv 1 \pmod{b+1}$, the identity modulus is $b+1$. Two facts are used by the search engine of Section 7.2 (L14.1):

- The integers with at most $j$ digits form an interval $[-m_j, M_j]$, where $M_j$ sums $(b-1)b^i$ over the even positions $i < j$ and $m_j$ over the odd ones. Fixing the top digits gives a translate of this interval. So the prefix tree of Section 4.1 works.
- A positive integer has an odd number of digits, because the leading term outweighs the rest and has the sign of $(-b)^{k-1}$. So Munchausen numbers in negative bases have odd length.

Example: in base $-10$, $404201091 = (4,0,5,8,1,9,1,1,1)$, and it equals $4^4 + 1 + 5^5 + 8^8 + 1 + 9^9 + 1 + 1 + 1$ under $0^0 = 1$ (L12.3).

## I.3 Complex bases and canonical number systems

**Definition I.2** (Canonical number system).
Let $\beta$ be an algebraic integer with minimal polynomial $P$ and let $N = |P(0)|$. The pair $(\beta, \{0, 1, ..., N-1\})$ is a canonical number system if every element of $\mathbb{Z}[\beta]$ has a finite expansion $\sum_i s_i \beta^i$ with digits in $\{0, ..., N-1\}$.

Expansions in a canonical number system are automatically unique, since the digits form a complete set of residues modulo $\beta$. Katai and Szabo [40] showed that the Gaussian integers $-n + i$, $n \geq 1$, are canonical number system bases, with digits $0, ..., n^2$. For quadratic bases there is a complete criterion.

**Theorem I.3** (Quadratic canonical number systems). CITED
A root $\beta$ of an irreducible polynomial $t^2 + Bt + C$ with integer coefficients is a canonical number system base exactly when $-1 \leq B \leq C$ and $C \geq 2$.

See Katai and Kovacs [41] and Gilbert [42]. The Eisenstein bases $\beta = -n + \omega$, with $\omega^2 + \omega + 1 = 0$, are roots of $t^2 + (2n+1)t + (n^2 + n + 1)$, which satisfies the criterion. Their identity modulus is $N(\beta - 1) = n^2 + 3n + 3$, an odd number for every $n$. That is why the parity effect is absent there (Section 7.3).

Imaginary bases $ci$ with $c \geq 2$ and digits $0, ..., c^2 - 1$ reduce to negative bases: $(ci)^2 = -c^2$, so the digits at even positions form an expansion in base $-c^2$, and a rational integer has zeros at all odd positions. Knuth's quater-imaginary base $2i$ is the case $c = 2$.

In complex bases the expansions of positive integers do not use every length. In base $-1 + i$ one has $(-1+i)^4 = -4$, so an integer is its base $-4$ expansion written in blocks of four binary digits, and its length is $\equiv 1$ or $4 \pmod 8$ (L23.4).

## I.4 Rational bases

Akiyama, Frougny and Sakarovitch [43] defined a number system with base $p/q$, for coprime $p > q \geq 1$.

**Theorem I.4** (Rational base expansions). CITED
Every positive integer $N$ has exactly one expansion

$$N = \sum_i \frac{a_i}{q} \left(\frac{p}{q}\right)^i, \quad 0 \leq a_i \leq p - 1.$$

The digits come from the recursion $N_0 = N$ and $qN_i = pN_{i+1} + a_i$, where $a_i$ is the residue of $qN_i$ modulo $p$.

The integers form a tree: the expansion of $N$ is that of its parent $(qN - a_0)/p$ followed by $a_0$. The set of all expansions is not a regular language, which makes the system subtle, but the tree makes it easy to search (L21.2). Reducing the recursion modulo $p - q$, where $p \equiv q$, gives $qN \equiv$ (digit sum) $(\mathrm{mod}\, p - q)$, so the identity modulus is $p - q$, unrelated to the digit range $p$. Example: $8 = (2, 1, 0, 1, 1)$ in base $3/2$, since $(2/2)(3/2)^4 + (1/2)(3/2)^3 + (1/2)(3/2) + 1/2 = 8$, and $2^2 + 1 + 1 + 1 + 1 = 8$ under $0^0 = 1$.

## I.5 Beta-expansions and Pisot bases

For a real $\beta > 1$, the greedy expansion of a positive real number takes the largest power $\beta^k$ below it, the largest digit $d$ with $d\beta^k$ below it, and continues with the remainder. The digits lie in $\{0, ..., \lceil \beta \rceil - 1\}$. Parry [44] characterized the digit strings that arise: a string is admissible when each of its tails is lexicographically smaller than the expansion of 1.

**Definition I.5** (Pisot number).
A Pisot number is a real algebraic integer $\beta > 1$ whose other conjugates all have absolute value less than 1.

The golden ratio and $\varepsilon_x = \left(x + \sqrt{x^2 + 4}\right)/2$, the root of $t^2 - xt - 1$, are Pisot numbers. Pisot bases behave like integer bases in one key way.

**Theorem I.6** (Frougny and Solomyak). CITED
If $\beta$ is a root of $t^d - a_1 t^{d-1} - ... - a_d$ with integers $a_1 \geq a_2 \geq ... \geq a_d \geq 1$, then every positive element of $\mathbb{Z}[1/\beta]$, in particular every positive integer, has a finite greedy expansion.

See [45]. For $\varepsilon_x$ the coefficients are $x \geq 1$, so every positive integer has a finite expansion, with digits $0, ..., x$ and every digit $x$ followed by a 0. Expansions of integers can have digits after the radix point: for the golden ratio, $2 = \varphi + \varphi^{-2}$. The norm of $\varepsilon_x - 1$ is $-x$, so the identity modulus is $x$ (L21.4).

## I.6 The factorial number system

Every positive integer has exactly one expansion $n = \sum_{i=1}^{L} d_i i!$ with $0 \leq d_i \leq i$ and $d_L \geq 1$. This is a mixed-radix system: the digit at position $i$ is the remainder after dividing by $i + 1$. Here no modulus is a function of the digit multiset: $(1, 0, 1)$ and $(1, 1, 0)$ have the same multiset and the values 7 and 8. A digit $v$ can only sit at a position $i \geq v$, so the large digits are confined to the top positions. Section 7.9 shows that this is exactly what makes the probabilistic model fail there (L21.5).

Appendix J

# Function fields over $\mathbb{F}_q$

The ring $\mathbb{F}_q[T]$ of polynomials over a finite field behaves in many ways like the integers. Section 7.8 uses it as a test world for Munchausen numbers: there the weights are additive, every structure can be computed exactly, and one can see precisely why the probabilistic model fails. This appendix gives the background. Rosen's book [98] develops the analogy between $\mathbb{F}_q[T]$ and $\mathbb{Z}$ in depth.

## J.1 The ring $\mathbb{F}_q[T]$

Let $q$ be a prime (the library also allows prime powers) and $\mathbb{F}_q = \mathbb{Z}/q\mathbb{Z}$. The ring $\mathbb{F}_q[T]$ has division with remainder by degree, so every ideal is principal and every nonzero polynomial factors uniquely into a constant times monic irreducible polynomials. These irreducibles play the role of primes. The degree measures size: put $|f| = q^{\deg f}$, the number of residues modulo $f$. A polynomial of degree $e$ has $q^e$ residues, so it is the analogue of an integer of size $q^e$.

## J.2 Expansions in a polynomial base

Fix a base $B$ of degree $e \geq 2$. The digits are the $q^e$ polynomials of degree less than $e$. Every polynomial $n$ has exactly one expansion $n = \sum_i d_i B^i$ with digits $d_i$: take $d_0$ the remainder of $n$ modulo $B$ and repeat with $(n - d_0)/B$.

The striking difference from the integers is that there are no carries. The sum of two digits has degree less than $e$, so it is again a digit, and the expansion of $n + n'$ is the digitwise sum. So the map from a polynomial to its digit vector is $\mathbb{F}_q$-linear.

## J.3 Frobenius

**Proposition J.1** (Frobenius is additive). PROVEN
In characteristic $q$, $(f+g)^q = f^q + g^q$ for all polynomials $f, g$. Since $c^q = c$ for every $c \in \mathbb{F}_q$, also $f(T)^q = f(T^q)$.

*Proof.* In $(f+g)^q = \sum_j \binom{q}{j} f^j g^{q-j}$, every middle binomial coefficient is divisible by the prime $q$. The second statement follows by applying the first to the monomials of $f$ and using Fermat's little theorem $c^q = c$. □

So the weight that Section 7.8 gives a digit $d$ of degree $j$,

$$w(d) = d^{q^j} = d\big(T^{q^j}\big),$$

the analogue of $x^x$ with the size $|d| = q^j$ as exponent, is additive on the digits of each degree. The Munchausen sum of $n$ therefore depends only on the sums $D_j$ of the digits of each degree $j$: $F(n) = \sum_j D_j\big(T^{q^j}\big) + mz$, with $z$ the number of zero digits. The problem becomes finite and linear, up to the split of the digits into degree classes (L21.1).

## J.4 The block structure

Frobenius also acts on expansions in a simple way. If $n = \sum_i d_i B^i$, then

$$n(T^q) = n^q = \sum_i d_i^q (B^q)^i = \sum_i d_i(T^q) B^{qi},$$

and $d_i(T^q)$ has degree at most $(e-1)q < qe$, so it has at most $q$ digits in base $B$. Each digit of $n$ becomes a block of $q$ digits of $n^q$, and the blocks do not overlap. Applied $j$ times, this shows that the digits of $F(n)$ are built from only $e$ digits, one for each degree class (Lemma 24.1 and Corollary 24.3 of L24.2). This rigid structure is what the probabilistic model, which treats the digits of $F$ as independent and uniform, cannot see.

> **Example J.2** (A small base over $\mathbb{F}_2$).
> Take $q = 2$ and $B = T^2 + T + 1$, so $e = 2$ and the digits are $0, 1, T, T+1$. Their weights under $0^0 = 0$ are $w(0) = 0$, $w(1) = 1$, $w(T) = T^2$ and $w(T+1) = (T+1)^2 = T^2 + 1$. In characteristic 2, $T^2 = B + (T+1)$, so $T^2$ has the digits $(1, T+1)$, and its Munchausen sum is $w(1) + w(T+1) = 1 + T^2 + 1 = T^2$. So $T^2$ is a solution. So is $T^2 + 1 = B + T$, with digits $(1, T)$ and sum $1 + T^2$. These are the $q(q-1) = 2$ solutions that Theorem 21.7 predicts, since $B - 1 = T(T+1)$ has two roots in $\mathbb{F}_2$ (Section 7.8).
>
> The block structure shows in squares. Take $n = T^2 + 1$ with digits $(1, T)$. Squaring sends the digit $T$ to $T^2$, the block $(1, T+1)$, and the digit 1 to the block $(0, 1)$. The blocks give $n^2$ the digits $(0, 1, 1, T+1)$, and without the leading zero $n^2 = B^2 + B + (T+1)$, which is indeed $T^4 + 1$.

## J.5 Traces and free residues

Let $P$ be an irreducible polynomial of degree $f$. The residue ring $\mathbb{F}_q[T]/(P)$ is a field with $q^f$ elements, and Frobenius $y \mapsto y^q$ is an automorphism of it of order $f$. The trace

$$\mathrm{Tr}(y) = y + y^q + y^{q^2} + \ldots + y^{q^{f-1}}$$

is an $\mathbb{F}_q$-linear map onto $\mathbb{F}_q$, and $\mathrm{Tr}(y^q) = \mathrm{Tr}(y)$.

The digit-sum identity holds here too: $n \equiv \sum_i d_i \pmod{B-1}$, because $B \equiv 1$. For a solution it reads $\sum_j \left(D_j\left(T^{q^j}\right) - D_j\right) + mz \equiv 0 \pmod{B-1}$. Reduce modulo an irreducible factor $P$ of $B - 1$ and apply the trace. Since $D_j\left(T^{q^j}\right) = D_j^{q^j}$ and the trace is invariant under Frobenius, every term $\mathrm{Tr}\left(D_j\left(T^{q^j}\right) - D_j\right)$ vanishes. So under $0^0 = 0$ the identity holds for free on a subspace of codimension at least the number of irreducible factors of $B - 1$ (Theorem 21.6 of L21.1). In the integers the only such free residue is $x^x \equiv x \pmod 2$; here every prime factor of $B - 1$ gives one. This is why the count grows by a factor of about $q$ per factor of $B - 1$, the analogue of the parity doubling.

Under $0^0 = 1$ the constant $mz$ contributes the trace $z \deg P$. If $\deg P$ is prime to $q$, a solution needs $q \mid z$. Chapter 24 of the library turns this into an exact statement: when $B - 1$ has such a factor, the solutions under $0^0 = 1$ are the solutions under $0^0 = 0$ whose number of zero digits is divisible by $q$ (Proposition 24.11 of L24.4). Because the block structure forces many zeros, this share is far from what the model expects.

## J.6 Frobenius on $\mathbb{F}_q[T]/(P^a)$

The exact value of the free residue dimension $k(B)$ (Theorem 7.55) needs Frobenius on the whole ring $R = \mathbb{F}_q[T]/(B-1)$, not only on its residue fields. Write $B - 1 = \prod_i P_i^{a_i}$ with distinct monic irreducible $P_i$ of degree $f_i$. By the Chinese remainder theorem $R$ is the product of the local rings $R_i = \mathbb{F}_q[T]/\left(P_i^{a_i}\right)$, and Frobenius $\varphi(x) = x^q$ acts on each factor separately. It is $\mathbb{F}_q$-linear, because

$(x+y)^q = x^q + y^q$ and $c^q = c$ for $c \in \mathbb{F}_q$. Four facts about one local ring $R_i$ do all the work. Write $f = f_i$, $a = a_i$, and $L = \mathbb{F}_{q^f}$ for the residue field $R_i/(P_i)$.

1. *Fixed points.* The solutions of $x^{q^j} = x$ in $R_i$ form a field with $q^{\gcd(j,f)}$ elements. Modulo $P_i$ they are the roots of $X^{q^j} - X$ in $L$, the subfield $\mathbb{F}_{q^{\gcd(j,f)}}$. Each such root lifts in exactly one way to $R_i$, by Hensel's lemma (Appendix C has the $p$-adic version), because the derivative of $X^{q^j} - X$ is $-1$ and never vanishes. In particular $\varphi - 1$ has a kernel of dimension 1 on each $R_i$, and of dimension $\omega(B-1)$ on $R$. This is the starting point of Berlekamp's factoring algorithm.
2. *The nilradical.* The multiples of $P_i$ form the nilradical $N$ of $R_i$: each of them has $x^a = 0$. Frobenius is nilpotent on $N$, since $x^{q^r} = 0$ once $q^r \geq a$. So $\varphi^j - 1$ is invertible on $N$, and the image of $\varphi^j - 1$ on $R_i$ contains $N$. It is the set of elements whose reduction lies in the image of $\varphi^j - 1$ on the field $L$.
3. *The image on the field.* On $L$, with $K = \mathbb{F}_{q^g}$ and $g = \gcd(j, f)$, the image of $\varphi^j - 1$ is the kernel of the trace from $L$ to $K$. The image lies in that kernel because the trace is invariant under $\varphi^j$. Both have dimension $f - g$ over $\mathbb{F}_q$, since the kernel of $\varphi^j - 1$ is $K$. This is the additive form of Hilbert's theorem 90.
4. *Sums of images.* If a set $S$ of exponents $j$ has $\gcd(S \cup \{f\}) = 1$, then the images of $\varphi^j - 1$ for $j \in S$ add up to the image of $\varphi - 1$, the kernel of the trace from $L$ to $\mathbb{F}_q$. Under the trace form $(x, y) \mapsto \mathrm{Tr}_{L/\mathbb{F}_q}(xy)$, the kernel of the trace to $K$ is the orthogonal complement of $K$. So the sum of the images is the orthogonal complement of the intersection of the fields $\mathbb{F}_{q^{\gcd(j,f)}}$, which is $\mathbb{F}_q$.

Two exponents $j$ and $j+1$ always satisfy the condition of fact 4, and so does $j = e - 1$ when $\gcd(e-1, f) = 1$. These are the two cases that the proof of Theorem 7.55 uses.

Appendix K

# Tables of solutions

This appendix lists every known solution in small bases, and the complete lists of short solutions. The tables are read from the CSV files in `docs/exposition/data`, which `tools/gen_data.py` copies from `output/` and `analysis/data/`. Digit strings are written most significant first, with the digits in decimal.

## K.1 Counts per base

Table K.1 gives the number of nontrivial solutions in each base from 2 to 41 and in base 43, from the exhaustive search of `fast` (L4.2). "Shared" counts the zero-free solutions, which are solutions under both conventions (Proposition 1.8). "Longest" is the largest number of digits under either convention. Base 42 was searched only for 42-digit numbers (none under either convention, L8.6).

Table K.1. Nontrivial solutions per base. $N_0$ and $N_1$ count the solutions under $0^0 = 0$ and $0^0 = 1$.

| base | $N_0$ | $N_1$ | shared | longest | base | $N_0$ | $N_1$ | shared | longest |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 0 | 1 | 0 | 2 | 23 | 1 | 0 | 0 | 22 |
| 3 | 2 | 2 | 2 | 2 | 24 | 1 | 1 | 1 | 23 |
| 4 | 3 | 2 | 2 | 3 | 25 | 5 | 2 | 2 | 24 |
| 5 | 2 | 0 | 0 | 4 | 26 | 1 | 2 | 0 | 25 |
| 6 | 2 | 2 | 2 | 5 | 27 | 0 | 0 | 0 | - |
| 7 | 1 | 1 | 1 | 5 | 28 | 0 | 3 | 0 | 27 |
| 8 | 2 | 0 | 0 | 3 | 29 | 1 | 0 | 0 | 28 |
| 9 | 6 | 3 | 3 | 8 | 30 | 0 | 1 | 0 | 29 |
| 10 | 2 | 1 | 1 | 9 | 31 | 3 | 2 | 1 | 30 |
| 11 | 5 | 2 | 2 | 8 | 32 | 4 | 0 | 0 | 31 |
| 12 | 1 | 1 | 1 | 10 | 33 | 3 | 2 | 1 | 32 |
| 13 | 5 | 4 | 4 | 12 | 34 | 1 | 1 | 1 | 33 |
| 14 | 1 | 4 | 1 | 13 | 35 | 0 | 0 | 0 | - |
| 15 | 7 | 5 | 2 | 15 | 36 | 0 | 1 | 0 | 35 |
| 16 | 4 | 2 | 2 | 16 | 37 | 3 | 3 | 3 | 36 |
| 17 | 4 | 4 | 3 | 16 | 38 | 2 | 3 | 2 | 37 |
| 18 | 2 | 3 | 1 | 17 | 39 | 2 | 2 | 1 | 38 |
| 19 | 0 | 0 | 0 | - | 40 | 3 | 1 | 1 | 39 |
| 20 | 2 | 2 | 2 | 17 | 41 | 1 | 1 | 1 | 40 |
| 21 | 1 | 2 | 1 | 20 | 43 | 1 | 2 | 1 | 42 |
| 22 | 1 | 1 | 1 | 20 | | | | | |

## K.2 All solutions in bases 2 to 43

Table K.2 and Table K.3 list every nontrivial solution in bases 2 to 41 and 43, from the exhaustive search (`output/b2-41-*.txt`, `output/b43-*.txt`). Bases 2 to 35 under $0^0 = 1$ agree with OEIS A166623 [6]. The lists end with three kinds of partial results, marked in the last column:

- "inexact": base 44 was searched with the inexact rules `--deficit 2 --band 2.5`, which can miss solutions (L9.3);

- "$\leq 30$": bases 44 to 127 were searched only for numbers with at most 30 digits (`analysis/data/b39-127-d30.txt`), so the lists there are complete only up to that length;
- the self-paying and swap families of Chapter 2 add infinitely many solutions in perfect-power bases beyond these ranges.

Table K.2. All known nontrivial solutions under $0^0 = 1$ in bases 2 to 127. Exhaustive for bases 2 to 41 and 43.

| base | $k$ | digits |
|---|---|---|
| 2 | 2 | (1, 0) |
| 3 | 2 | (1, 2) |
| 3 | 2 | (2, 2) |
| 4 | 3 | (1, 3, 1) |
| 4 | 3 | (3, 1, 3) |
| 6 | 5 | (2, 2, 3, 5, 2) |
| 6 | 5 | (2, 3, 4, 5, 2) |
| 7 | 5 | (1, 3, 4, 5, 4) |
| 9 | 2 | (3, 1) |
| 9 | 6 | (1, 5, 6, 2, 6, 2) |
| 9 | 7 | (1, 6, 5, 6, 5, 4, 7) |
| 10 | 4 | (3, 4, 3, 5) |
| 11 | 8 | (1, 8, 4, 5, 3, 2, 7, 8) |
| 11 | 8 | (1, 8, 4, 5, 3, 4, 8, 7) |
| 12 | 10 | (3, 10, 6, 7, 10, 5, 4, 8, 3, 2) |
| 13 | 5 | (3, 3, 6, 6, 1) |
| 13 | 10 | (2, 10, 10, 8, 3, 4, 6, 6, 8, 10) |
| 13 | 12 | (4, 12, 10, 9, 2, 10, 2, 3, 3, 5, 1, 8) |
| 13 | 12 | (4, 12, 10, 9, 2, 10, 2, 3, 3, 5, 3, 8) |
| 14 | 2 | (2, 3) |
| 14 | 5 | (2, 6, 0, 3, 6) |
| 14 | 12 | (4, 5, 10, 0, 10, 0, 4, 5, 1, 3, 12, 12) |
| 14 | 13 | (10, 9, 9, 2, 11, 5, 13, 9, 6, 7, 2, 0, 13) |
| 15 | 13 | (4, 11, 1, 6, 4, 8, 4, 2, 0, 13, 12, 13, 0) |
| 15 | 14 | (5, 10, 9, 9, 14, 5, 3, 8, 3, 3, 9, 10, 4, 3) |
| 15 | 14 | (5, 10, 12, 11, 12, 4, 1, 12, 1, 9, 14, 3, 3, 3) |
| 15 | 14 | (5, 10, 12, 11, 12, 4, 1, 12, 1, 9, 14, 4, 0, 0) |
| 15 | 14 | (5, 13, 0, 11, 1, 9, 7, 12, 2, 5, 14, 0, 5, 6) |
| 16 | 15 | (12, 4, 14, 15, 7, 2, 2, 11, 7, 8, 2, 12, 2, 6, 15) |
| 16 | 15 | (12, 7, 6, 7, 1, 2, 15, 15, 12, 3, 1, 1, 14, 6, 14) |
| 17 | 2 | (3, 3) |
| 17 | 15 | (2, 10, 4, 2, 2, 10, 2, 8, 13, 12, 5, 15, 2, 8, 6) |
| 17 | 16 | (6, 7, 10, 9, 11, 7, 14, 10, 10, 0, 4, 8, 5, 12, 0, 16) |
| 17 | 16 | (6, 10, 4, 15, 5, 12, 11, 16, 3, 3, 14, 14, 2, 7, 3, 6) |
| 18 | 11 | (2, 9, 0, 0, 10, 7, 8, 6, 8, 12, 2) |
| 18 | 15 | (1, 3, 10, 11, 15, 4, 5, 14, 3, 13, 8, 6, 5, 6, 13) |
| 18 | 16 | (2, 14, 7, 8, 1, 3, 0, 12, 1, 4, 10, 3, 16, 2, 2, 15) |
| 20 | 4 | (6, 5, 3, 4) |
| 20 | 17 | (1, 5, 16, 3, 9, 6, 17, 6, 3, 11, 14, 6, 3, 8, 13, 7, 12) |
| 21 | 19 | (3, 2, 19, 2, 9, 10, 11, 11, 5, 5, 17, 8, 10, 8, 9, 13, 13, 17, 7) |
| 21 | 20 | (8, 1, 9, 18, 8, 4, 0, 18, 7, 11, 0, 7, 17, 12, 10, 19, 10, 20, 9, 14) |
| 22 | 20 | (3, 5, 20, 17, 4, 17, 3, 10, 10, 11, 14, 11, 6, 1, 12, 16, 5, 3, 4, 8) |
| 24 | 23 | (18, 4, 10, 16, 23, 21, 2, 14, 16, 4, 22, 2, 19, 15, 23, 16, 15, 5, 5, 16, 11, 15, 8) |
| 25 | 2 | (1, 3) |
| 25 | 3 | (5, 1, 3) |
| 26 | 25 | (9, 19, 15, 1, 10, 9, 7, 25, 13, 21, 12, 13, 0, 12, 7, 13, 9, 12, 7, 6, 9, 5, 21, 12, 5) |
| 26 | 25 | (10, 5, 0, 2, 11, 4, 15, 10, 13, 14, 2, 24, 21, 0, 24, 10, 4, 24, 8, 18, 0, 25, 21, 22, 10) |
| 28 | 16 | (15, 12, 12, 13, 1, 15, 18, 0, 18, 9, 10, 16, 12, 16, 10, 14) |
| 28 | 27 | (20, 27, 9, 27, 5, 23, 20, 16, 17, 22, 3, 22, 5, 2, 24, 4, 1, 22, 16, 0, 1, 10, 10, 14, 22, 13, 4) |
| 28 | 27 | (21, 3, 12, 3, 0, 7, 0, 1, 27, 8, 23, 4, 11, 27, 18, 5, 21, 21, 12, 4, 5, 23, 21, 10, 10, 1, 26) |
| 30 | 29 | (11, 11, 1, 29, 10, 22, 7, 4, 22, 19, 11, 8, 0, 2, 12, 8, 12, 13, 11, 14, 5, 28, 18, 23, 11, 6, 23, 8, 19) |

| base | $k$ | digits | |
|---|---|---|---|
| 31 | 30 | (11, 22, 25, 18, 25, 8, 19, 7, 7, 0, 29, 7, 10, 7, 0, 23, 2, 1, 30, 17, 17, 23, 18, 19, 23, 17, 15, 27, 24, 1) | |
| 31 | 30 | (11, 22, 29, 5, 8, 19, 20, 16, 11, 6, 25, 6, 25, 6, 17, 19, 20, 19, 28, 18, 28, 12, 26, 8, 16, 24, 23, 26, 21, 30) | |
| 33 | 32 | (12, 15, 21, 32, 30, 17, 8, 17, 25, 28, 23, 5, 9, 0, 29, 22, 3, 31, 9, 15, 30, 11, 14, 12, 17, 26, 10, 25, 1, 11, 10, 0) | |
| 33 | 32 | (24, 31, 3, 12, 21, 11, 24, 24, 7, 5, 31, 32, 13, 28, 10, 15, 12, 6, 19, 20, 19, 23, 8, 6, 11, 16, 16, 32, 5, 3, 12, 22) | |
| 34 | 33 | (12, 28, 17, 21, 19, 15, 10, 22, 26, 7, 24, 12, 4, 24, 30, 18, 24, 4, 15, 20, 33, 12, 1, 2, 29, 29, 2, 24, 6, 29, 14, 23, 32) | |
| 36 | 35 | (27, 0, 7, 32, 14, 18, 24, 31, 34, 5, 28, 16, 35, 10, 33, 30, 15, 35, 18, 6, 7, 16, 10, 19, 7, 16, 27, 23, 2, 18, 24, 32, 0, 31, 25) | |
| 37 | 34 | (4, 7, 12, 11, 25, 10, 19, 29, 34, 6, 5, 19, 26, 25, 6, 16, 27, 31, 24, 30, 22, 20, 32, 16, 34, 15, 3, 5, 15, 30, 17, 15, 33, 6) | |
| 37 | 35 | (5, 14, 34, 22, 29, 23, 6, 2, 27, 23, 1, 29, 28, 23, 26, 13, 32, 16, 14, 11, 29, 35, 15, 9, 29, 4, 16, 6, 3, 20, 13, 30, 34, 24, 12) | |
| 37 | 36 | (27, 22, 3, 4, 25, 5, 22, 4, 24, 22, 9, 32, 5, 9, 29, 36, 32, 16, 32, 29, 26, 16, 25, 14, 18, 25, 36, 32, 14, 13, 32, 9, 21, 33, 14, 17) | |
| 38 | 33 | (33, 23, 32, 14, 23, 2, 21, 18, 2, 33, 28, 27, 7, 9, 16, 19, 3, 19, 34, 30, 21, 24, 25, 5, 31, 6, 13, 4, 18, 18, 9, 21, 28) | |
| 38 | 37 | (14, 6, 12, 2, 11, 1, 1, 21, 24, 20, 9, 20, 16, 25, 1, 21, 3, 28, 26, 7, 28, 8, 31, 4, 17, 30, 16, 30, 31, 33, 10, 24, 34, 37, 31, 16, 3) | |
| 38 | 37 | (14, 17, 14, 35, 35, 3, 0, 14, 19, 30, 14, 36, 14, 0, 8, 6, 30, 5, 24, 4, 6, 22, 36, 15, 18, 21, 12, 37, 2, 28, 35, 33, 2, 5, 0, 35, 24) | |
| 39 | 38 | (14, 26, 19, 20, 31, 29, 21, 5, 28, 31, 30, 23, 28, 22, 30, 19, 9, 37, 32, 11, 14, 3, 10, 21, 35, 32, 6, 15, 22, 6, 8, 36, 1, 38, 9, 14, 30, 36) | |
| 39 | 38 | (14, 37, 22, 3, 21, 36, 34, 29, 16, 14, 0, 15, 34, 27, 12, 29, 9, 0, 37, 28, 19, 26, 6, 15, 22, 38, 7, 23, 17, 35, 16, 8, 10, 11, 20, 24, 37, 24) | |
| 40 | 36 | (1, 32, 18, 23, 5, 36, 16, 22, 7, 33, 28, 30, 2, 4, 21, 10, 7, 9, 12, 23, 21, 28, 20, 26, 25, 32, 36, 30, 6, 2, 24, 9, 30, 3, 21, 35) | |
| 41 | 40 | (15, 11, 2, 12, 35, 10, 24, 3, 32, 32, 34, 30, 1, 2, 25, 31, 1, 31, 36, 40, 14, 37, 3, 32, 18, 34, 17, 10, 13, 32, 33, 35, 21, 33, 33, 31, 32, 6, 5, 16) | |
| 43 | 42 | (16, 6, 16, 24, 24, 1, 6, 16, 40, 1, 8, 39, 2, 22, 18, 27, 39, 39, 36, 17, 42, 12, 31, 29, 15, 17, 21, 41, 23, 15, 10, 15, 34, 29, 18, 34, 3, 37, 30, 8, 35, 35) | |
| 43 | 42 | (32, 12, 30, 27, 38, 34, 34, 23, 39, 5, 27, 3, 32, 33, 19, 21, 32, 37, 24, 20, 21, 15, 41, 42, 25, 42, 25, 0, 33, 38, 6, 21, 38, 36, 15, 3, 41, 0, 6, 16, 40, 2) | |
| 44 | 43 | (32, 32, 36, 30, 0, 0, 37, 23, 32, 33, 4, 5, 17, 4, 30, 7, 31, 25, 43, 25, 15, 21, 38, 19, 18, 7, 35, 10, 34, 39, 4, 30, 25, 33, 24, 43, 27, 26, 13, 1, 38, 24, 22) | inexact |
| 48 | 5 | (3, 8, 6, 2, 0) | $\leq 30$ |
| 64 | 2 | (4, 1) | $\leq 30$ |
| 64 | 11 | (16, 0, 0, 0, 0, 0, 1, 0, 0, 4, 8) | $\leq 30$ |
| 70 | 2 | (4, 3) | $\leq 30$ |
| 91 | 3 | (6, 1, 5) | $\leq 30$ |
| 93 | 2 | (3, 4) | $\leq 30$ |
| 127 | 2 | (4, 4) | $\leq 30$ |

Table K.3. All known nontrivial solutions under $0^0 = 0$ in bases 2 to 127. Exhaustive for bases 2 to 41 and 43.

| base | $k$ | digits |
|---|---|---|
| 3 | 2 | (1, 2) |
| 3 | 2 | (2, 2) |
| 4 | 3 | (1, 3, 0) |
| 4 | 3 | (1, 3, 1) |
| 4 | 3 | (3, 1, 3) |
| 5 | 3 | (1, 0, 3) |
| 5 | 4 | (2, 0, 2, 4) |
| 6 | 5 | (2, 2, 3, 5, 2) |
| 6 | 5 | (2, 3, 4, 5, 2) |
| 7 | 5 | (1, 3, 4, 5, 4) |
| 8 | 3 | (4, 0, 0) |
| 8 | 3 | (4, 0, 1) |
| 9 | 2 | (3, 0) |
| 9 | 2 | (3, 1) |
| 9 | 6 | (1, 5, 6, 2, 6, 2) |
| 9 | 7 | (1, 6, 4, 7, 0, 6, 3) |
| 9 | 7 | (1, 6, 5, 6, 5, 4, 7) |
| 9 | 8 | (3, 4, 6, 6, 4, 0, 8, 4) |
| 10 | 4 | (3, 4, 3, 5) |
| 10 | 9 | (4, 3, 8, 5, 7, 9, 0, 8, 8) |
| 11 | 5 | (6, 6, 5, 0, 0) |

| base | $k$ | digits |
|---|---|---|
| 11 | 5 | (6, 6, 5, 0, 1) |
| 11 | 6 | (5, 1, 7, 5, 0, 3) |
| 11 | 8 | (1, 8, 4, 5, 3, 2, 7, 8) |
| 11 | 8 | (1, 8, 4, 5, 3, 4, 8, 7) |
| 12 | 10 | (3, 10, 6, 7, 10, 5, 4, 8, 3, 2) |
| 13 | 5 | (3, 3, 6, 6, 0) |
| 13 | 5 | (3, 3, 6, 6, 1) |
| 13 | 10 | (2, 10, 10, 8, 3, 4, 6, 6, 8, 10) |
| 13 | 12 | (4, 12, 10, 9, 2, 10, 2, 3, 3, 5, 1, 8) |
| 13 | 12 | (4, 12, 10, 9, 2, 10, 2, 3, 3, 5, 3, 8) |
| 14 | 2 | (2, 3) |
| 15 | 13 | (4, 11, 1, 6, 4, 8, 4, 2, 0, 13, 12, 13, 2) |
| 15 | 14 | (5, 10, 9, 9, 14, 5, 3, 8, 3, 3, 9, 10, 4, 3) |
| 15 | 14 | (5, 10, 9, 10, 14, 2, 8, 11, 1, 0, 3, 11, 0, 0) |
| 15 | 14 | (5, 10, 9, 10, 14, 2, 8, 11, 1, 0, 3, 11, 0, 1) |
| 15 | 14 | (5, 10, 12, 11, 12, 4, 1, 12, 1, 9, 14, 3, 3, 3) |
| 15 | 14 | (5, 10, 12, 11, 12, 4, 1, 12, 1, 9, 14, 4, 0, 2) |
| 15 | 15 | (1, 2, 4, 3, 14, 14, 13, 3, 4, 1, 9, 1, 1, 0, 14) |
| 16 | 15 | (12, 4, 14, 15, 7, 2, 2, 11, 7, 8, 2, 0, 12, 2, 15) |
| 16 | 15 | (12, 4, 14, 15, 7, 2, 2, 11, 7, 8, 2, 12, 2, 6, 15) |
| 16 | 15 | (12, 7, 6, 7, 1, 2, 15, 15, 12, 3, 1, 1, 14, 6, 14) |
| 16 | 16 | (1, 2, 3, 11, 2, 3, 0, 9, 15, 15, 5, 7, 2, 15, 6, 11) |
| 17 | 2 | (3, 3) |
| 17 | 14 | (1, 2, 1, 4, 1, 3, 10, 6, 8, 0, 11, 14, 1, 3) |
| 17 | 15 | (2, 10, 4, 2, 2, 10, 2, 8, 13, 12, 5, 15, 2, 8, 6) |
| 17 | 16 | (6, 10, 4, 15, 5, 12, 11, 16, 3, 3, 14, 14, 2, 7, 3, 6) |
| 18 | 15 | (1, 3, 10, 11, 15, 4, 5, 14, 3, 13, 8, 6, 5, 6, 13) |
| 18 | 17 | (6, 17, 6, 5, 13, 5, 1, 14, 7, 16, 7, 8, 0, 12, 5, 12, 5) |
| 20 | 4 | (6, 5, 3, 4) |
| 20 | 17 | (1, 5, 16, 3, 9, 6, 17, 6, 3, 11, 14, 6, 3, 8, 13, 7, 12) |
| 21 | 19 | (3, 2, 19, 2, 9, 10, 11, 11, 5, 5, 17, 8, 10, 8, 9, 13, 13, 17, 7) |
| 22 | 20 | (3, 5, 20, 17, 4, 17, 3, 10, 10, 11, 14, 11, 6, 1, 12, 16, 5, 3, 4, 8) |
| 23 | 22 | (8, 18, 9, 12, 10, 0, 14, 13, 6, 2, 12, 21, 16, 14, 1, 16, 9, 12, 20, 4, 22, 9) |
| 24 | 23 | (18, 4, 10, 16, 23, 21, 2, 14, 16, 4, 22, 2, 19, 15, 23, 16, 15, 5, 5, 16, 11, 15, 8) |
| 25 | 2 | (1, 3) |
| 25 | 3 | (5, 0, 0) |
| 25 | 3 | (5, 0, 1) |
| 25 | 3 | (5, 1, 3) |
| 25 | 24 | (9, 9, 17, 8, 8, 12, 14, 11, 13, 0, 18, 24, 22, 1, 18, 10, 18, 19, 3, 7, 3, 7, 4, 12) |
| 26 | 24 | (3, 21, 0, 3, 21, 4, 24, 0, 7, 10, 7, 14, 3, 3, 18, 4, 16, 6, 0, 16, 17, 15, 8, 16) |
| 29 | 28 | (10, 24, 25, 24, 4, 8, 17, 20, 8, 20, 13, 7, 11, 26, 25, 11, 28, 22, 20, 13, 22, 22, 12, 20, 17, 0, 14, 19) |
| 31 | 29 | (8, 29, 27, 0, 10, 1, 0, 13, 29, 1, 23, 24, 14, 9, 17, 1, 12, 5, 7, 8, 22, 15, 11, 5, 19, 23, 16, 13, 12) |
| 31 | 30 | (11, 22, 25, 18, 25, 8, 19, 7, 7, 0, 29, 7, 10, 7, 0, 23, 2, 1, 30, 17, 17, 23, 18, 19, 23, 17, 15, 27, 24, 2) |
| 31 | 30 | (11, 22, 29, 5, 8, 19, 20, 16, 11, 6, 25, 6, 25, 6, 17, 19, 20, 19, 28, 18, 28, 12, 26, 8, 16, 24, 23, 26, 21, 30) |
| 32 | 13 | (16, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) |
| 32 | 13 | (16, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1) |
| 32 | 30 | (9, 11, 3, 30, 22, 26, 8, 14, 15, 13, 10, 16, 7, 9, 0, 9, 24, 8, 14, 16, 26, 29, 11, 30, 5, 13, 29, 2, 17, 25) |
| 32 | 31 | (23, 29, 18, 20, 13, 27, 29, 27, 31, 5, 1, 20, 31, 20, 16, 13, 1, 14, 14, 28, 11, 0, 28, 12, 10, 12, 17, 6, 9, 29, 21) |
| 33 | 32 | (12, 10, 26, 13, 12, 27, 32, 19, 23, 2, 8, 2, 23, 27, 19, 6, 10, 21, 17, 0, 25, 12, 17, 28, 5, 28, 25, 20, 18, 14, 8, 9) |
| 33 | 32 | (12, 15, 21, 32, 30, 17, 8, 17, 25, 28, 23, 5, 9, 0, 29, 22, 3, 31, 9, 15, 30, 11, 14, 12, 17, 26, 10, 25, 1, 11, 10, 2) |
| 33 | 32 | (24, 31, 3, 12, 21, 11, 24, 24, 7, 5, 31, 32, 13, 28, 10, 15, 12, 6, 19, 20, 19, 23, 8, 6, 11, 16, 16, 32, 5, 3, 12, 22) |
| 34 | 33 | (12, 28, 17, 21, 19, 15, 10, 22, 26, 7, 24, 12, 4, 24, 30, 18, 24, 4, 15, 20, 33, 12, 1, 2, 29, 29, 2, 24, 6, 29, 14, 23, 32) |
| 37 | 34 | (4, 7, 12, 11, 25, 10, 19, 29, 34, 6, 5, 19, 26, 25, 6, 16, 27, 31, 24, 30, 22, 20, 32, 16, 34, 15, 3, 5, 15, 30, 17, 15, 33, 6) |
| 37 | 35 | (5, 14, 34, 22, 29, 23, 6, 2, 27, 23, 1, 29, 28, 23, 26, 13, 32, 16, 14, 11, 29, 35, 15, 9, 29, 4, 16, 6, 3, 20, 13, 30, 34, 24, 12) |
| 37 | 36 | (27, 22, 3, 4, 25, 5, 22, 4, 24, 22, 9, 32, 5, 9, 29, 36, 32, 16, 32, 29, 26, 16, 25, 14, 18, 25, 36, 32, 14, 13, 32, 9, 21, 33, 14, 17) |
| 38 | 33 | (33, 23, 32, 14, 23, 2, 21, 18, 2, 33, 28, 27, 7, 9, 16, 19, 3, 19, 34, 30, 21, 24, 25, 5, 31, 6, 13, 4, 18, 18, 9, 21, 28) |

| base | $k$ | digits | |
|---|---|---|---|
| 38 | 37 | (14, 6, 12, 2, 11, 1, 1, 21, 24, 20, 9, 20, 16, 25, 1, 21, 3, 28, 26, 7, 28, 8, 31, 4, 17, 30, 16, 30, 31, 33, 10, 24, 34, 37, 31, 16, 3) | |
| 39 | 37 | (5, 26, 12, 5, 31, 29, 35, 27, 0, 6, 6, 12, 37, 6, 16, 30, 35, 5, 28, 24, 11, 5, 4, 7, 34, 36, 32, 19, 36, 35, 27, 12, 16, 23, 9, 17, 5) | |
| 39 | 38 | (14, 26, 19, 20, 31, 29, 21, 5, 28, 31, 30, 23, 28, 22, 30, 19, 9, 37, 32, 11, 14, 3, 10, 21, 35, 32, 6, 15, 22, 6, 8, 36, 1, 38, 9, 14, 30, 36) | |
| 40 | 36 | (1, 32, 18, 23, 5, 36, 16, 22, 7, 33, 28, 30, 2, 4, 21, 10, 7, 9, 12, 23, 21, 28, 20, 26, 25, 32, 36, 30, 6, 2, 24, 9, 30, 3, 21, 35) | |
| 40 | 39 | (14, 36, 7, 17, 28, 28, 1, 39, 23, 27, 33, 37, 7, 15, 0, 23, 23, 23, 17, 1, 34, 35, 10, 24, 32, 7, 7, 33, 23, 15, 31, 19, 31, 13, 23, 1, 37, 32, 5) | |
| 40 | 39 | (15, 1, 30, 31, 13, 27, 29, 25, 15, 27, 13, 38, 32, 3, 33, 31, 0, 26, 10, 10, 35, 34, 7, 11, 23, 34, 33, 12, 13, 1, 35, 11, 4, 23, 21, 12, 29, 39, 29) | |
| 41 | 40 | (15, 11, 2, 12, 35, 10, 24, 3, 32, 32, 34, 30, 1, 2, 25, 31, 1, 31, 36, 40, 14, 37, 3, 32, 18, 34, 17, 10, 13, 32, 33, 35, 21, 33, 33, 31, 32, 6, 5, 16) | |
| 43 | 42 | (16, 6, 16, 24, 24, 1, 6, 16, 40, 1, 8, 39, 2, 22, 18, 27, 39, 39, 36, 17, 42, 12, 31, 29, 15, 17, 21, 41, 23, 15, 10, 15, 34, 29, 18, 34, 3, 37, 30, 8, 35, 35) | |
| 49 | 4 | (7, 0, 0, 0) | $\leq 30$ |
| 49 | 4 | (7, 0, 0, 1) | $\leq 30$ |
| 64 | 2 | (4, 0) | $\leq 30$ |
| 64 | 2 | (4, 1) | $\leq 30$ |
| 64 | 11 | (16, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) | $\leq 30$ |
| 64 | 11 | (16, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1) | $\leq 30$ |
| 64 | 11 | (16, 0, 0, 0, 0, 0, 0, 0, 0, 4, 0) | $\leq 30$ |
| 64 | 11 | (16, 0, 0, 0, 0, 0, 0, 0, 0, 4, 1) | $\leq 30$ |
| 70 | 2 | (4, 3) | $\leq 30$ |
| 81 | 5 | (9, 0, 0, 0, 0) | $\leq 30$ |
| 81 | 5 | (9, 0, 0, 0, 1) | $\leq 30$ |
| 81 | 21 | (3, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 27) | $\leq 30$ |
| 81 | 21 | (3, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 9, 0, 0, 0, 27) | $\leq 30$ |
| 91 | 3 | (6, 1, 5) | $\leq 30$ |
| 93 | 2 | (3, 4) | $\leq 30$ |
| 121 | 6 | (11, 0, 0, 0, 0, 0) | $\leq 30$ |
| 121 | 6 | (11, 0, 0, 0, 0, 1) | $\leq 30$ |
| 125 | 17 | (25, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) | $\leq 30$ |
| 125 | 17 | (25, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1) | $\leq 30$ |
| 127 | 2 | (4, 4) | $\leq 30$ |

## K.3 Short solutions in all bases

The lists in this section come from `kdigit`, which fixes a digit string and solves for the base (Section 4.8). A list for $k$ digits with digits up to $M$ is complete for every base $b$ with $b^k \leq (M+1)^{M+1}$ (Proposition 1.5). Table K.4 gives the ranges. Beyond them, the binomial families of Theorem 2.21 give solutions of every length in very large bases.

Table K.4. Ranges of the complete lists of short solutions (L3.1, L11.5, L19.1, L25.2). For 3 digits the sieve of L25.2 extends the list of `kdigit` to every base up to $10^{3301}$; it found no new solution.

| $k$ | digits up to | complete for bases up to | source |
|---|---|---|---|
| 3 | 80; 2000 | $10^{51.5}$; $10^{3301}$ | `analysis/data/k3.txt`; `r5f_sieve3.txt` |
| 4 | 300 | $10^{186.5}$ | `analysis/data/k4.txt` |
| 5 | 80 | $10^{30.9}$ | `analysis/data/k5.txt` |
| 6 | 40 | $10^{11.0}$ | `analysis/data/k6.txt` |
| 7 | 32 | $10^{7.2}$ | `analysis/data/r3c_k7_M32.txt` |
| 8 | 24 | $10^{4.4}$ (base 23,364) | `analysis/data/r3c_k8_M24.txt` |

Two-digit solutions are described completely by Proposition 2.14, so they are not listed. Under $0^0 = 0$ the lists also contain the self-paying families $(x, 0, ..., 0)$ and $(x, 0, ..., 0, 1)$ with $x^{x-1} = b^{k-1}$ (Section 2.5). Table K.5 lists everything under $0^0 = 1$. Table K.6 lists the solutions under $0^0 = 0$ that are not in these families, and gives the number of family members in each list.

Table K.5. Every solution with 3 to 8 digits under $0^0 = 1$, in the ranges of Table K.4.

| $k$ | base | digits |
|---|---|---|
| 3 | 4 | (1, 3, 1) |
| 3 | 4 | (3, 1, 3) |
| 3 | 25 | (5, 1, 3) |
| 3 | 91 | (6, 1, 5) |
| 3 | 215 | (1, 2, 6) |
| 3 | 904 | (1, 7, 1) |
| 4 | 10 | (3, 4, 3, 5) |
| 4 | 20 | (6, 5, 3, 4) |
| 5 | 6 | (2, 2, 3, 5, 2) |
| 5 | 6 | (2, 3, 4, 5, 2) |
| 5 | 7 | (1, 3, 4, 5, 4) |
| 5 | 13 | (3, 3, 6, 6, 1) |
| 5 | 14 | (2, 6, 0, 3, 6) |
| 5 | 48 | (3, 8, 6, 2, 0) |
| 6 | 9 | (1, 5, 6, 2, 6, 2) |
| 7 | 9 | (1, 6, 5, 6, 5, 4, 7) |
| 8 | 11 | (1, 8, 4, 5, 3, 2, 7, 8) |
| 8 | 11 | (1, 8, 4, 5, 3, 4, 8, 7) |

Table K.6. Every solution with 3 to 8 digits under $0^0 = 0$ that is not a self-paying family member $(x, 0, ..., 0, e)$, $e \in \{0, 1\}$, in the ranges of Table K.4. Family members per length: 3: 84, 4: 204, 5: 38, 6: 14, 7: 12, 8: 4.

| $k$ | base | digits |
|---|---|---|
| 3 | 4 | (1, 3, 0) |
| 3 | 4 | (1, 3, 1) |
| 3 | 4 | (3, 1, 3) |
| 3 | 5 | (1, 0, 3) |
| 3 | 25 | (5, 1, 3) |
| 3 | 91 | (6, 1, 5) |
| 3 | 215 | (1, 2, 6) |
| 3 | 904 | (1, 7, 0) |
| 3 | 904 | (1, 7, 1) |
| 3 | $3^{40}$ | (3, 0, 27) |
| 4 | 5 | (2, 0, 2, 4) |
| 4 | 10 | (3, 4, 3, 5) |
| 4 | 20 | (6, 5, 3, 4) |
| 4 | 128 | (8, 0, 2, 4) |
| 4 | $2^{682}$ | (4, 0, 0, 256) |
| 5 | 6 | (2, 2, 3, 5, 2) |
| 5 | 6 | (2, 3, 4, 5, 2) |
| 5 | 7 | (1, 3, 4, 5, 4) |
| 5 | 11 | (6, 6, 5, 0, 0) |
| 5 | 11 | (6, 6, 5, 0, 1) |
| 5 | 13 | (3, 3, 6, 6, 0) |
| 5 | 13 | (3, 3, 6, 6, 1) |
| 5 | $3^{20}$ | (3, 0, 0, 0, 27) |
| 6 | 9 | (1, 5, 6, 2, 6, 2) |
| 6 | 11 | (5, 1, 7, 5, 0, 3) |
| 6 | $3^{16}$ | (3, 0, 0, 0, 0, 27) |
| 6 | $3^{16}$ | (3, 0, 0, 0, 9, 27) |
| 7 | 9 | (1, 6, 4, 7, 0, 6, 3) |
| 7 | 9 | (1, 6, 5, 6, 5, 4, 7) |
| 8 | 9 | (3, 4, 6, 6, 4, 0, 8, 4) |
| 8 | 11 | (1, 8, 4, 5, 3, 2, 7, 8) |

| $k$ | base | digits |
|---|---|---|
| 8 | 11 | (1, 8, 4, 5, 3, 4, 8, 7) |

## K.4 Negative bases

Table K.7 lists every nontrivial solution in the bases $-2$ to $-38$ (L12.3, L14.3, L23.2). A string $(s_{k-1}, ..., s_0)$ is a solution in base $-b$ when $\sum_i s_i(-b)^i = F(s)$. The census is complete: `worlds brute` and `worlds lanes` search every length allowed by the length bound of negative bases (Section 7.2). Bases not in the table have no solution under either convention: $-3$, $-6$, $-9$, $-15$, $-18$, $-20$, $-30$, $-34$ and $-38$.

Table K.7. All nontrivial solutions in the bases $-2$ to $-38$.

| base | convention | $k$ | digits |
|---|---|---|---|
| -2 | $0^0 = 0$ | 3 | (1, 1, 0) |
| -2 | $0^0 = 0$ | 3 | (1, 1, 1) |
| -2 | $0^0 = 1$ | 3 | (1, 1, 1) |
| -4 | $0^0 = 0$ | 5 | (1, 3, 0, 2, 3) |
| -5 | $0^0 = 0$ | 3 | (1, 0, 3) |
| -5 | $0^0 = 0$ | 5 | (1, 3, 4, 3, 3) |
| -5 | $0^0 = 1$ | 5 | (1, 3, 4, 3, 3) |
| -7 | $0^0 = 0$ | 3 | (1, 3, 0) |
| -7 | $0^0 = 0$ | 3 | (1, 3, 1) |
| -7 | $0^0 = 0$ | 7 | (1, 2, 6, 6, 2, 5, 0) |
| -7 | $0^0 = 0$ | 7 | (1, 2, 6, 6, 2, 5, 1) |
| -7 | $0^0 = 1$ | 3 | (1, 3, 1) |
| -7 | $0^0 = 1$ | 7 | (1, 2, 6, 6, 2, 5, 1) |
| -7 | $0^0 = 1$ | 7 | (1, 4, 0, 1, 0, 6, 5) |
| -8 | $0^0 = 0$ | 3 | (4, 0, 0) |
| -8 | $0^0 = 0$ | 3 | (4, 0, 1) |
| -8 | $0^0 = 1$ | 7 | (7, 5, 7, 6, 2, 0, 3) |
| -10 | $0^0 = 0$ | 9 | (4, 0, 5, 8, 1, 9, 4, 3, 0) |
| -10 | $0^0 = 0$ | 9 | (4, 0, 5, 8, 1, 9, 4, 3, 1) |
| -10 | $0^0 = 0$ | 9 | (5, 8, 1, 1, 8, 2, 4, 9, 2) |
| -10 | $0^0 = 0$ | 11 | (1, 9, 2, 1, 7, 4, 5, 8, 9, 8, 9) |
| -10 | $0^0 = 1$ | 9 | (4, 0, 5, 8, 1, 9, 1, 1, 1) |
| -10 | $0^0 = 1$ | 9 | (5, 8, 1, 1, 8, 2, 4, 9, 2) |
| -10 | $0^0 = 1$ | 11 | (1, 9, 2, 1, 7, 4, 5, 8, 9, 8, 9) |
| -11 | $0^0 = 0$ | 11 | (1, 3, 8, 9, 2, 10, 2, 5, 10, 7, 0) |
| -11 | $0^0 = 0$ | 11 | (1, 3, 8, 9, 2, 10, 2, 5, 10, 7, 1) |
| -11 | $0^0 = 1$ | 9 | (2, 2, 0, 9, 0, 3, 7, 6, 3) |
| -11 | $0^0 = 1$ | 11 | (1, 3, 8, 9, 2, 10, 2, 5, 10, 7, 1) |
| -12 | $0^0 = 1$ | 11 | (10, 8, 8, 9, 11, 2, 4, 7, 6, 0, 11) |
| -13 | $0^0 = 0$ | 7 | (4, 5, 7, 0, 1, 6, 8) |
| -13 | $0^0 = 0$ | 13 | (1, 3, 0, 8, 7, 5, 12, 10, 0, 12, 3, 3, 4) |
| -13 | $0^0 = 0$ | 13 | (1, 8, 6, 2, 7, 1, 11, 11, 9, 12, 5, 11, 2) |
| -13 | $0^0 = 0$ | 13 | (2, 12, 12, 0, 6, 12, 0, 0, 2, 9, 3, 6, 2) |
| -13 | $0^0 = 1$ | 13 | (1, 8, 6, 2, 7, 1, 11, 11, 9, 12, 5, 11, 2) |
| -14 | $0^0 = 0$ | 15 | (1, 13, 3, 12, 9, 13, 4, 5, 8, 8, 5, 13, 5, 7, 8) |
| -14 | $0^0 = 1$ | 13 | (1, 12, 3, 2, 8, 9, 10, 0, 10, 4, 0, 7, 4) |
| -14 | $0^0 = 1$ | 15 | (1, 13, 3, 12, 9, 13, 4, 5, 8, 8, 5, 13, 5, 7, 8) |
| -16 | $0^0 = 0$ | 15 | (7, 15, 5, 4, 10, 2, 10, 0, 11, 4, 0, 6, 7, 7, 13) |
| -17 | $0^0 = 0$ | 17 | (1, 11, 13, 5, 16, 4, 13, 1, 10, 15, 0, 3, 7, 1, 11, 6, 15) |
| -17 | $0^0 = 0$ | 17 | (1, 11, 8, 8, 8, 7, 5, 16, 3, 11, 0, 0, 5, 12, 0, 10, 9) |
| -17 | $0^0 = 1$ | 17 | (2, 15, 11, 2, 16, 8, 2, 13, 0, 16, 15, 3, 16, 10, 0, 9, 14) |
| -19 | $0^0 = 0$ | 19 | (1, 12, 7, 13, 11, 17, 18, 10, 7, 12, 7, 5, 14, 14, 12, 13, 16, 13, 15) |
| -19 | $0^0 = 1$ | 19 | (1, 12, 7, 13, 11, 17, 18, 10, 7, 12, 7, 5, 14, 14, 12, 13, 16, 13, 15) |
| -21 | $0^0 = 0$ | 21 | (1, 14, 20, 17, 9, 12, 3, 1, 6, 16, 11, 10, 10, 9, 4, 15, 13, 3, 15, 5, 11) |

| base | convention | $k$ | digits |
|---|---|---|---|
| -21 | $0^0 = 1$ | 21 | (1, 14, 20, 17, 9, 12, 3, 1, 6, 16, 11, 10, 10, 9, 4, 15, 13, 3, 15, 5, 11) |
| -21 | $0^0 = 1$ | 21 | (1, 5, 0, 9, 20, 18, 10, 13, 20, 7, 10, 12, 5, 8, 19, 2, 0, 6, 8, 16, 4) |
| -22 | $0^0 = 0$ | 21 | (9, 16, 5, 8, 2, 5, 1, 5, 17, 5, 4, 17, 4, 7, 21, 1, 1, 9, 1, 0, 5) |
| -23 | $0^0 = 0$ | 21 | (4, 10, 12, 2, 20, 2, 6, 9, 18, 21, 11, 14, 5, 14, 14, 18, 8, 20, 20, 0, 10) |
| -23 | $0^0 = 0$ | 23 | (1, 15, 16, 20, 2, 7, 12, 11, 15, 20, 2, 20, 10, 13, 2, 22, 19, 15, 10, 16, 6, 11, 6) |
| -23 | $0^0 = 1$ | 23 | (1, 15, 16, 20, 2, 7, 12, 11, 15, 20, 2, 20, 10, 13, 2, 22, 19, 15, 10, 16, 6, 11, 6) |
| -24 | $0^0 = 0$ | 23 | (1, 21, 18, 17, 1, 3, 5, 5, 22, 16, 4, 4, 7, 12, 21, 19, 11, 2, 21, 4, 6, 0, 17) |
| -24 | $0^0 = 0$ | 23 | (10, 17, 17, 15, 22, 8, 21, 20, 15, 6, 23, 22, 21, 8, 21, 19, 14, 7, 1, 0, 14, 1, 18) |
| -24 | $0^0 = 0$ | 23 | (10, 20, 1, 11, 18, 1, 9, 12, 12, 4, 8, 13, 21, 0, 14, 2, 22, 3, 13, 3, 8, 1, 23) |
| -25 | $0^0 = 0$ | 3 | (5, 0, 0) |
| -25 | $0^0 = 0$ | 3 | (5, 0, 1) |
| -26 | $0^0 = 1$ | 27 | (1, 25, 13, 0, 10, 6, 13, 19, 25, 20, 18, 12, 25, 25, 4, 6, 21, 22, 16, 11, 8, 23, 0, 17, 7, 3, 3) |
| -27 | $0^0 = 0$ | 25 | (4, 2, 13, 25, 6, 6, 21, 18, 5, 11, 0, 3, 1, 19, 8, 12, 21, 17, 4, 16, 2, 21, 5, 15, 22) |
| -27 | $0^0 = 0$ | 27 | (1, 17, 4, 20, 1, 4, 4, 2, 0, 3, 7, 0, 10, 22, 2, 1, 17, 15, 20, 22, 9, 14, 18, 19, 12, 26, 16) |
| -27 | $0^0 = 1$ | 25 | (1, 24, 6, 18, 5, 19, 0, 19, 13, 3, 1, 16, 13, 5, 6, 3, 9, 0, 24, 22, 8, 4, 16, 22, 16) |
| -28 | $0^0 = 0$ | 23 | (20, 9, 17, 23, 9, 1, 1, 11, 8, 18, 18, 12, 3, 19, 1, 5, 8, 9, 24, 2, 18, 22, 5) |
| -28 | $0^0 = 1$ | 23 | (20, 9, 17, 23, 9, 1, 1, 11, 8, 18, 18, 12, 3, 19, 1, 5, 8, 9, 24, 2, 18, 22, 5) |
| -28 | $0^0 = 1$ | 27 | (1, 20, 6, 1, 22, 0, 6, 26, 15, 23, 26, 5, 19, 17, 25, 8, 15, 6, 10, 4, 16, 13, 2, 4, 14, 9, 24) |
| -29 | $0^0 = 0$ | 27 | (5, 23, 5, 24, 9, 0, 11, 4, 27, 17, 10, 2, 8, 2, 21, 2, 7, 22, 16, 8, 2, 0, 6, 18, 22, 22, 3) |
| -31 | $0^0 = 0$ | 29 | (10, 29, 15, 25, 4, 24, 20, 17, 5, 11, 28, 23, 0, 25, 29, 5, 6, 28, 15, 2, 18, 14, 10, 9, 24, 10, 18, 27, 12) |
| -31 | $0^0 = 0$ | 29 | (9, 0, 22, 21, 24, 5, 27, 8, 29, 21, 10, 4, 28, 14, 8, 10, 13, 15, 22, 5, 29, 9, 13, 2, 17, 7, 27, 25, 12) |
| -31 | $0^0 = 0$ | 31 | (1, 20, 28, 18, 3, 4, 20, 17, 14, 10, 18, 16, 29, 18, 12, 4, 19, 30, 28, 7, 2, 25, 10, 9, 17, 9, 16, 7, 25, 3, 29) |
| -31 | $0^0 = 0$ | 31 | (1, 20, 28, 20, 11, 30, 9, 2, 12, 12, 17, 4, 19, 29, 16, 21, 24, 5, 20, 22, 14, 29, 4, 23, 5, 8, 27, 12, 8, 4, 8) |
| -31 | $0^0 = 0$ | 31 | (1, 8, 20, 26, 29, 5, 28, 21, 22, 9, 15, 24, 8, 11, 27, 29, 5, 30, 30, 16, 15, 22, 25, 10, 7, 27, 6, 29, 13, 13, 14) |
| -31 | $0^0 = 1$ | 31 | (1, 20, 28, 18, 3, 4, 20, 17, 14, 10, 18, 16, 29, 18, 12, 4, 19, 30, 28, 7, 2, 25, 10, 9, 17, 9, 16, 7, 25, 3, 29) |
| -31 | $0^0 = 1$ | 31 | (1, 20, 28, 20, 11, 30, 20, 27, 25, 20, 0, 1, 1, 6, 4, 3, 15, 1, 11, 14, 25, 6, 16, 23, 1, 21, 29, 29, 15, 0, 26) |
| -31 | $0^0 = 1$ | 31 | (1, 20, 28, 20, 11, 30, 9, 2, 12, 12, 17, 4, 19, 29, 16, 21, 24, 5, 20, 22, 14, 29, 4, 23, 5, 8, 27, 12, 8, 4, 8) |
| -31 | $0^0 = 1$ | 31 | (1, 8, 20, 26, 29, 5, 28, 21, 22, 9, 15, 24, 8, 11, 27, 29, 5, 30, 30, 16, 15, 22, 25, 10, 7, 27, 6, 29, 13, 13, 14) |
| -32 | $0^0 = 0$ | 13 | (16, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) |
| -32 | $0^0 = 0$ | 13 | (16, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1) |
| -32 | $0^0 = 0$ | 31 | (1, 28, 20, 8, 18, 12, 2, 24, 25, 7, 28, 11, 23, 26, 9, 9, 21, 5, 15, 2, 13, 0, 20, 8, 5, 30, 10, 10, 20, 26, 0) |
| -32 | $0^0 = 0$ | 31 | (1, 28, 20, 8, 18, 12, 2, 24, 25, 7, 28, 11, 23, 26, 9, 9, 21, 5, 15, 2, 13, 0, 20, 8, 5, 30, 10, 10, 20, 26, 1) |
| -32 | $0^0 = 1$ | 31 | (1, 23, 12, 27, 17, 30, 1, 11, 11, 30, 9, 20, 23, 28, 28, 20, 29, 8, 2, 0, 29, 3, 23, 0, 3, 27, 25, 11, 26, 14, 0) |
| -32 | $0^0 = 1$ | 31 | (25, 30, 1, 6, 20, 28, 28, 1, 22, 16, 9, 8, 0, 27, 12, 5, 16, 24, 17, 12, 14, 5, 6, 18, 31, 20, 26, 13, 4, 0, 31) |
| -33 | $0^0 = 0$ | 33 | (1, 21, 11, 7, 15, 28, 18, 24, 0, 3, 2, 22, 15, 32, 2, 7, 0, 28, 4, 20, 20, 26, 16, 6, 20, 15, 1, 22, 12, 17, 12, 6, 29) |
| -33 | $0^0 = 0$ | 33 | (1, 21, 11, 7, 15, 28, 7, 18, 26, 19, 22, 32, 2, 29, 23, 12, 19, 21, 1, 25, 9, 13, 4, 14, 13, 27, 2, 3, 13, 10, 20, 16, 13) |
| -33 | $0^0 = 0$ | 33 | (1, 21, 11, 7, 15, 3, 18, 2, 29, 0, 8, 11, 23, 8, 18, 16, 4, 12, 19, 11, 5, 21, 3, 22, 21, 9, 22, 25, 10, 29, 16, 14, 32) |
| -33 | $0^0 = 1$ | 33 | (1, 21, 11, 7, 15, 28, 7, 18, 26, 19, 22, 32, 2, 29, 23, 12, 19, 21, 1, 25, 9, 13, 4, 14, 13, 27, 2, 3, 13, 10, 20, 16, 13) |
| -35 | $0^0 = 0$ | 33 | (6, 31, 27, 18, 31, 18, 32, 33, 12, 28, 2, 28, 18, 32, 25, 25, 23, 2, 30, 5, 11, 12, 10, 0, 10, 30, 24, 14, 2, 1, 28, 9, 15) |
| -36 | $0^0 = 0$ | 35 | (27, 0, 6, 7, 13, 26, 29, 2, 25, 2, 28, 3, 16, 0, 19, 11, 35, 8, 24, 4, 34, 12, 28, 17, 20, 35, 18, 5, 10, 20, 22, 18, 1, 7, 2) |
| -36 | $0^0 = 1$ | 33 | (5, 33, 22, 0, 27, 22, 8, 20, 5, 26, 32, 12, 2, 5, 5, 31, 17, 23, 21, 21, 4, 24, 17, 17, 27, 16, 12, 25, 2, 33, 31, 24, 16) |
| -36 | $0^0 = 1$ | 35 | (27, 0, 6, 7, 13, 26, 29, 2, 25, 2, 28, 3, 16, 0, 19, 11, 35, 8, 24, 4, 34, 12, 28, 17, 20, 35, 18, 5, 10, 20, 22, 18, 1, 7, 1) |
| -36 | $0^0 = 1$ | 35 | (27, 5, 1, 27, 32, 31, 35, 15, 27, 0, 32, 13, 12, 3, 12, 28, 23, 13, 28, 32, 8, 13, 8, 35, 14, 14, 3, 29, 8, 30, 23, 14, 31, 15, 6) |
| -37 | $0^0 = 0$ | 37 | (1, 24, 35, 5, 34, 28, 25, 23, 5, 22, 14, 9, 11, 3, 11, 4, 30, 31, 17, 25, 17, 5, 5, 15, 15, 36, 15, 4, 12, 12, 15, 22, 29, 11, 8, 21, 14) |
| -37 | $0^0 = 1$ | 37 | (1, 24, 35, 5, 34, 28, 25, 23, 5, 22, 14, 9, 11, 3, 11, 4, 30, 31, 17, 25, 17, 5, 5, 15, 15, 36, 15, 4, 12, 12, 15, 22, 29, 11, 8, 21, 14) |

## K.5 Other number systems

The solutions in the other number systems of Chapter 7 are short lists. The chapter states them where they are discussed:

- imaginary bases $ci$ and Gaussian bases $-n + i$: Section 7.3 (L14.4, L14.5, L23.4);
- Eisenstein bases $-n + \omega$: Section 7.3 (L21.3);

- rational bases $p/q$: Section 7.4 (L16.5, L21.2);
- the Pisot bases $\left(x+\sqrt{x^2+4}\right)/2$: Section 7.5 (L21.4);
- the factorial number system: Section 7.9 (L21.5);
- function fields $\mathbb{F}_q[T]$: Section 7.8 (L21.1, L24.3).

Appendix L

# Software and data

Every computation in this document can be repeated with the programs in the repository. This appendix says what each program does, how the results were checked, and where the data lives. The library cites the script behind each number; the badges $Lx.y$ lead to it.

## L.1 The repository

Table L.1. The parts of the repository.

| path | contents |
|---|---|
| `src/` | the original Python search over digit multisets, and the output format (`formatutil.py`) |
| `fast/` | the Rust programs: the exact search `fast`, `kdigit` for short solutions, and engines for other number systems |
| `output/` | the exhaustive lists of solutions: `b2-41-0.txt`, `b2-41-1.txt`, `b43-*.txt` |
| `analysis/` | Python, PARI/GP, Sage and C scripts for the model, the number theory and the experiments |
| `analysis/data/` | the output of every script, cited by file name in the library |
| `docs/algorithm.md` | the search algorithm in detail |
| `docs/library/` | the library: 35 chapters and the open questions |
| `docs/exposition/` | this document |
| `lean/` | the Lean 4 formalization (Appendix M) |

## L.2 The exact search

`fast search` finds every solution in a base under both conventions in one run. It is the prefix-pruned tree of Section 4.1: it fixes the counts of the digits from the largest down, keeps an interval for the sum, and uses the leading digits that the interval shares. Numbers are stored as one byte per base-$b$ digit in AVX2 registers, so the program needs no big-number library and handles bases up to 127 and numbers up to 62 digits. A carry-lookahead identity on bit masks adds two such vectors in about 15 instructions.

Large runs are split into about $10^5$ work units in a fixed order. With `--journal` the program records each finished unit and each solution at once, so a stopped run resumes where it stopped. `--shard i/n` splits a base over machines, and `fast merge` combines the outputs. `--progress` prints the node count every 10 seconds and an estimate of the share of the work done; the estimate weights each unit by the size its type predicts and is off by at most 0.10, against 0.49 for a plain count of units (L27.5). `--unit-stats <file>` writes one line per finished unit, for studies of the work law: a few unit types hold most of the work, and scheduling on one machine is within 0.1% of ideal. `--shard-by size` gives each shard units of about equal predicted size instead of every $n$-th unit (L31.5). The search drops a node with one free slot when no digit completes its residue modulo $b-1$ (Theorem 4.4); `--no-residue` turns this off, to compare or to resume a journal of an older build. With `-m 0` or `-m 1` the search reports one convention, and the rule tests only that one.

With `--features gpu` the program also runs on a GPU through Vulkan 1.3. `--gpu` runs every unit on the GPU, and `--hybrid` shares the units between the GPU and the CPU threads. Both give the same solutions and node counts as the CPU search, and both support the `--zscore` rule and `--shard`

(L28.3). `--journal` does not work with the GPU. At base 38 with `--zscore 3.4 --zlaw cond`, `--hybrid` took 24.7 s against 70.5 s on 8 CPU threads.

Other options change the weights for experiments on the model: `--zero`, `--shift`, `--factorial`, `--power`, `--perturb`, `--keep-residues` and `--set-residue` (Section 3.4, Section 7.6). The inexact rules `--cap`, `--band`, `--deficit`, `--ratio` and `--zscore` skip nodes that the model calls unlikely, and print the expected number of missed solutions (Section 4.3). The z-score rule compares $\ln E$ with a law chosen by `--zlaw`: `poisson`, the law of L10.3, overstates the standard deviation by up to 3.8 times; `cond` conditions the cells on their total and is within 5% of a simulation (L28.1). `--zhist <file>` records, per length, the nodes by the smallest z-score on their path, so one run gives the cost of every threshold below the one used. With `--offset D` the search counts the near-solutions $n - F(n) = D$ (Section 3.5).

Table L.2. Recommended settings for large bases (L28.3).

| goal | setting |
|---|---|
| miss about 2% of solutions | `--zscore 3.4 --zlaw cond` |
| miss about 1% of solutions | `--zscore 3.8 --zlaw cond` |
| exact search | no inexact rule; `--journal` for long runs |
| a machine with a GPU | add `--hybrid` (2.6 to 2.9 times faster); not above base 64 |
| several machines | `--shard i/n` on each, then `fast merge`; for 16 shards or more add `--unit-depth 6 --shard-by size` |
| one convention only | `-m 0` or `-m 1`; the residue rule then removes more |
| length | do not restrict the length; windows do not pay (L28.2) |

The miss rates hold from base 30 to 200 in the model. `fast search 10 41 --zscore 3.4 --zlaw cond --hybrid` finds all 122 nontrivial solutions of bases 10 to 41.

Base 37 takes about 205 seconds on an 8-thread laptop, and each further base costs about 1.9 times more.

## L.3 Short solutions

`fast/target/release/kdigit k M` lists every $k$-digit solution in every base $b$ with $b^k \le (M+1)^{M+1}$. It enumerates digit strings, tests the polynomial $Q_s$ for a root modulo a list of prime powers (default 32, 27, 25, 49 and the primes 11 to 127), and solves $P_s(b) = F(s)$ exactly for the few strings that pass (Section 4.8, Section 5.2). Options: `--offset D`, `--moduli`, `--stats`, `--survivors`. The Python version `analysis/kdigit.py` is the slow reference.

## L.4 Engines for other systems

Table L.3. The engines in `fast/src/bin/`.

| program | what it does | used in |
|---|---|---|
| `worlds` | negative bases ($-2$ to $-38$), imaginary bases, factorions; engines `brute`, `tree`, `lanes` | Section 7.2 |
| `gauss` | Gaussian bases $-n + i$ | Section 7.3 |
| `r3t_eisen` | Eisenstein bases $-n + \omega$, with a Gaussian control | Section 7.3 |
| `r3t_afs` | rational bases $p/q$ by a tree of parents | Section 7.4 |
| `r3t_pisot` | the Pisot bases $\left(x + \sqrt{x^2+4}\right)/2$ in exact arithmetic of $\mathbb{Z}[\varepsilon]$ | Section 7.5 |
| `r3t_factorial` | the factorial number system up to 34! | Section 7.9 |
| `r3t_ffield`, `r4f_ffield` | function fields $\mathbb{F}_q[T]$; the second uses the Frobenius blocks | Section 7.8 |
| `dyn_cycles`, `r2c_cycles2` | cycles and near-2-cycles of $n \mapsto F(n)$ on digit multisets | Section 3.10 |

## L.5 Analysis scripts

The scripts in `analysis/` carry a prefix for the round of work that wrote them. Each library chapter ends with a table of its scripts and data files.

Table L.4. Prefixes of the analysis scripts.

| prefix | topic | library |
|---|---|---|
| (none) | model, counts, tree measurements, inexact rules, defects, local tests | L1 to L12 |
| `thy_` | the NP-completeness reduction and the Poisson theorem | L13 |
| `wld_` | negative, imaginary and Gaussian bases, factorions | L14 |
| `dyn_` | cycles, two-digit bases, coverings, constructions, several bases | L15 |
| `alg_` | Aurifeuillian factors, reducibility, Galois groups, rational roots | L16 |
| `wgt_` | other weights | L17 |
| `r2nt_` | the equation $b(b+x) = x^n + 1$ | L18 |
| `r3c_` | the prime-power sieve, short-string spectra, narcissistic numbers | L19 |
| `r3nt_` | multisets, residues of $x^x - x$, the even case, $c = 3, 25$ | L20 |
| `r3t_` | five more number systems | L21 |
| `r4b_` | blocks and binomial families | L22 |
| `r2c_` | factorions, empty negative bases, cycles, scatter | L23 |
| `r4f_` | function fields: why the model misses | L24 |
| `r5f_` | identity families and 3-digit solutions | L25 |
| `r5e_` | near-solutions as evidence | L26 |
| `r5l_` | limits of search: solvers, orders, certificates, work units | L27 |
| `r5p_` | the z-score law, length windows and the GPU | L28 |
| `r5m_` | theorems about the model: length shares, $k(B)$, $\rho(b)$, block and top-digit models | L29 |
| `r6a_` | anti-concentration, collisions of $x^x - x$, fixed points, 2-digit narcissistic numbers | L30 |
| `r6s_` | the residue rule, nodes by free slots, cycle counts, shard balance | L31 |
| `r6b_` | the 2-adic bound for 3-digit families and the sieve below it | L32 |
| `r6f_` | least inert primes, 3-digit factorions, the Brocard sieve, 4-digit factorions, identity census, explicit abc bounds, chains | L33 |
| `r6q_` | the exact refined model by zero class, the length matrix and its trace, characters, perturbed worlds, the ledger of P values, neighbouring bases | L34 |
| `r6h_` | arrangements modulo $q$, the root heuristic against the refined model, invariant ideals of 67 systems, the averaged multiset count, near-solutions in every base | L35 |

The number theory uses PARI/GP [75] for class groups (`bnfinit`, with `bnfcertify` where a result must not depend on GRH), regulators and integral points, and Sage [99] for elliptic curves.

## L.6 How the results were checked

- `fast` recomputes $F(n)$ with arbitrary-precision integers before it prints a solution. `cargo test` compares it with a brute force (bases 2 to 9) and with a plain `u128` reference search (bases 10 to 22).
- The lists of bases 2 to 41 agree with the older Python search for bases 2 to 19, and with OEIS A166623 for bases 2 to 35 under $0^0 = 1$.
- Every option for other weights is tested against a brute force over all digit multisets in bases 2 to 8.
- `kdigit` reproduces the Python lists line for line, and the prime-power sieve reproduces the prime sieve.
- The engines for other systems agree with each other and with brute force where both run, as the library records for each one.
- The script `analysis/verify_constructions.py` checks the families of Chapter 2 with exact integers.

## L.7 Data in this directory

`docs/exposition/data/` holds CSV copies of the lists that Appendix K prints: `counts-by-base.csv`, `solutions-by-base.csv`, `short-solutions.csv` and `negative-bases.csv`. Run

```
python3 docs/exposition/tools/gen_data.py
```

after the lists in `output/` or `analysis/data/` change, then rebuild the document.

Appendix M

# Formalization in Lean

A formalization of the library in Lean 4 with Mathlib [100] is being built in the directory `lean/` of the repository. Its record, `lean/STATUS.md`, maps every proven result of the library to a Lean declaration or gives the reason why it is not formalized: usually a heuristic statement, a large computation, or a deep cited theorem such as Chebotarev's or Laurent's. A statement of this document that has a formal proof carries a green Lean badge, and its box ends with the name of the Lean declaration. The table below lists all of them, and the index of results (Appendix N) marks them.

## M.1 Status

61 statements of this document are formalized, some of them in part; the table names what a partial formalization covers. All Lean names live in the namespace `Munchausen`. The formalization also states several conjectures of the library as propositions and proves the negation of four refuted ones, among them the finiteness conjecture for short solutions under $0^0 = 1$ (Corollary 2.22).

| Statement | Name | Lean declaration |
|---|---|---|
| Proposition 1.3 | Length bound | `length_digits_le` |
| Proposition 1.5 | Digit bound | `pow_self_lt_of_mem_digits` |
| Proposition 1.7 | One base per string | `IsSolution.base_unique` |
| Proposition 1.8 | The two conventions | `F_one_eq` |
| Proposition 1.9 | Digit-sum identity | `IsSolution.Fs_modEq_sum` |
| Proposition 1.11 | Rules from the identity | `IsSolution.even_count_zero`, `IsSolution.even_countP_mod_four`, `IsSolution.three_dvd_countP_mod_six`, `IsMunchausen.last_digit_parity` |
| Proposition 1.12 | The only invariant residue | `residue_invariant_iff` |
| Theorem 1.13 | Parity invariant | `F_modEq_self_iff` |
| Proposition 2.1 | Twins | `isSolution_twin` |
| Proposition 2.2 | Which digits pay for themselves | `selfPaying_of_dvd`, `SelfPaying.exists_pow` |
| Proposition 2.3 | Bases with a self-paying digit | `prop15_8` |
| Proposition 2.4 | Self-paying digits are far apart | `SelfPaying.position_gt`, `card_selfPaying_lt_log` |
| Proposition 2.5 | Structure under $0^0 = 0$ | `isSolution_selfPaying_iff`, `isMunchausen_iff_core` |
| Corollary 2.6 | Solutions from self-paying digits | `corollary_2_4` |
| Proposition 2.7 | A self-paying digit on top | `isSolution_selfPaying_cons_iff` |
| Theorem 2.13 | Swaps are cores in self-paying bases | `theorem15_9` |
| Proposition 2.14 | Two-digit solutions | `isSolution_two_iff` |
| Theorem 2.19 | Two 2-digit solutions | `theorem15_7 (partial: the case 0^0 = 1)` |
| Proposition 2.20 | Every length under $0^0 = 0$ | `isSolution_prop27` |
| Theorem 2.21 | Binomial families | `isSolution_binomial` |

| Statement | Name | Lean declaration |
|---|---|---|
| Corollary 2.22 | Every length under $0^0 = 1$ | `isSolution_corollary_22_7, infinite_solutions_of_length` |
| Proposition 2.24 | When the remainder vanishes | `prop22_8` |
| Theorem 2.27 | The large digit is last | `theorem25_2` |
| Theorem 2.30 | A second infinite family | `theorem25_5` |
| Lemma 2.41 | Splits | `split_iff` |
| Proposition 2.45 | Unbounded counts under $0^0 = 0$ | `theorem9_2, prop4_2` |
| Proposition 2.46 | Rich bases are rare | `selfPaying_base_pow, card_selfPaying_le (partial: the shape r^t with t >= c + 1)` |
| Proposition 2.50 | Two-digit representations under $0^0 = 1$ | `prop15_16` |
| Lemma 3.43 | Swap lemma | `swap_lemma, swap_lemma_perm, swap_last_two` |
| Proposition 3.74 | Cycles live on multisets | `F_lt_pow_self (partial: F maps numbers below b^b to themselves)` |
| Proposition 3.77 | Coincident sums | `prop26_10_one, prop26_10_two` |
| Proposition 4.26 | Few solutions of a fixed length | `prop4_1, prop10_4` |
| Proposition 5.9 | Local obstruction | `not_isSolution_of_no_root_mod` |
| Proposition 5.10 | The test modulo 2 | `exists_root_mod_two_iff` |
| Lemma 5.12 | Small moduli | `T_prime_eq_univ, T_three, pow_self_modEq_27, multiple_mem_T (partial: part 1 modulo 2^e only for multiples of 2)` |
| Theorem 5.13 | Covering by 27 | `covering_27` |
| Lemma 5.16 | Lifting | `lemma20_7` |
| Theorem 5.17 | Primes from 7 on | `theorem20_8` |
| Theorem 5.18 | The 3-adic law | `pow_self_modEq_81 (partial: the inclusion modulo 81)` |
| Corollary 5.20 | 81 is the last useful power of 3 | `covering_81` |
| Lemma 5.37 | Blocks are inside the heuristic | `P_sub_Fs_append_one, P_sub_Fs_append_zero` |
| Theorem 5.45 | Large leading digit | `theorem8_3, theorem8_3_one` |
| Theorem 5.46 | Three digits with one large digit | `theorem25_7_ii_last, theorem25_7_ii_middle, theorem25_7_ii_leading, prop22_8 (partial: case (ii), and case (i) for even x)` |
| Proposition 5.56 | Few short solutions per base | `prop4_1` |
| Proposition 5.57 | Few bases per length | `prop10_4` |
| Proposition 5.67 | Odd length | `IsNegSolution.odd_length` |
| Proposition 5.68 | Partners | `IsSolution.negPartner` |
| Lemma 5.75 | Two-digit bases are large | `lemma18_14` |
| Lemma 5.78 | The largest digit | `lemma25_13` |
| Corollary 5.79 | Two consecutive values | `corollary25_14` |
| Proposition 5.81 | One member per base | `prop25_16_even (partial: members with even x)` |
| Proposition 6.7 | Elementary exclusions | `prop18_1_a, prop18_1_b, prop18_1_c, prop18_1_d, prop16_5` |
| Proposition 6.34 | The family $(1, 2, x)$ | `prop20_20` |
| Proposition 7.7 | Odd length | `IsNegSolution.odd_length` |
| Proposition 7.10 | The only invariant residue | `neg_residue_invariant_iff, IsNegSolution.dvd` |
| Theorem 7.22 | Base $-1 + i$ is base $-4$ in blocks | `gaussian_blocks (partial: the block form; existence and uniqueness cited)` |
| Proposition 7.52 | Linearity | `ffWeight_eq_expand, ffWeight_sum_of_degree` |
| Proposition 7.53 | Families and self-paying digits | `ffWeight_add_C (partial: the weight identity)` |

| Statement | Name | Lean declaration |
|---|---|---|
| Lemma 7.61 | Frobenius blocks | `expand_ringVal (partial: the identity before uniqueness of expansions)` |
| Proposition 7.74 | No invariant residue | `factorial_no_invariant_residue` |
| Proposition 7.75 | Self-paying digits | `factorial_selfPaying` |

## M.2 How the marks are kept up to date

`lean/STATUS.md` in the repository is the record of what is formalized. The file `docs/exposition/lean-status.typ` maps the id of each statement of this document to its Lean declaration. When a result is formalized, add a line such as

```
"thm-length": "Munchausen.length_le_base",
```

and rebuild. The badge, the name in the box and the tables of this appendix and of Appendix N follow from that one line.

A formal proof covers exactly the statement in Lean, which can be narrower than the statement in the box: for example one convention, or a bound with explicit constants. Where the two differ, the box should say so in a remark.

Appendix N

# Index of results

This index lists every statement of the document that makes a claim, in order, with its status, the library sections that hold the details, and the page. Definitions, remarks and examples are left out. The index builds itself from the statement boxes, so it is always complete.

The statuses mean:

| status | meaning |
|---|---|
| PROVEN | a proof is in the library, and usually a proof or sketch is here |
| CITED | a published theorem, used without proof |
| CONDITIONAL | proven under a stated conjecture or assumption, such as abc |
| COMPUTATION | an exhaustive computation; the range is part of the statement |
| HEURISTIC | a prediction of the probabilistic model or the root heuristic, tested against data |
| CONJECTURE | a conjecture, with its evidence in the text |
| OPEN | an open problem |

Totals in this version:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| PROVEN | 200 | CITED | 35 | COMPUTATION | 103 | HEURISTIC | 32 |
| CONJECTURE | 13 | CONDITIONAL | 7 | OPEN | 38 | LEAN | 61 |

| Statement | Name | Status | Library | Page | Lean |
|---|---|---|---|---|---|
| Proposition 1.3 | Length bound | PROVEN | L1.2 | 9 | LEAN |
| Proposition 1.5 | Digit bound | PROVEN | L1.2 | 9 | LEAN |
| Proposition 1.7 | One base per string | PROVEN | L1.4 | 10 | LEAN |
| Proposition 1.8 | The two conventions | PROVEN | L1.5 | 10 | LEAN |
| Proposition 1.9 | Digit-sum identity | PROVEN | L1.6 | 11 | LEAN |
| Proposition 1.11 | Rules from the identity | PROVEN | L1.6 | 11 | LEAN |
| Proposition 1.12 | The only invariant residue | PROVEN | L10.4 | 11 | LEAN |
| Theorem 1.13 | Parity invariant | PROVEN | L15.1 | 12 | LEAN |
| Proposition 1.14 | Solutions with $b$ digits | PROVEN | L1.8 | 12 | |
| Computation 1.15 | Solutions with $b$ digits | COMPUTATION | L7.1, L8.6 | 12 | |
| Proposition 2.1 | Twins | PROVEN | L2.1 | 15 | LEAN |
| Proposition 2.2 | Which digits pay for themselves | PROVEN | L2.2 | 15 | LEAN |
| Proposition 2.3 | Bases with a self-paying digit | PROVEN | L15.3 | 16 | LEAN |
| Proposition 2.4 | Self-paying digits are far apart | PROVEN | L8.3 | 16 | LEAN |
| Proposition 2.5 | Structure under $0^0 = 0$ | PROVEN | L2.2 | 16 | LEAN |
| Corollary 2.6 | Solutions from self-paying digits | PROVEN | L2.2 | 16 | LEAN |
| Proposition 2.7 | A self-paying digit on top | PROVEN | L2.2 | 17 | LEAN |
| Theorem 2.10 | Chains and cycles | PROVEN | L15.5 | 17 | |
| Computation 2.11 | Chain solutions | COMPUTATION | L15.5, L33.5 | 18 | |
| Proposition 2.12 | One-step chains | PROVEN | L33.5 | 18 | |

| Statement | Name | Status | Library | Page | Lean |
|---|---|---|---|---|---|
| Theorem 2.13 | Swaps are cores in self-paying bases | PROVEN | L15.3 | 18 | LEAN |
| Proposition 2.14 | Two-digit solutions | PROVEN | L2.4 | 18 | LEAN |
| Computation 2.15 | Two-digit solutions with small digits | COMPUTATION | L2.4 | 19 | |
| Proposition 2.16 | Density of the last digits | PROVEN | L30.2 | 19 | |
| Corollary 2.17 | Small last digits | CITED | L30.2 | 19 | |
| Computation 2.18 | Fixed points and the census | COMPUTATION | L30.2 | 20 | |
| Theorem 2.19 | Two 2-digit solutions | PROVEN | L15.2 | 20 | LEAN |
| Proposition 2.20 | Every length under $0^0 = 0$ | PROVEN | L2.5 | 20 | LEAN |
| Theorem 2.21 | Binomial families | PROVEN | L22.3 | 21 | LEAN |
| Corollary 2.22 | Every length under $0^0 = 1$ | PROVEN | L22.3 | 21 | LEAN |
| Proposition 2.24 | When the remainder vanishes | PROVEN | L22.3 | 22 | LEAN |
| Theorem 2.27 | The large digit is last | PROVEN | L25.1 | 23 | LEAN |
| Corollary 2.28 | One large digit | PROVEN | L25.1 | 23 | |
| Theorem 2.29 | Linear identities | PROVEN | L25.1 | 23 | |
| Theorem 2.30 | A second infinite family | PROVEN | L25.1 | 23 | LEAN |
| Proposition 2.33 | Nonlinear identities | PROVEN | L25.1 | 24 | |
| Computation 2.34 | Census of identities | COMPUTATION | L25.1 | 25 | |
| Conjecture 2.35 | Families explain the rest | CONJECTURE | L25.1 | 25 | |
| Theorem 2.36 | The next 3-digit member | PROVEN | L25.2 | 25 | |
| Theorem 2.37 | A sufficient condition for families | PROVEN | L33.3 | 25 | |
| Proposition 2.38 | Factorials are almost never powers | PROVEN | L33.3 | 26 | |
| Computation 2.39 | Identity members for five weights | COMPUTATION | L33.3 | 26 | |
| Theorem 2.40 | One dominant digit under abc | CONDITIONAL | L33.4 | 26 | |
| Lemma 2.41 | Splits | PROVEN | L22.1 | 27 | LEAN |
| Computation 2.43 | All blocks with small digits | COMPUTATION | L22.2 | 27 | |
| Heuristic 2.44 | Rate of blocks | HEURISTIC | L22.2 | 28 | |
| Proposition 2.45 | Unbounded counts under $0^0 = 0$ | PROVEN | L4.1, L9.8 | 28 | LEAN |
| Proposition 2.46 | Rich bases are rare | PROVEN | L9.8 | 29 | LEAN |
| Proposition 2.47 | Self-paying digits under $0^0 = 1$ | PROVEN | L8.4 | 29 | |
| Proposition 2.48 | Many bases under $0^0 = 0$ | PROVEN | L15.6 | 30 | |
| Proposition 2.50 | Two-digit representations under $0^0 = 1$ | PROVEN | L15.6 | 30 | LEAN |
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Appendix O

# Notation and glossary

## O.1 Notation

| symbol | meaning | see |
|---|---|---|
| $b$ | the base, an integer $b \geq 2$ unless a chapter says otherwise | Section 1.1 |
| $m$ | the convention: the value of $0^0$, which is 0 or 1 | Section 1.4 |
| $w(x)$ | the weight of a digit: $x^x$ for $x \geq 1$, and $w(0) = m$ | Section 1.1 |
| $F(n)$, $F(s)$ | the Munchausen sum: the sum of the weights of the digits | Section 1.1 |
| $F_0$, $F_1$ | the sum under $0^0 = 0$ and under $0^0 = 1$ | Section 1.4 |
| $(s_{k-1}, ..., s_0)$ | a digit string, most significant digit first; position 0 is the last digit | Section 1.1 |
| $k$ | the number of digits of a string or a solution | |
| $J$ | $b - k$, how far the length of a solution is below the base | Section 3.9 |
| $c_x$ | the number of copies of the digit $x$; the vector $c$ is the digit multiset | Section 1.1 |
| $z(n)$ | the number of zero digits of $n$ | Section 1.4 |
| $N_m(b)$ | the number of nontrivial solutions in base $b$ under convention $m$ | Section 3.1 |
| $N_{m,k}(b)$ | the number of those with $k$ digits | |
| $X_k(b)$ | the largest digit $x$ with $x^x < b^k$ | Section 1.2 |
| $P_s(r)$ | the value polynomial $\sum_i s_i r^i$ of a digit string | Section 1.3 |
| $Q_s(r)$ | the Munchausen polynomial $P_s(r) - F(s)$; $s$ is a solution in base $b$ exactly when $Q_s(b) = 0$ and $b$ exceeds every digit | Section 1.3 |
| $g(c)$ | $w(c) - c$ | Section 2.4 |
| $\sigma(c)$ | the digit sum of a multiset or string $c$; the library writes $S(c)$ for it in chapter 20 and for the weight sum $F(c)$ in chapter 5 | Section 5.5.4 |
| $N_s(q)$ | the number of residues $r$ modulo $q$ with $P_s(r) \equiv F(s)$ | Section 5.5.4 |
| $\gamma$ | how concentrated a multiset of digits is modulo $q$: the largest $\lvert 1/k \sum e(hx_i/q)\rvert$ over $h \not\equiv 0$ | Section 5.5.4 |
| $T(q)$ | the set of residues of $x^x - x$ modulo $q$ | Section 5.3 |
| $h(\alpha)$ | the absolute logarithmic height of an algebraic number | Section E.2 |
| $\eta$ | the fundamental unit $> 1$ of a real quadratic field | Section 5.7 |
| $\rho_s(p)$ | the number of roots of $Q_s$ modulo $p$ | Section 5.5 |
| $D$ | an offset: near-solutions satisfy $n - F(n) = D$ | Section 3.5 |
| $\lambda_m(b)$ | the mean number of near-solutions per offset, an estimate of the expected $N_m(b)$ | Section 3.5 |
| $s_j$ | the limit share of the length $b - j$ in the basic model | Section 3.2.1 |
| $M_b$ | the length matrix: entry $(d, d')$ is the chance that $F$ of a uniform $d$-digit string has $d'$ digits | Section 3.2.2 |
| $C$ | the first-order constant of the trace, $\operatorname{tr} M_b - E_1 = 1 + C/b + o(1/b)$, $C = 1.104636$ | Section 3.2.2 |
| $\tau_t(b)$ | the part of the refined count that comes from the character $t$ modulo $b - 1$ | Section 3.3.1 |
| $d_{\mathrm{TV}}$ | total variation distance between two laws | Section A.1 |
| $e(t)$ | $e^{2\pi i t}$, an additive character | Section 3.7.2 |
| $\rho(b)$ | the largest normalized exponential sum of $x^x - x$ modulo $b - 1$; Assumption R holds when it is below $1 - 3 \ln b / b$ | Section 3.7.2 |
| $\rho^*(b)$ | the largest share of one residue class modulo $b - 1$ among the strings of the refined coupled model | Section 3.7.3 |
| $\tau(n)$ | the number of divisors of $n$ | Section D.8 |

| symbol | meaning | see |
|---|---|---|
| $L(b)$ | the number of digits of $(b-1)!$, for factorions | Section 7.6 |
| $k(B)$ | the free residue dimension of a base $B$ in $\mathbb{F}_q[T]$; equal to $\omega(B-1)$ in most cases | Section 7.8 |
| $\omega(B-1)$ | the number of distinct monic irreducible factors of $B-1$ | Section 7.8 |
| ln, log | natural logarithm; $\log_2$ and $\log_b$ carry their base | |
| $a \mid b$ | $a$ divides $b$ | |
| $a \equiv b \pmod q$ | $q$ divides $a-b$ | |
| $\|y\|$ | the distance from the real number $y$ to the nearest integer | Section 5.6 |

## O.2 Glossary

**Solution**. A Munchausen number with at least two digits, so not 0 or 1. Short for "nontrivial solution". (Section 1.1)

**Shared solution**. A solution without the digit 0. It is a solution under both conventions. (Proposition 1.8)

**Twins**. Under $0^0 = 0$, a solution that ends in 0 and the number one larger, which is also a solution. (Proposition 2.1)

**Balance**. The amount $xb^i - w(x)$ that a digit $x$ at position $i$ adds to $n - F(n)$. A number is a solution when its balances add up to 0. (Section 1.5)

**Self-paying digit**. A digit $a \geq 2$ at a position $k \geq 1$ with $ab^k = a^a$. It adds the same amount to $n$ and to $F(n)$. Only perfect-power bases have them. (Section 2.2)

**Core**. What remains of a solution under $0^0 = 0$ after removing its self-paying digits and a final 1. (Proposition 2.5)

**Swap**. Two digits that pay for each other: $a$ at position $k$ and $a^a$ at position 0, with $ab^k = (a^a)^{a^a}$. (Section 2.3)

**Self-paying block**. A string $t$ with $b^m P_t(b) = F(t)$ for some shift $m \geq 1$. A tail that pays for itself completes it to a solution. (Section 2.8)

**Binomial family**. The solutions of Theorem 2.21, whose leading digits are the binomial coefficients of $(cr+e)^{k-1}$. They give infinitely many solutions of every length under both conventions. (Section 2.6)

**Free slot**. A position of a search node that will hold a digit below the current value, still unknown. Nodes with one free slot are 43% of the exact tree in base 28, and the residue rule acts on them. (Section 4.1.6)

**Height**. A measure of how complicated an algebraic number is, roughly the number of digits needed to write it down. Lower bounds for linear forms in logarithms depend on it. (Section E.2)

**Digit-sum identity**. $n \equiv$ (digit sum of $n$) $\bmod (b-1)$, which for a solution becomes a condition on the digit multiset. (Proposition 1.9)

**Identity modulus**. The modulus of the digit-sum identity in a number system: $b-1$ for base $b$, $b+1$ for base $-b$, $N(\beta-1)$ for a complex base $\beta$, $p-q$ for the rational base $p/q$. (Section 7.1)

**Parity effect**. In odd bases under $0^0 = 0$ the digit-sum identity holds modulo 2 for free, which doubles the expected number of solutions. (Section 3.3)

**Basic model, refined model**. The probabilistic model that treats the digits of $F(c)$ as random, without and with the digit-sum identity. (Section 3.2)

**Near-solution, offset, defect spectrum**. A number with $n - F(n) = D$ for a small offset $D$. The counts over many offsets measure each base's own rate of coincidences. (Section 3.5)

**Forced top digits**. The leading digits of $F(c)$, which the counts of the largest digits fix. They are the main known correction to the model. (Section 3.5)

**Root heuristic**. The heuristic for short solutions: the chance that the real root of $Q_s$ is an integer, times local factors. (Section 5.5)

**Arrangement factor**. For a modulus $q$ prime to $b(b-1)$, $q$ times the share of the arrangements of a multiset whose value is congruent to its weight sum. It has mean 1 and tends to 1 with the length. (Section 5.5.4)

**Invariant ideal**. The ideal generated by the differences of the values of strings with the same digit multiset. It gives the identity modulus of a number system. (Section 7.1.1)

**Local obstruction**. A modulus $q$ for which $Q_s$ has no root modulo $q$. Then $s$ is a solution in no base. (Proposition 5.9)

**Covering**. A finite set of moduli that excludes every value of one varying digit of a string family. (Section 5.3)

**Square class**. The case where the dominant term of the equation of a short string is a perfect power, so that the equation is close to a difference of powers. (Section 5.6)

**Pure and mixed solutions**. For $b(b+x) = x^n + 1$ with odd $x$: whether the ideal $B$ with $(b+\varepsilon) = B^n$ takes all its primes from one of the two conjugate ideals above $x$ or mixes them. (Section 6.4)

**Aurifeuillian factors**. The two algebraic factors $L < M$ into which $x^x + 1$ or $x^x - 1$ splits for squarefree odd $x$ of the right residue modulo 4. (Section 6.2)

**Prefix-pruned tree**. The exact search: it fixes the digit counts from the largest digit down and prunes with the leading digits that the interval of possible sums shares. (Section 4.1)

**Inexact search**. A search that skips nodes the model calls unlikely, with a measured miss rate. (Section 4.3)

**Tally language**. A set of words over a one-letter alphabet, such as the bases written in unary that have a solution. (Section 4.7)

**Stochastic matrix**. A square matrix with nonnegative entries whose rows sum to 1, the transition matrix of a random process on finitely many states. Its trace is 1 plus the eigenvalues other than 1. (Section A.5)

**Fisher's method**. A way to combine independent P values: $-2\sum \ln p_i$ has the chi-square law with $2k$ degrees of freedom when all null hypotheses hold. (Section A.9)

**Named in advance**. A test whose hypothesis was written down before its data were seen. Its P value needs no correction for selection. (Section 3.6.3)

**Factorion**. A number equal to the sum of the factorials of its digits. (Section 7.6)

**Inert prime**. A prime that stays prime in a quadratic field $\mathbb{Q}\left(\sqrt{D}\right)$; for odd primes not dividing the discriminant, one modulo which the discriminant is not a square. (Section D.4)

**Brocard's problem**. The question whether $n! + 1 = m^2$ has solutions other than $n = 4, 5, 7$. The leading case of 3-digit factorions is a problem of the same kind. (Section 7.7)

**Narcissistic number**. A $k$-digit number equal to the sum of the $k$-th powers of its digits. (Section 7.6)

**Frobenius block**. In $\mathbb{F}_q[T]$, the $q$ digits that one base-$B$ digit becomes under $n \mapsto n^q$. (Section 7.8)

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